# ASUS ZS630KL
# RF Hardware Trouble Shooting Guide

# ZS630KL Supported Bands (TW)

**2G: GSM** GSM850/EGSM/DCS/PCS

**3G: WCDMA** B1/B2/B3/B4/B5(B6/B19)/B8

**4G: LTE-FDD** B1/B2/B3/B4/B5/B7/B8/B18/B19/
B26/B28A/B28B/B12/B17(NA)

**LTE-TDD** B38/B39/B41(partial)

**4*4MIMO** B3/B7

**LAA** B46

# ZS630KL Supported Bands (WW)

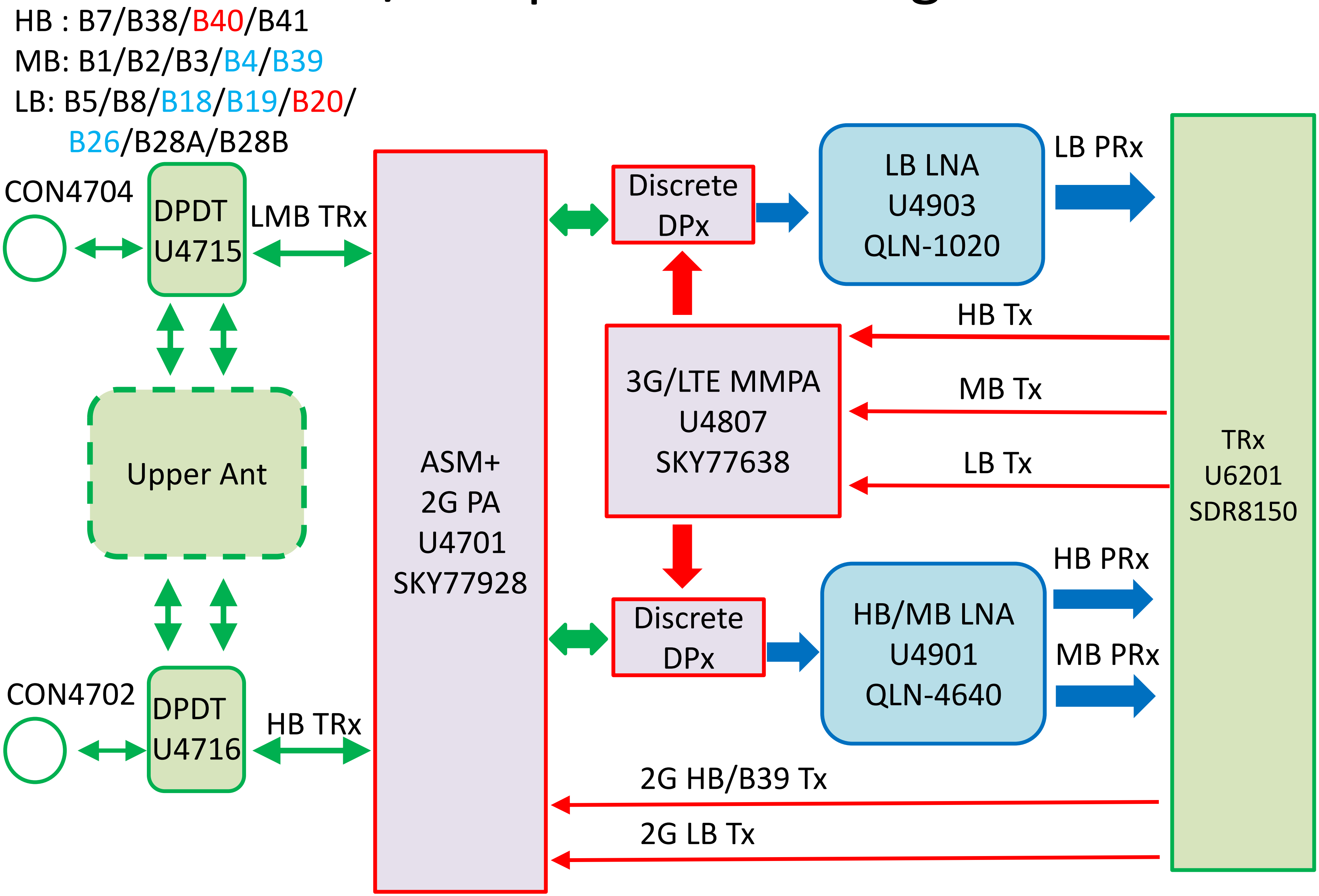
**2G:** **GSM** GSM850/EGSM/DCS/PCS

**3G:** **WCDMA** B1/B2/B5/B8

**4G:** **LTE-FDD** B1/B2/B3/B5/B7/B8/B20/B28A/B28B

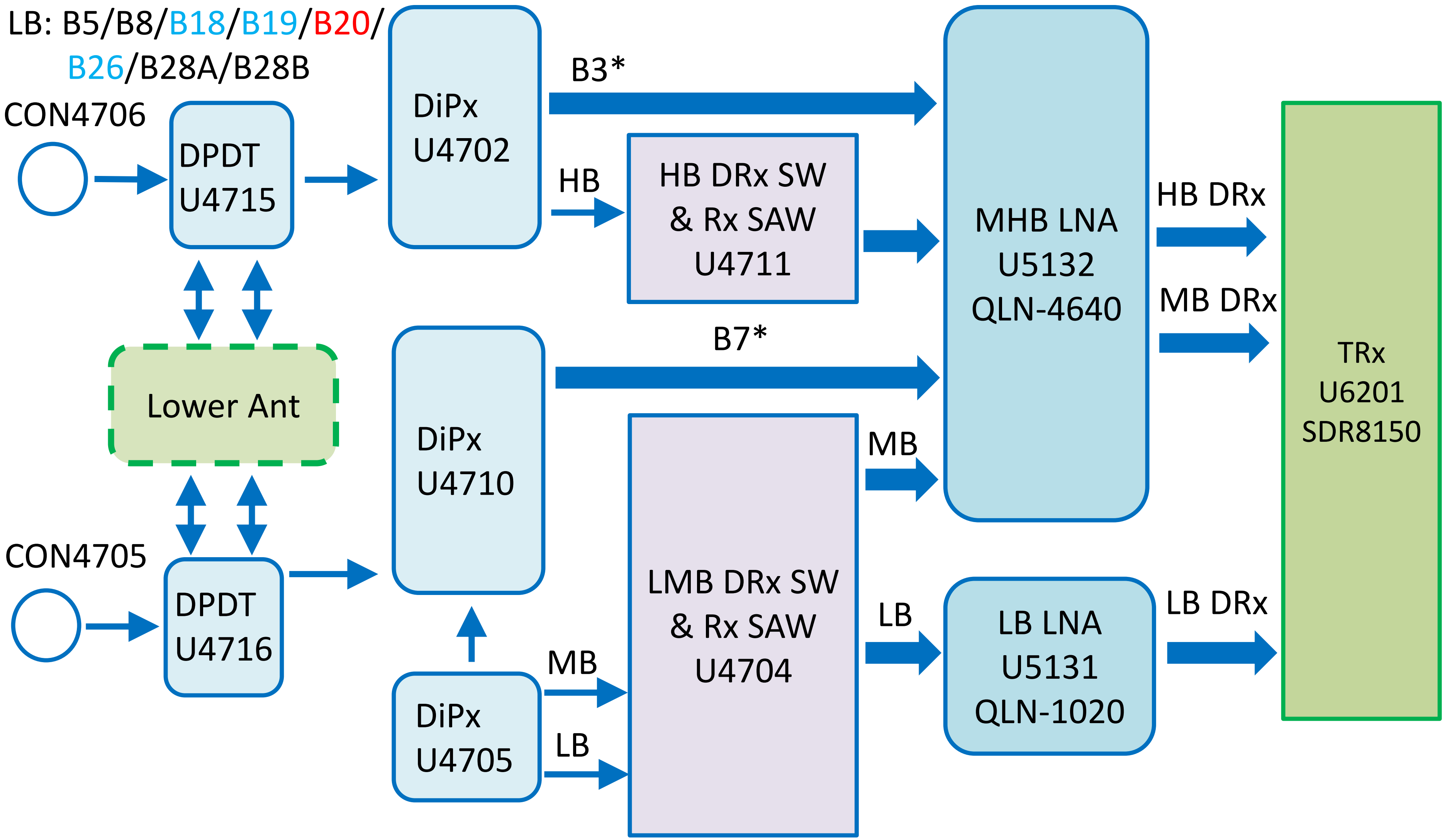
**LTE-TDD** B38/B40/B41(full)

# Tx/PRx path block diagram



# DRx path block diagram

HB : B7/B38/B40/B41
MB: B1/B2/B3/B4/B39
LB: B5/B8/B18/B19/B20/
B26/B28A/B28B



QFDT

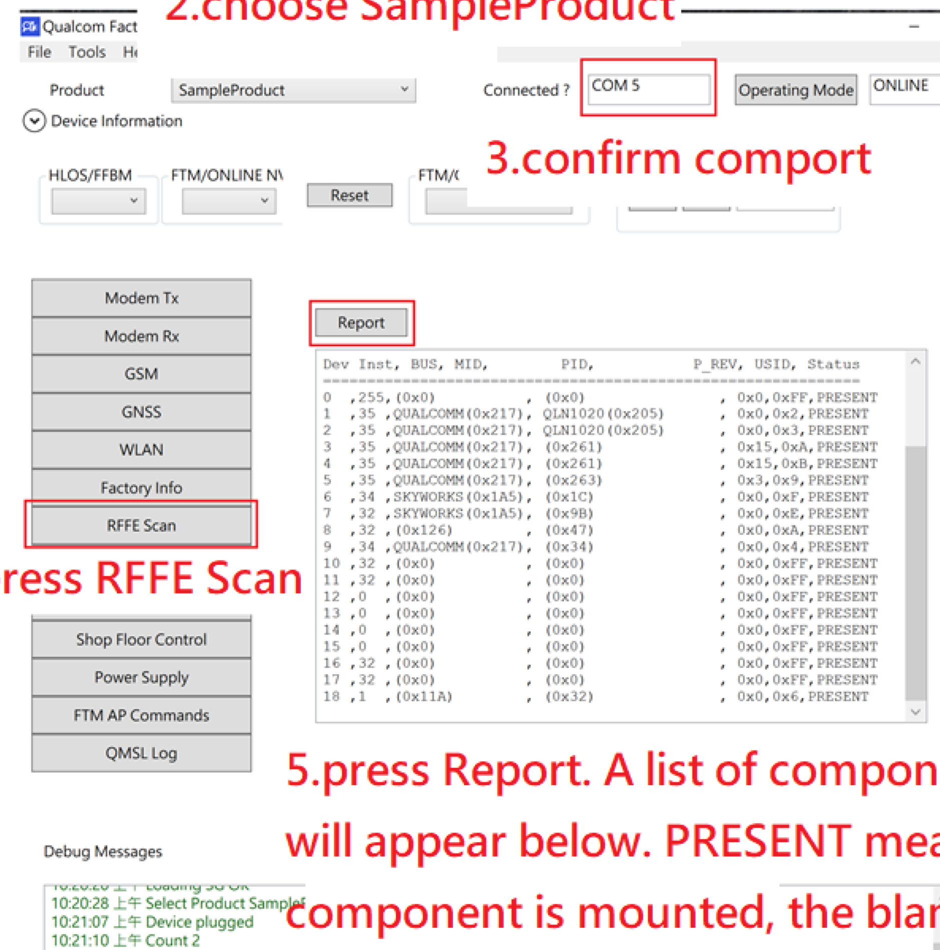
1. FIND QFDT

# SN/BT Error Code : 3921

2.choose SampleProduct

3.confirm comport

4.press RFFE Scan




5.press Report. A list of component will appear below. PRESENT means component is mounted, the blank is not. Right is check form.

| DEV | Component | Position |
|---|---|---|
| 0 | SDR8150 | U6201 |
| 1 | QLN1020(LB PRx) | U4903 |
| 2 | QLN1020(LB DRx) | U5131 |
| 3 | QLN4640(MHB PRx) | U4901 |
| 4 | QLN4640(MHB DRx) | U5132 |
| 5 | QLN4650(LAA) | U6132 |
| 6 | SKY77638 | U4807 |
| 7 | SKY77928 | U4701 |
| 8 | RFASWR681CTF03 | U4704 |
| 9 | QET4101 | U4603 |

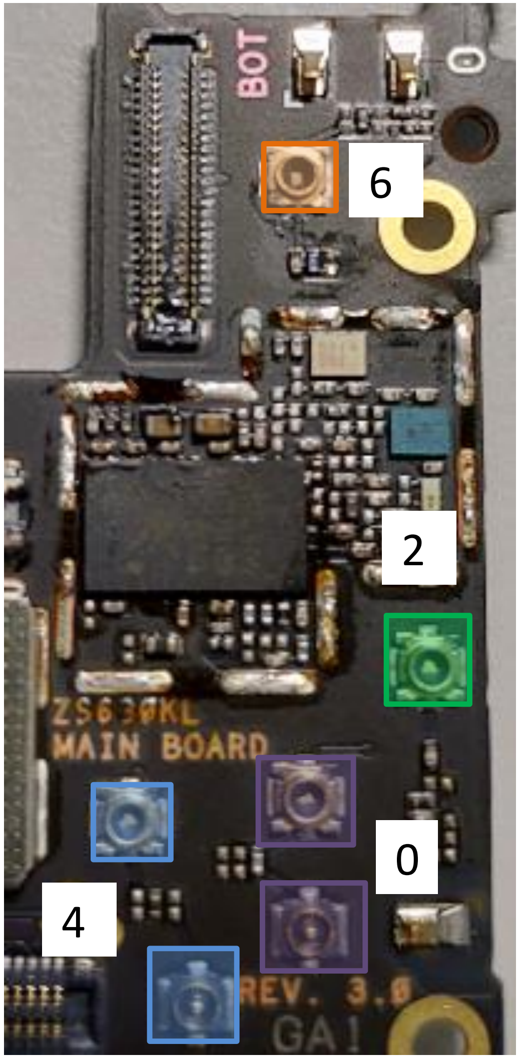
# Cal log

| | | |
|---|---|---|
| LTE_B39_IDC_RxRSB | PASS | 0.73 |
| LTE_B41_IDC_RxRSB | PASS | 1.15 |
| LTE_B1_Connected_Primary | PASS | 0.68 |
| LTE_B2_Connected_Primary | PASS | 0.42 |
| LTE_B3_Connected_Primary | PASS | 0.96 |
| LTE_B4_Connected_Primary | PASS | 0.43 |
| LTE_B5_Connected_Primary | PASS | 0.30 |
| LTE_B7_Connected_Primary | PASS | 0.67 |
| LTE_B8_Connected_Primary | PASS | 0.29 |
| LTE_B18_Connected_Primary | PASS | 0.29 |
| LTE_B19_Connected_Primary | PASS | 0.29 |
| LTE_B26_Connected_Primary | PASS | 0.29 |
| LTE_B28A_Connected_Primary | PASS | 0.29 |
| LTE_B28B_Connected_Primary | PASS | 0.29 |
| LTE_B38_Connected_Primary | PASS | 0.42 |
| LTE_B39_Connected_Primary | PASS | 0.42 |
| LTE_B41_Connected_Primary | PASS | 0.52 |
| LTE_B46_Connected_Primary | PASS | 0.62 |
| Lock_Resource | PASS | 0.00 |
| LTE_B1_V2 | PASS | 0.26 |
| LTE_B2_V2 | PASS | 0.26 |
| LTE_B3_V2 | PASS | 0.88 |
| LTE_B4_V2 | PASS | 0.50 |
| LTE_B5_V2 | PASS | 0.29 |
| LTE_B7_V2 | PASS | 0.51 |
| LTE_B8_V2 | PASS | 0.25 |
| LTE_B18_V2 | PASS | 0.27 |
| LTE_B19_V2 | PASS | 0.27 |
| LTE_B26_V2 | PASS | 0.27 |
| LTE_B28A_V2 | PASS | 0.27 |
| LTE_B28B_V2 | PASS | 0.28 |
| LTE_B38_V2 | PASS | 0.49 |
| LTE_B39_V2 | PASS | 0.50 |
| LTE_B41_V2 | PASS | 0.49 |
| LTE_B1_Connected_Diversity | PASS | 0.65 |
| LTE_B2_Connected_Diversity | PASS | 0.38 |
| LTE_B3_Connected_Diversity | PASS | 0.83 |
| LTE_B4_Connected_Diversity | PASS | 0.38 |
| LTE_B5_Connected_Diversity | PASS | 0.28 |
| LTE_B8_Connected_Diversity | PASS | 0.28 |
| LTE_B18_Connected_Diversity | PASS | 0.28 |

You can find cal log on your PC. According to error band, find the hardware path and repair it. Left is cal log sketch.

E0_2K2130702100119_20190216_170706.html

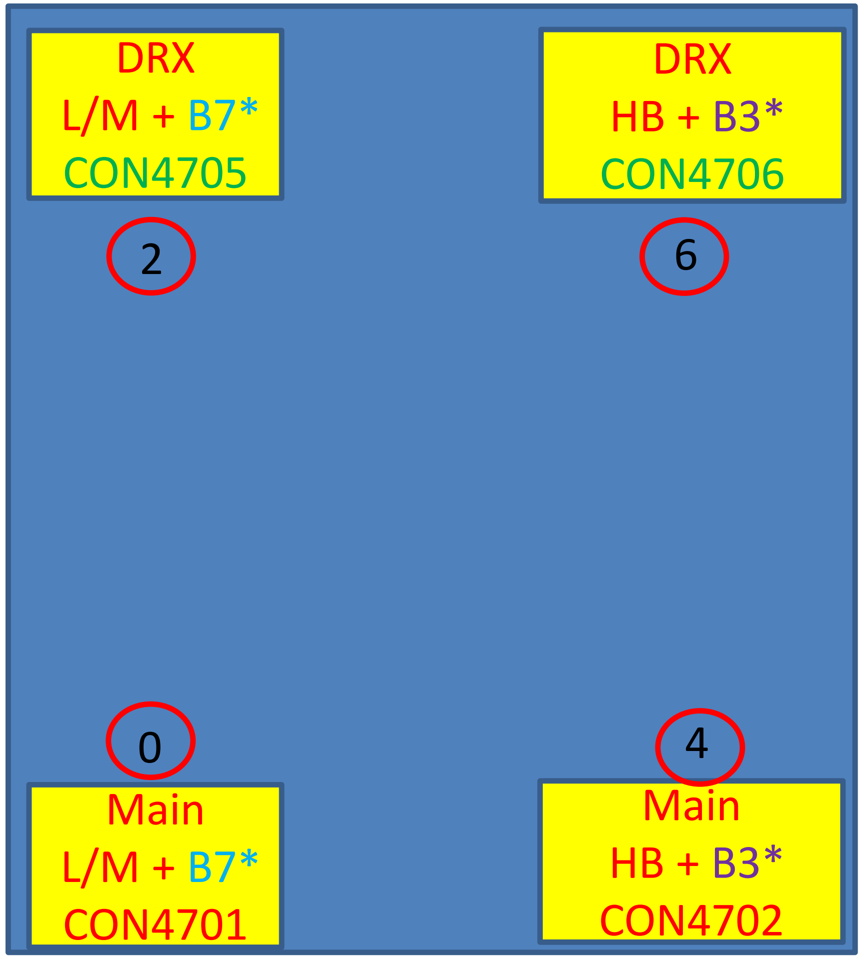
bottom




# Antenna Port

**PARAMETERS: FBRx Gain**

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path |
|---|---|---|---|---|---|
| LTE | 1 | 0 | 0 | 8 | 0 |



# TW B3 MIMO

## B3 TX

### B1/B3/B4 TriPx U4809 — P19

| Signal Path | Tx Device | Antenna Path | Status | Status Max | Time (s) |
|---|---|---|---|---|---|
| 114 | 14 | 0 | 0 | 0 | |

### B3/B7 MIMO Path — P20

| Signal Path | Tx Device | Antenna Path | Status | Status Max | Time (s) |
|---|---|---|---|---|---|
| 115 | 14 | 0 | 0 | 0 | |

## B3 PRx

### B1/B3/B4 TriPx U4809 — P38

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 103 | 10 | 0 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 106 | 2 | 0 | 0 | |

### B3/DCS (B3 DPX B7 MIMO Path) — P39

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 112 | 10 | 0 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 113 | 2 | 0 | 0 | |

### B3 MIMO (B7 DPX B3 MIMO Path) — P53

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 108 | 10 | 4 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 110 | 2 | 4 | 0 | |

## B3 DRx

### B1/B3/B4 Rx SAW U5114 — P67

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 104 | 11 | 2 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 107 | 3 | 2 | 0 | |

### B3 MIMO — P77

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 109 | 11 | 6 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 111 | 3 | 6 | 0 | |

# TW B7 MIMO

## B7 PRx

### B7 DPX B3 MIMO Path

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 35 | 2 | 4 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 37 | 10 | 4 | 4 | |

P49

### B3 DPX B7 MIMO Path

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 39 | 2 | 0 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 41 | 10 | 0 | 4 | |

P47

## B7 DRx

### B7

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 36 | 3 | 6 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 38 | 11 | 6 | 4 | |

P75

### B7 MIMO

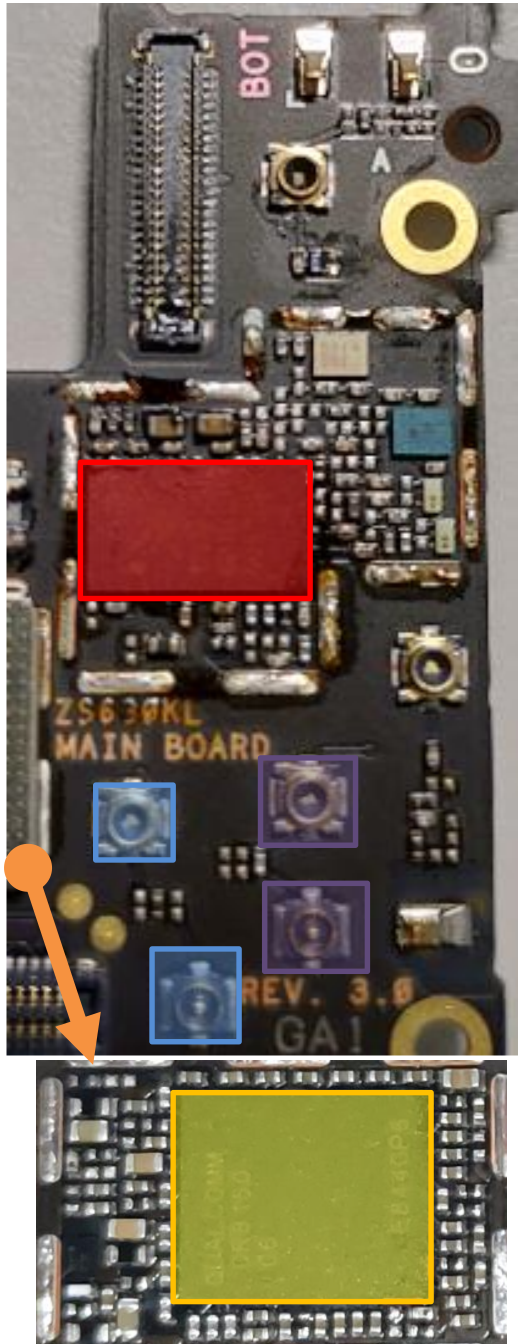
| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 40 | 3 | 2 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 42 | 11 | 2 | 4 | |

P69

# Key Components : Tx/PRx

bottom

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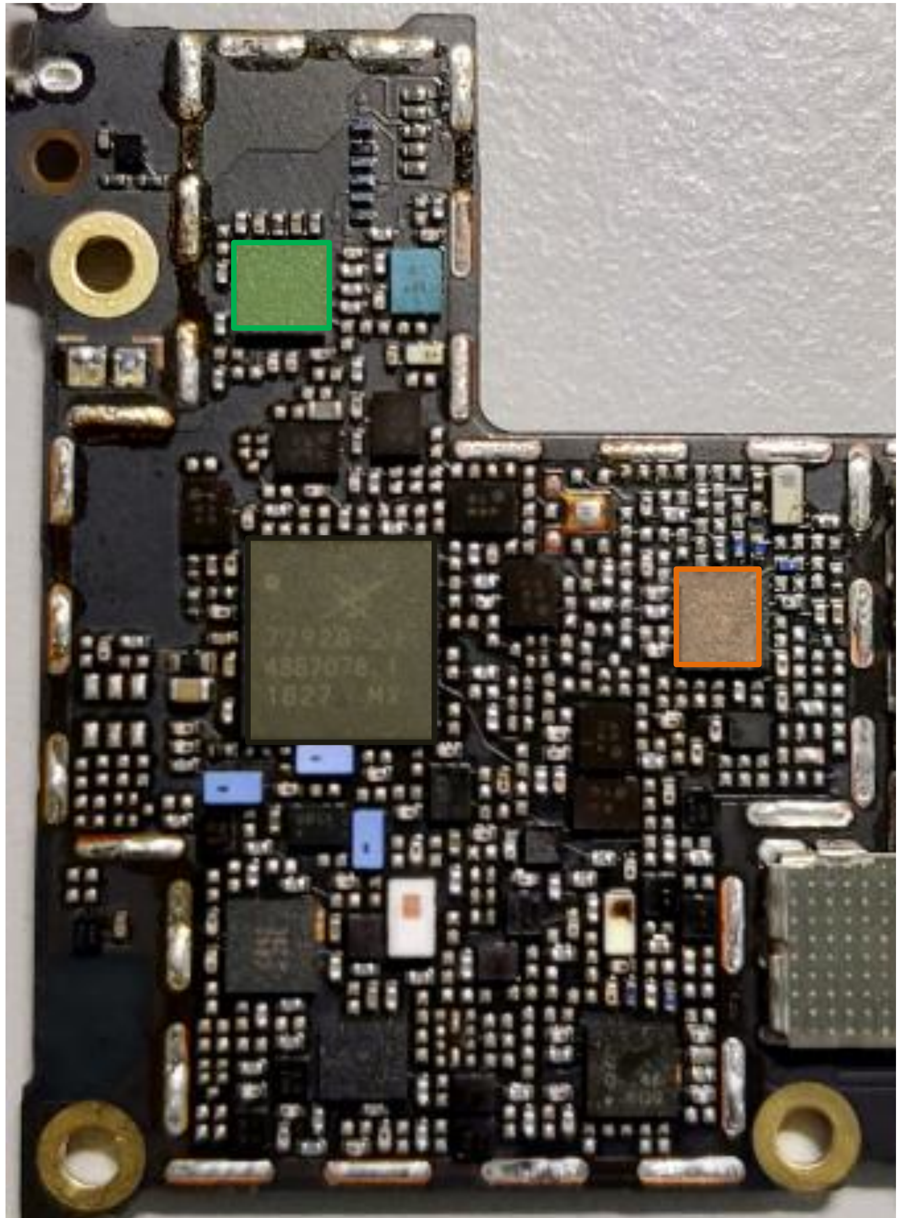


RF Connector HB
**CON4702 CON4703**

RF Connector LB/MB
**CON4701 CON4704**

3G/4G MMPA
**U4807**

ASM+2G PA
**U4701**

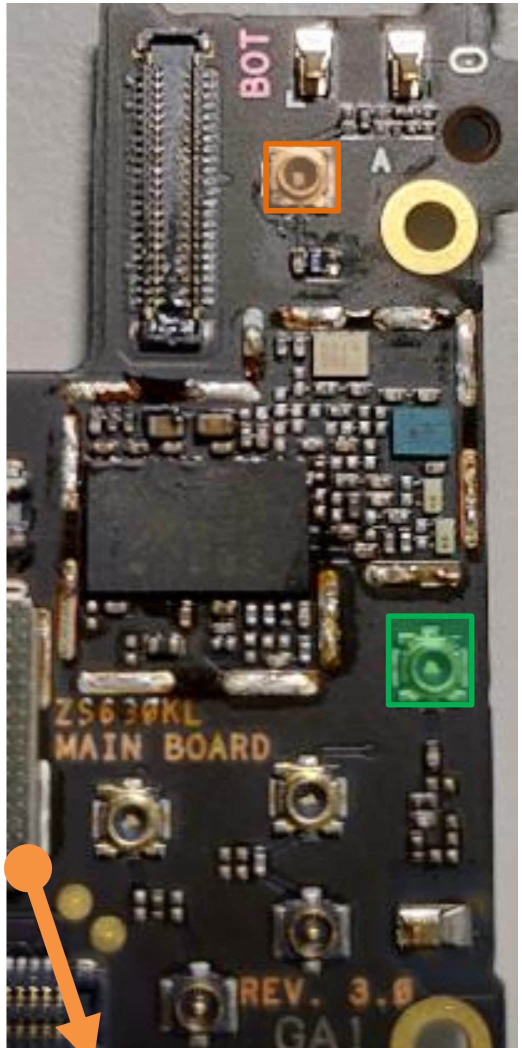
MB/HB QLN4640
**U4901**

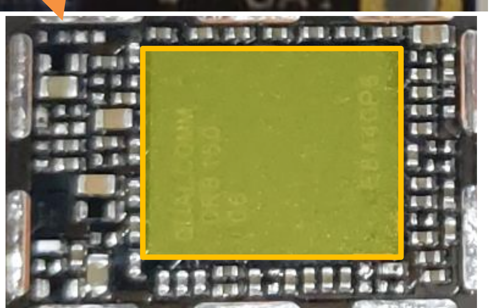
LB QLN1020
**U4903**

SDR8150
**U6201**

# Key Components : DRx

bottom



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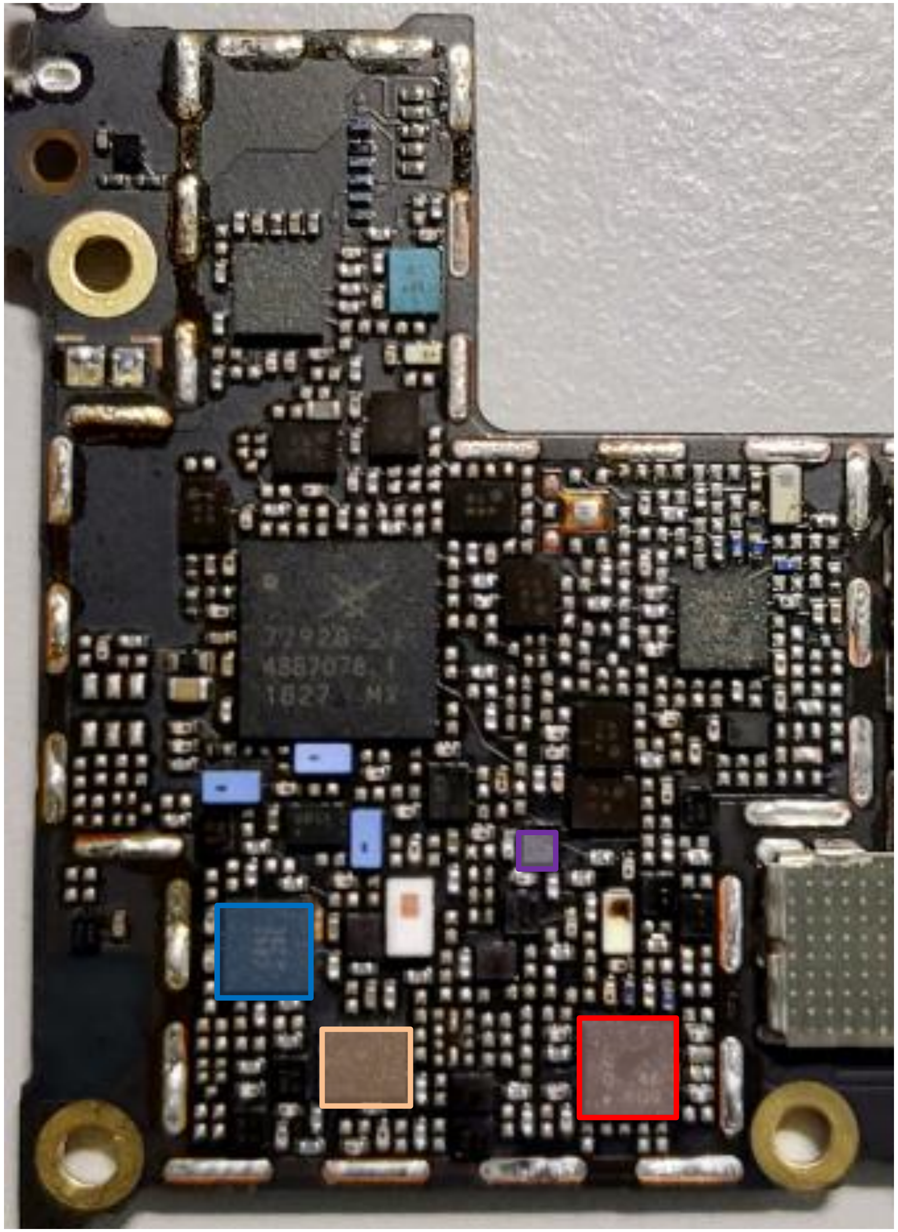


DRX MB/HB QLN4640
**U5132**

DRX LB QLN1020
**U5131**

Div. RF connector LB/MB
**CON4705**

Div. RF connector HB
**CON4706**

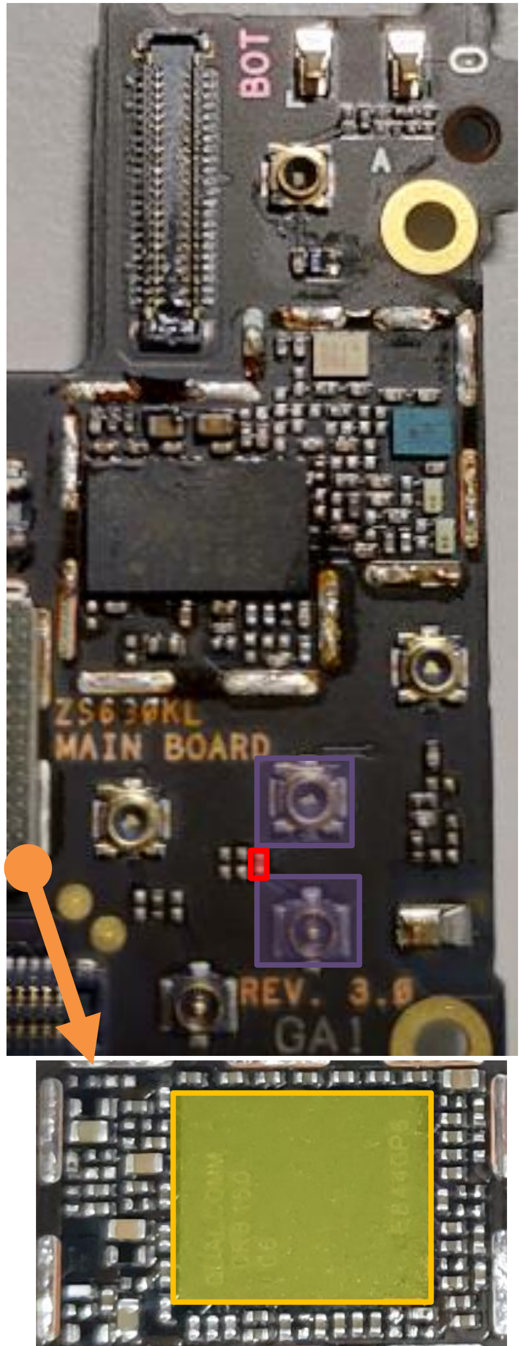
L/MB Div. RX SW
**U4704**

HB Div. RX SW
**U4711**

SDR8150
**U6201**

# TX Path : GSM 850/900

bottom

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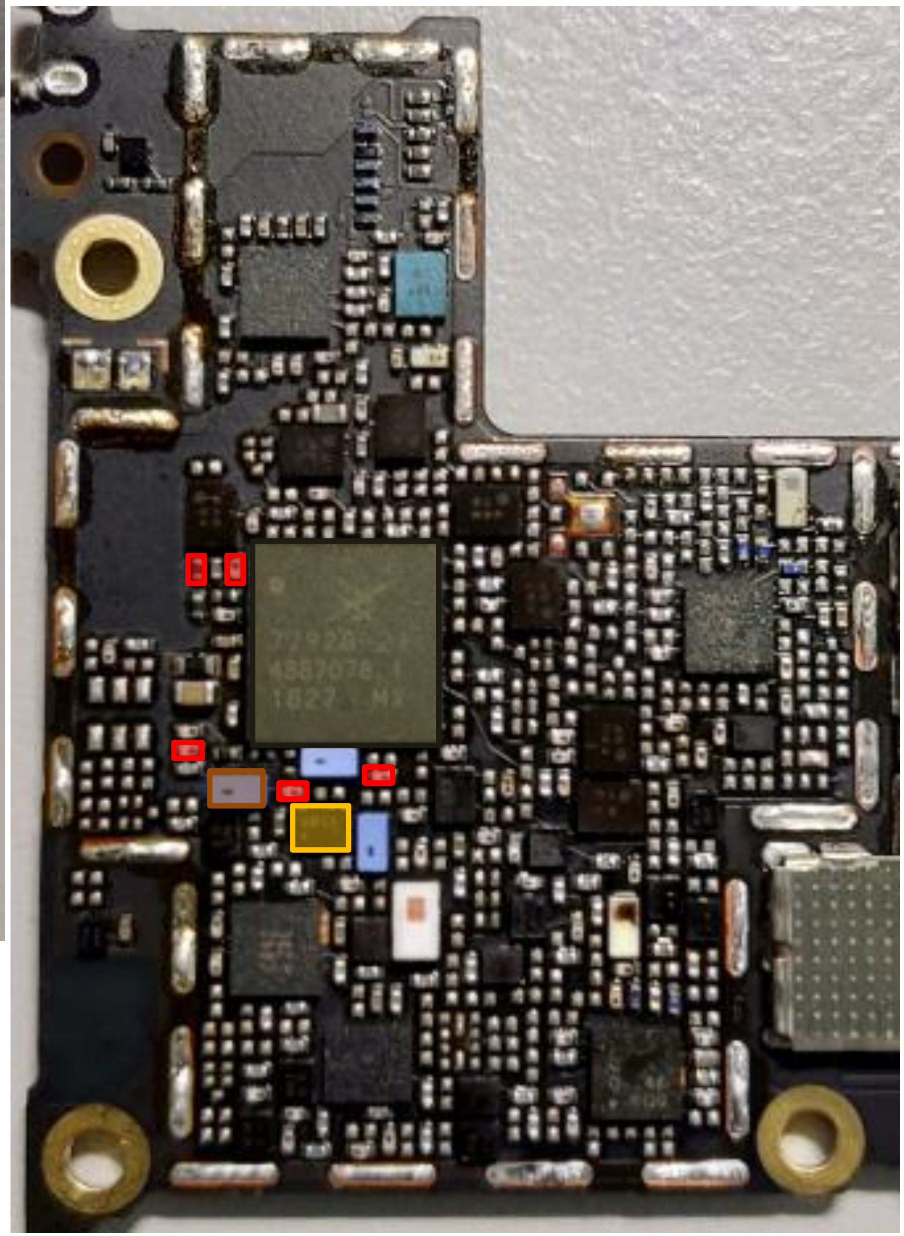


RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

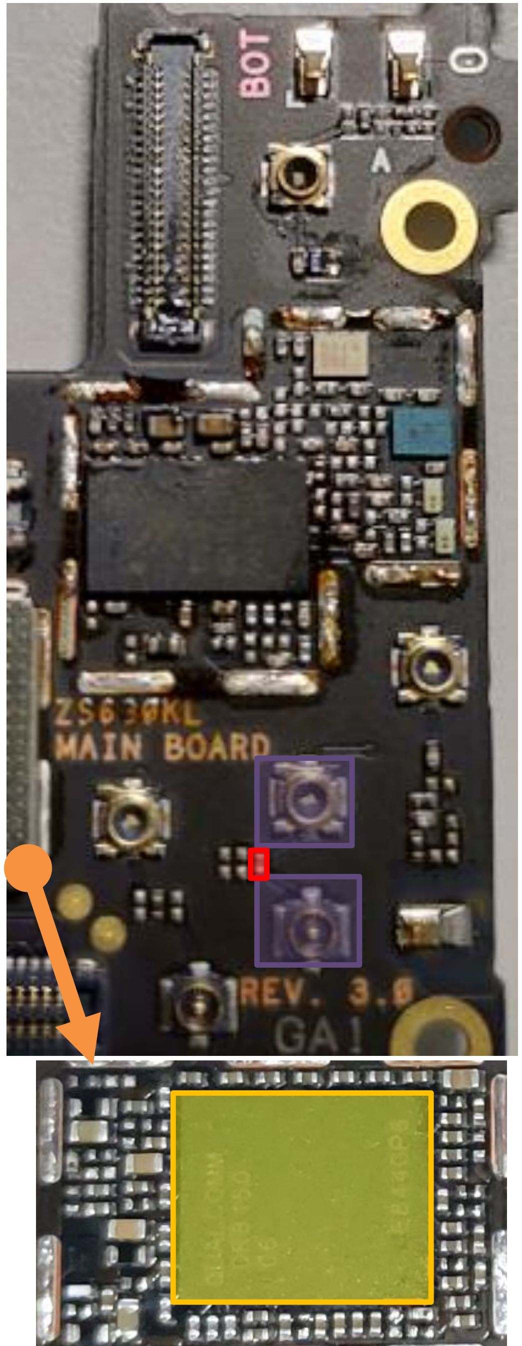
DPDT **U4715**

ASM+2G PA **U4701**

SDR8150 **U6201**

# TX Path : DCS/PCS/B39

bottom

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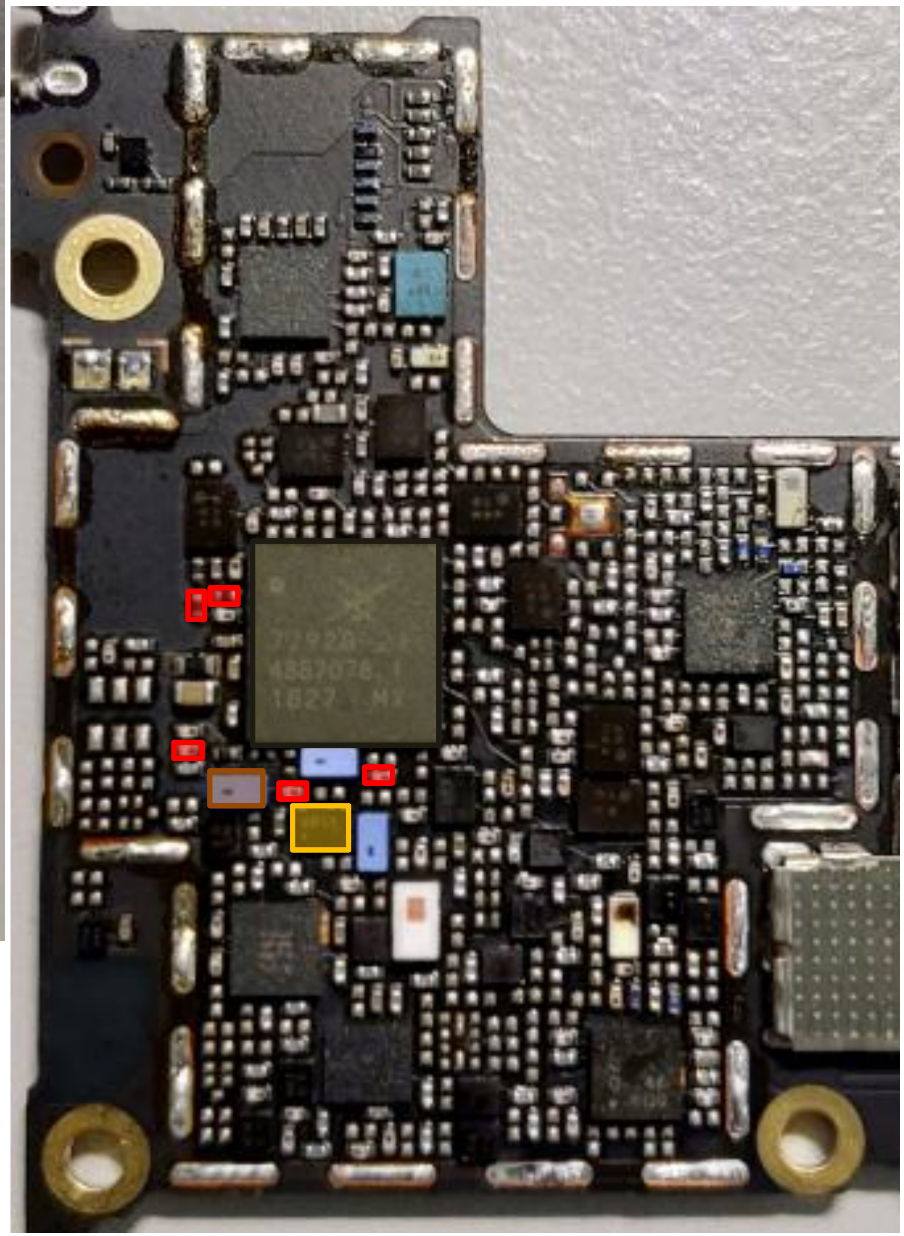


RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

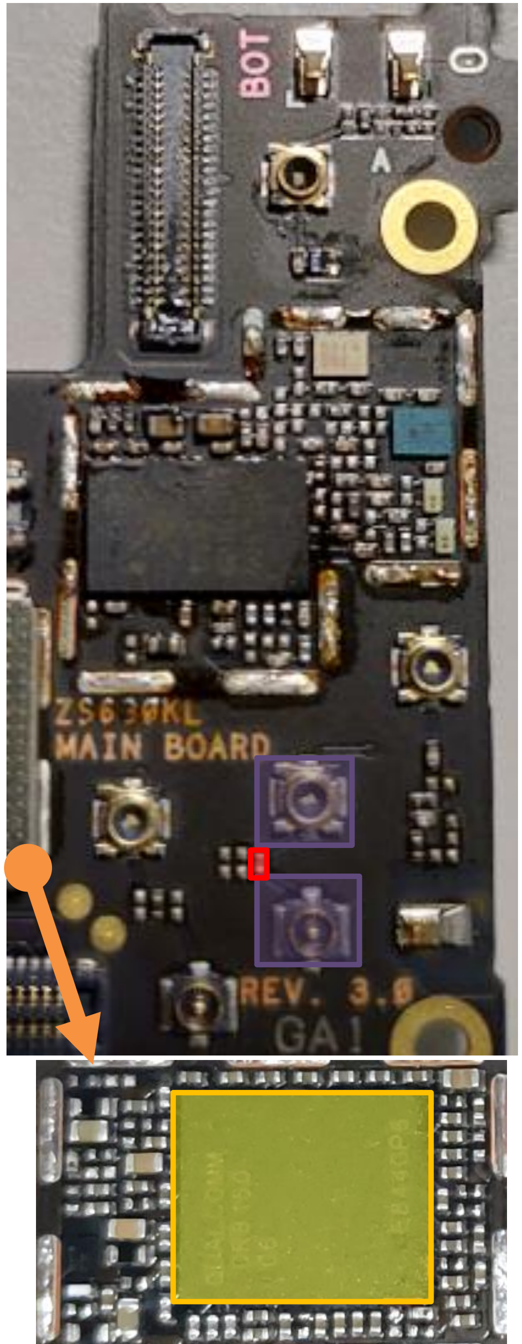
DPDT **U4715**

ASM+2G PA
**U4701**

SDR8150
**U6201**

# LMB Tx Common Path

bottom

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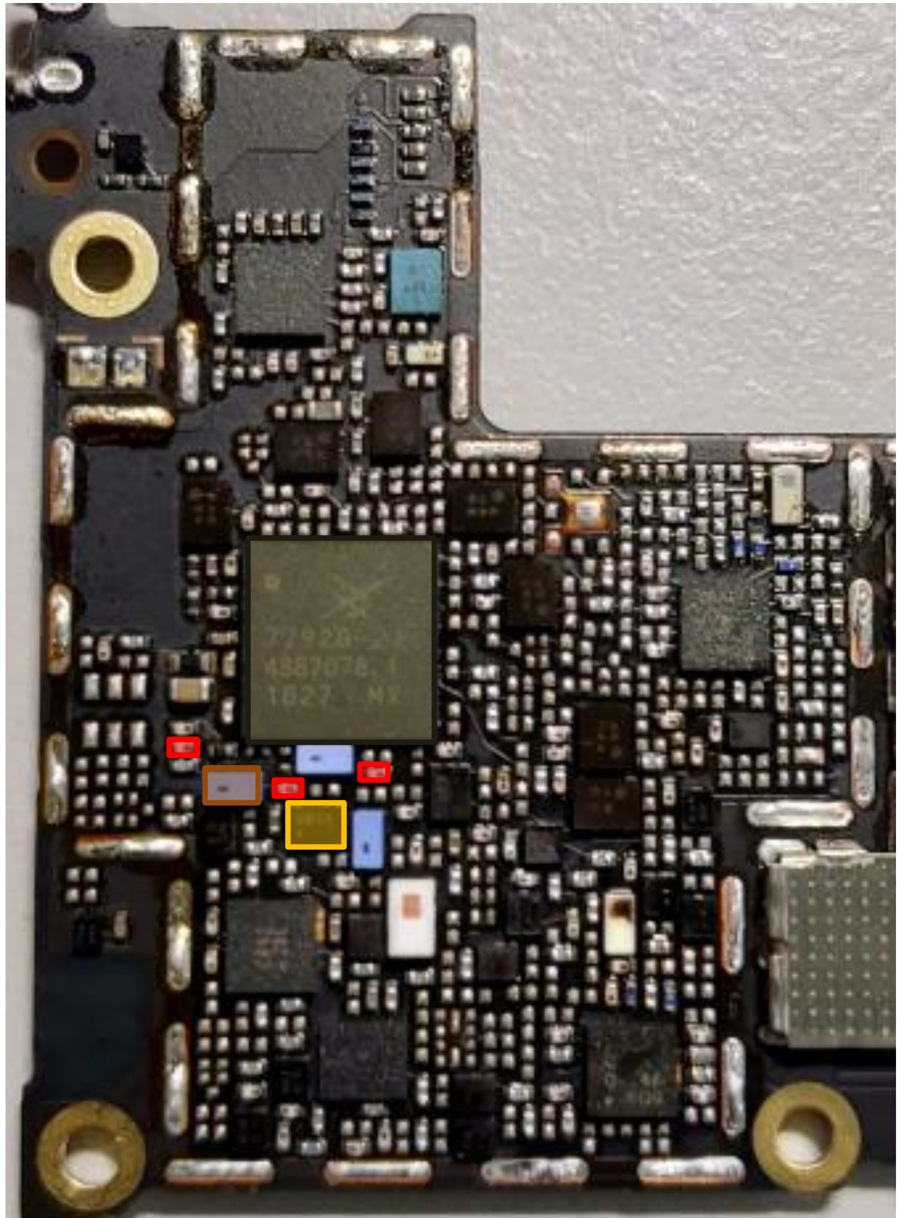


MB: B1/B2/B3/B4/B39

LB: B5/B8/B18/B19/B20/B26/B28A/B28B

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

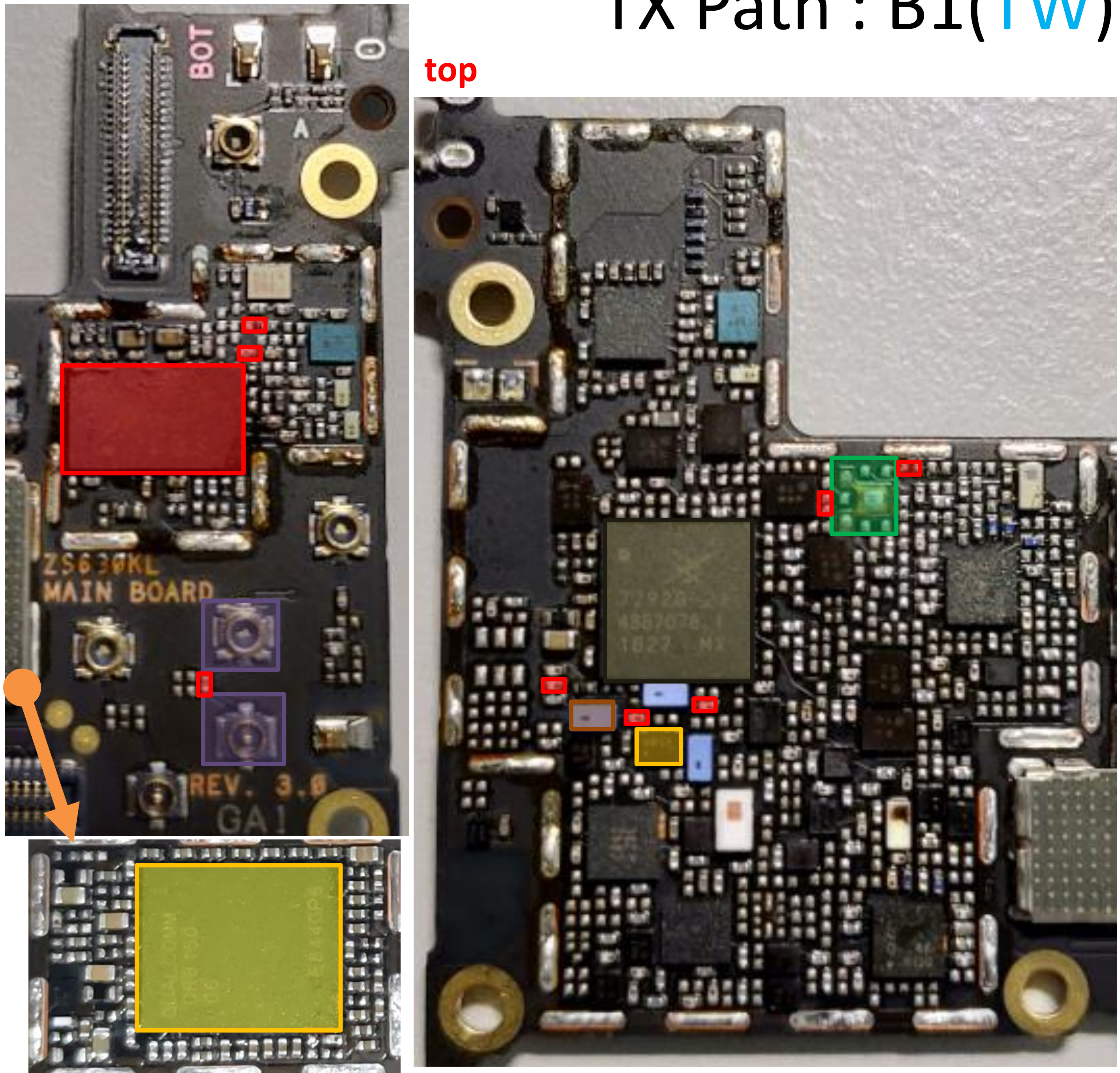
ASM+2G PA **U4701**

SDR8150 **U6201**

# TX Path : B1(TW)

bottom

top




RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

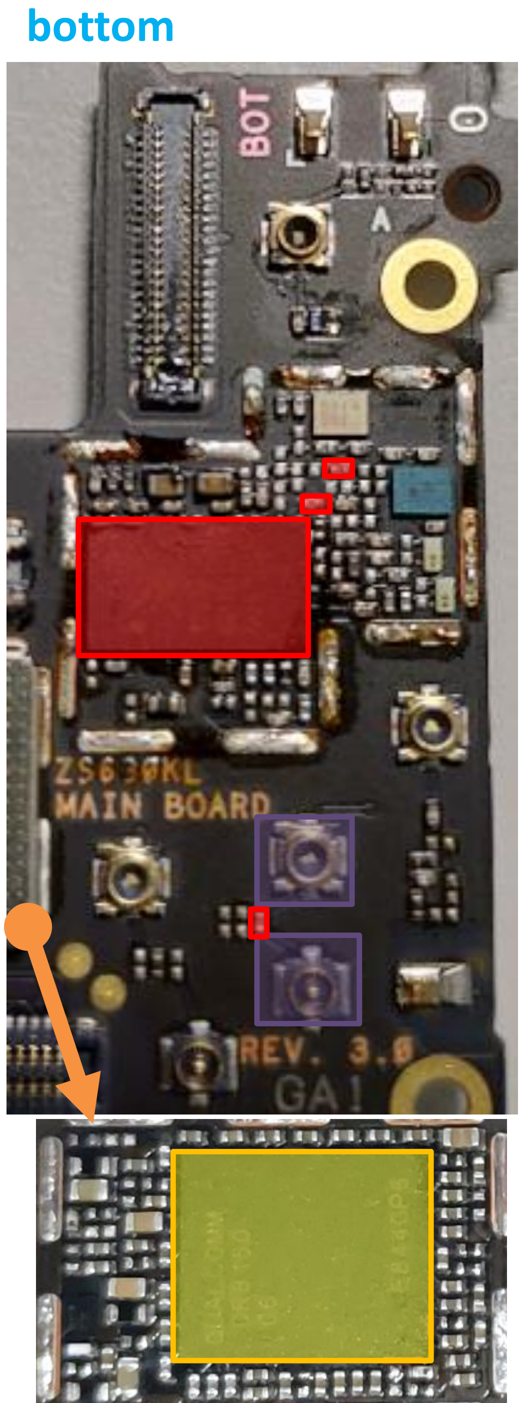
ASM+2G PA **U4701**

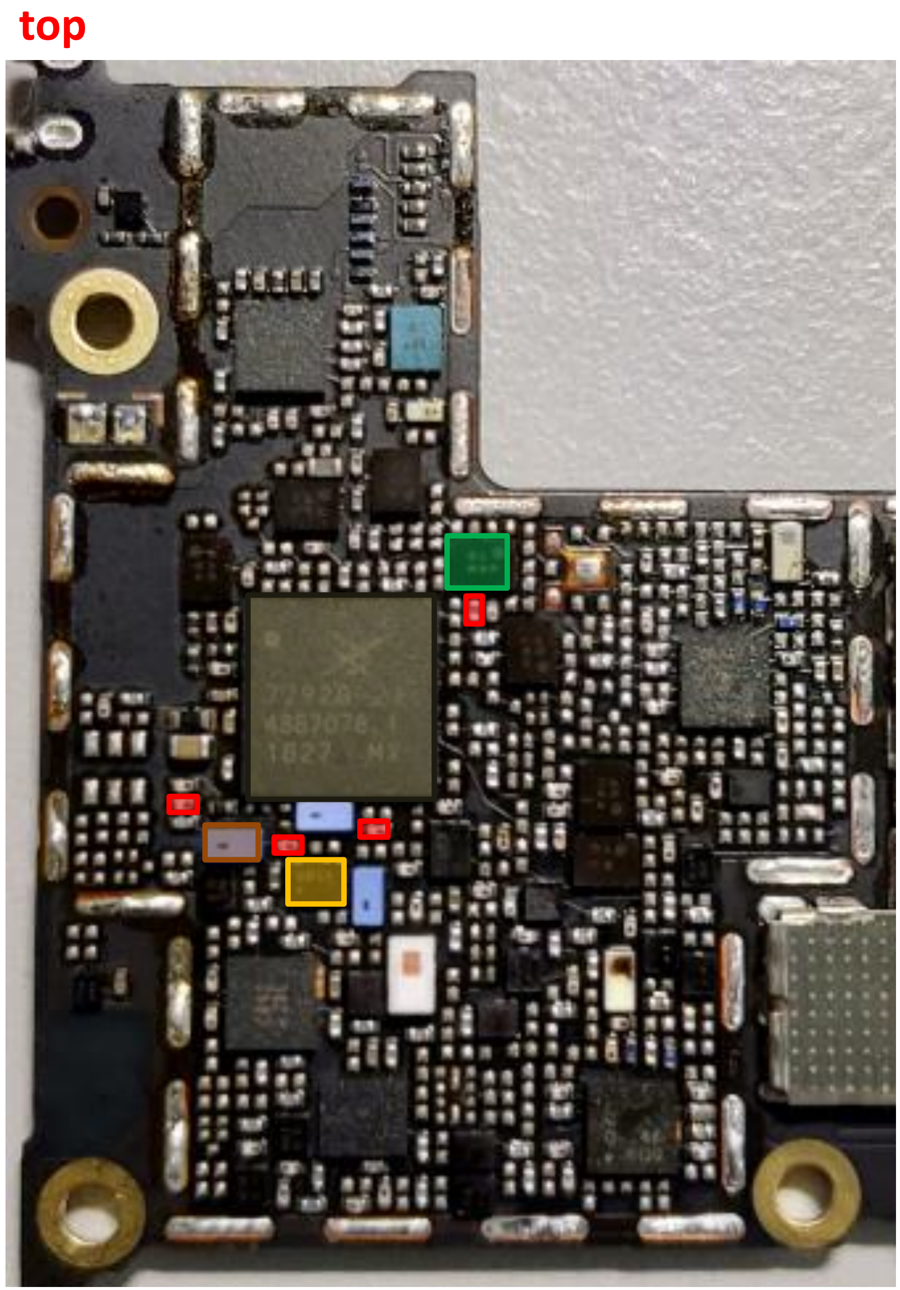
SDR8150 **U6201**

3G/4G MMPA **U4807**

TriPX **U4809**

# TX Path : B1(WW)




RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

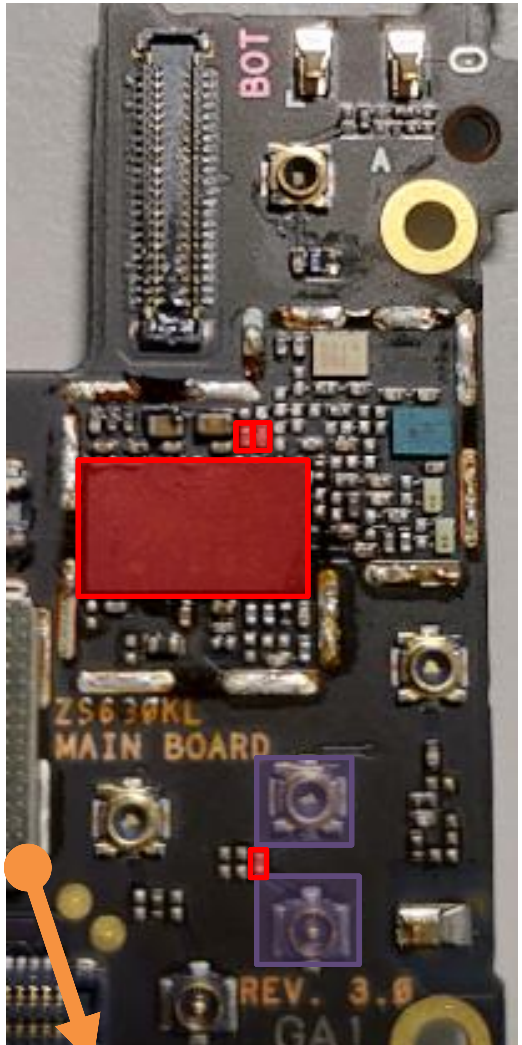
SDR8150 **U6201**

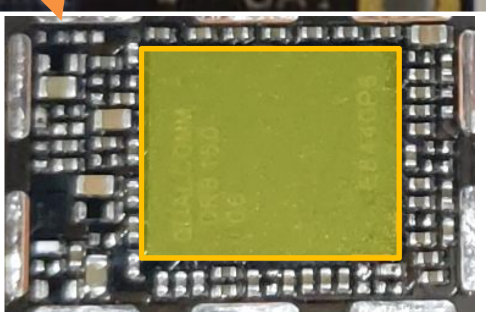
3G/4G MMPA **U4807**

DPX **U4820**

# TX Path : B2

bottom



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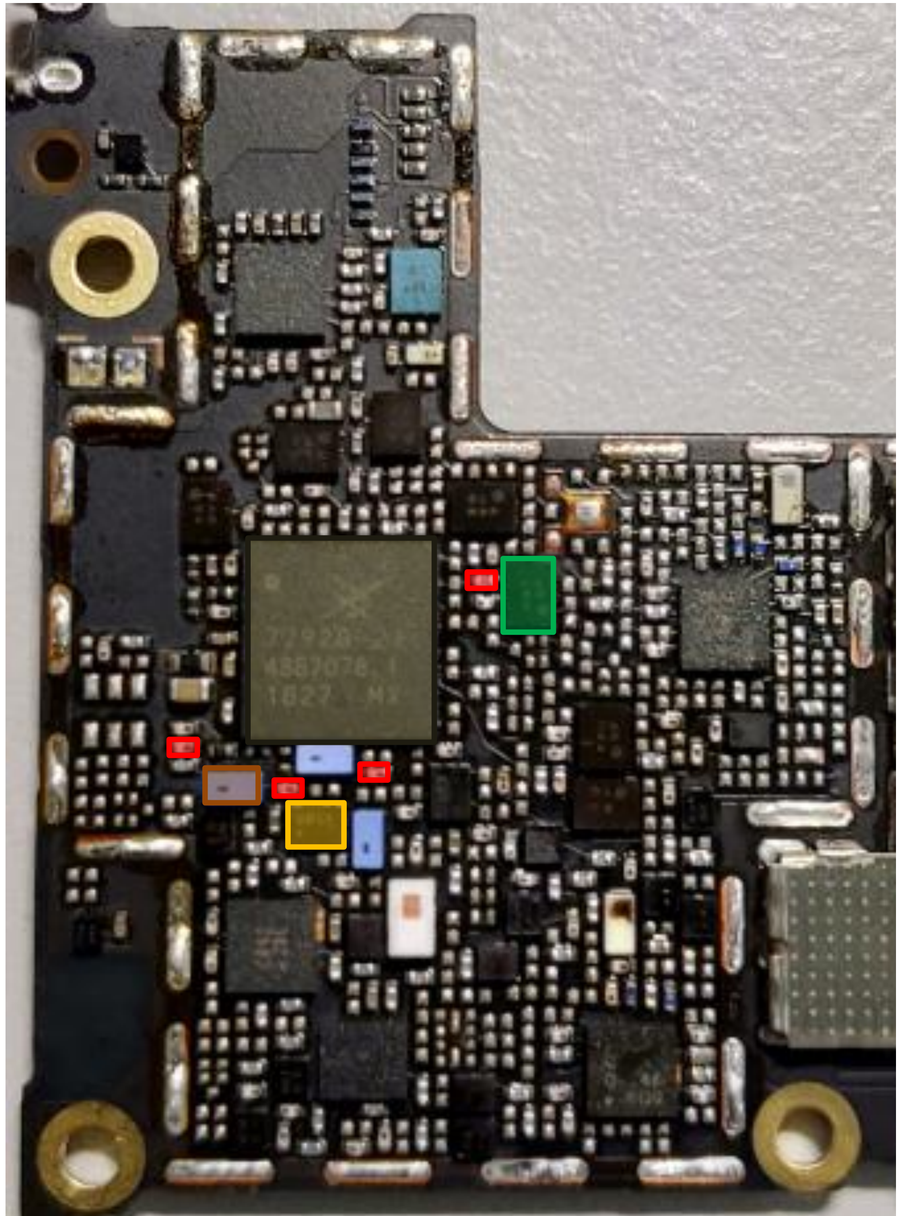


3G/4G MMPA
**U4807**

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

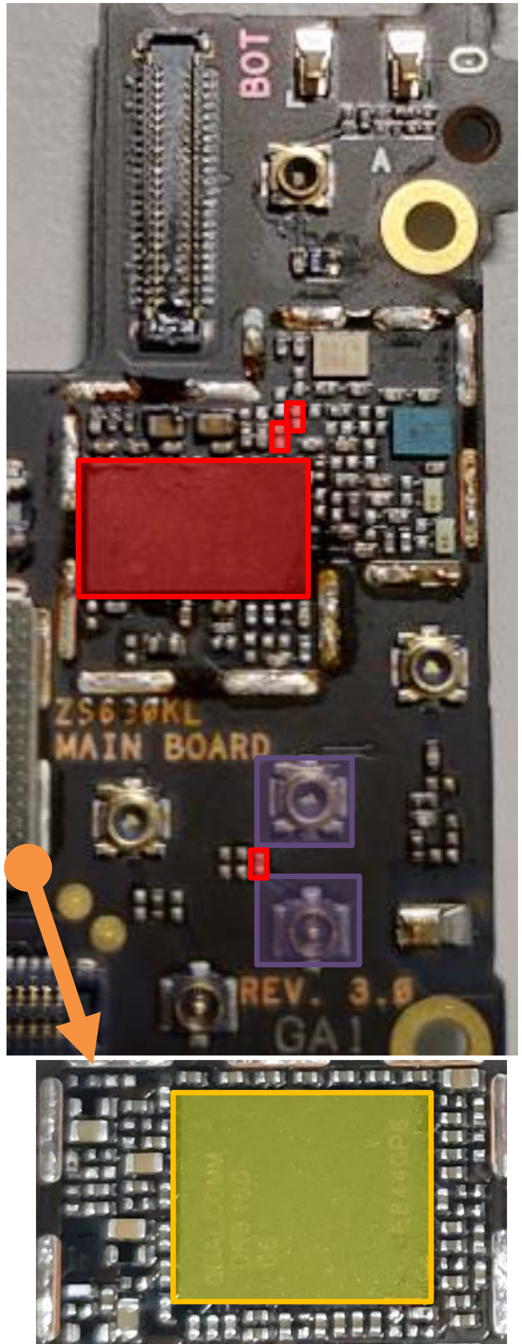
ASM+2G PA
**U4701**

SDR8150
**U6201**

DPX
**U4806**

# TX Path : B3/B4(TW)

bottom

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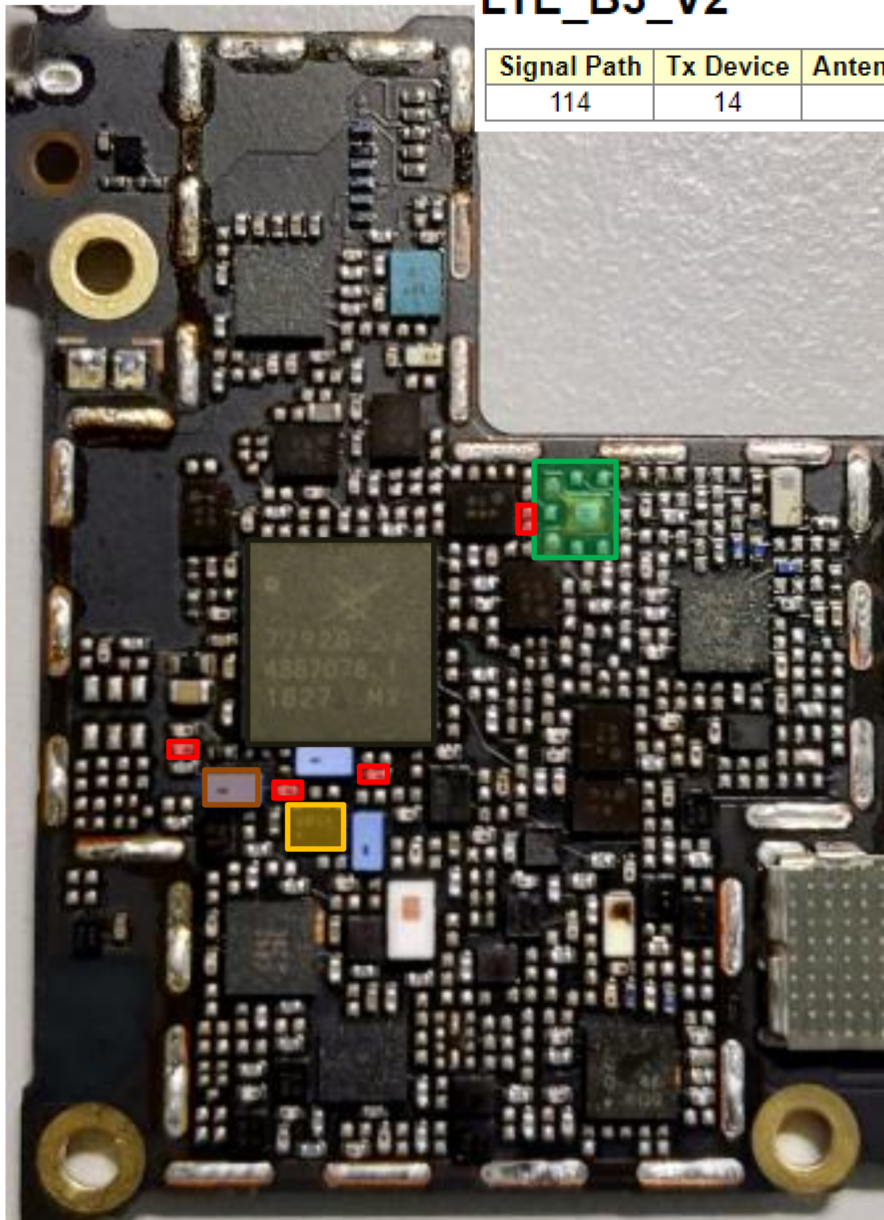


LTE_B3_V2

| Signal Path | Tx Device | Antenna Path | Status | Status Max | Time (s) |
|---|---|---|---|---|---|
| 114 | 14 | 0 | 0 | 0 | |

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

SDR8150 **U6201**

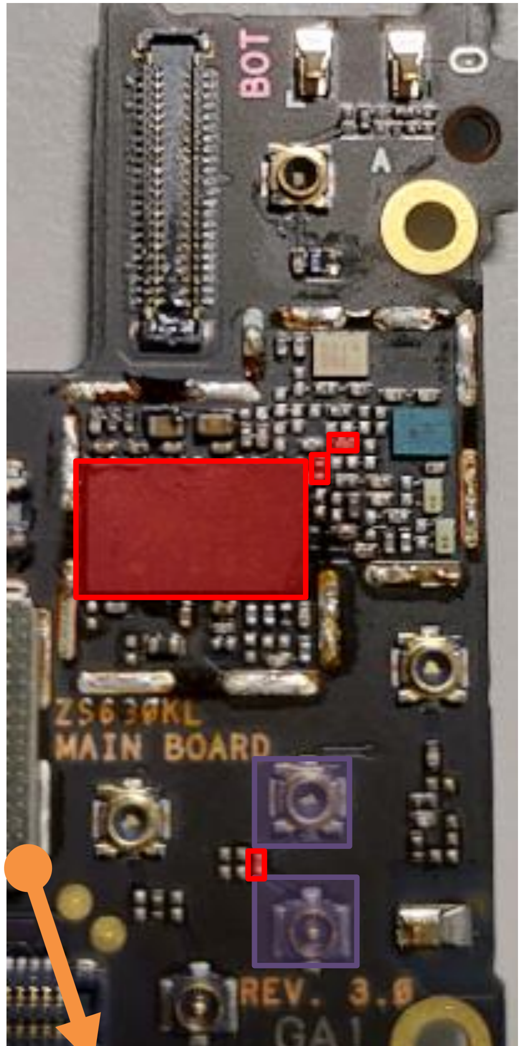
3G/4G MMPA **U4807**

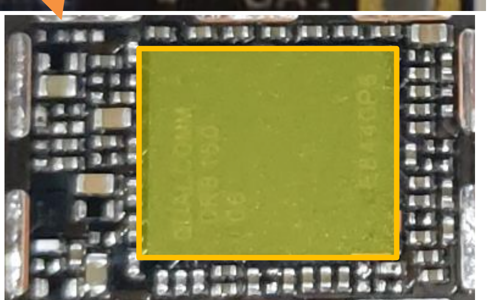
TriPX **U4809**

# TX Path : B3(TW)

| Signal Path | Tx Device | Antenna Path | Status | Status Max | Time (s) |
|---|---|---|---|---|---|
| 115 | 14 | 0 | 0 | 0 | |

bottom



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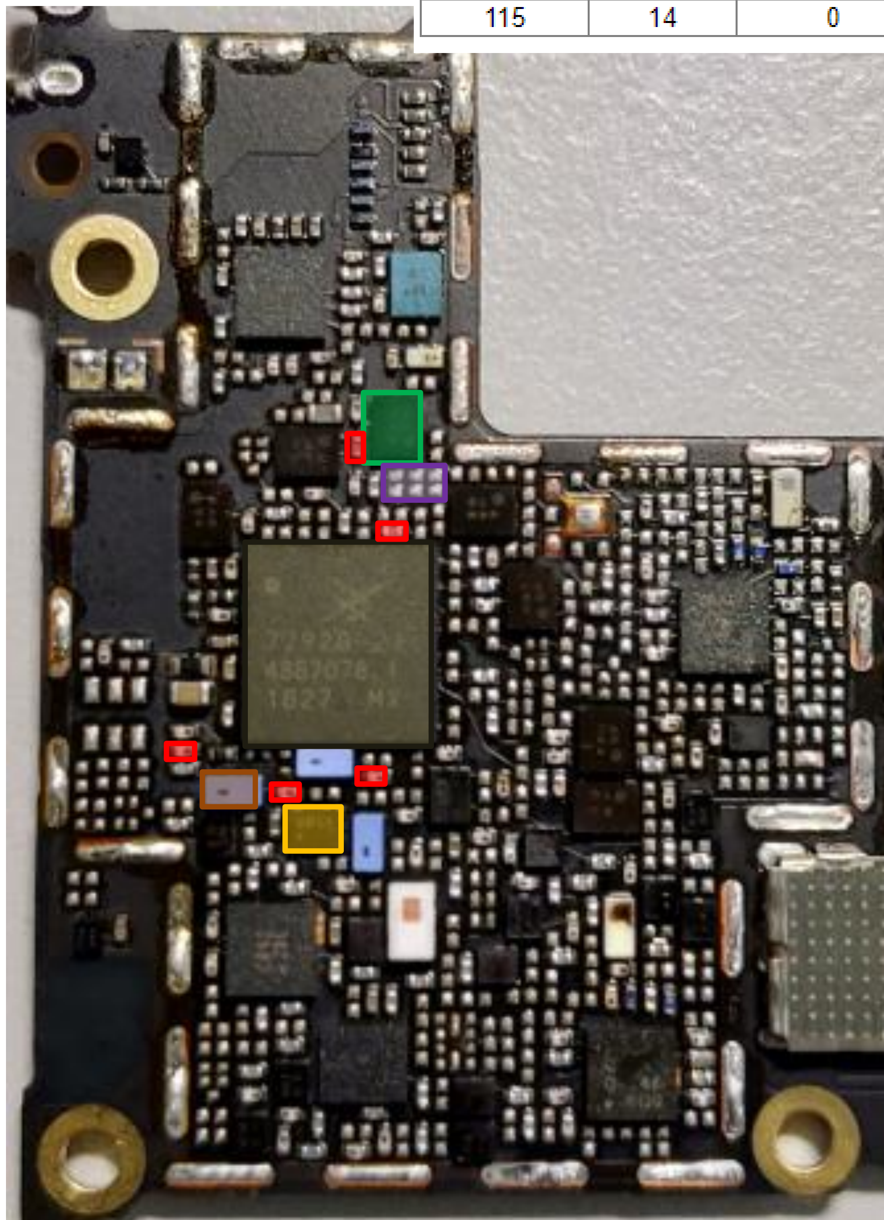


3G/4G MMPA **U4807**

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

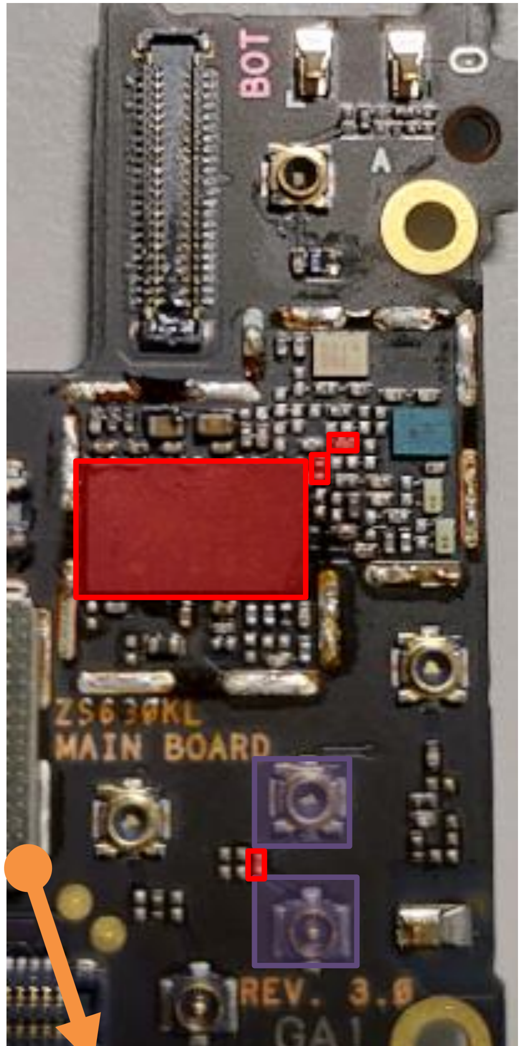
SDR8150 **U6201**

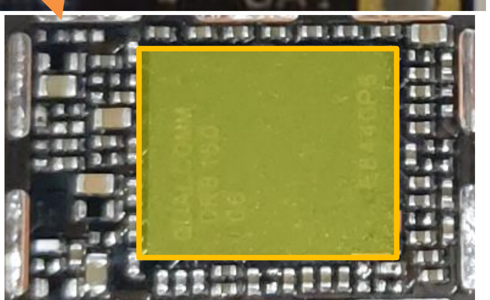
DPX **U4821**

DiPx **U4804**

# TX Path : B3(WW)

bottom



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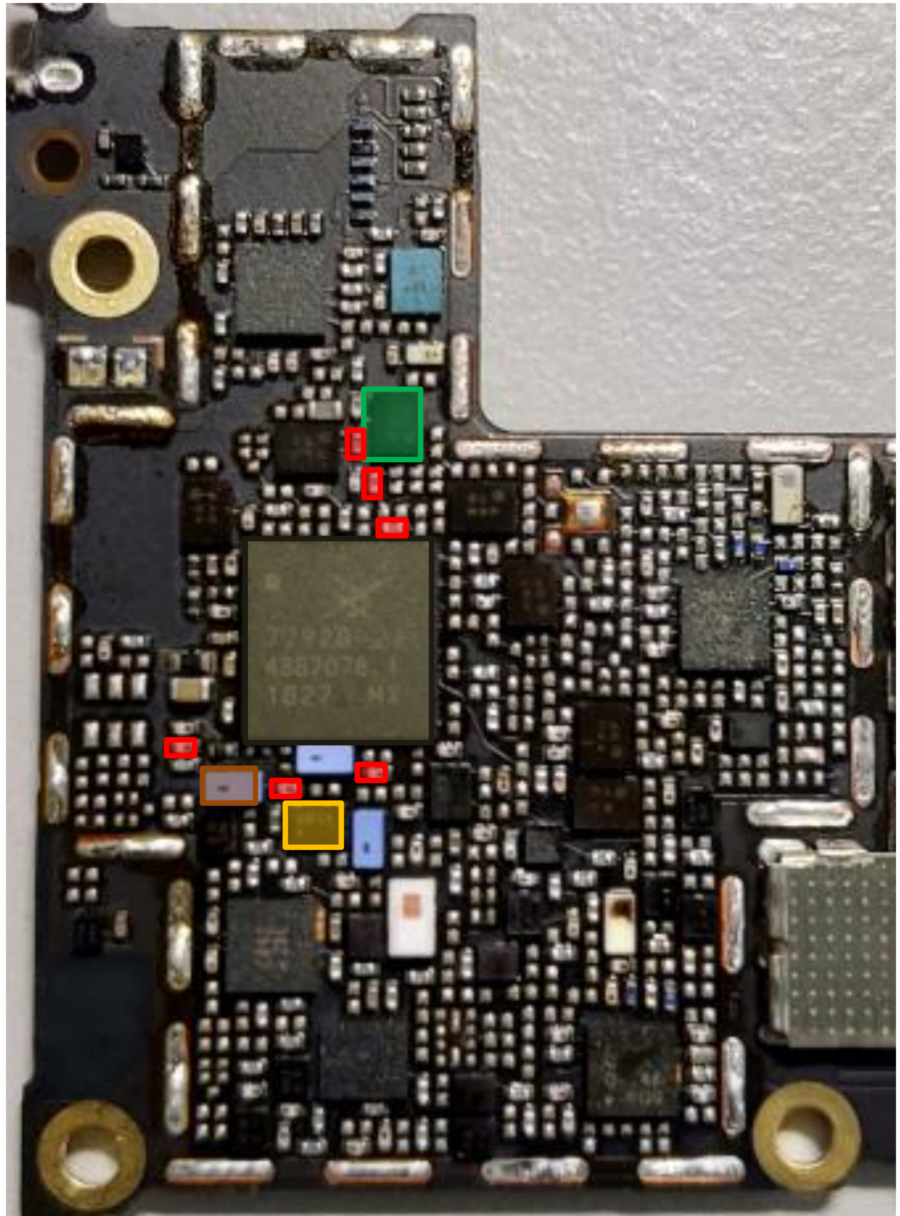


3G/4G MMPA
**U4807**

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

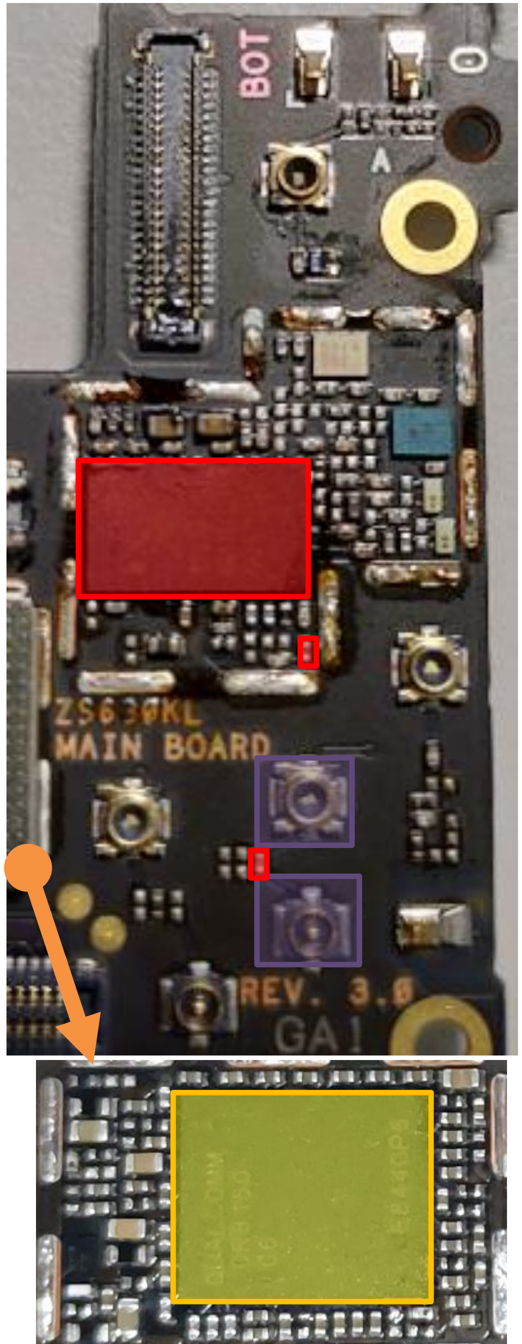
ASM+2G PA
**U4701**

SDR8150
**U6201**

DPX
**U4821**

# TX Path : B5/B6/B18/B19/B26(TW)
# B5(WW)

bottom

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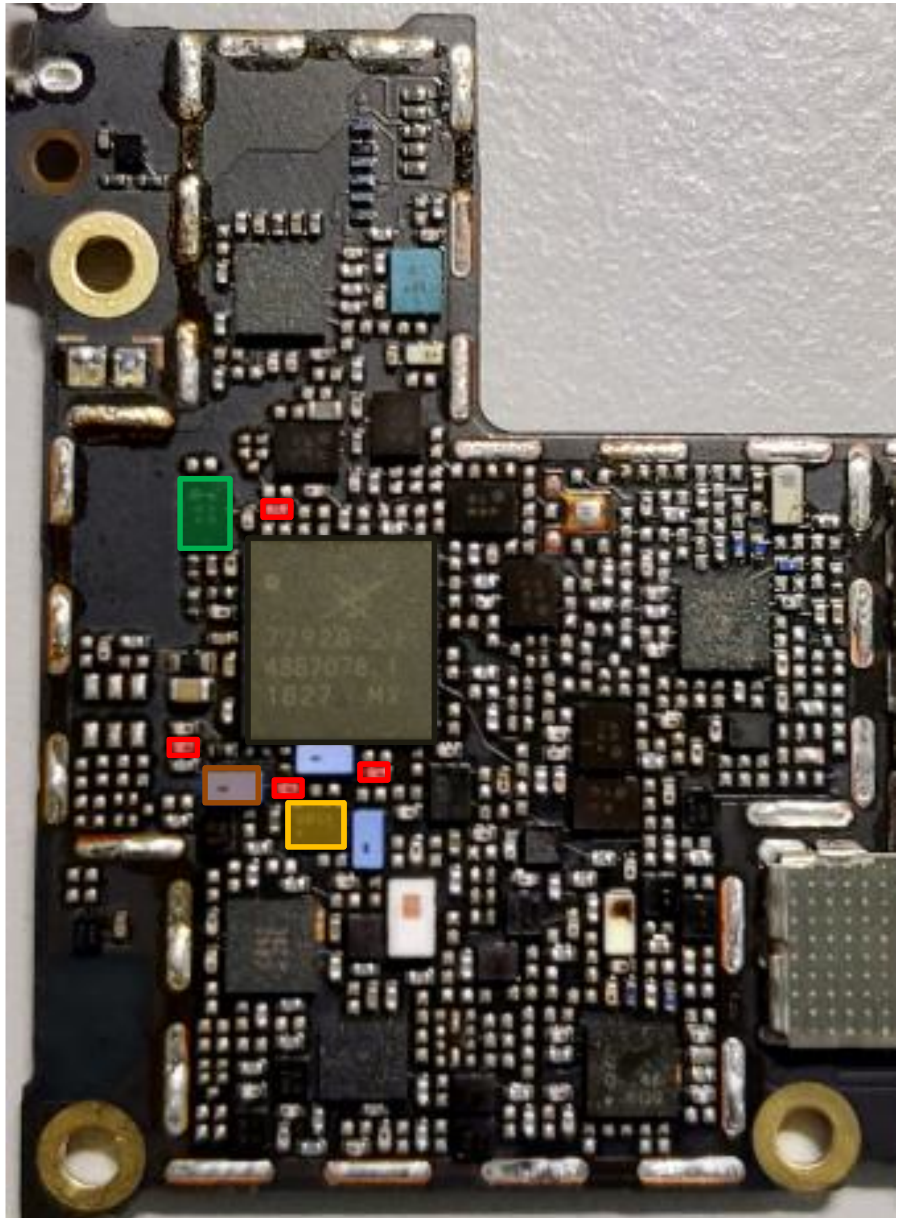
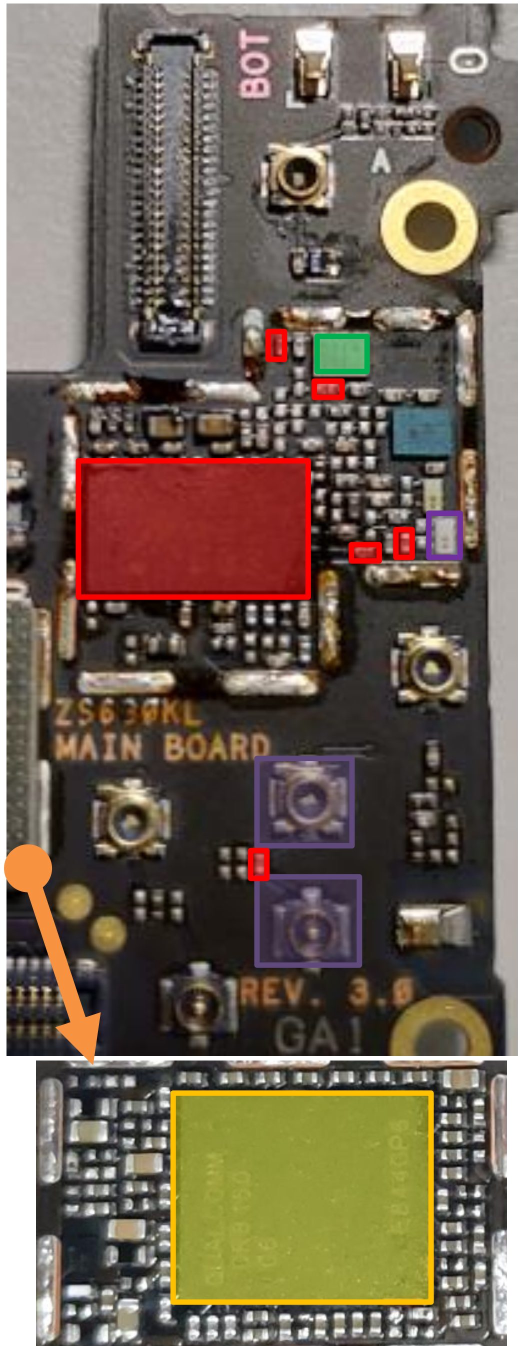


- 3G/4G MMPA **U4807**
- RF Connector LB/MB **CON4701 CON4704**
- SAW **U4708**
- DPDT **U4715**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- DPX **U4813**

***different DPx between TW & WW**

# TX Path : B8

bottom

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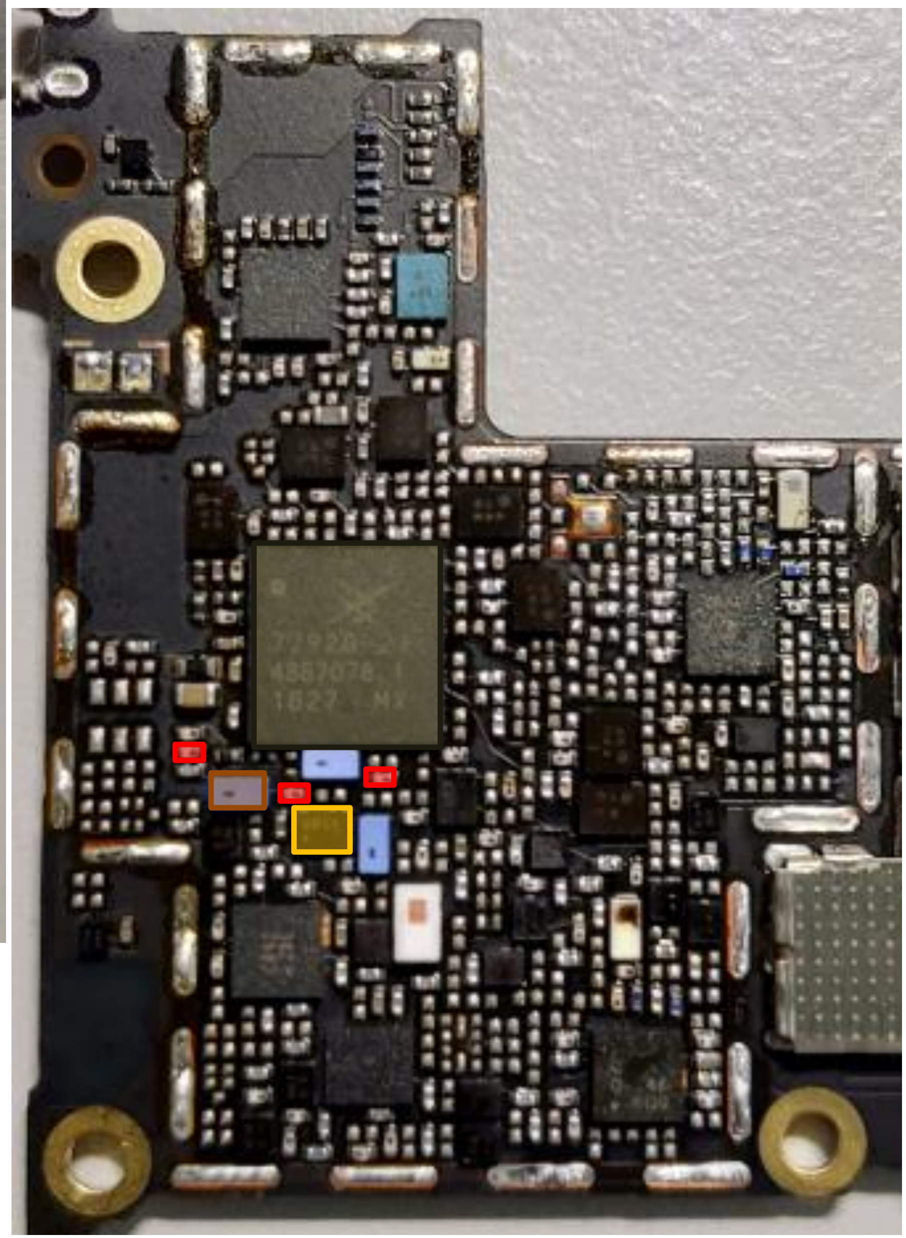
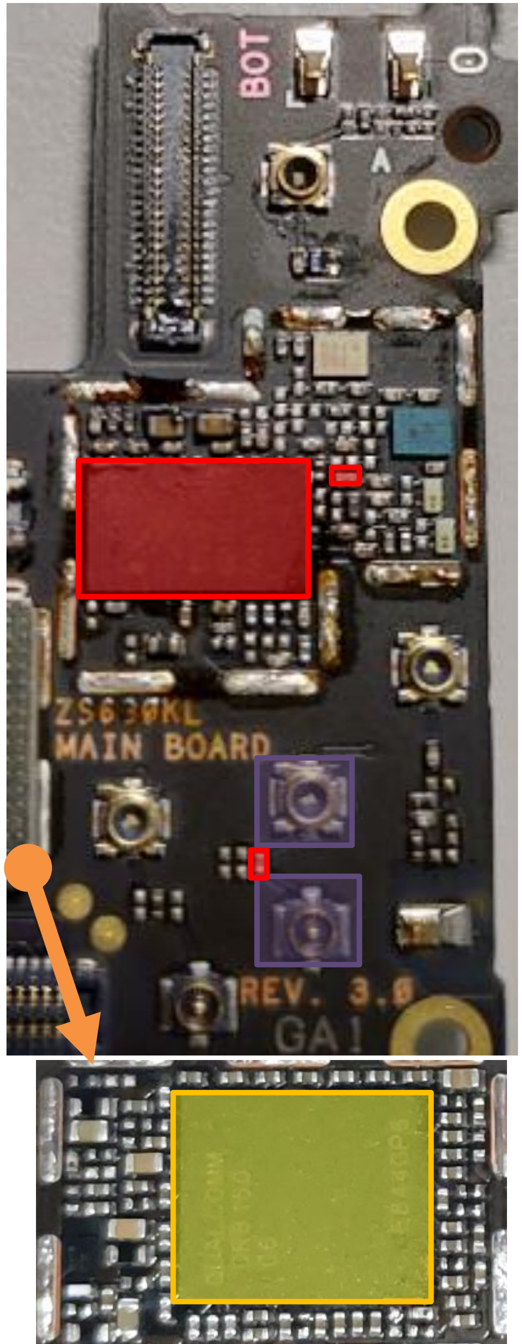


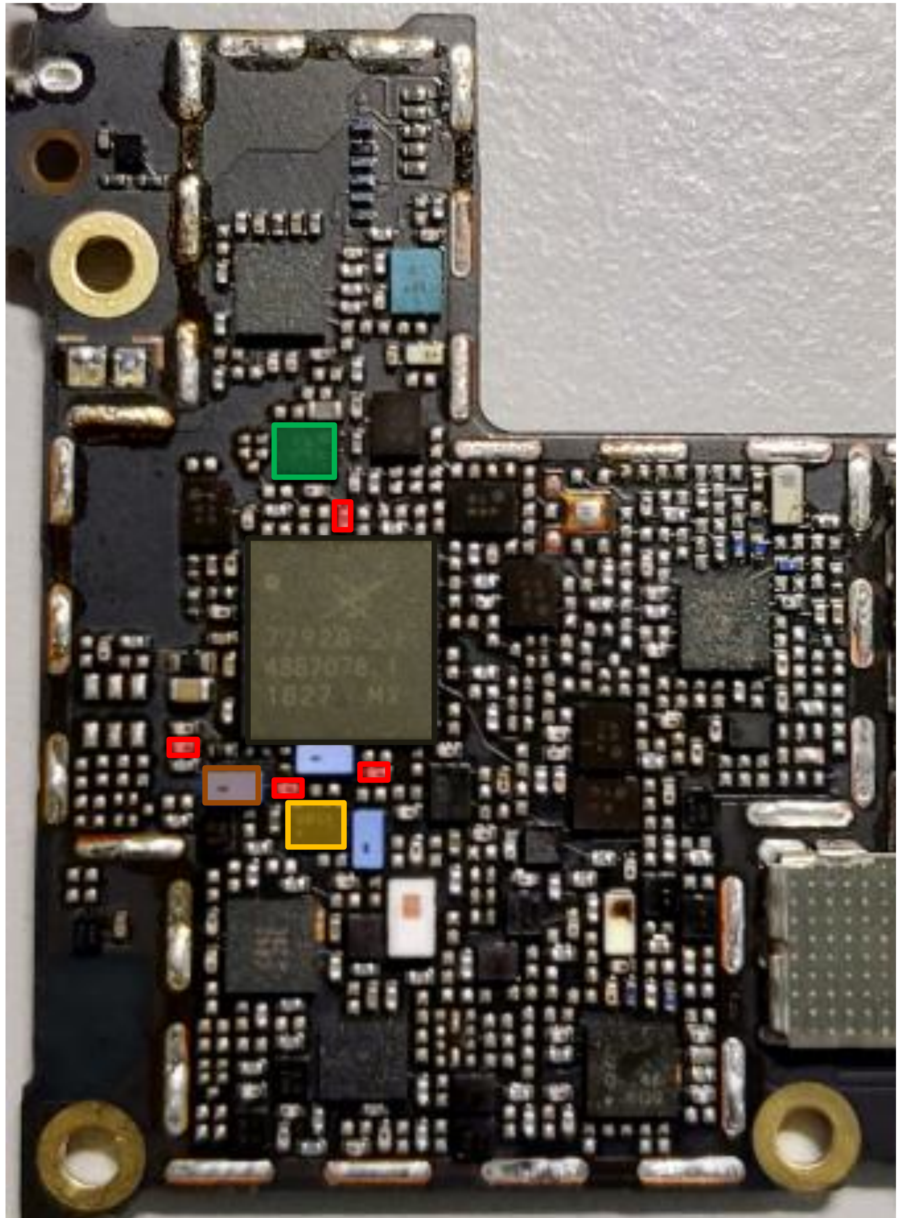
- 3G/4G MMPA **U4807**
- RF Connector LB/MB **CON4701 CON4704**
- SAW **U4708**
- DPDT **U4715**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- DPX **U4812**
- SAW **U4825**

# TX Path : B20(WW)、B12/B17(NA)

bottom




top

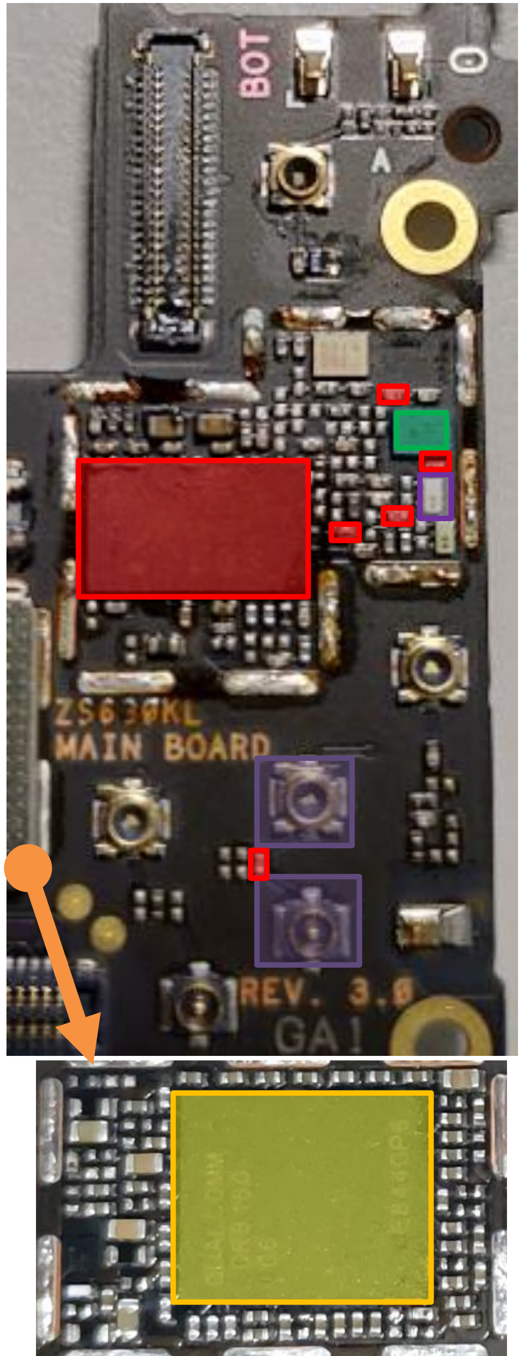


- 3G/4G MMPA **U4807**
- RF Connector LB/MB **CON4701 CON4704**
- SAW **U4708**
- DPDT **U4715**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- DPX **U4819**

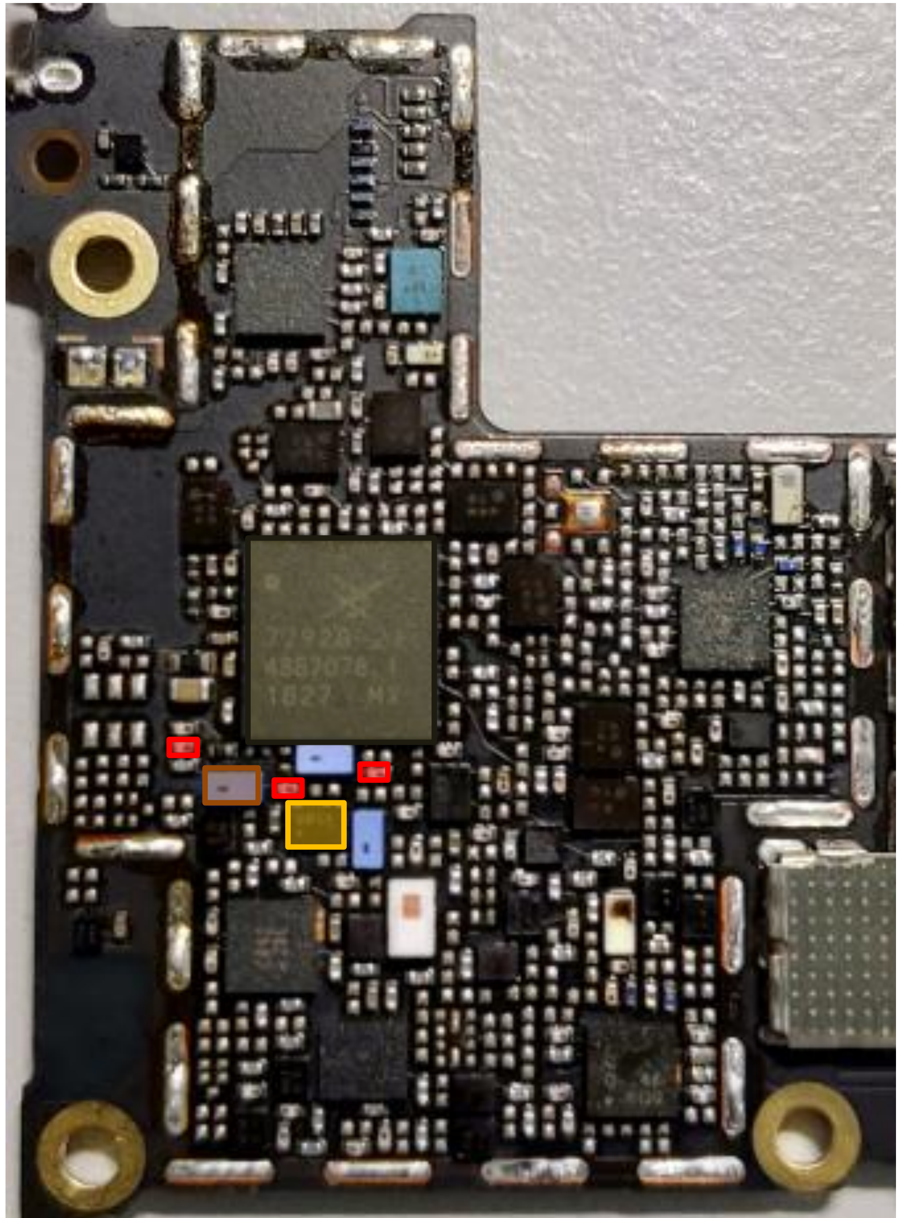
***different DPx between NA & WW**

# TX Path : B28a

bottom




top



3G/4G MMPA
**U4807**

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

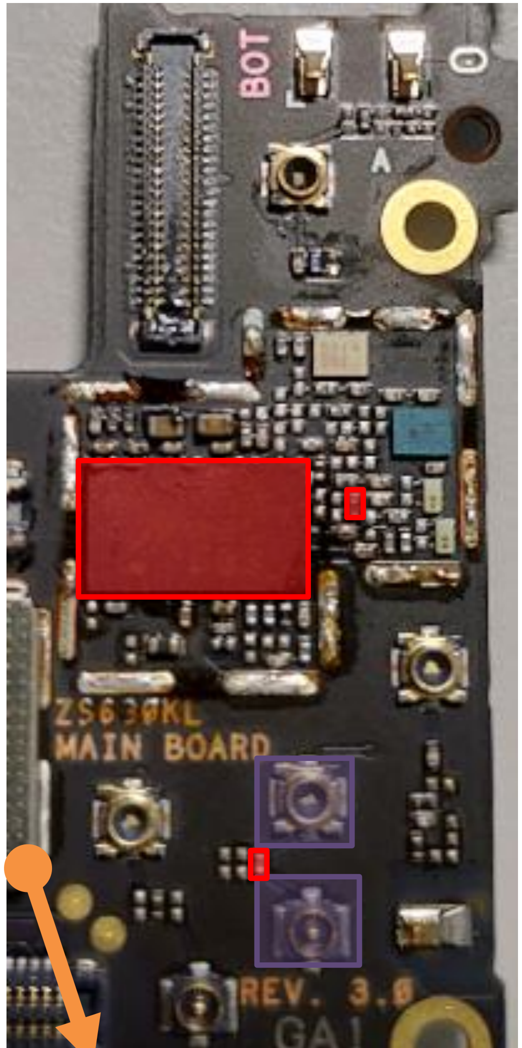
SDR8150
**U6201**

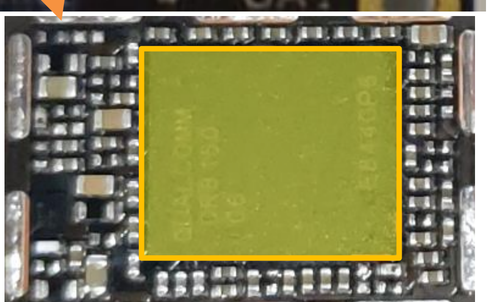
DPX
**U4811**

SAW **U4827**

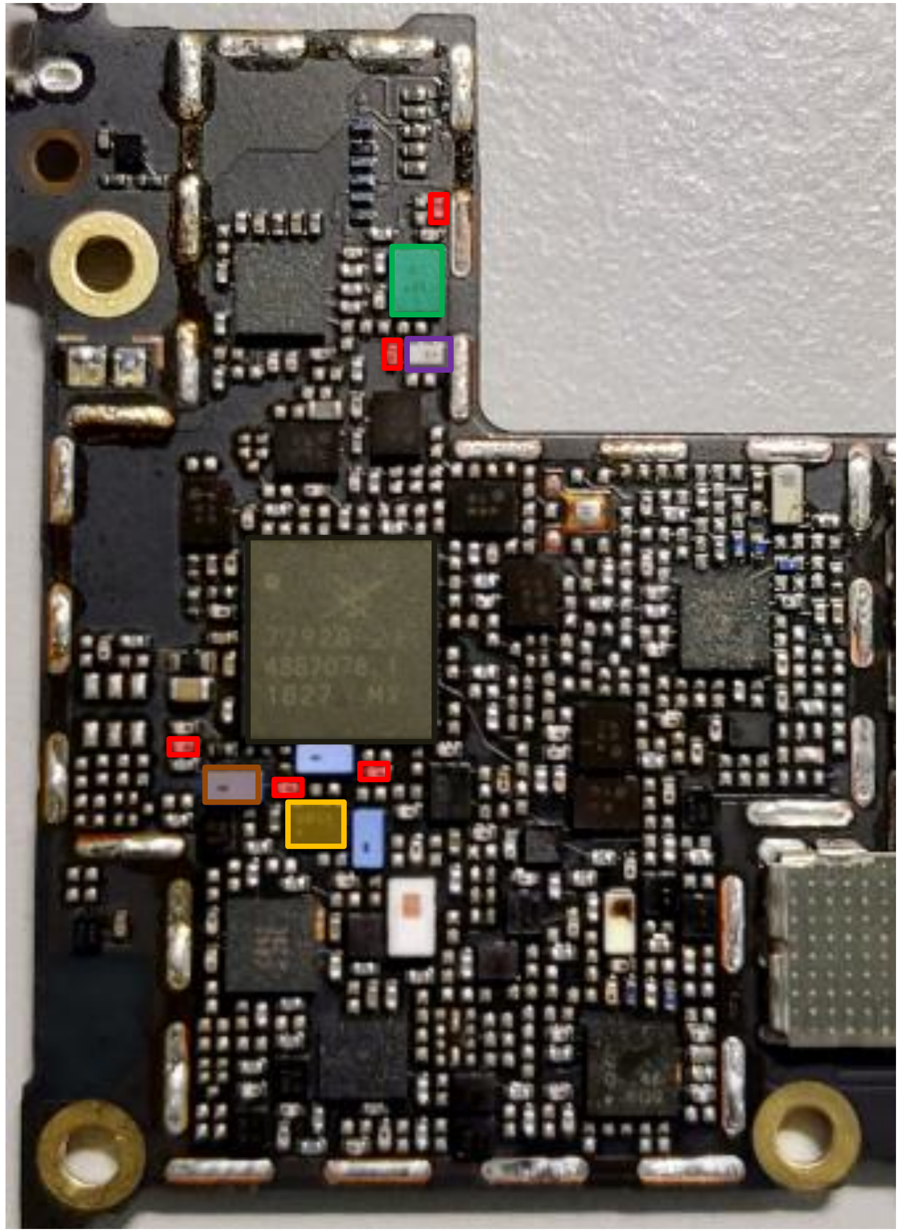
# TX Path : B28b

bottom



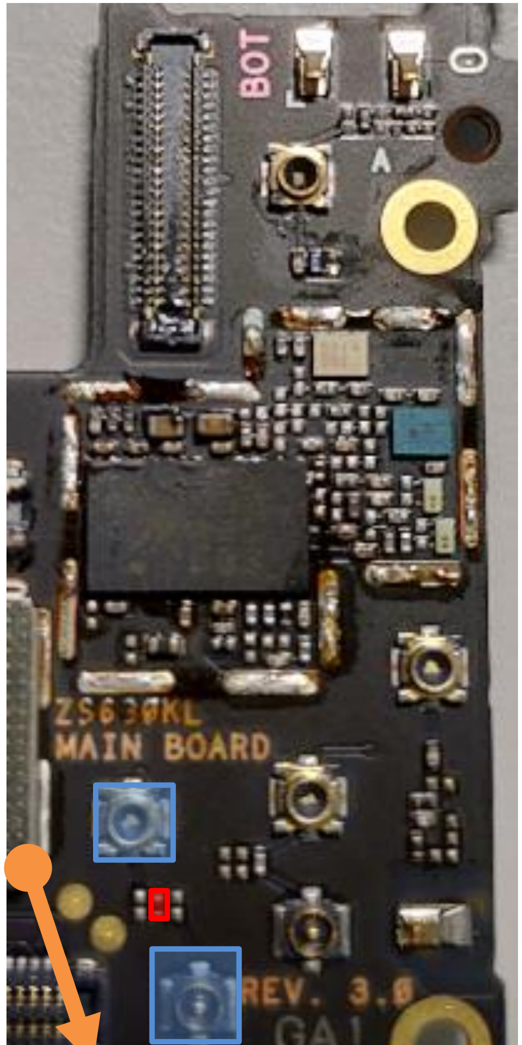
top

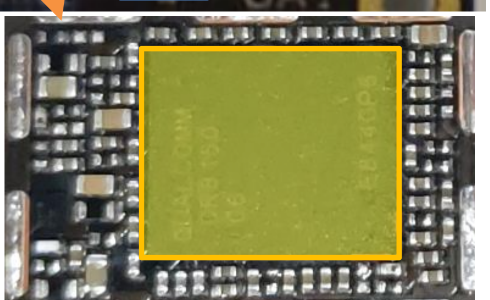


- 3G/4G MMPA **U4807**
- RF Connector LB/MB **CON4701 CON4704**
- SAW **U4708**
- DPDT **U4715**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- DPX **U4808**
- SAW **U4826**

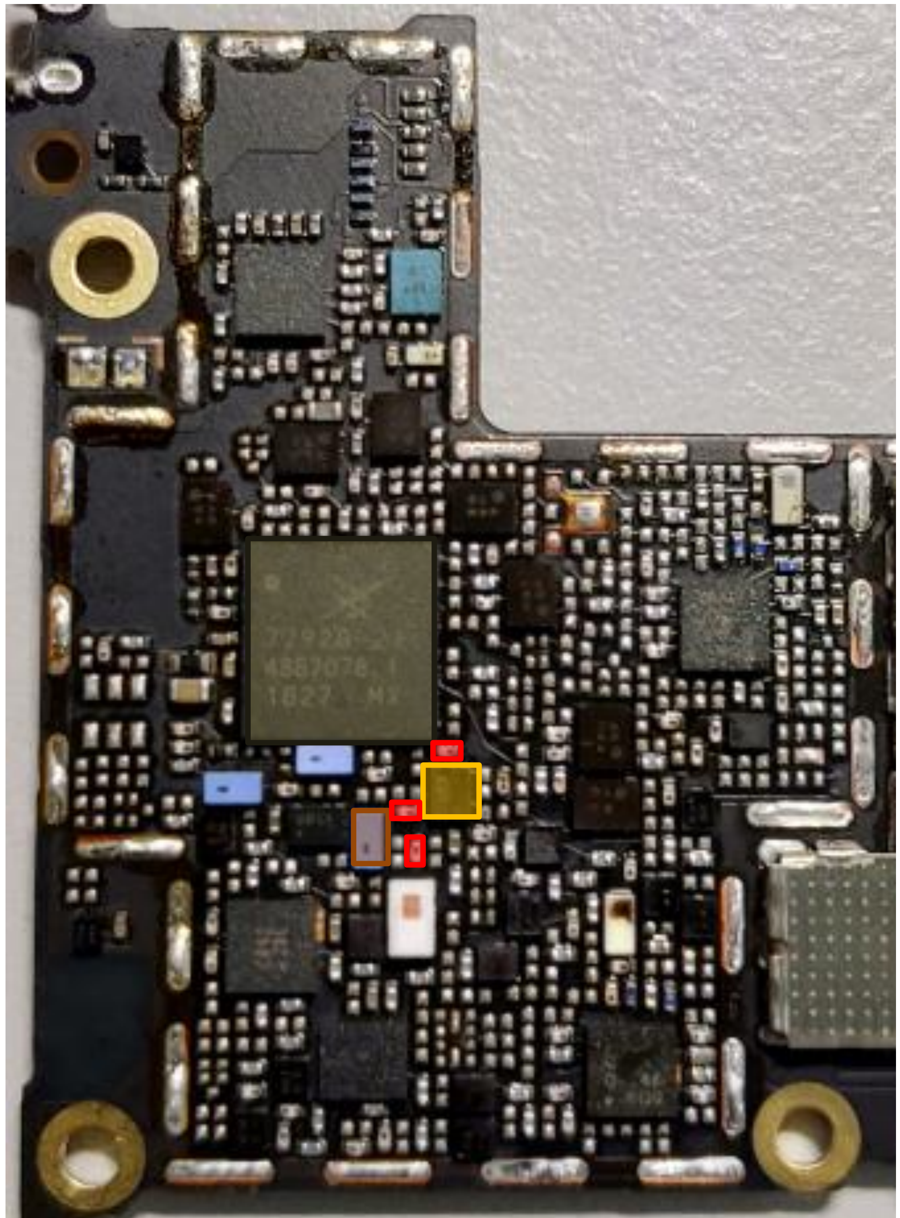
# HB Tx Common Path

bottom



top



HB : B7/B38/B40/B41

RF Connector HB
**CON4702 CON4703**

SAW **U4709**

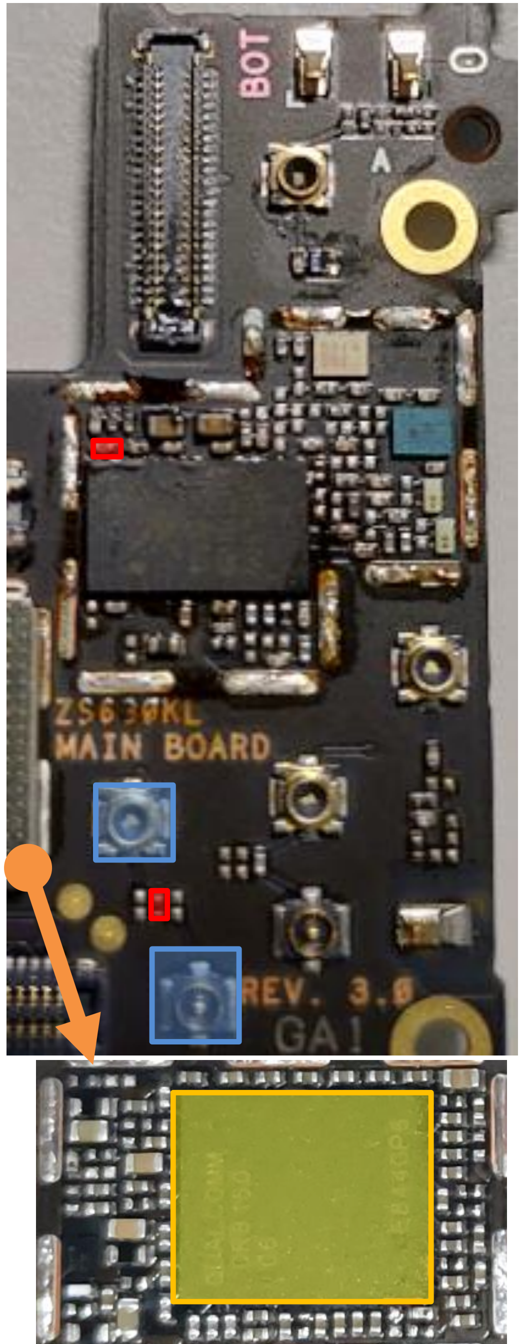
DPDT **U4716**

ASM+2G PA
**U4701**

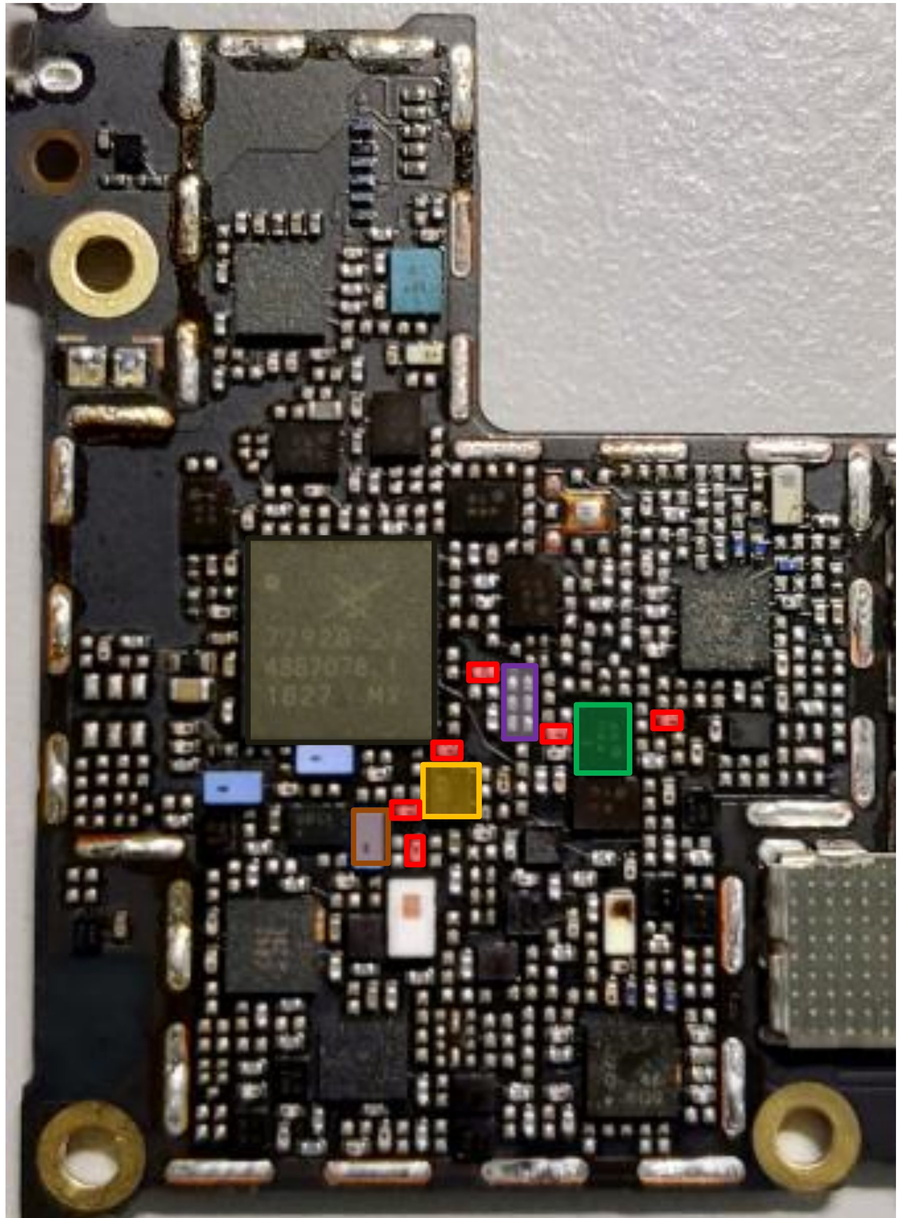
SDR8150
**U6201**

# TX Path : B7(TW)

bottom




top



RF Connector HB **CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

ASM+2G PA **U4701**

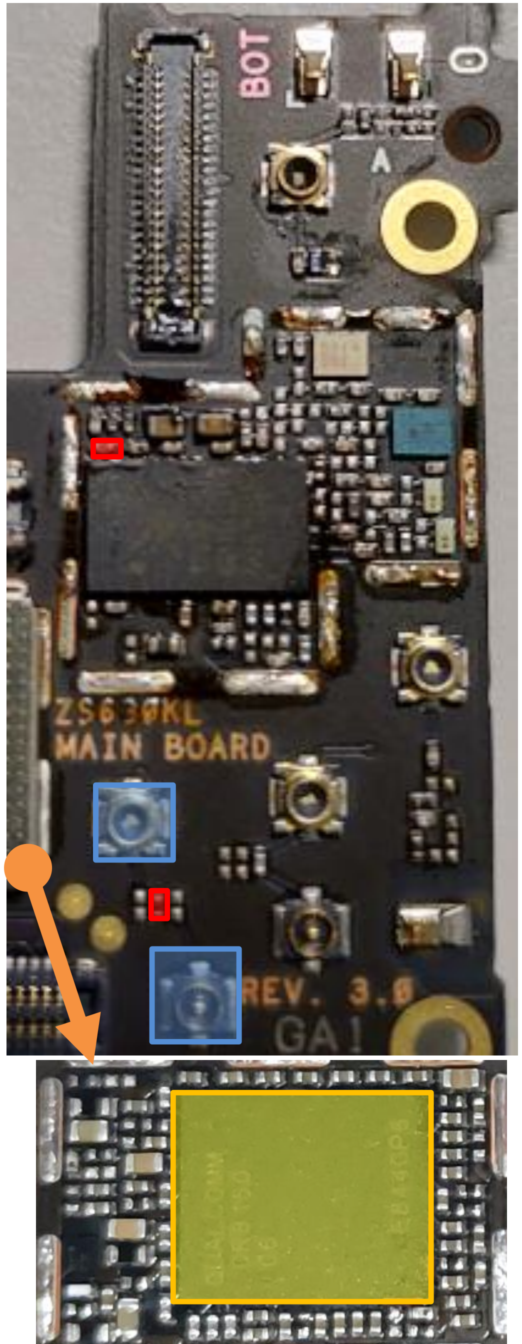
SDR8150 **U6201**

DPX **U4824**

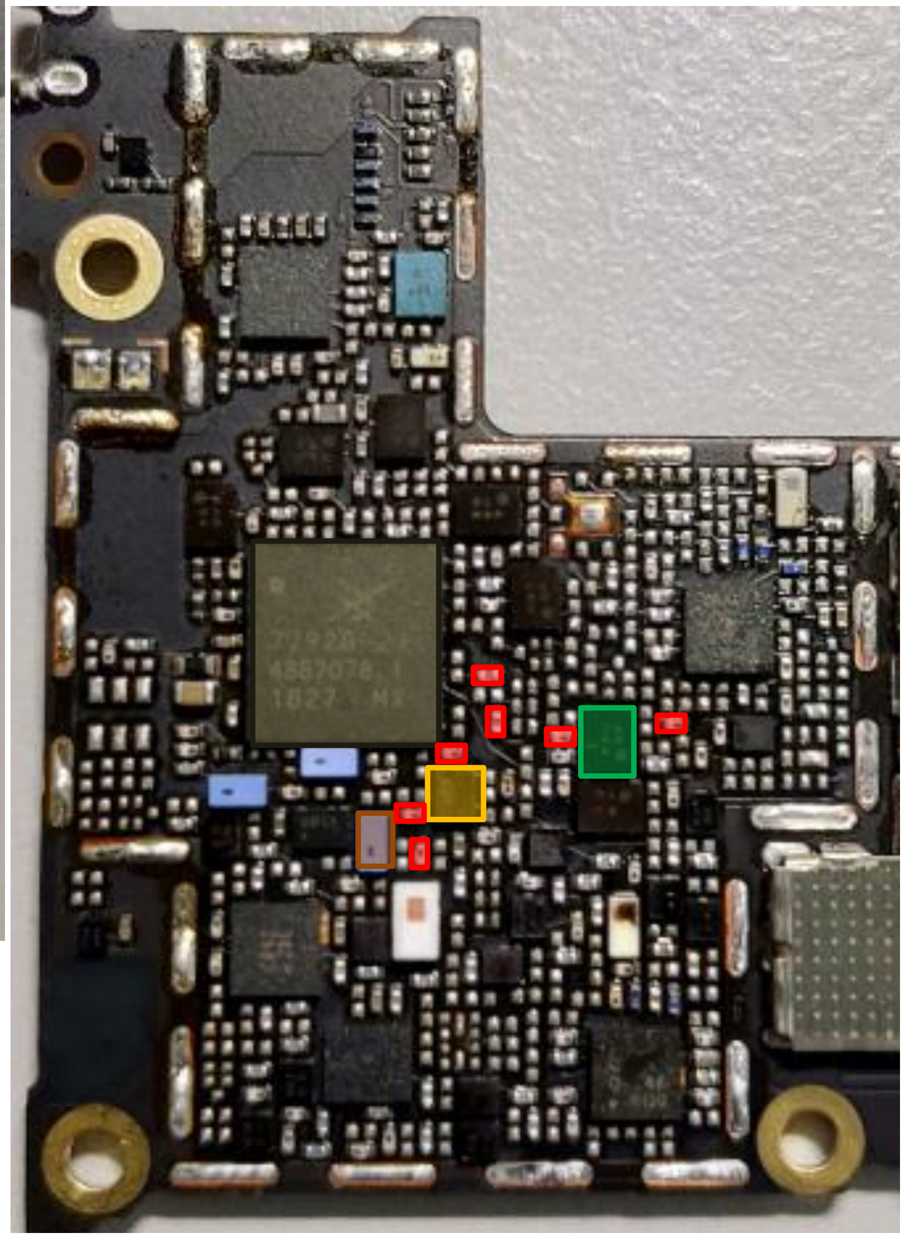
DiPx **U4802**

# TX Path : B7(WW)

bottom




top



RF Connector HB
**CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

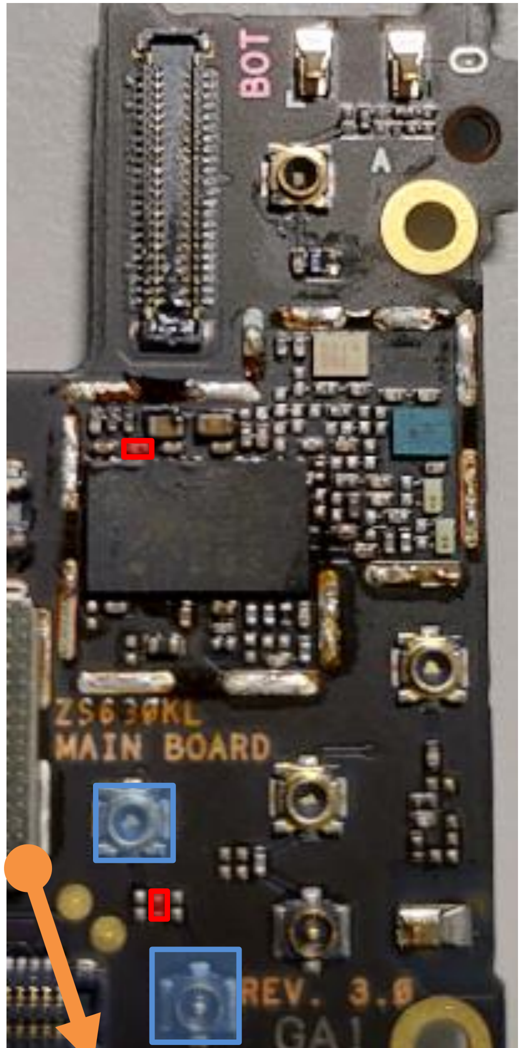
ASM+2G PA
**U4701**

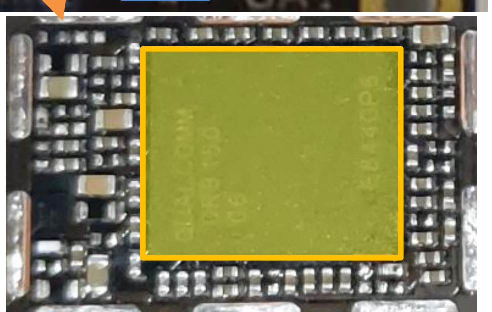
SDR8150
**U6201**

DPX
**U4824**

# TX Path : B38/B41(TW) 、 B40(WW)

bottom



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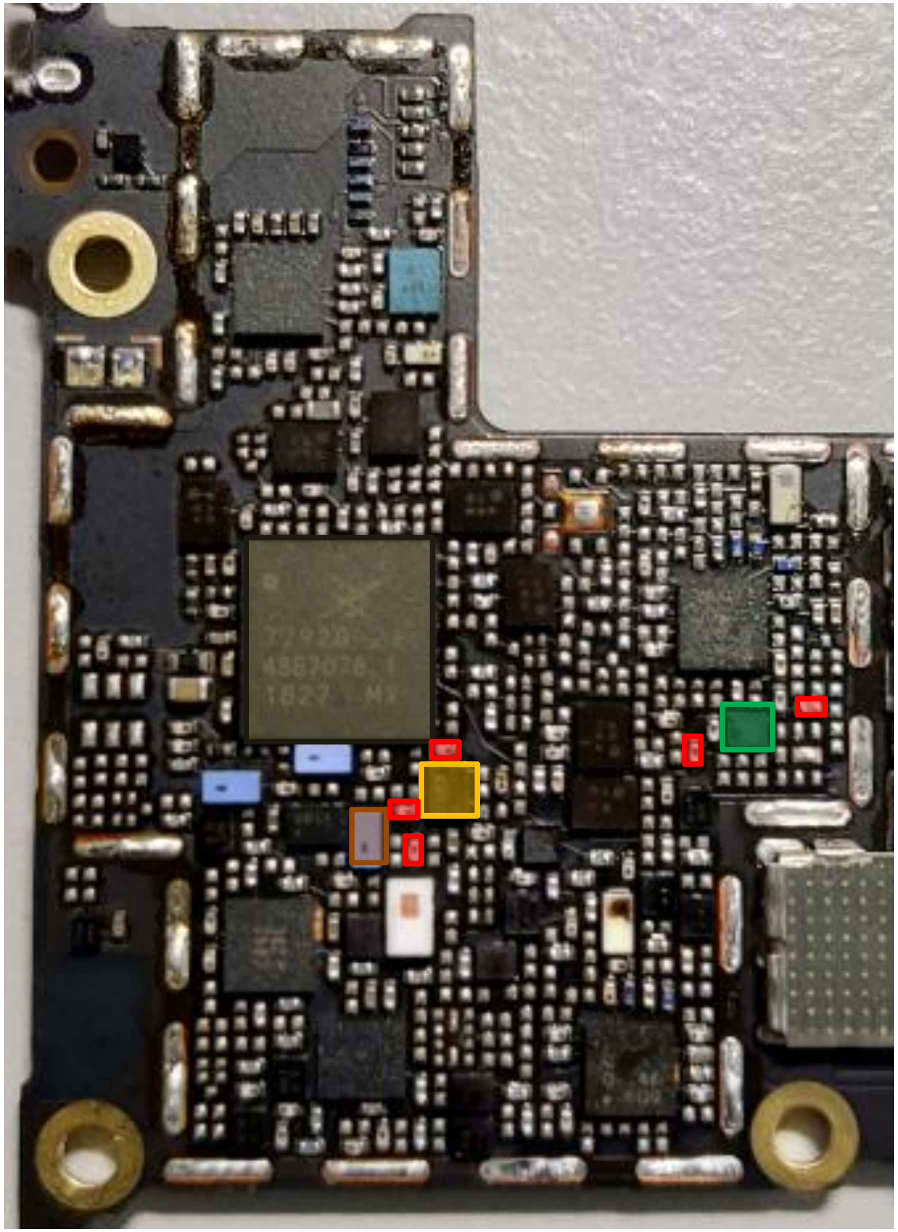


RF Connector HB
**CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

ASM+2G PA
**U4701**

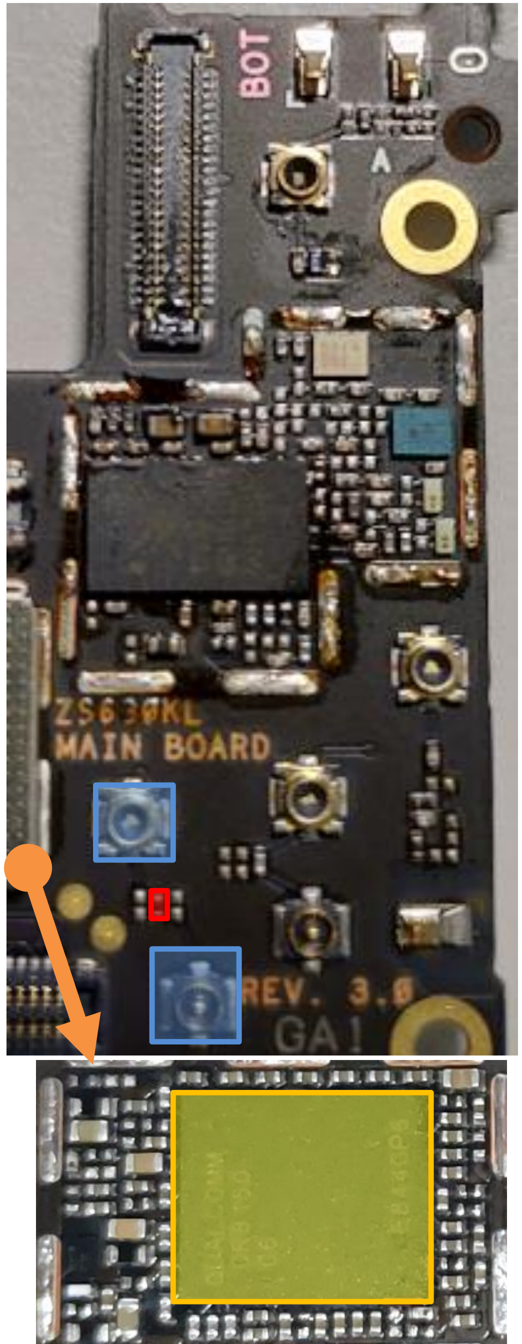
SDR8150
**U6201**

SAW
**U4816**

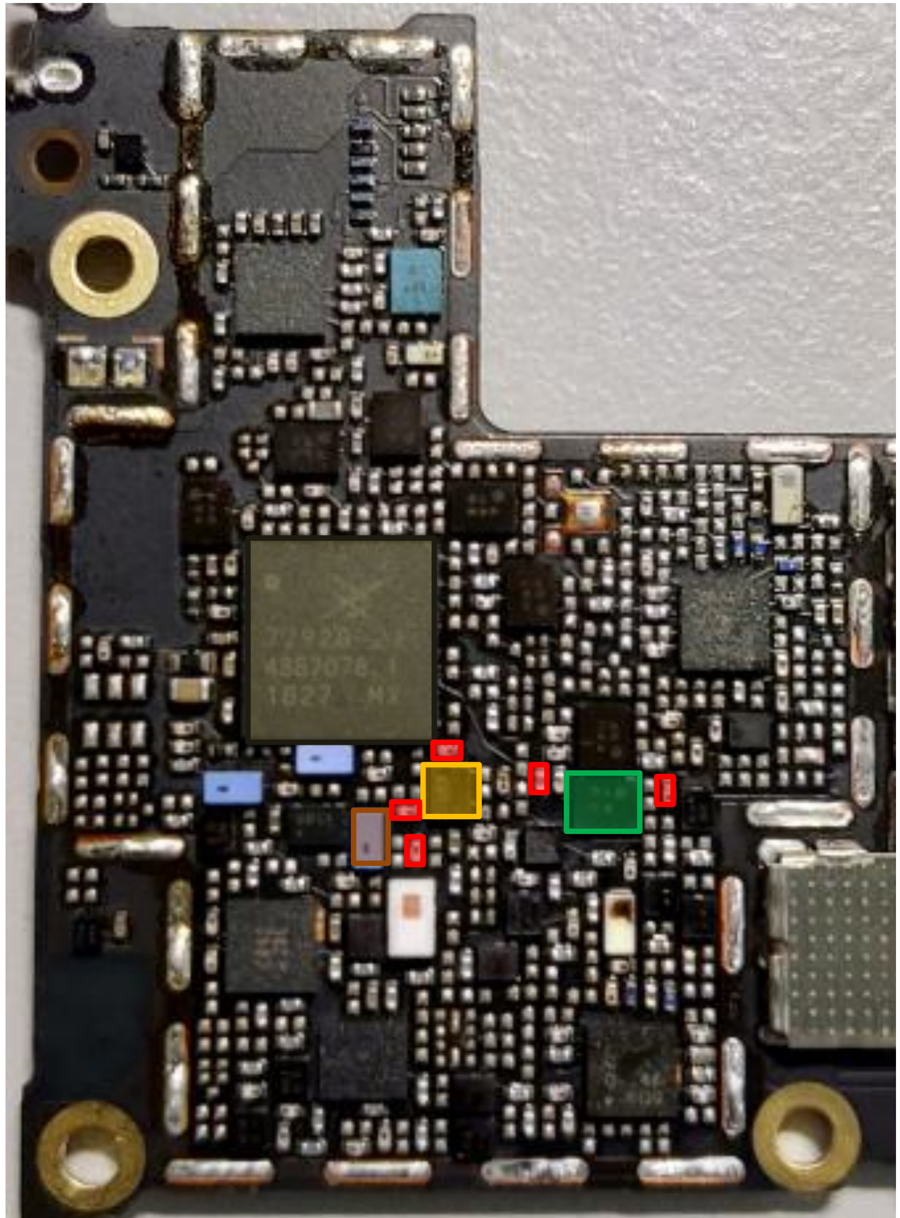
***different SAW between TW & WW**

# TX Path : B38/B41(WW)

bottom




top



RF Connector HB **CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

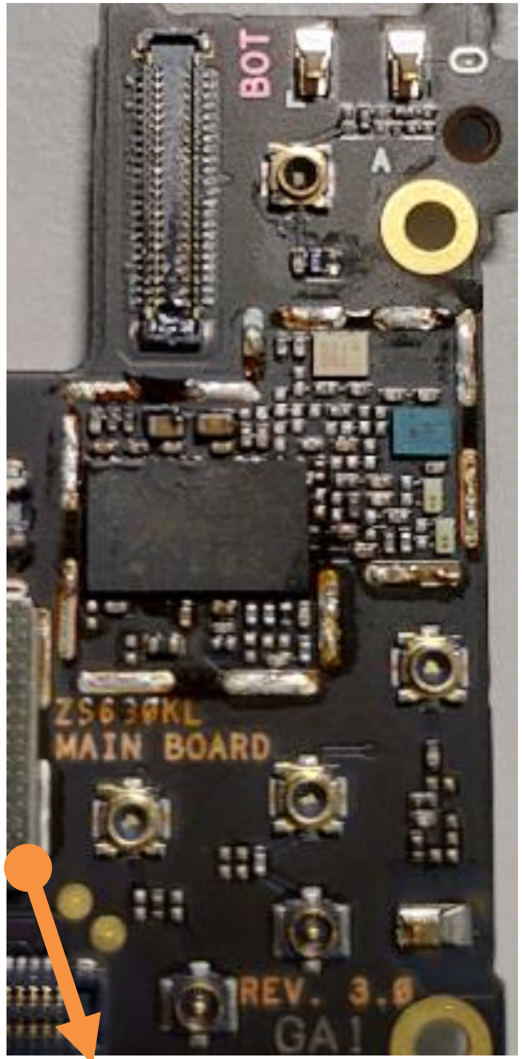
ASM+2G PA **U4701**

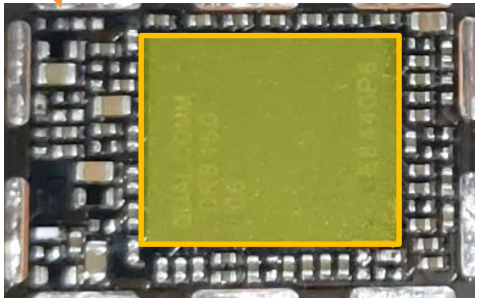
SDR8150 **U6201**

SAW **U4801**

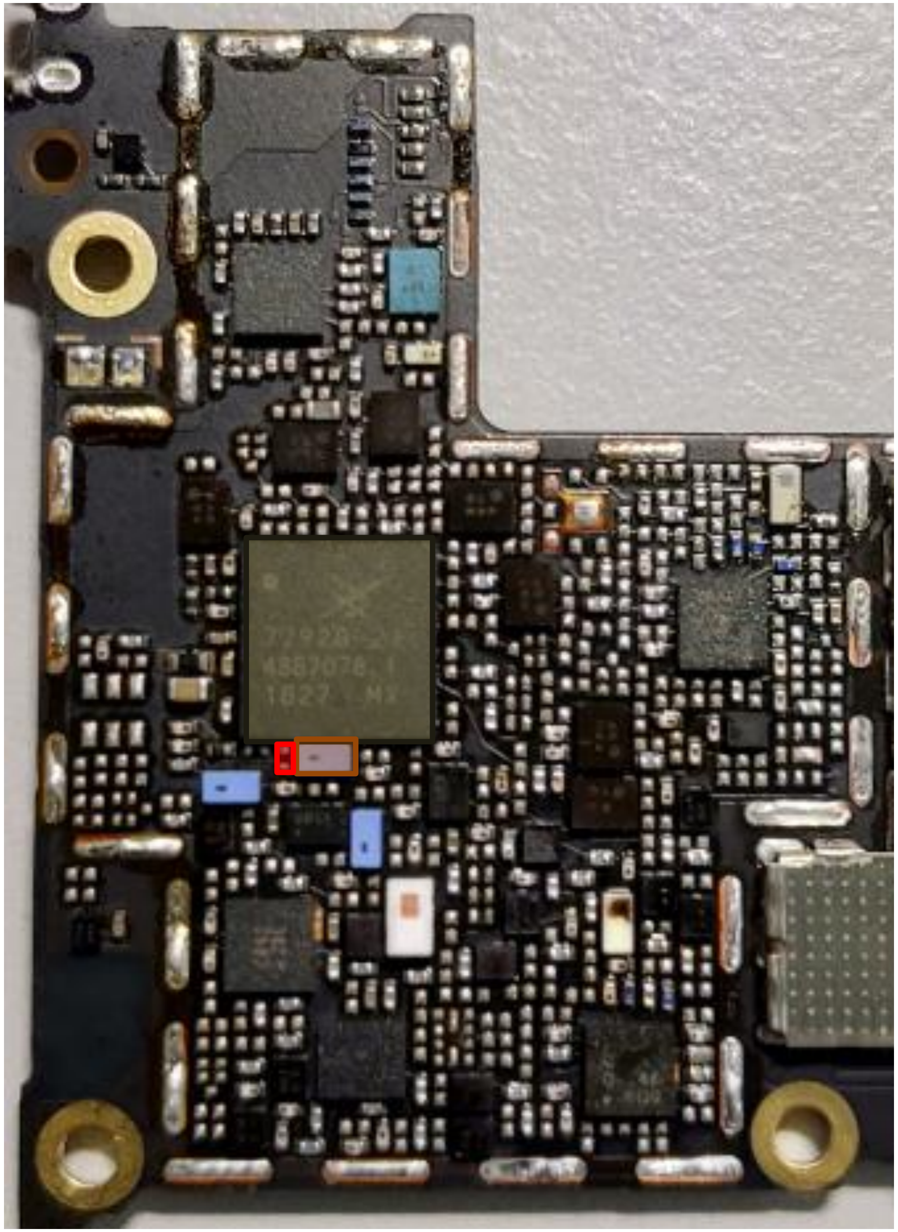
# FBRx Path

bottom



top



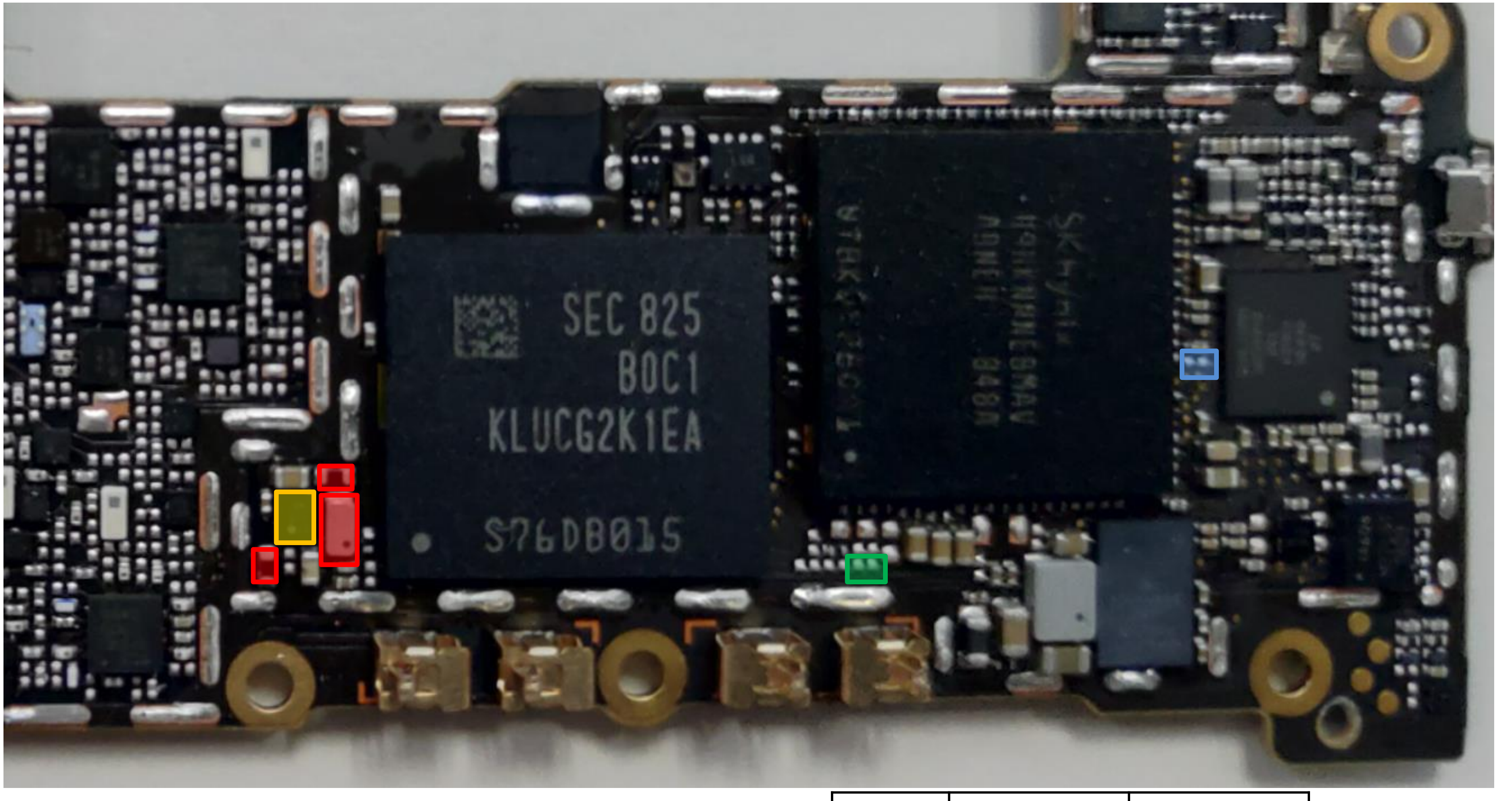
SAW **U4712**

ASM+2G PA
**U4701**

SDR8150
**U6201**

# DCDC for LMHB PA SKY77638 & SKU

top

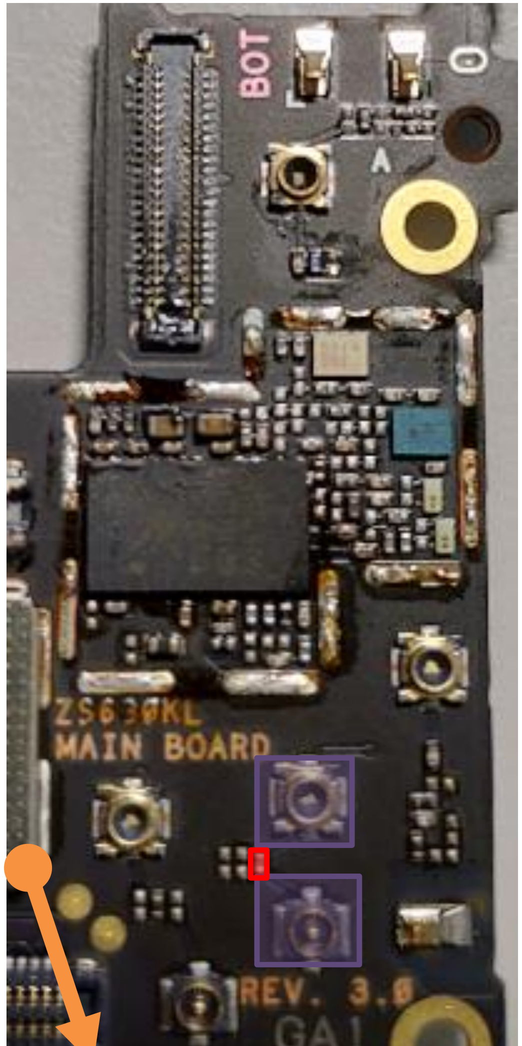


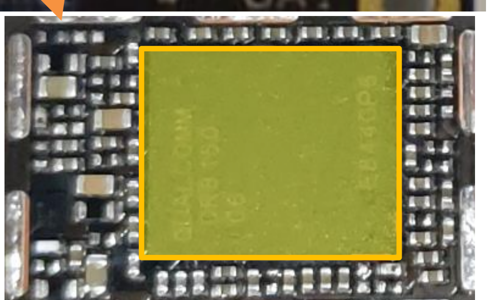

| SKU | (blue) | (green) |
|---|---|---|
| TW | NC | NC |
| WW | MOUNT | NC |
| NA | NC | MOUNT |

QET4101 **U4603**

# LMB PRx Common Path

bottom



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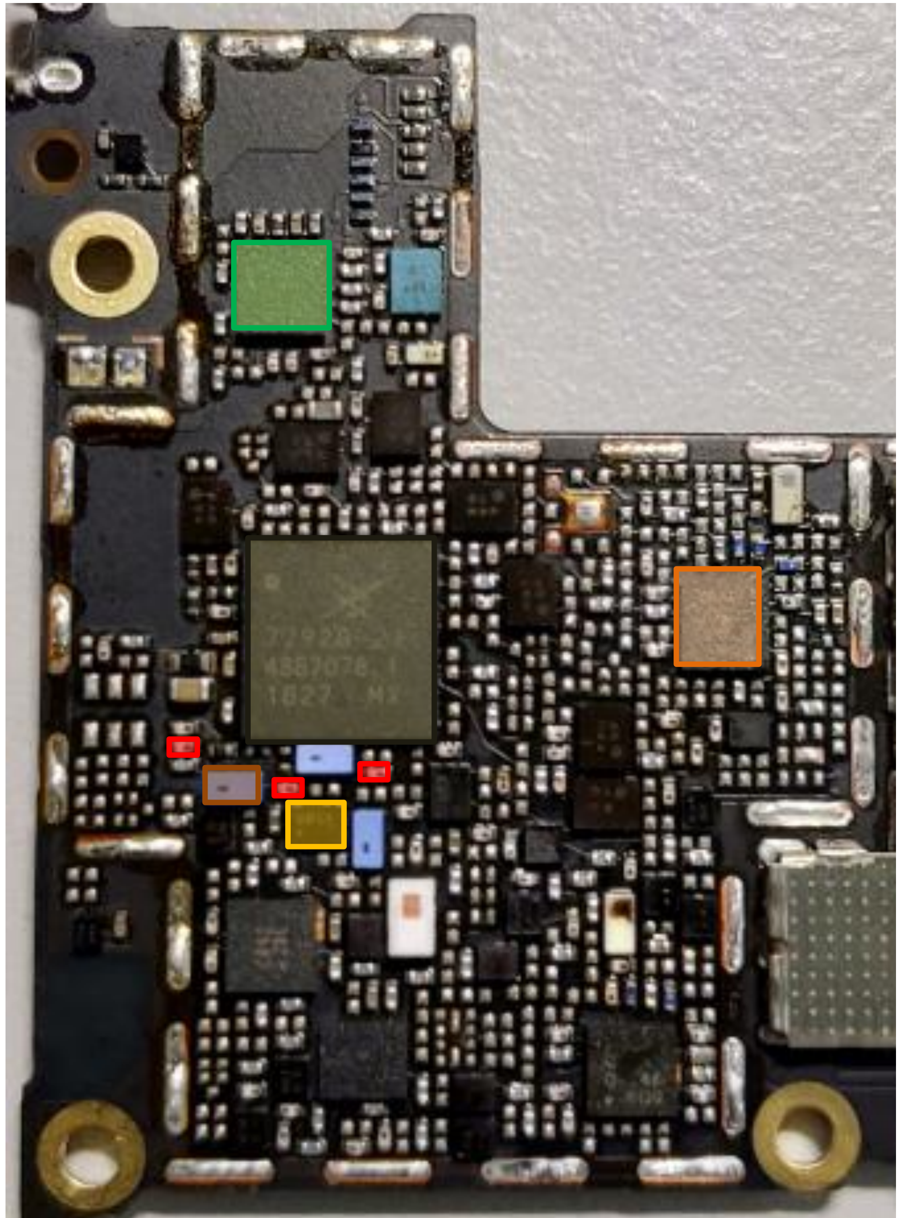


MB: B1/B2/B3/B4/B39
LB: B5/B8/B18/B19/B20/
B26/B28A/B28B

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

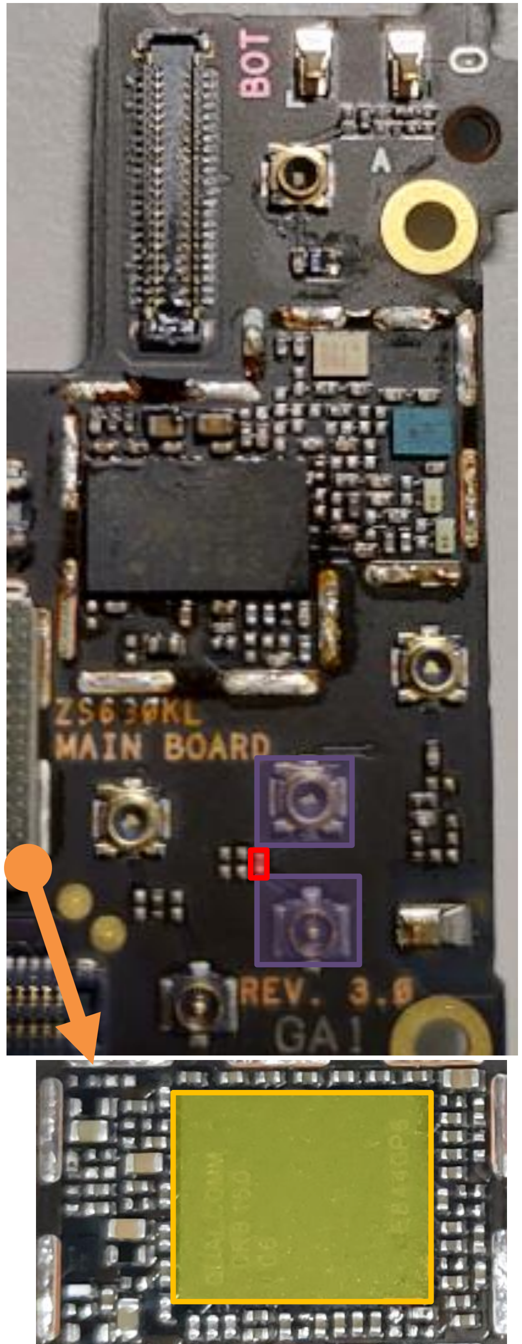
SDR8150
**U6201**

LB QLN1020
**U4903**

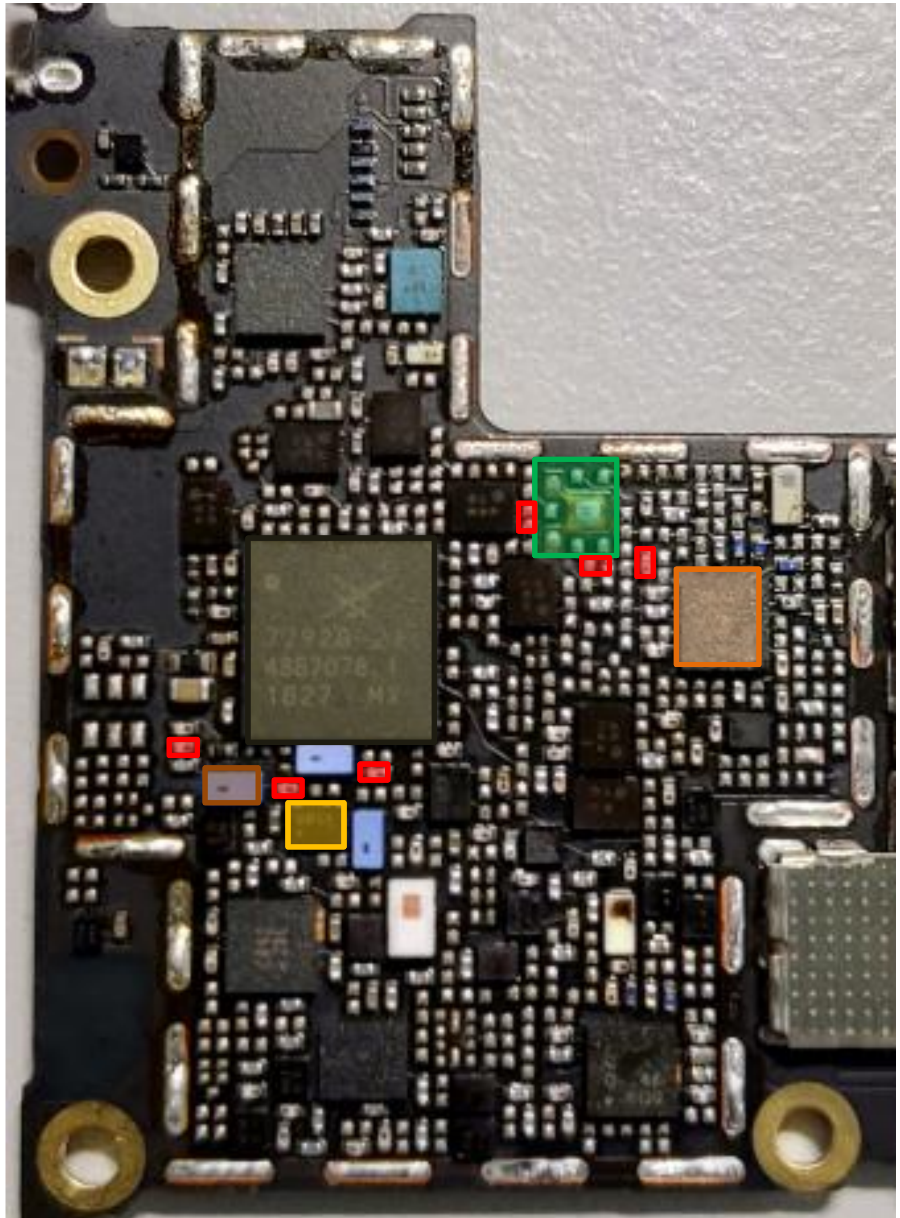
MB/HB QLN4640
**U4901**

# PRX Path : B1/B4(TW)

bottom




top



RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

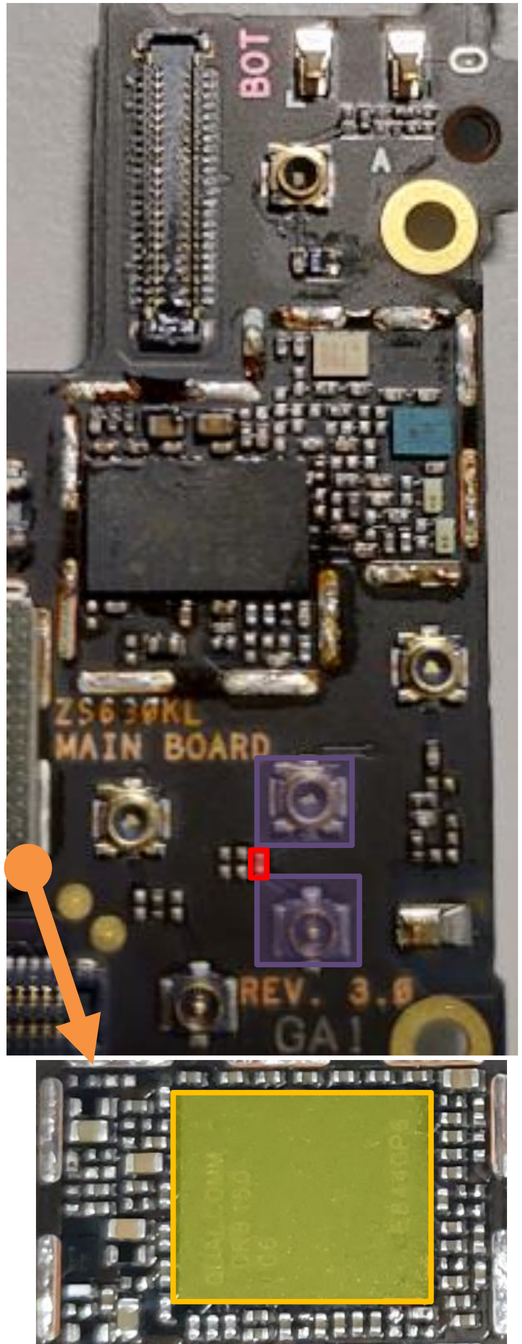
SDR8150
**U6201**

MB/HB QLN4640
**U4901**

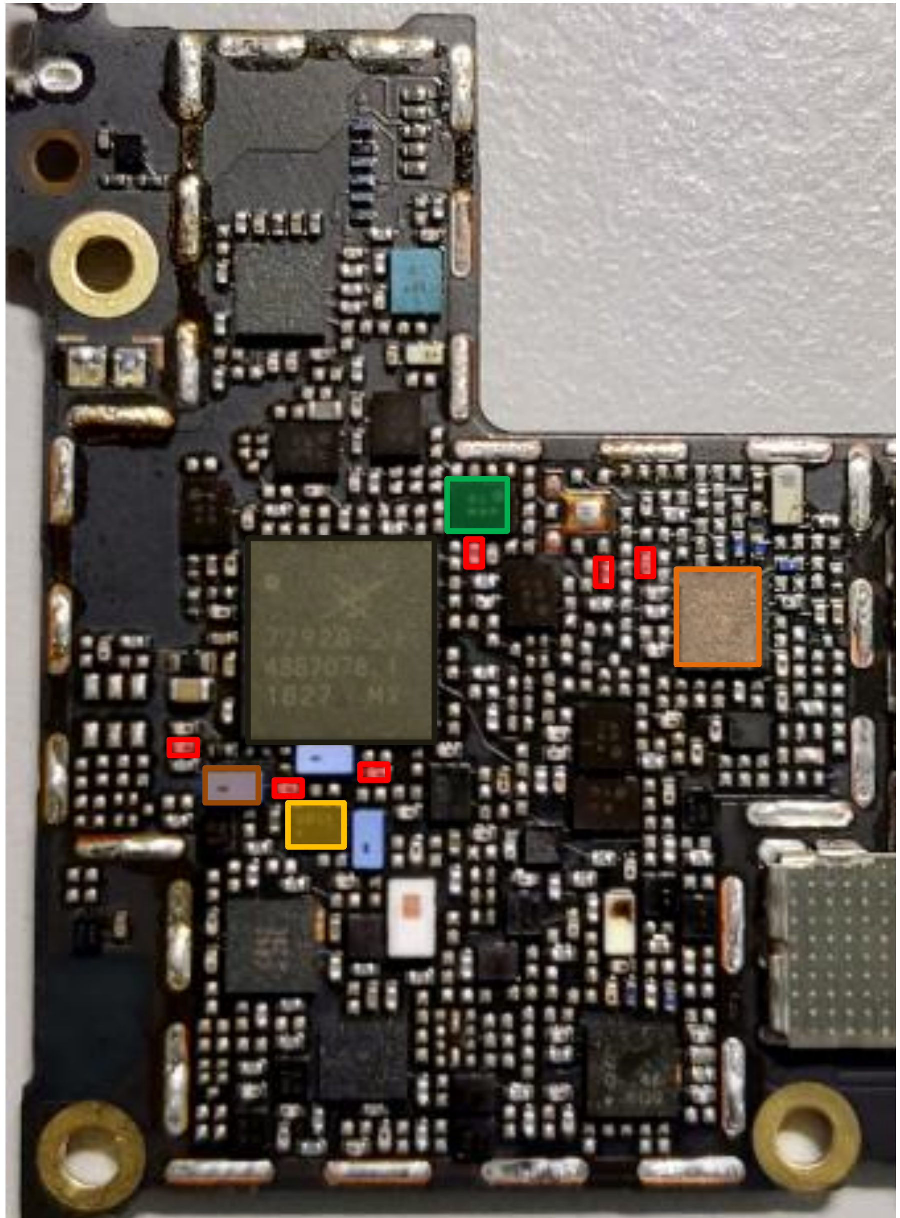
TriPX
**U4809**

# PRX Path : B1(WW)

bottom




top



RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

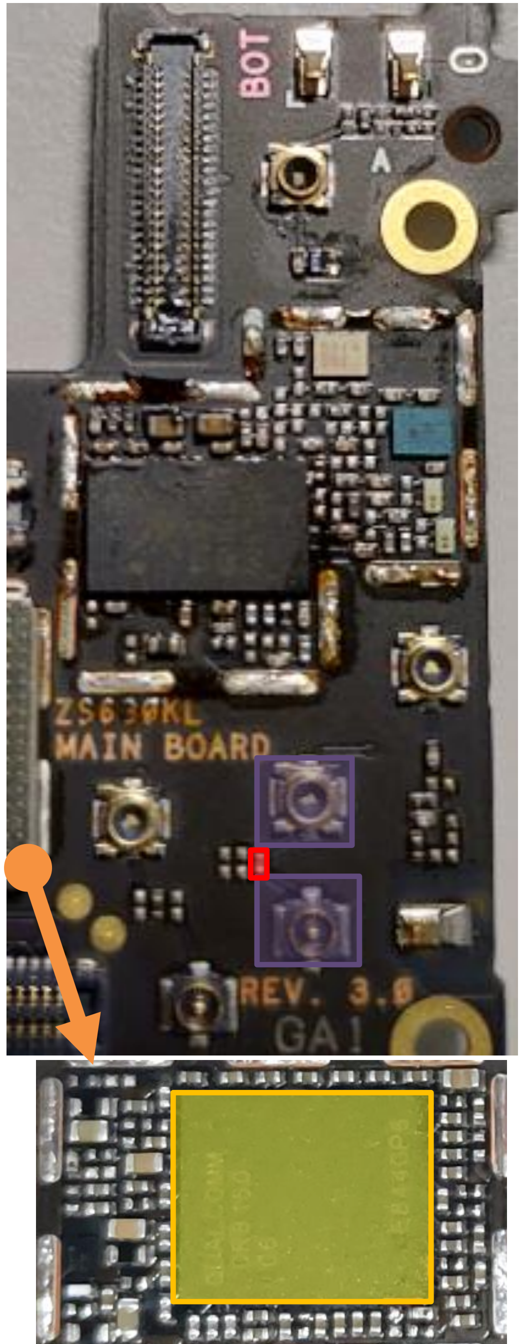
SDR8150
**U6201**

DPX
**U4820**

MB/HB QLN4640
**U4901**

# PRX Path : B2/PCS

bottom

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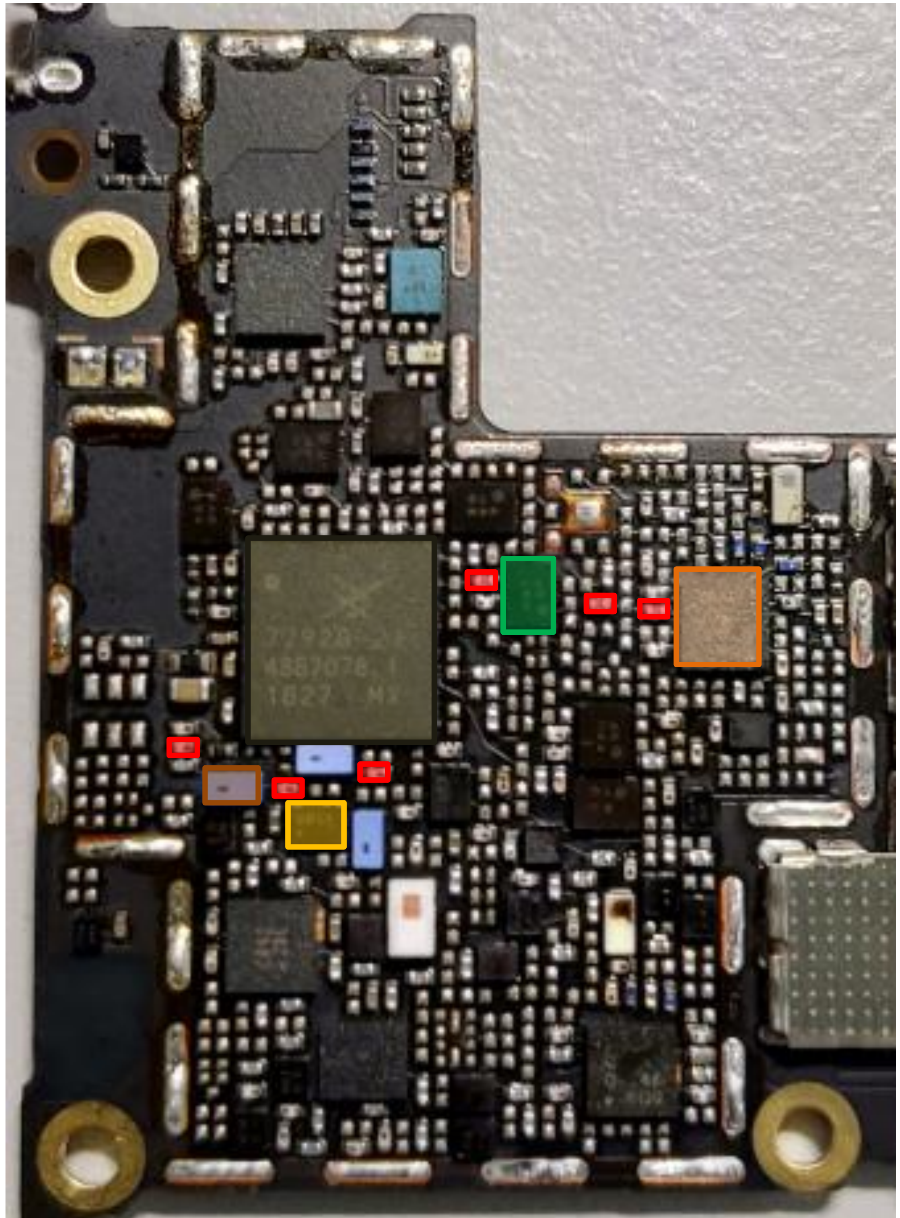


3G/4G MMPA
U4807

RF Connector LB/MB
CON4701 CON4704

SAW U4708

DPDT U4715

ASM+2G PA
U4701

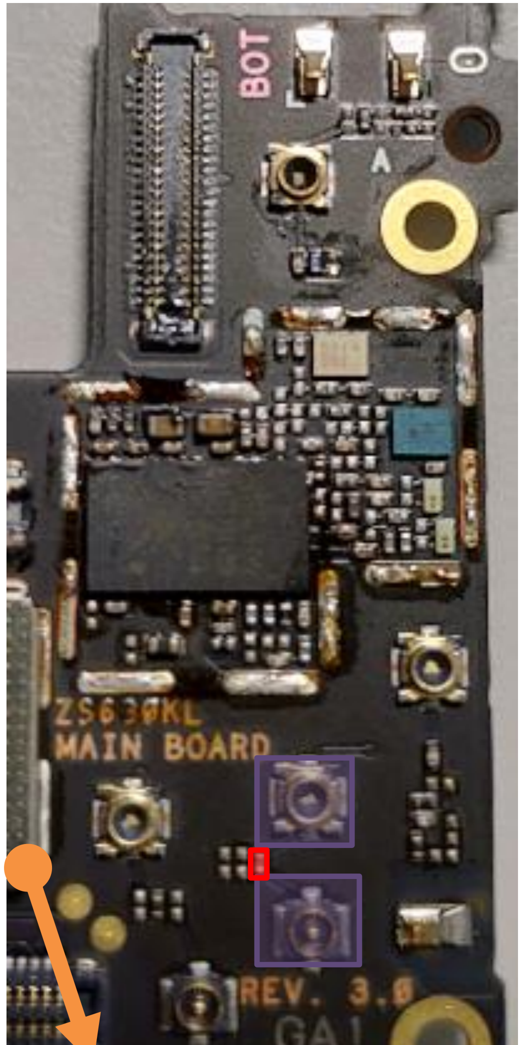
SDR8150
U6201

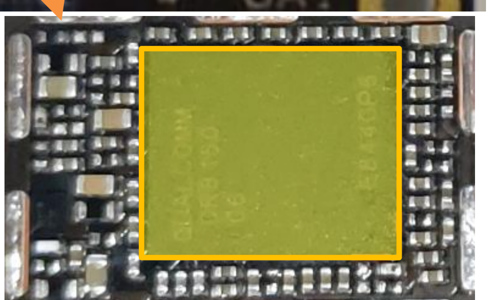
DPX
U4806

MB/HB QLN4640
U4901

# PRX Path : B3(TW)TriPx

bottom

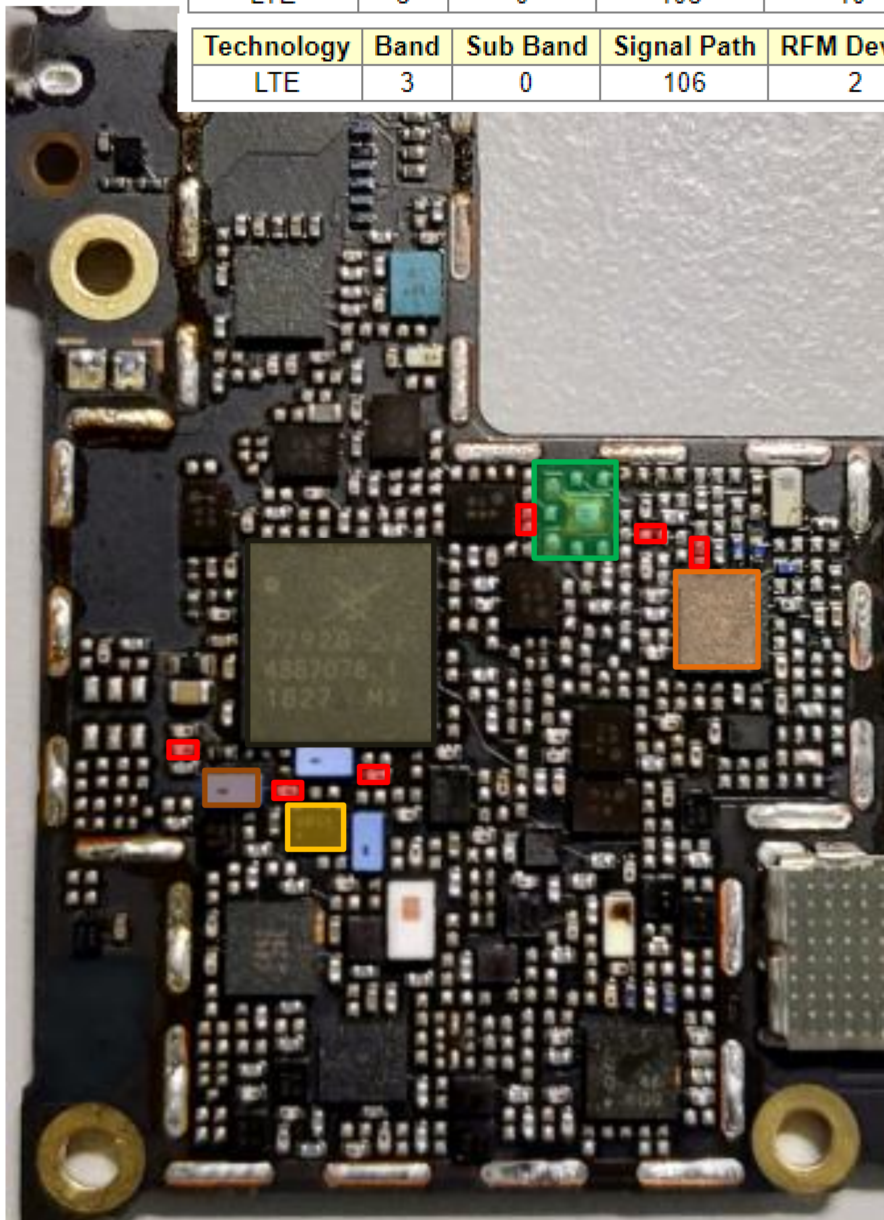


top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 103 | 10 | 0 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 106 | 2 | 0 | 0 | |



RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

SDR8150
**U6201**

MB/HB QLN4640
**U4901**

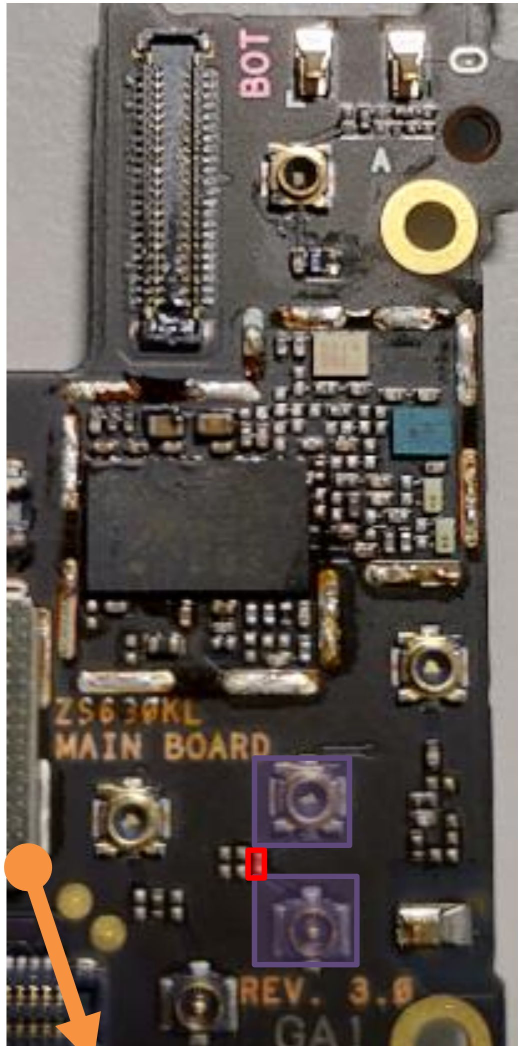
TriPX
**U4809**

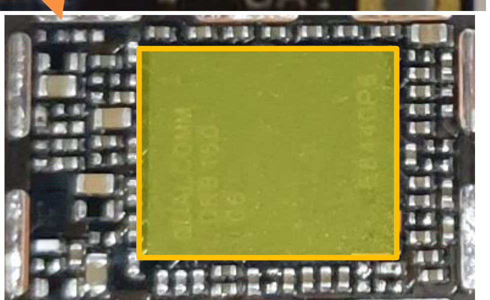
# PRX Path : B3/DCS(TW)

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 112 | 10 | 0 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 113 | 2 | 0 | 0 | |

bottom



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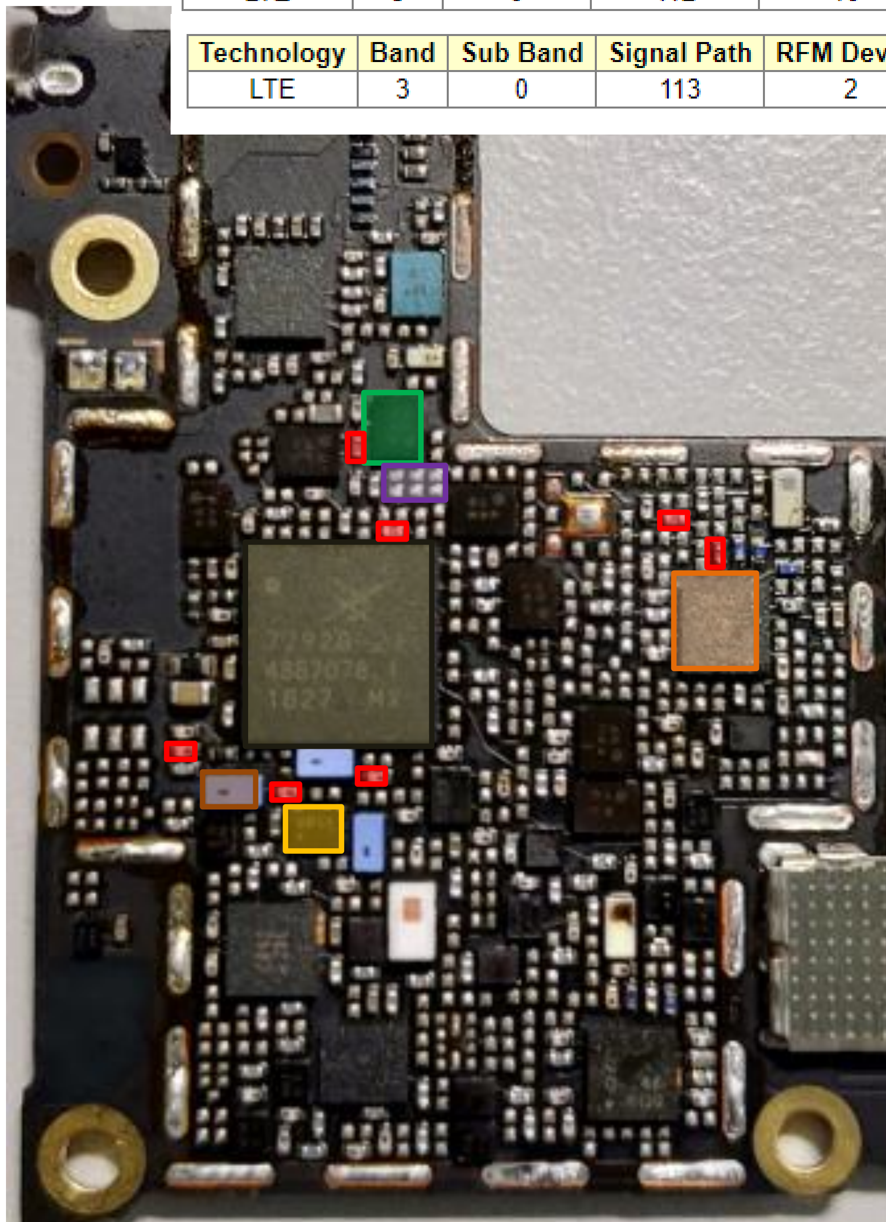


MB/HB QLN4640 **U4901**

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

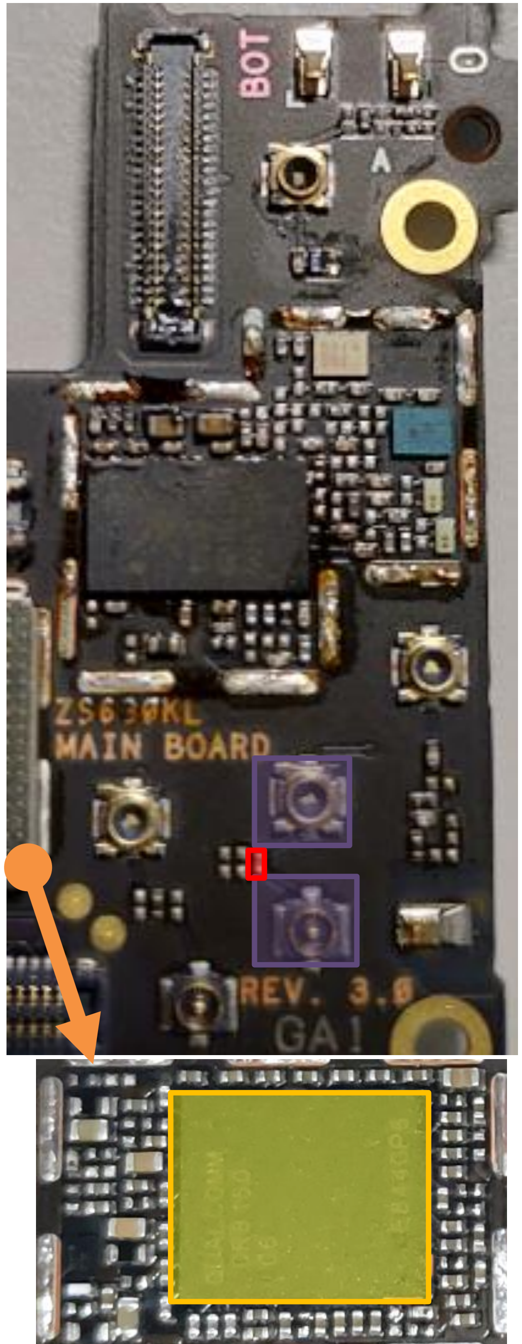
SDR8150 **U6201**

DPX **U4821**

DiPx **U4804**

# PRX Path : B3/DCS(WW)

bottom

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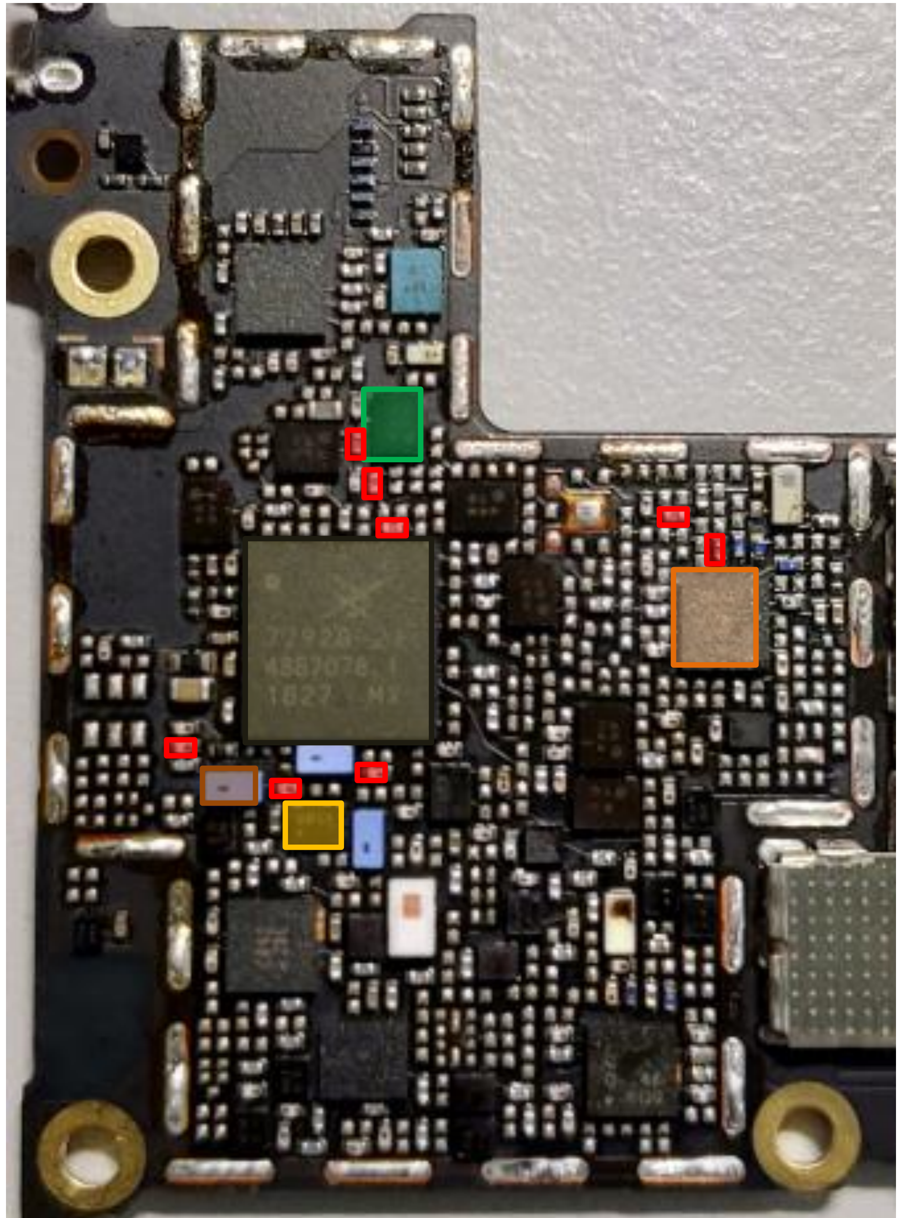


RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

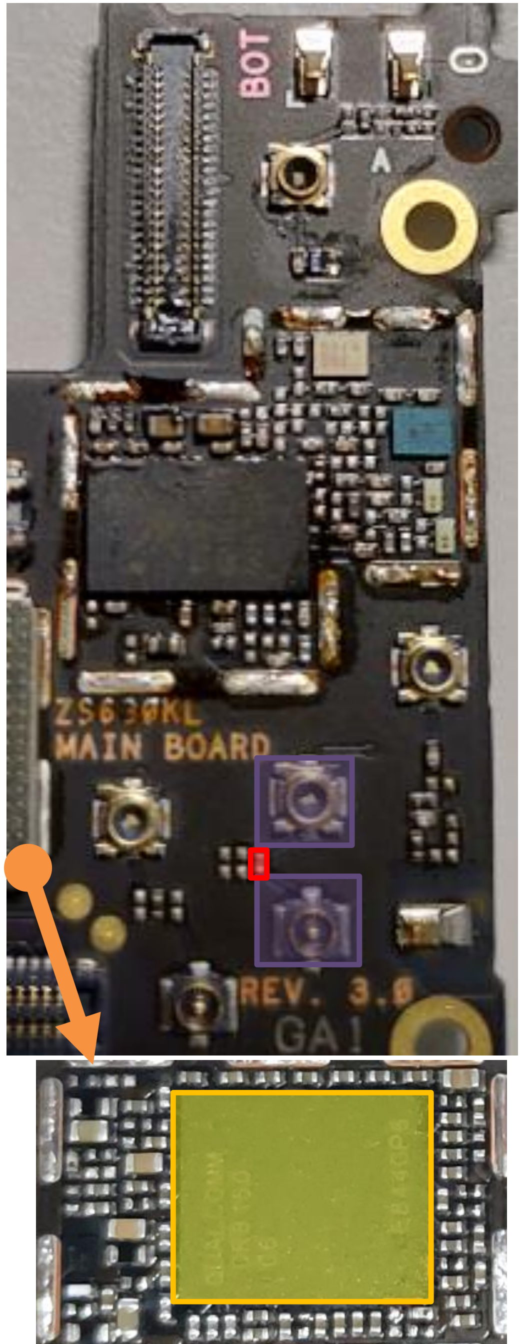
SDR8150
**U6201**

DPX
**U4821**

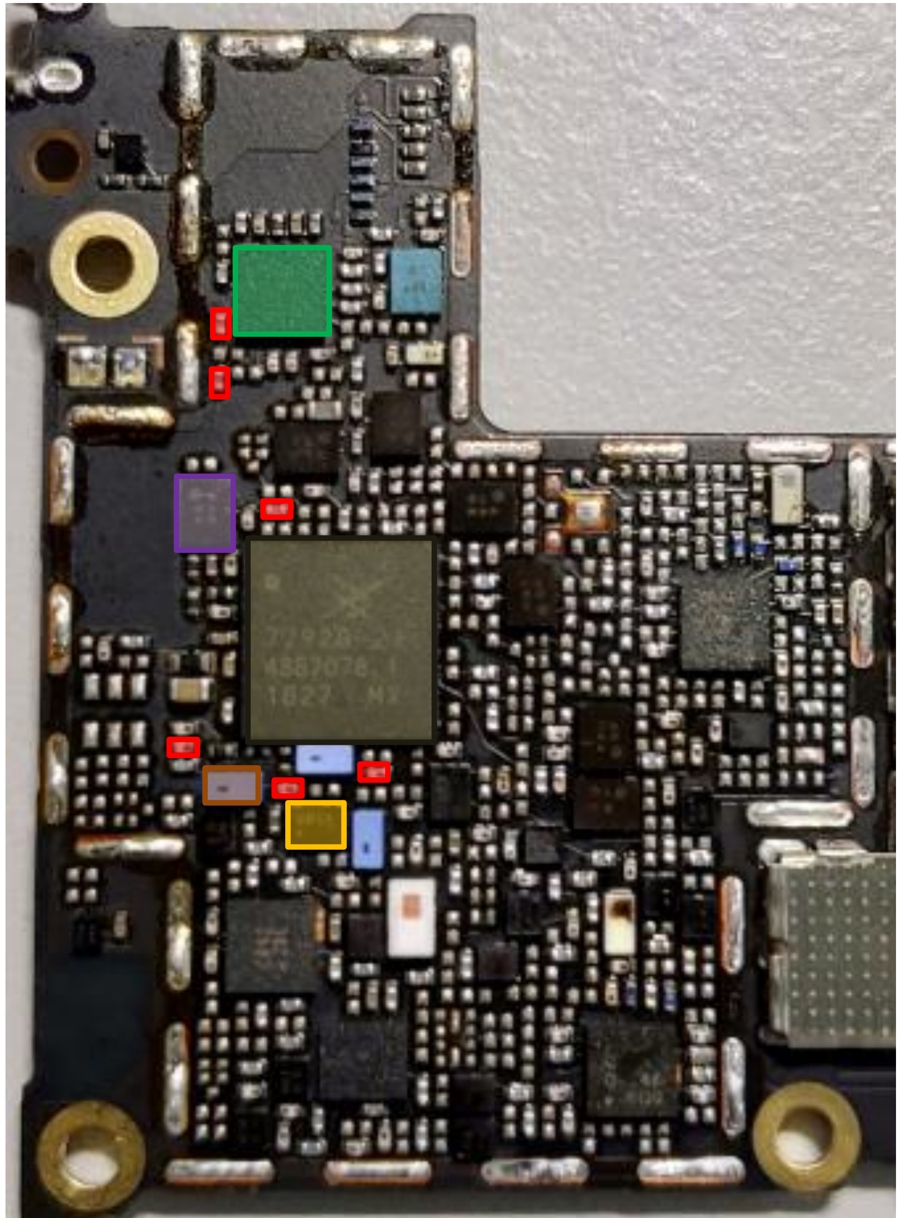
MB/HB QLN4640
**U4901**

# PRX Path : B5/B6/B18/B19/B26/GSM850(TW)
# B5/GSM850(WW)

bottom




top



LB QLN1020
**U4903**

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

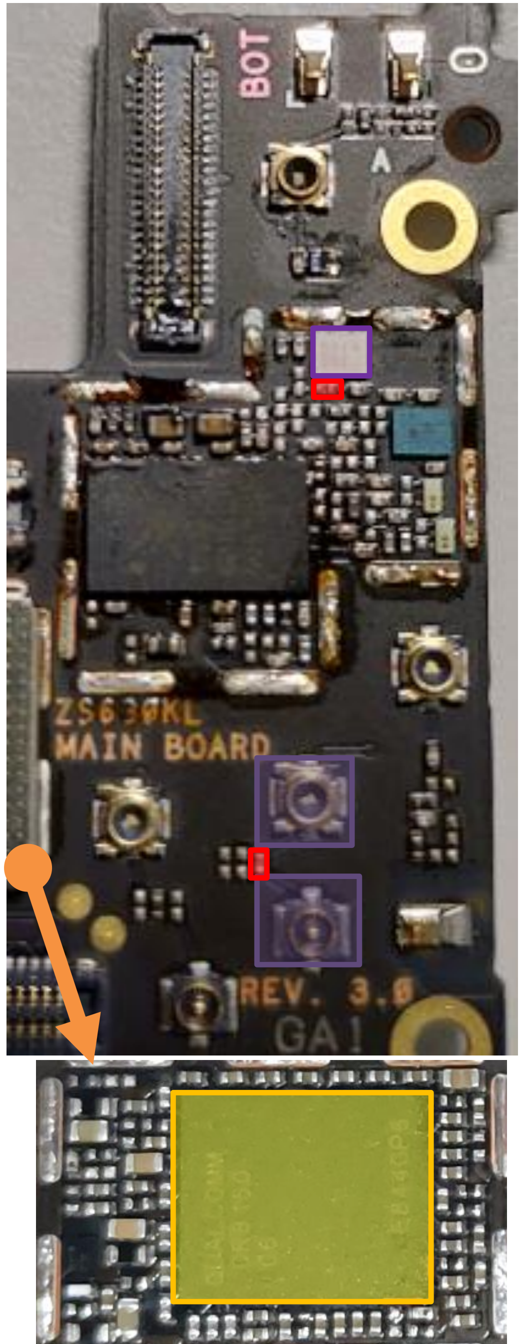
SDR8150
**U6201**

DPX
**U4813**

***different DPx between TW & WW**

# PRx Path : B8/GSM900

bottom

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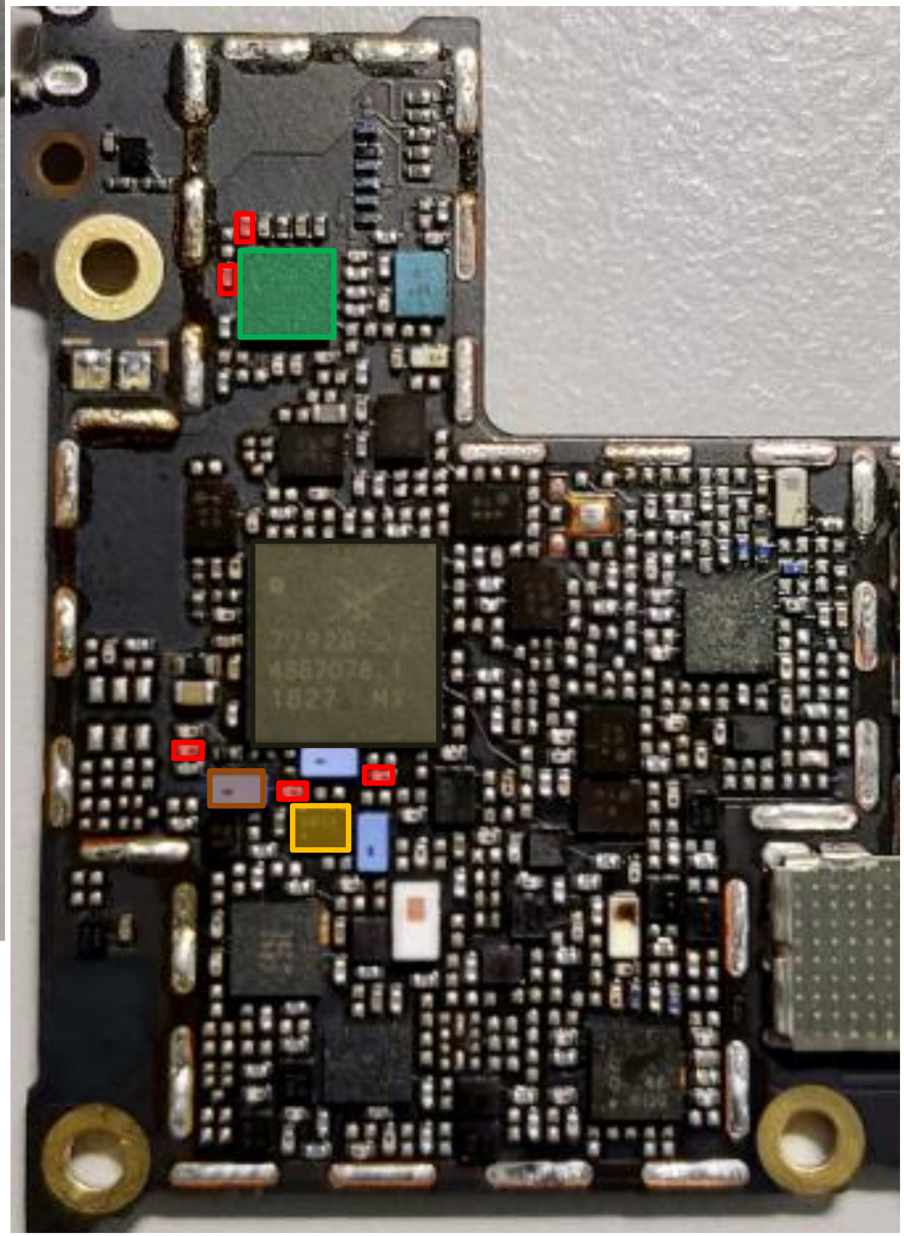


LB QLN1020 **U4903**

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

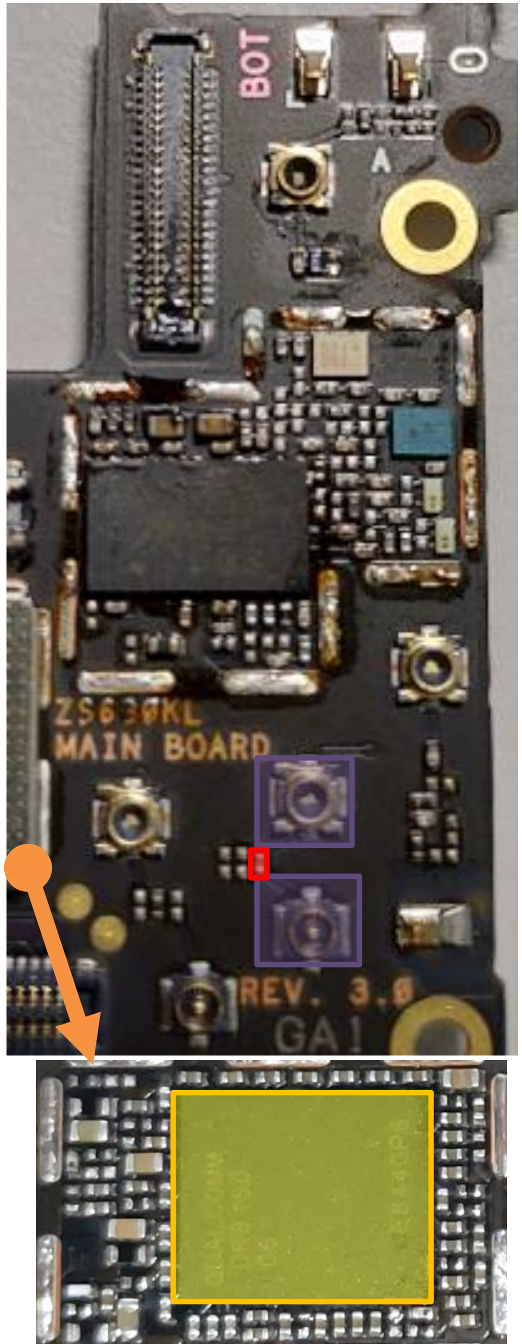
ASM+2G PA **U4701**

SDR8150 **U6201**

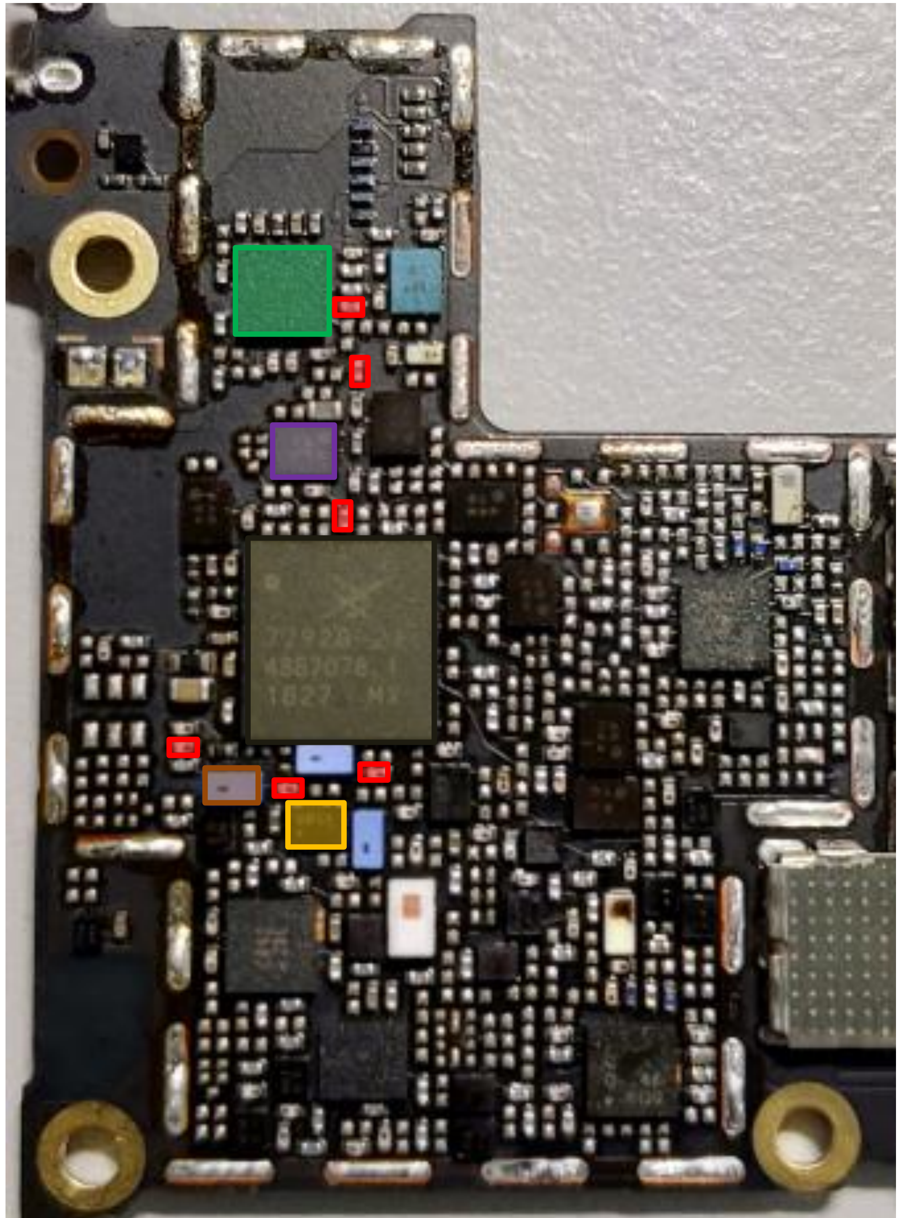
DPX **U4812**

# PRx Path : B20(WW)、B12/B17(NA)

bottom




top



LB QLN1020 **U4903**

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

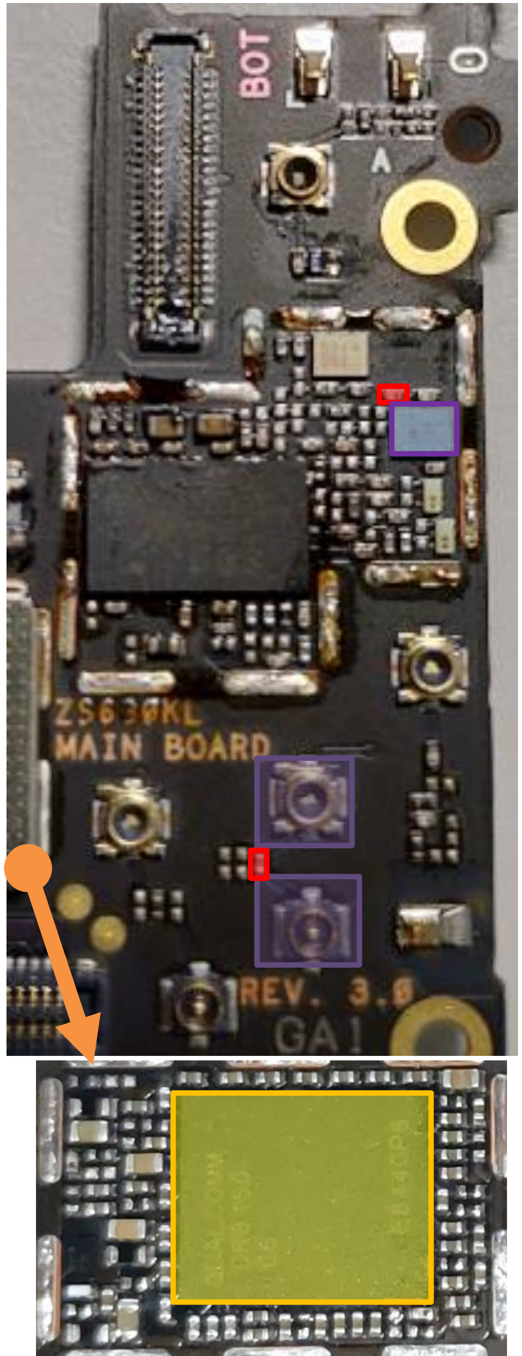
SDR8150 **U6201**

DPX **U4819**

***different DPx between NA & WW**

# PRx Path : B28a

bottom

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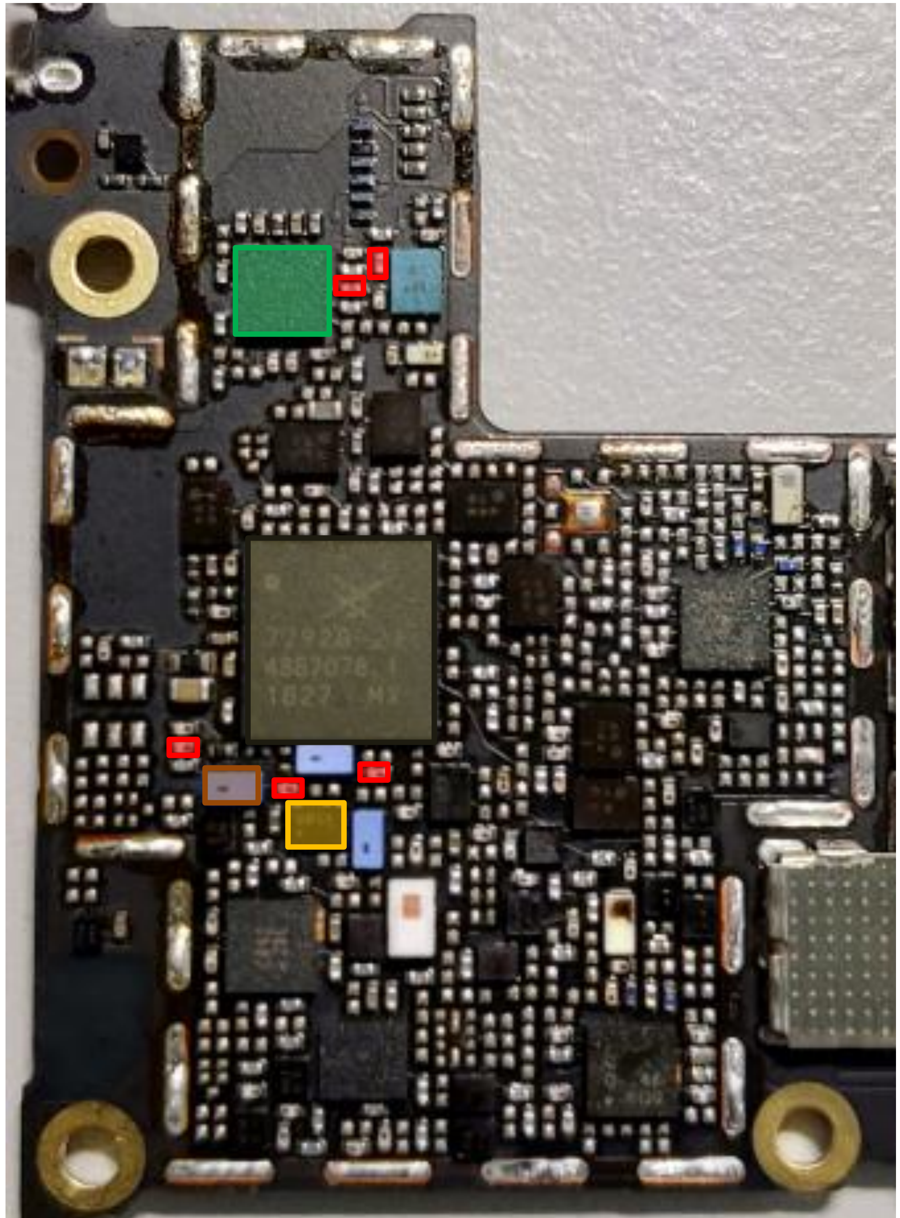


LB QLN1020
**U4903**

RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

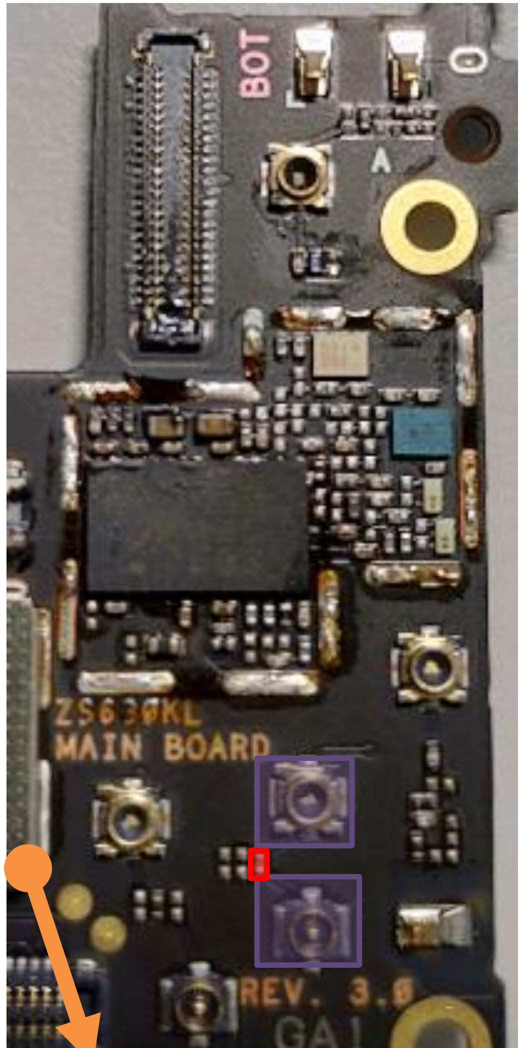
ASM+2G PA
**U4701**

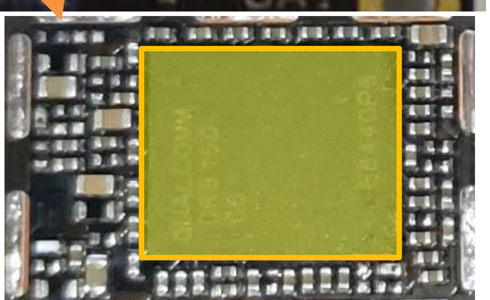
SDR8150
**U6201**

DPX
**U4811**

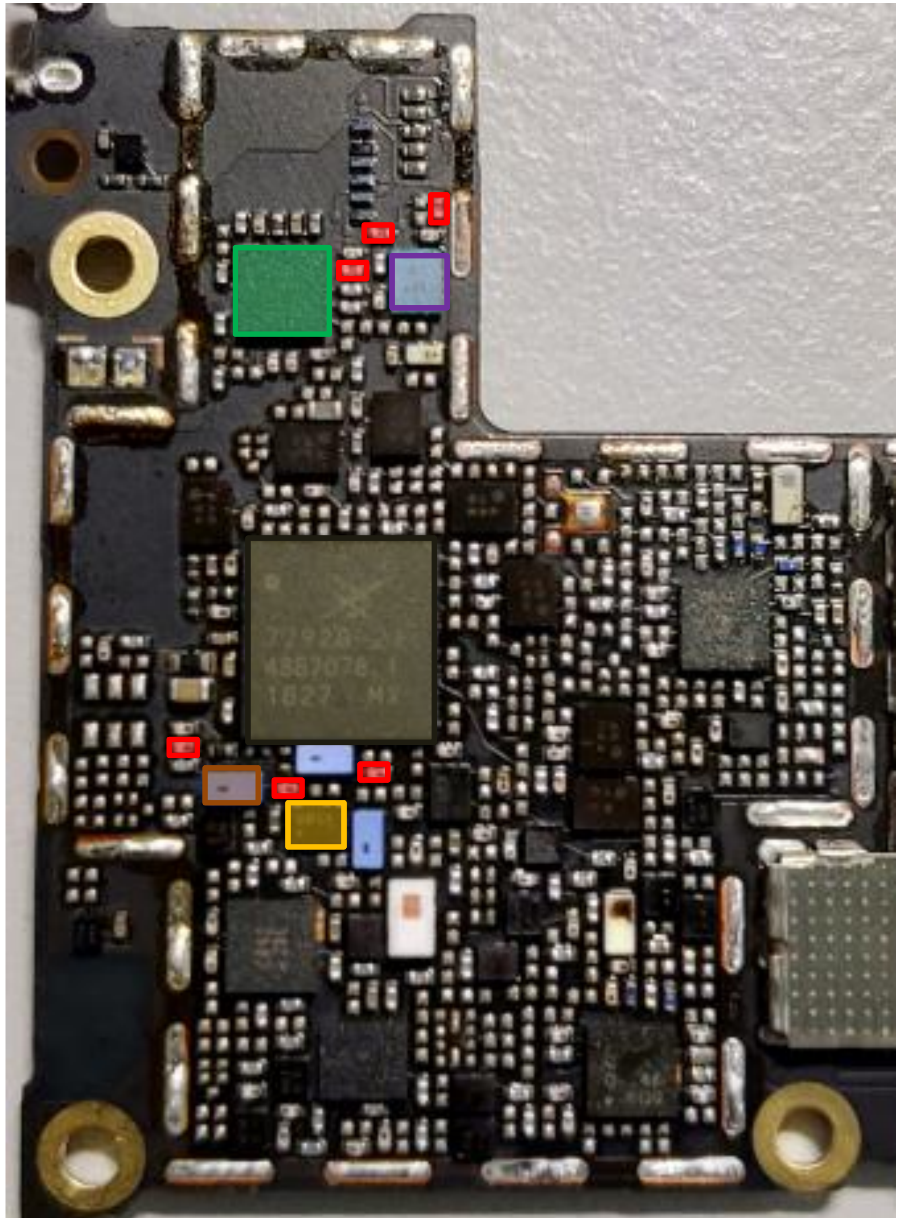
# PRx Path : B28b

bottom



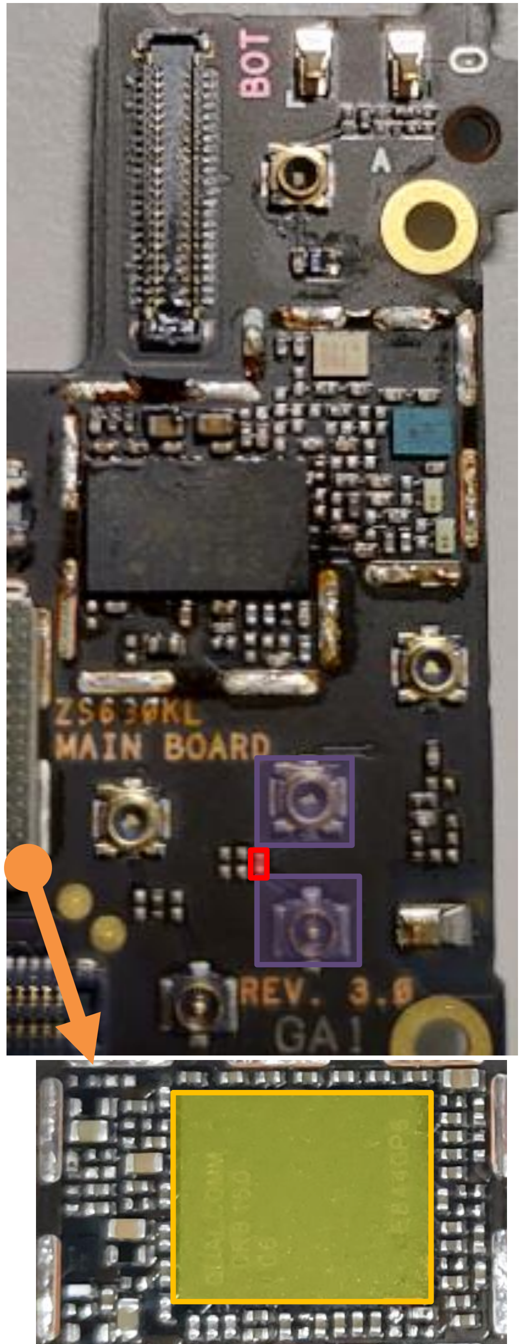
top



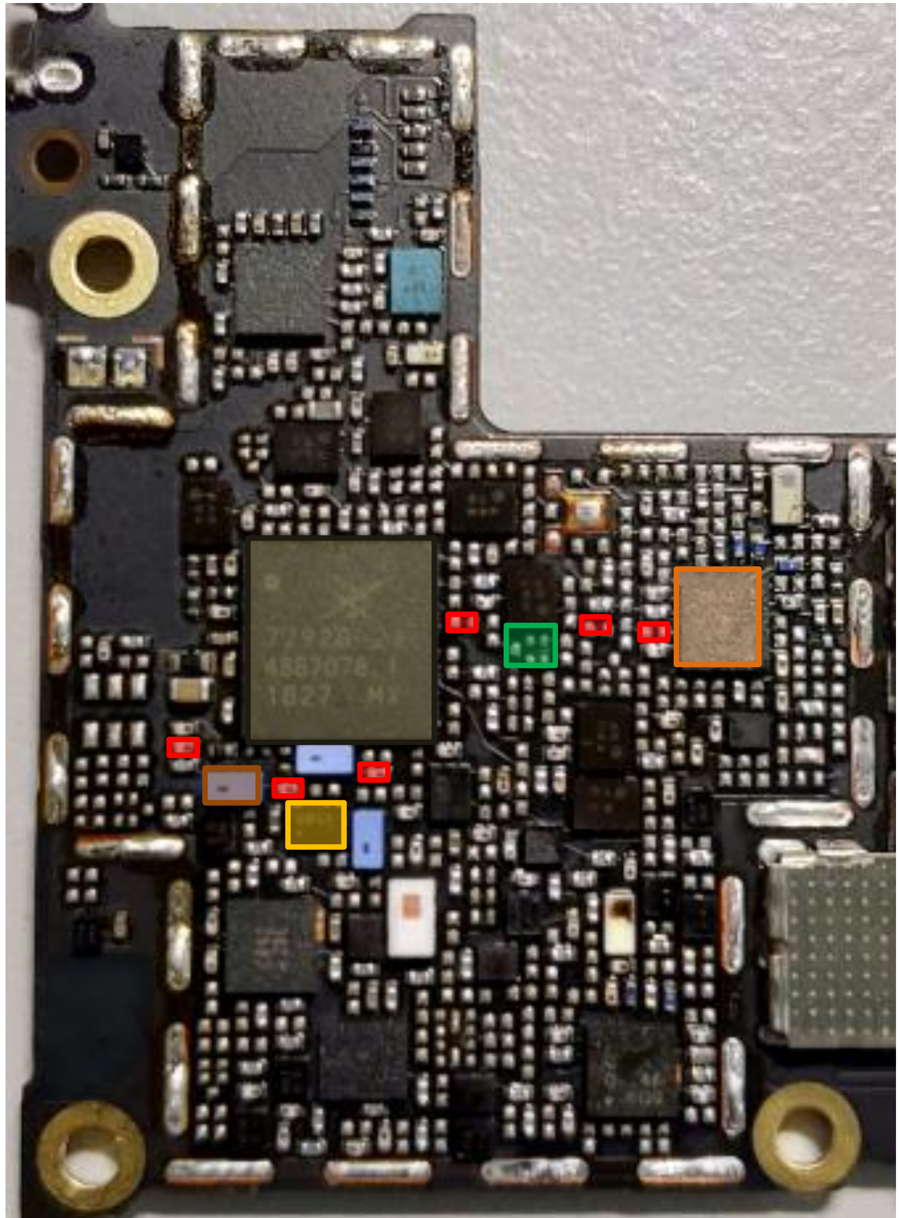
- LB QLN1020 **U4903**
- RF Connector LB/MB **CON4701 CON4704**
- SAW **U4708**
- DPDT **U4715**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- DPX **U4808**

# PRX Path : B39(TW)

bottom




top



RF Connector LB/MB
**CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA
**U4701**

SDR8150
**U6201**

MB/HB QLN4640
**U4901**

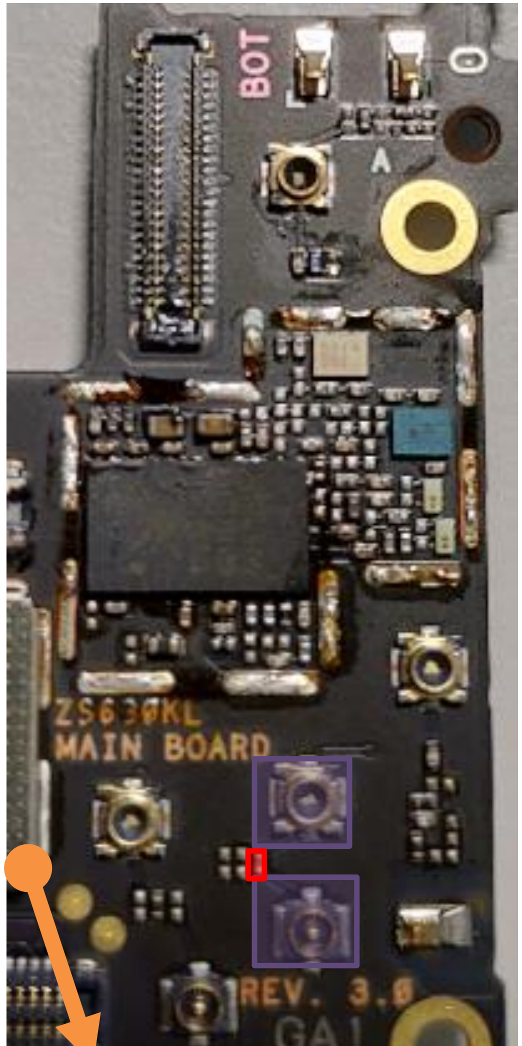
SAW
**U4902**

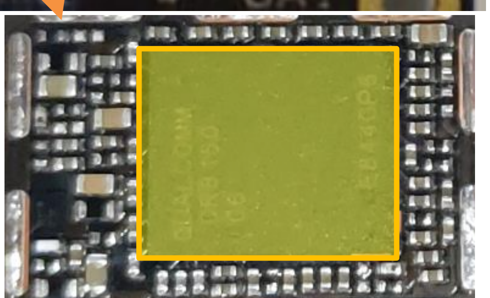
# PRX Path : B7 MIMO(TW)

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 39 | 2 | 0 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 41 | 10 | 0 | 4 | |

bottom



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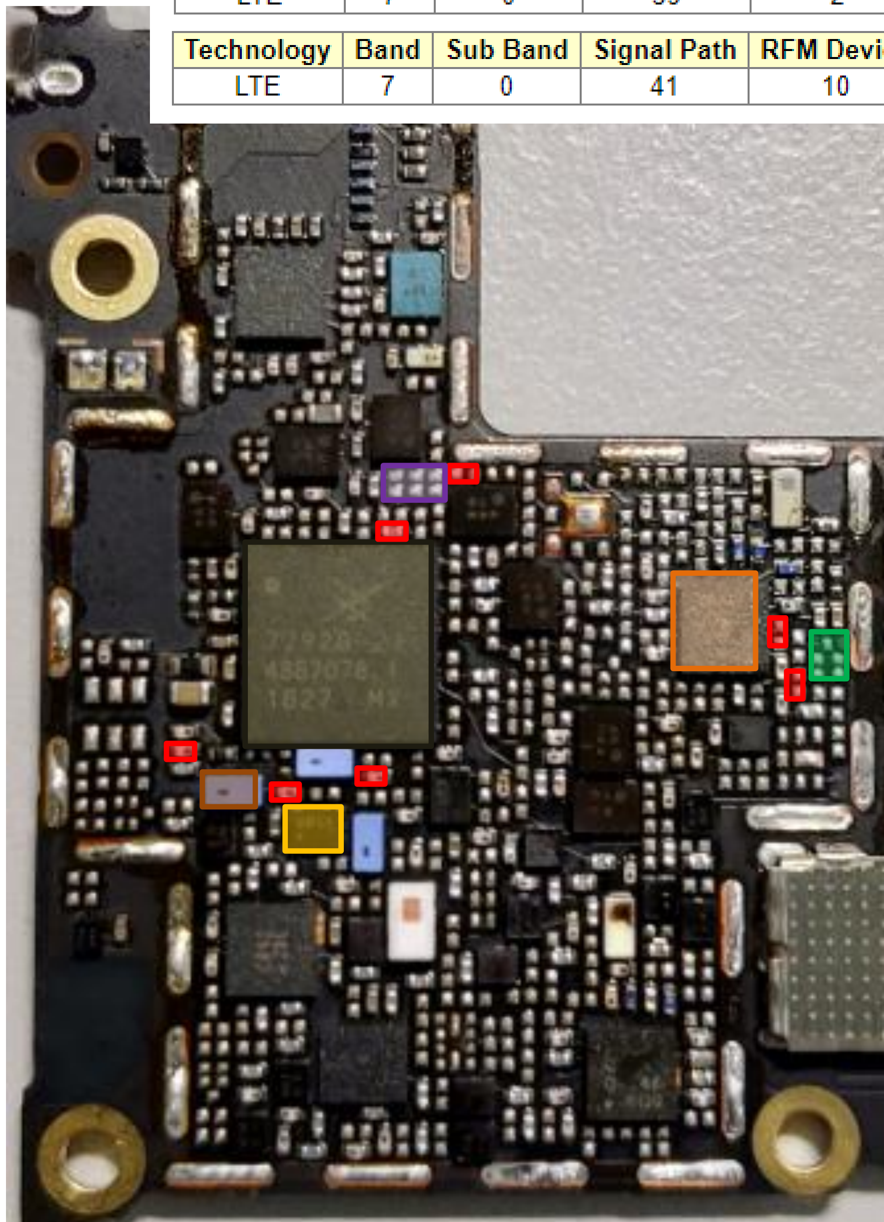


MB/HB QLN4640 **U4901**

RF Connector LB/MB **CON4701 CON4704**

SAW **U4708**

DPDT **U4715**

ASM+2G PA **U4701**

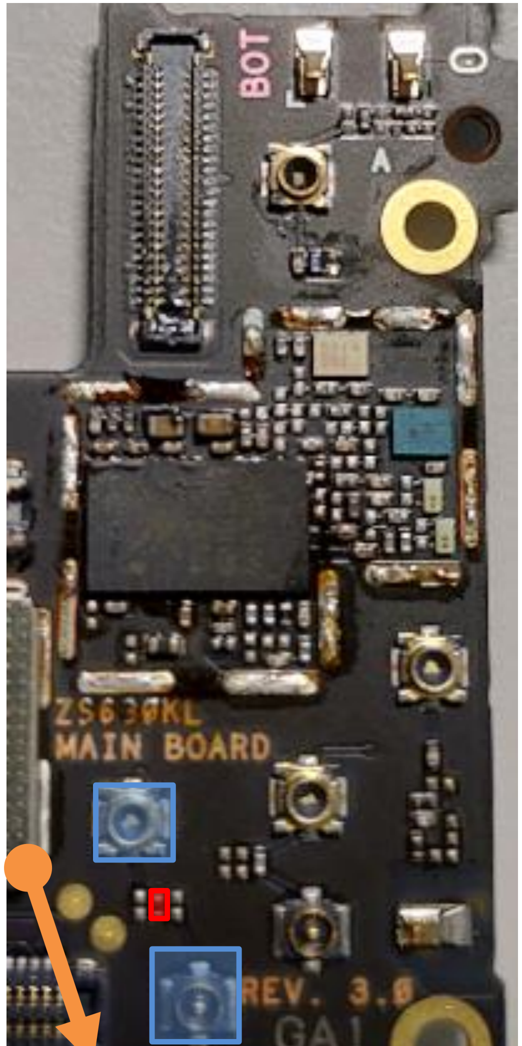
SDR8150 **U6201**

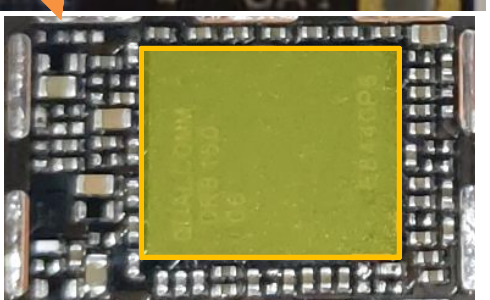
SAW **U4805**

DiPx **U4804**

# HB PRx Common Path

bottom



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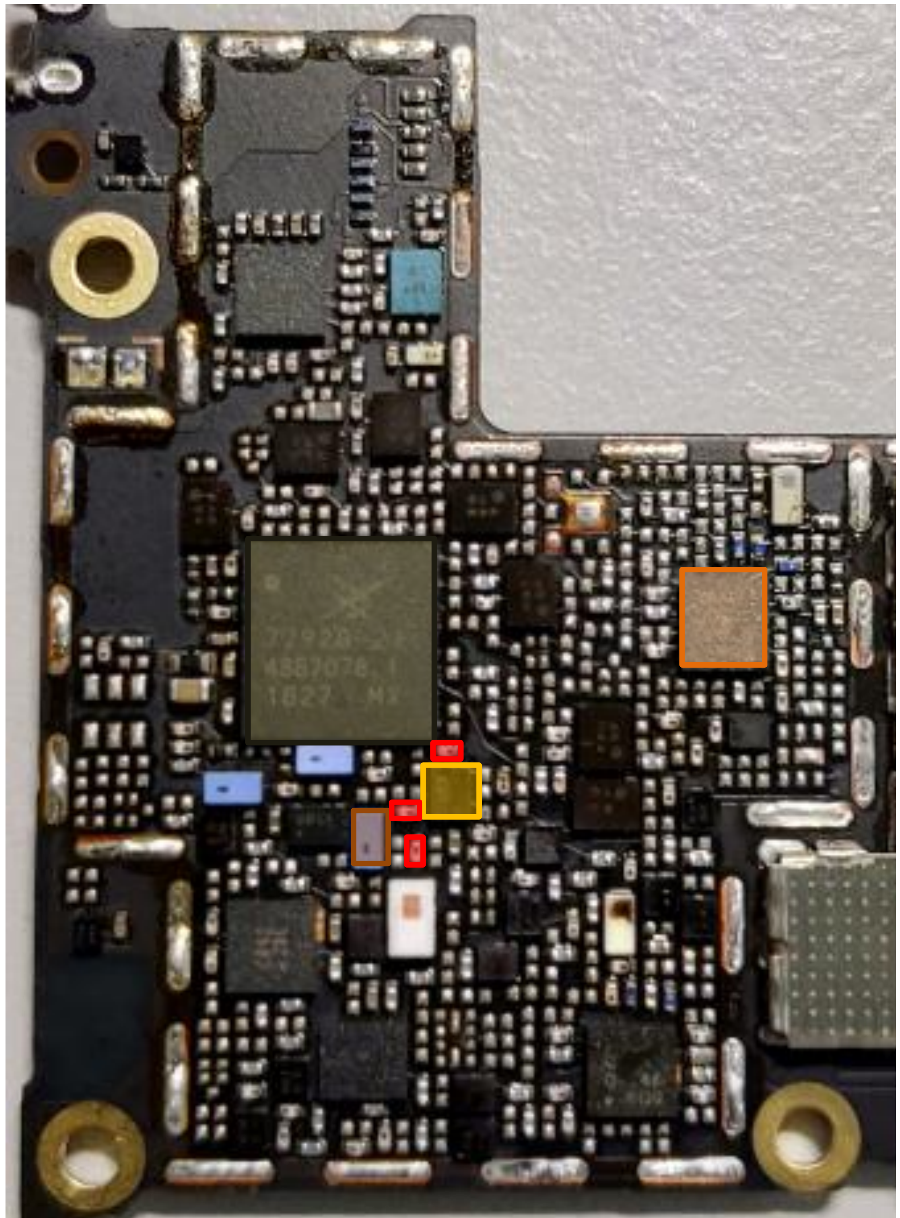


HB : B7/B38/B40/B41

RF Connector HB
**CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

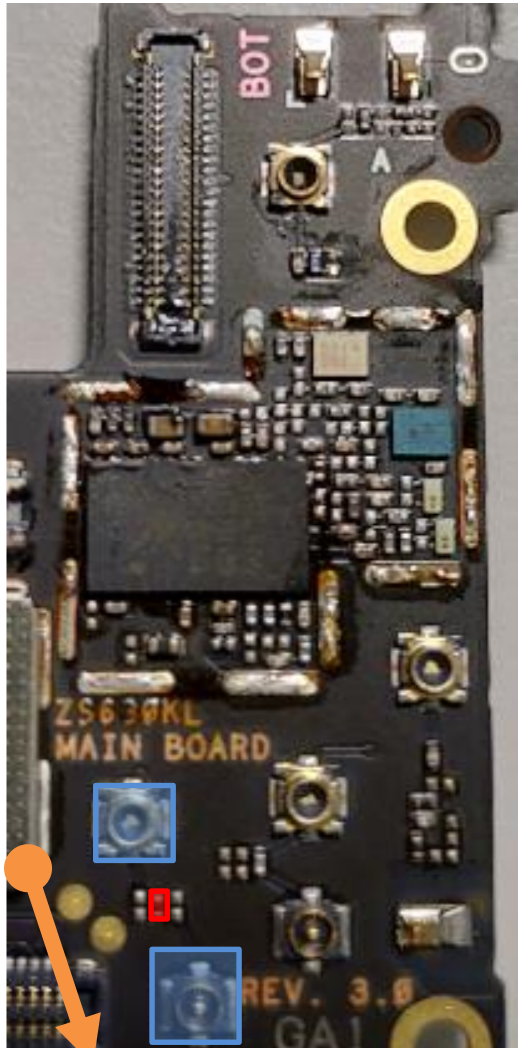
ASM+2G PA
**U4701**

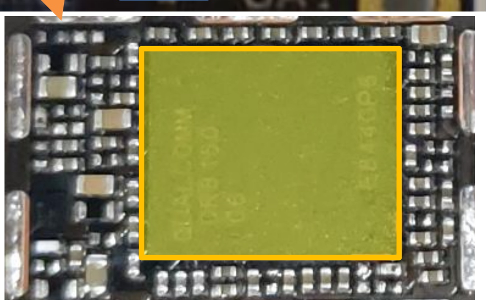
SDR8150
**U6201**

MB/HB QLN4640
**U4901**

# PRX Path : B7(TW)

bottom

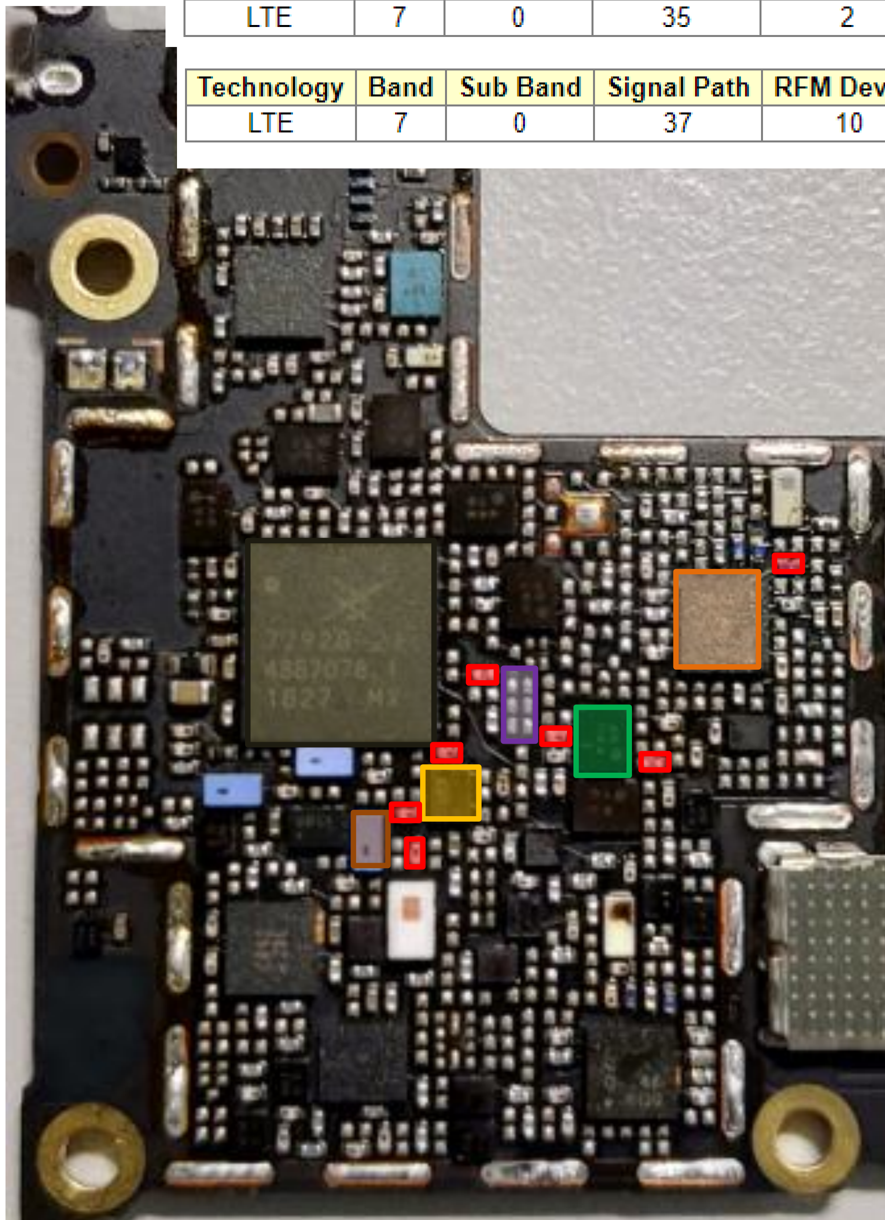


top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 35 | 2 | 4 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 37 | 10 | 4 | 4 | |



MB/HB QLN4640 **U4901**

RF Connector HB **CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

ASM+2G PA **U4701**

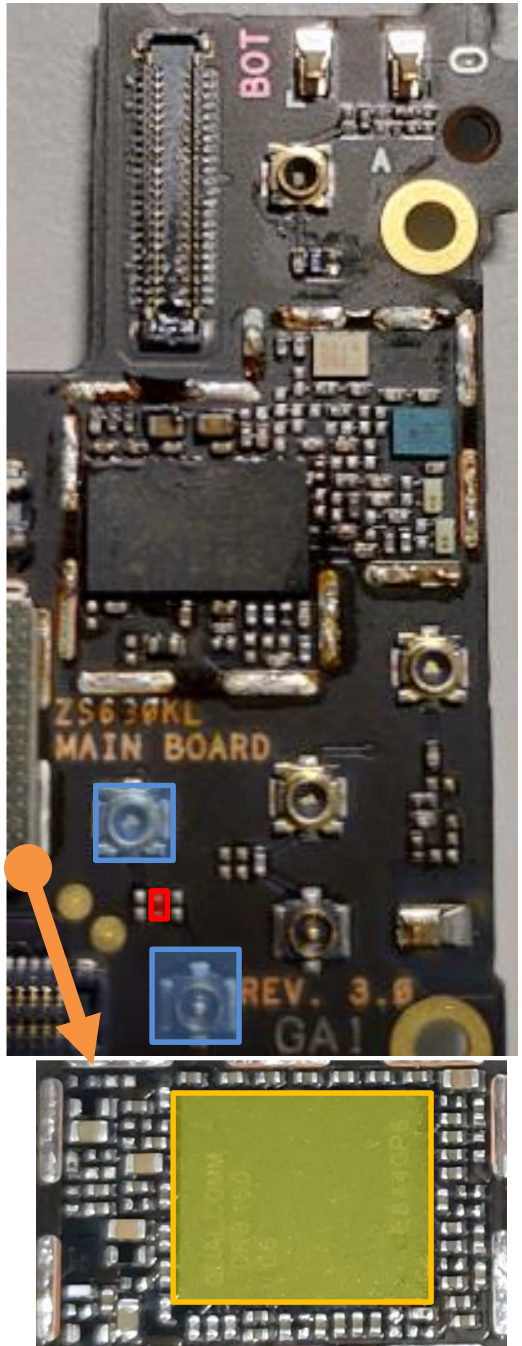
SDR8150 **U6201**

DPX **U4824**

DiPx **U4802**

# PRX Path : B7(WW)

bottom

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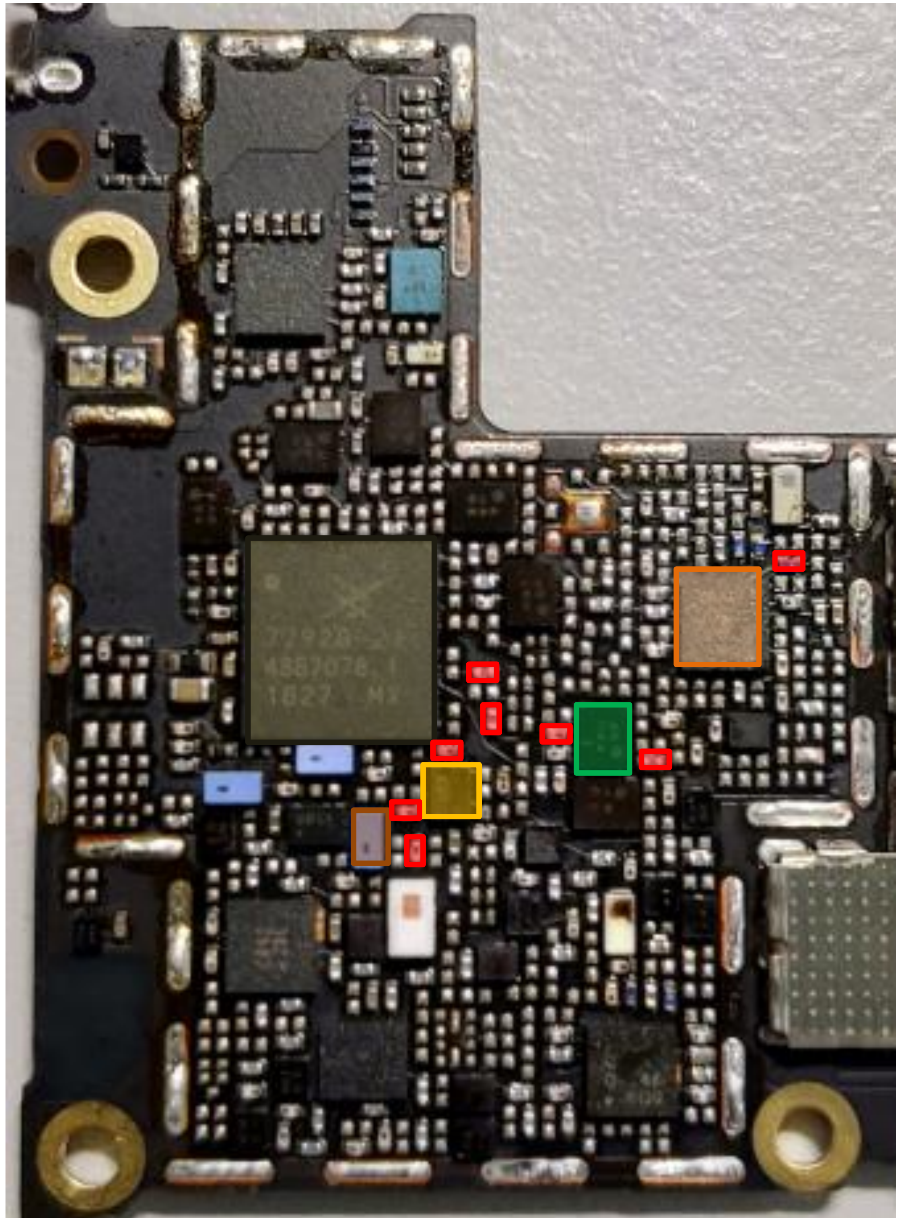


MB/HB QLN4640
**U4901**

RF Connector HB
**CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

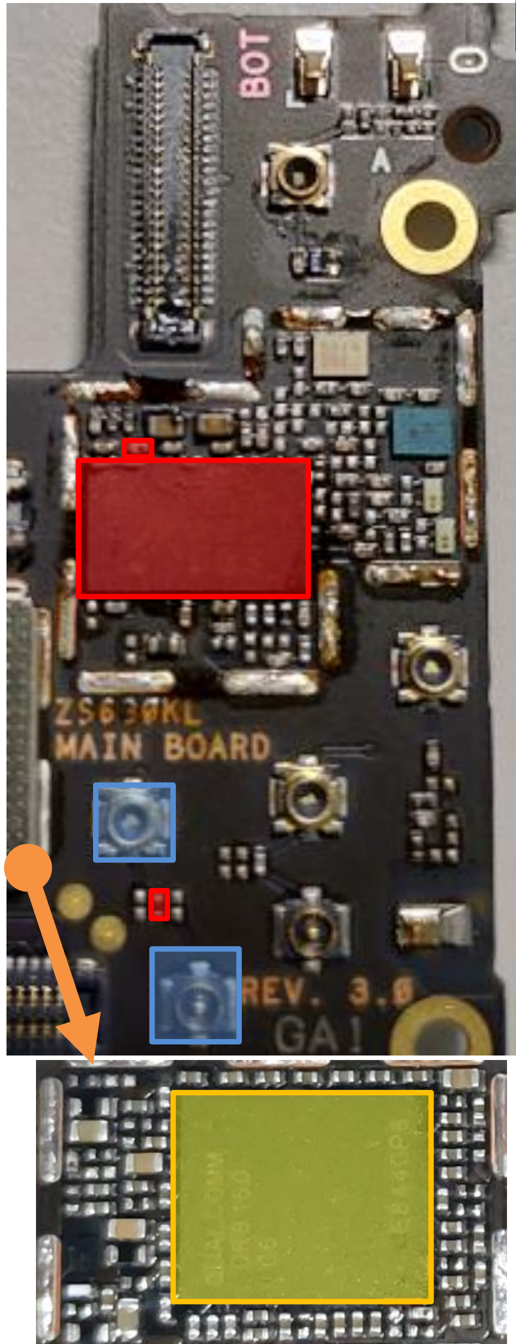
ASM+2G PA
**U4701**

SDR8150
**U6201**

DPX
**U4824**

# PRX Path : B38/B41(TW)、B40(WW)

bottom

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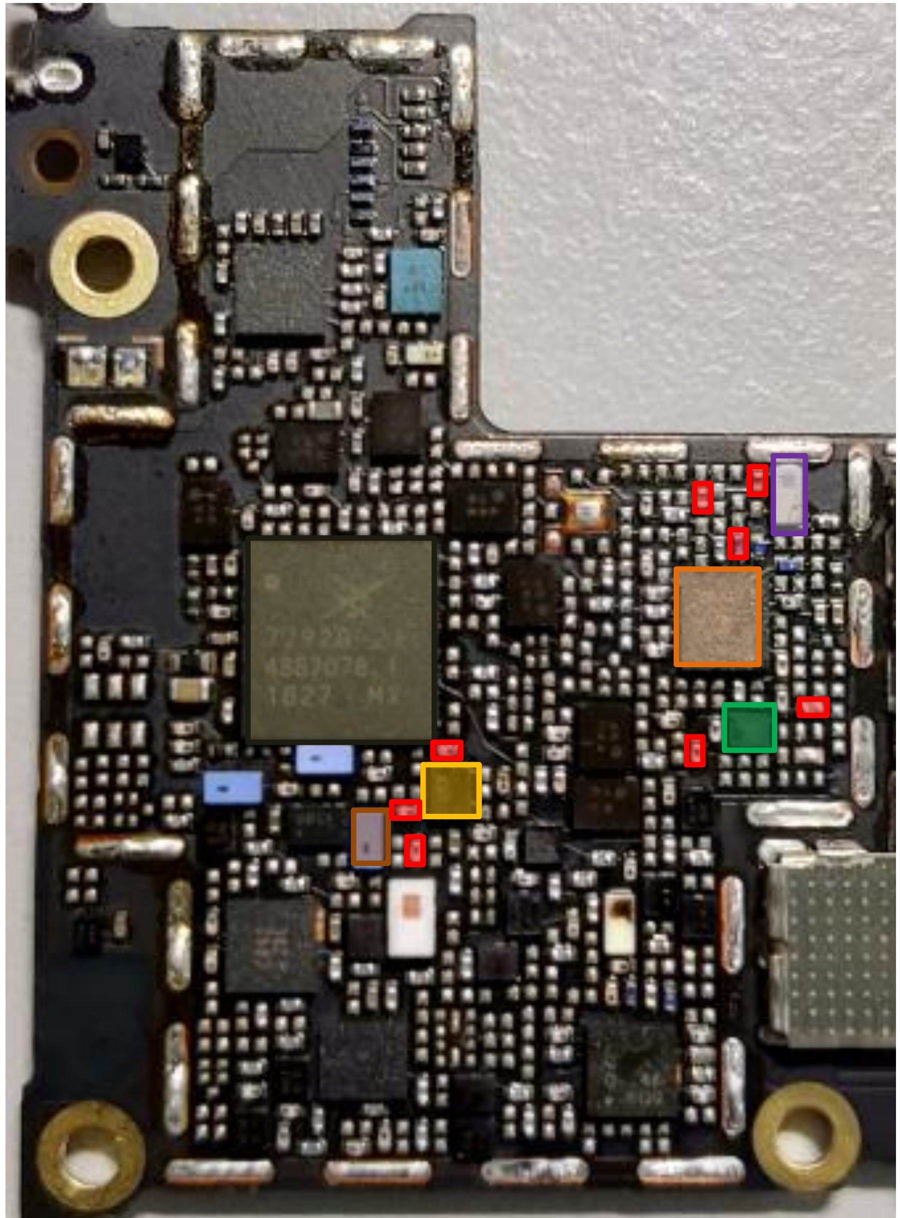


3G/4G MMPA **U4807**

MB/HB QLN4640 **U4901**

RF Connector HB **CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

ASM+2G PA **U4701**

SDR8150 **U6201**

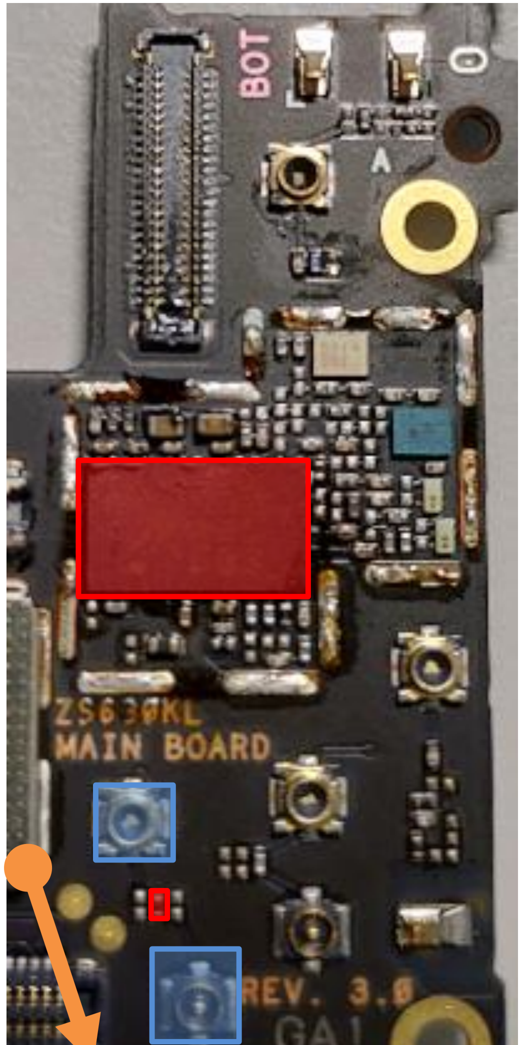
SAW **U4816**

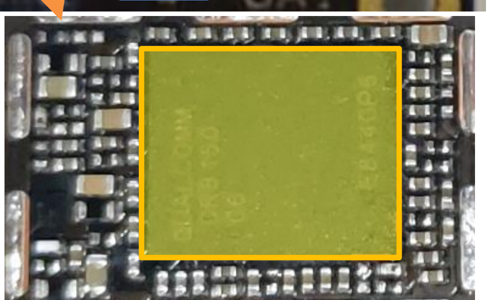
Rx SAW **U4905**

***different SAW between TW & WW**

# PRX Path : B38/B41(WW)

bottom



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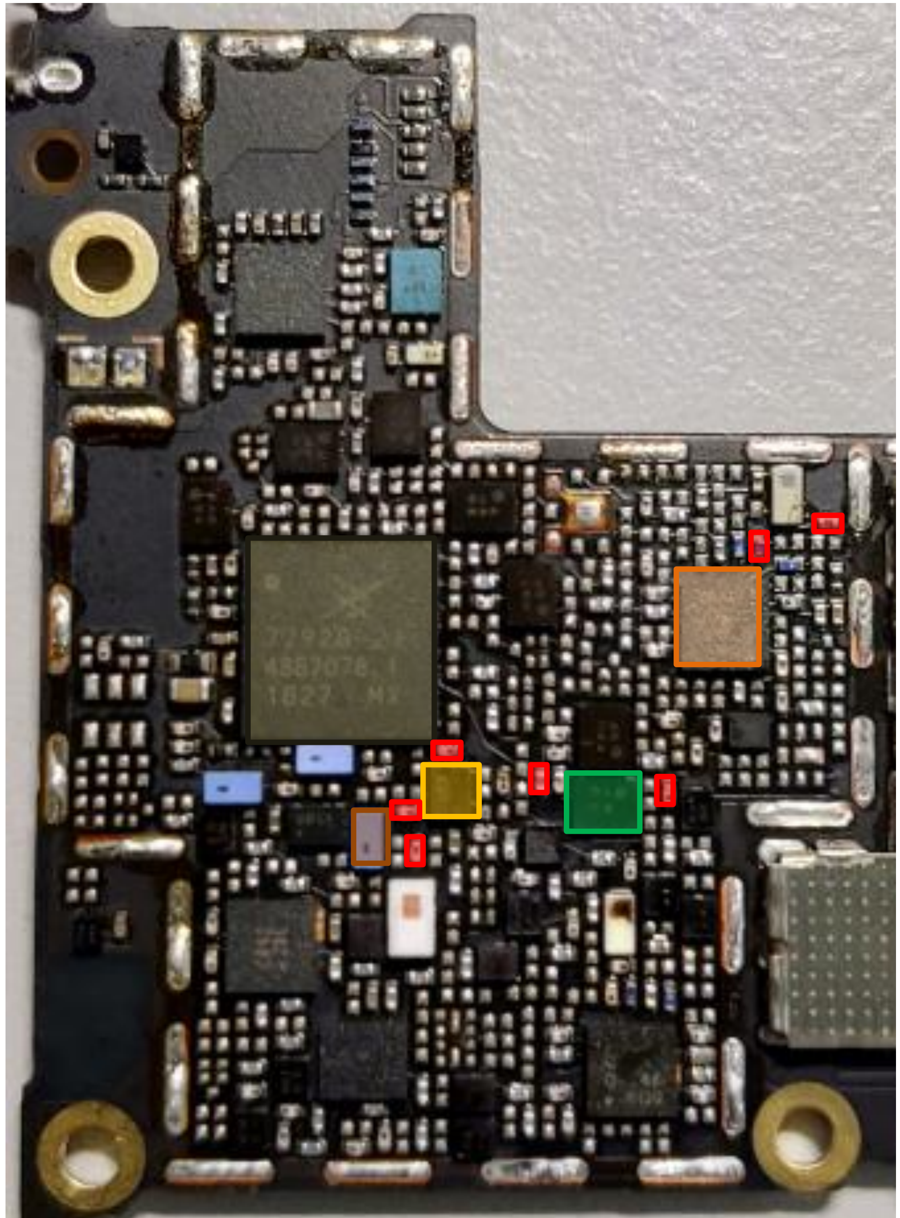
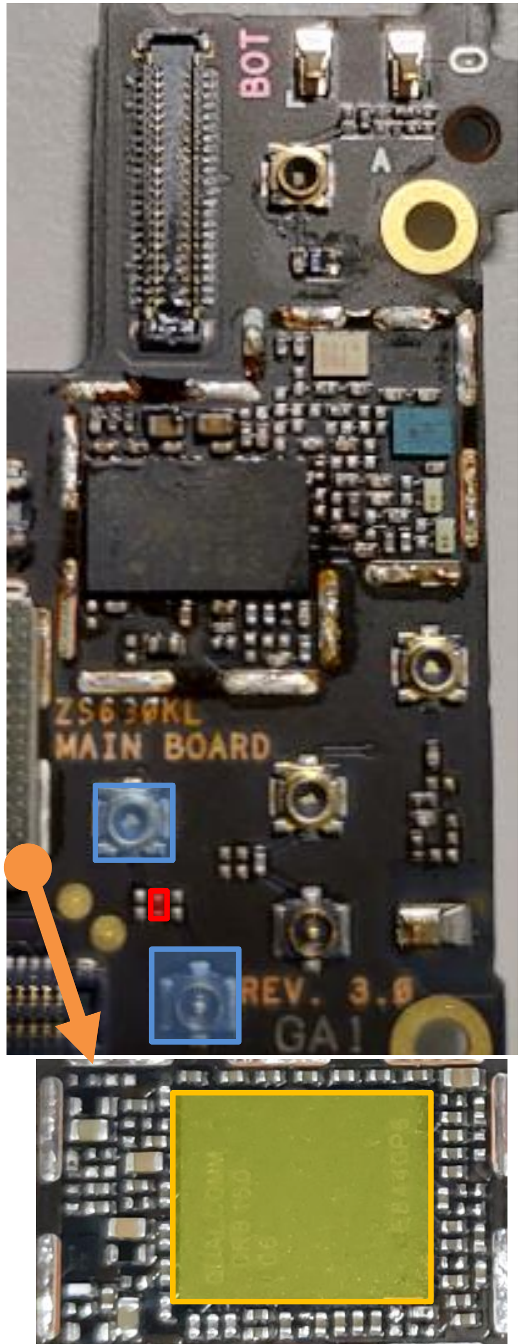


- 3G/4G MMPA **U4807**
- MB/HB QLN4640 **U4901**
- RF Connector HB **CON4702 CON4703**
- SAW **U4709**
- DPDT **U4716**
- ASM+2G PA **U4701**
- SDR8150 **U6201**
- SAW **U4801**

# PRX Path : B3 MIMO(TW)

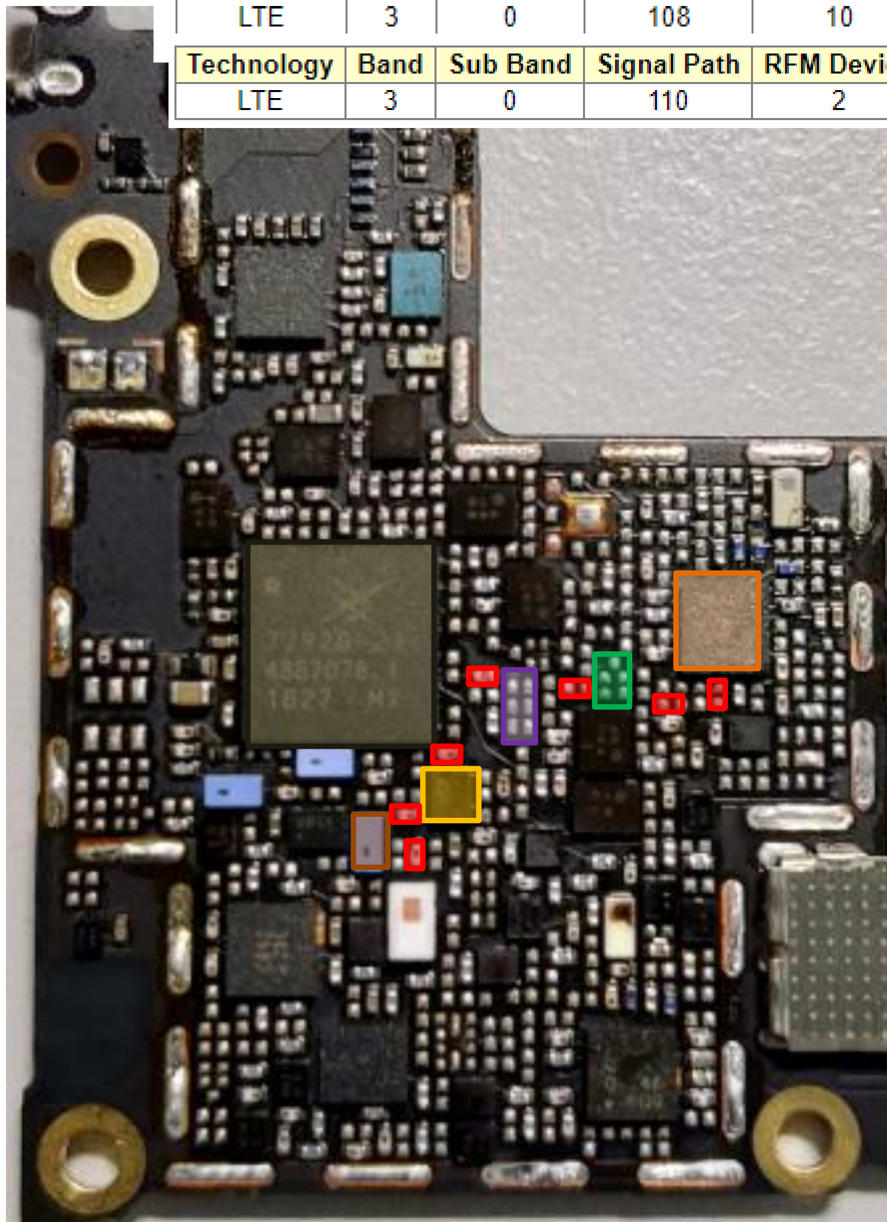
bottom



top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 108 | 10 | 4 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 110 | 2 | 4 | 0 | |



MB/HB QLN4640 **U4901**

RF Connector HB **CON4702 CON4703**

SAW **U4709**

DPDT **U4716**

ASM+2G PA **U4701**

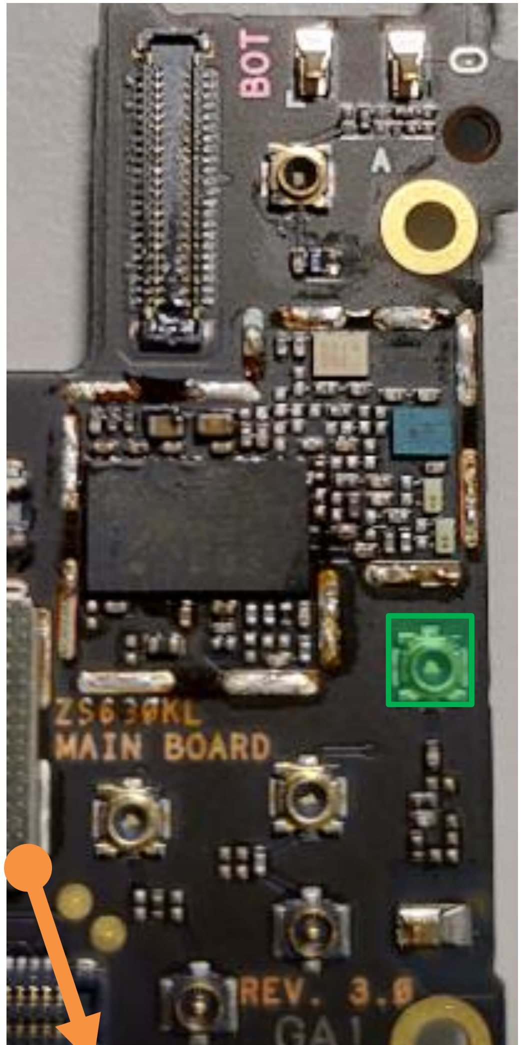
SDR8150 **U6201**

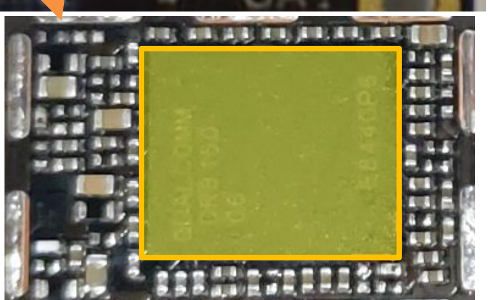
SAW **U4803**

DiPx **U4802**

# LB DRx Common Path(TW)

bottom



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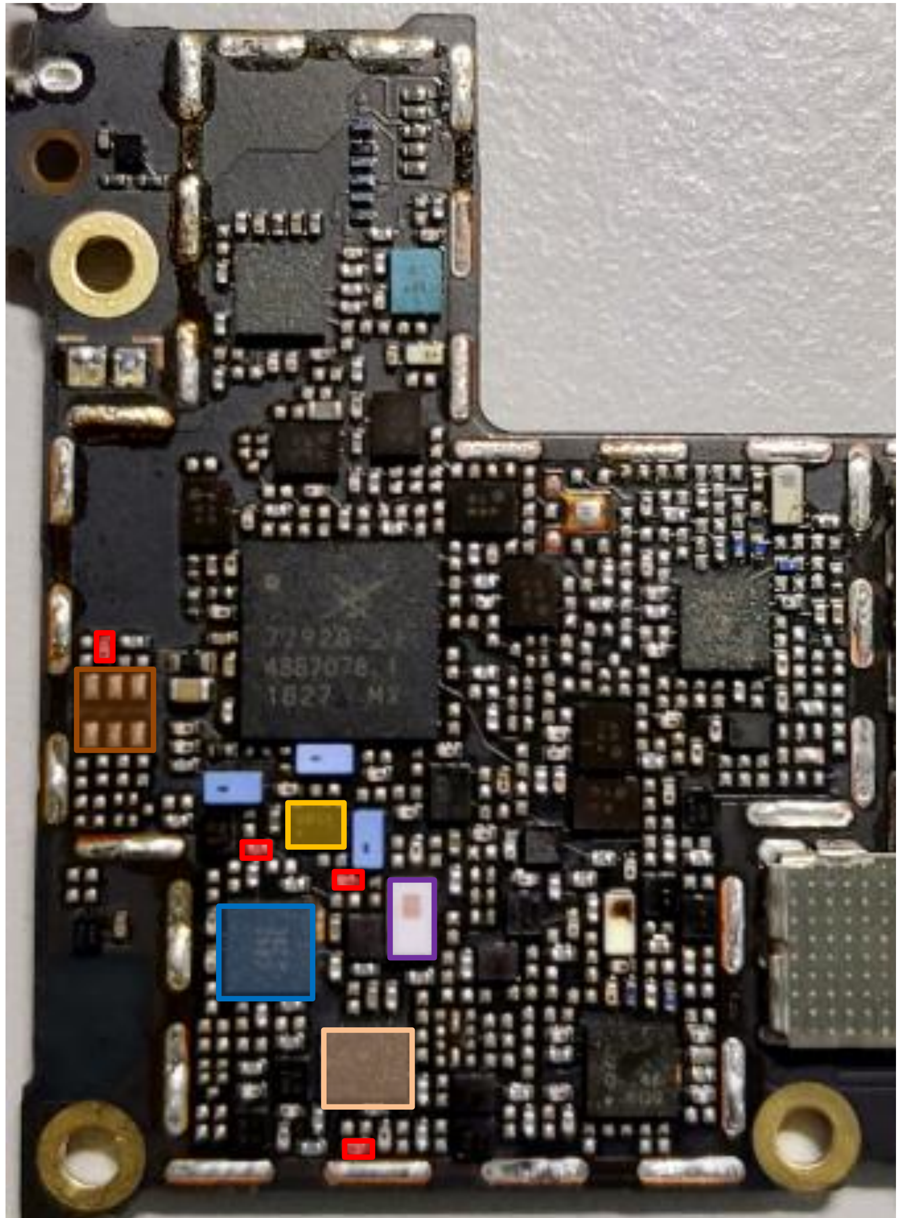


LB: B5/B8/B18/B19/
B26/B28A/B28B

DRX LB QLN1020
**U5131**

Div. RF connector
LB/MB
**CON4705**

DPDT **U4715**

L/MB Div. RX SW
**U4704**

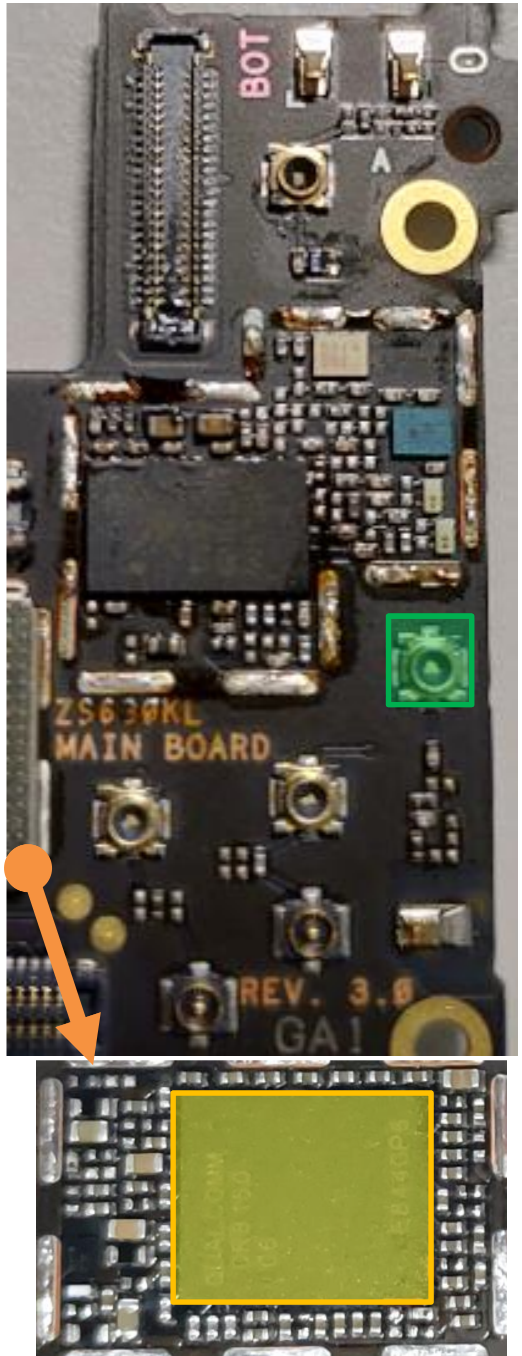
SDR8150
**U6201**

DiPx **U4705**

DiPx(LMB&B7MIMO)
**U4710**

# DRx Path : B5/B6/B18/B19/B26/GSM850(TW)

bottom

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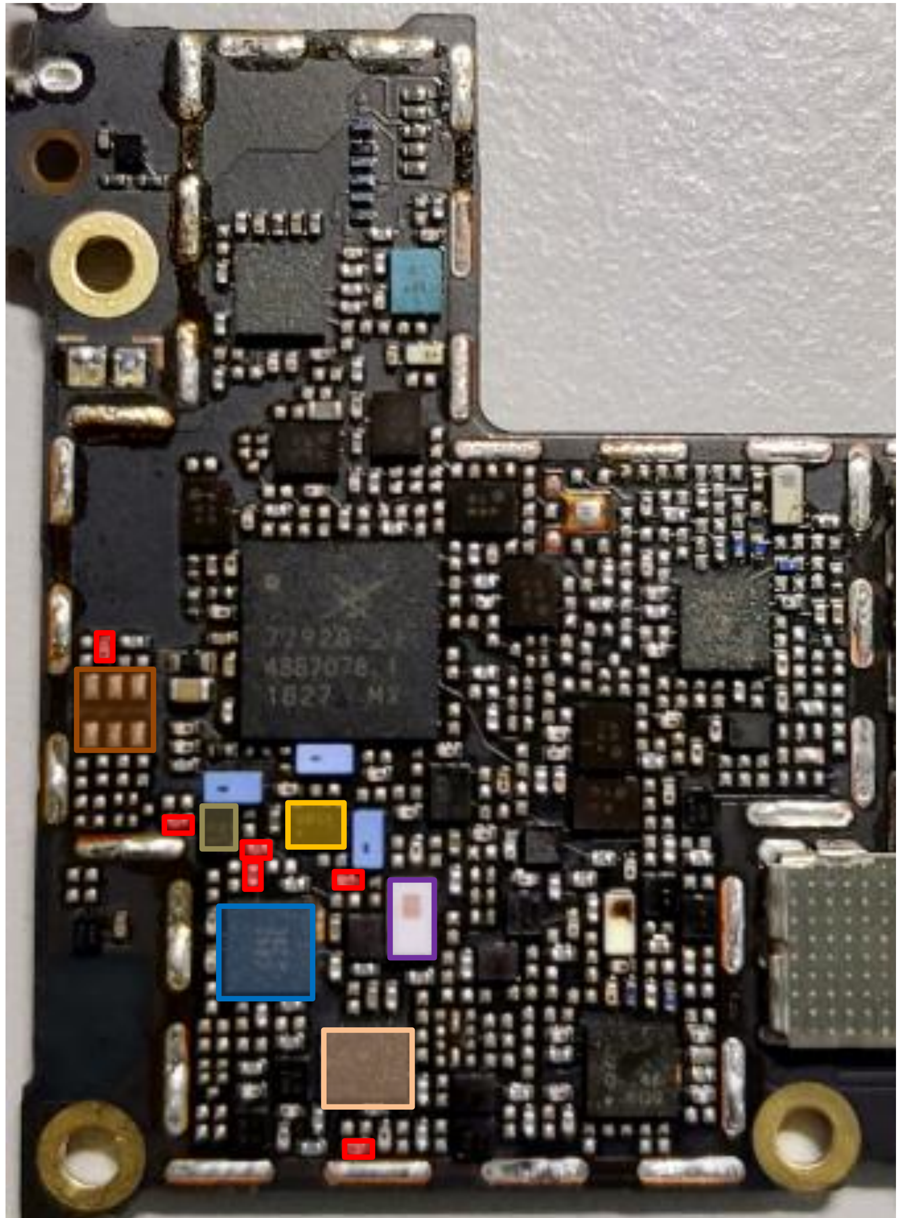
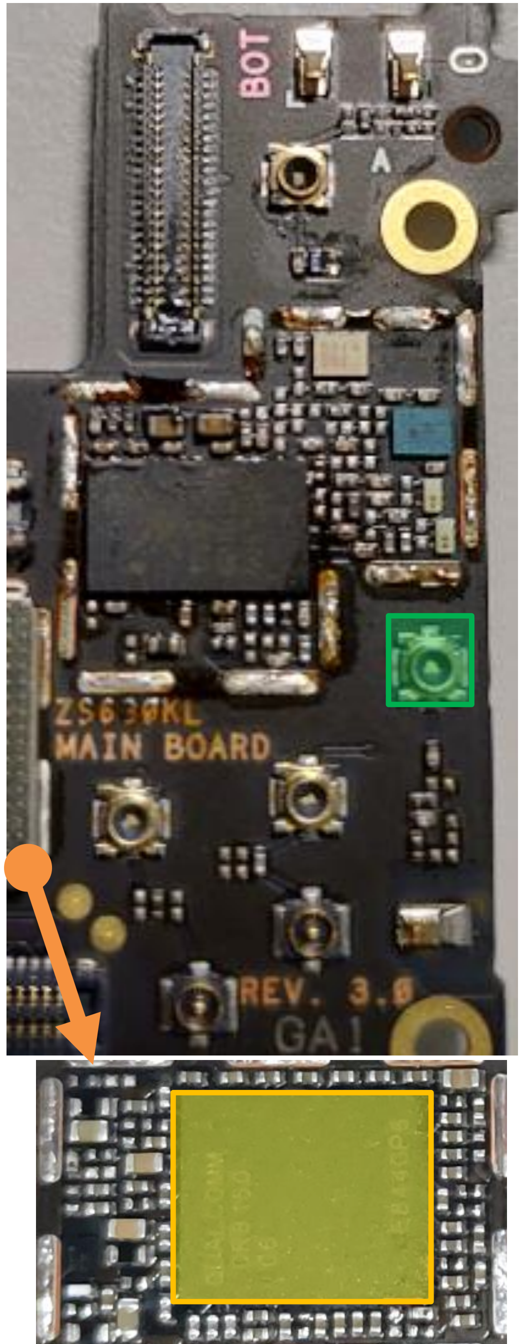


- DRX LB QLN1020 **U5131**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- DiPx(LMB&B7MIMO) **U4710**
- SAW **U5124**

# DRx Path : B8/GSM900(TW)

bottom

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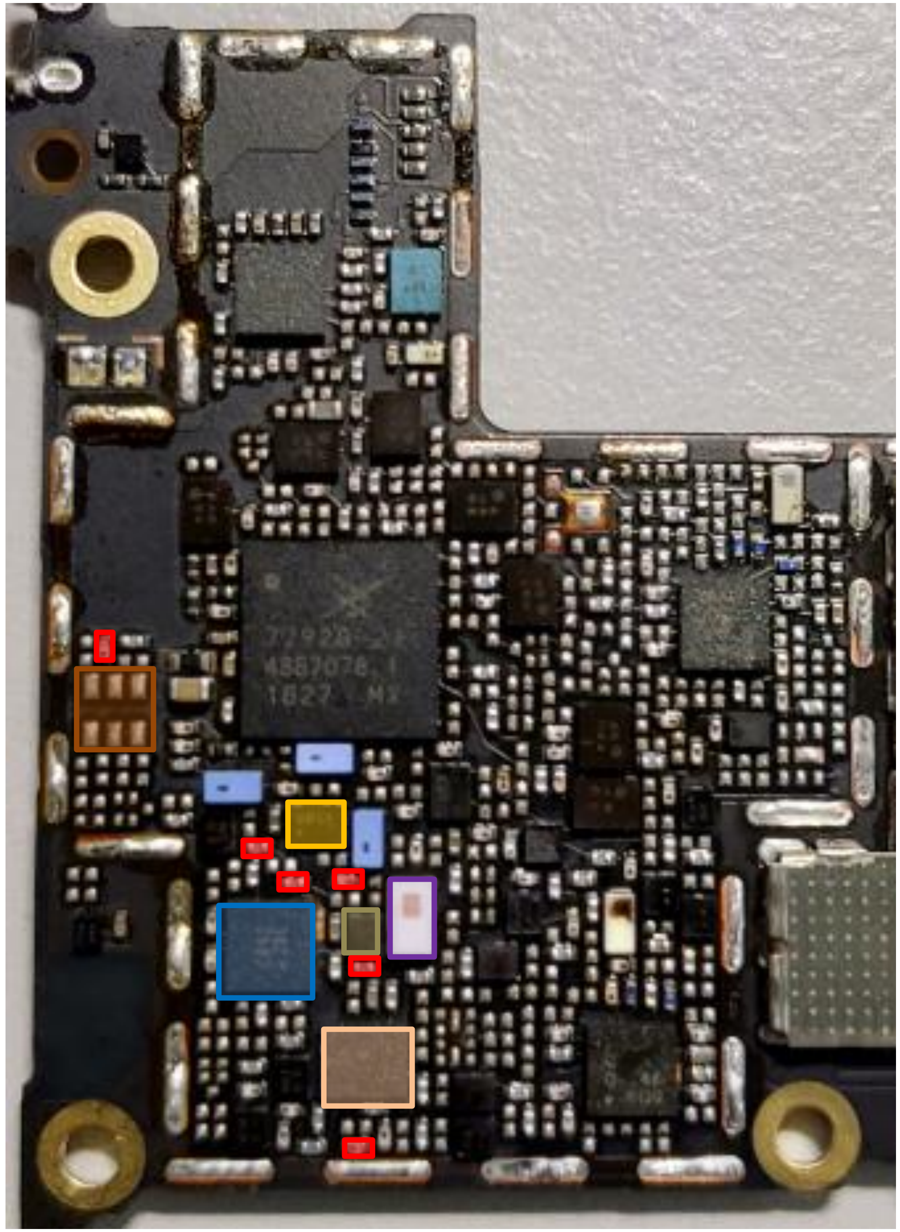


DRX LB QLN1020 **U5131**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

SDR8150 **U6201**

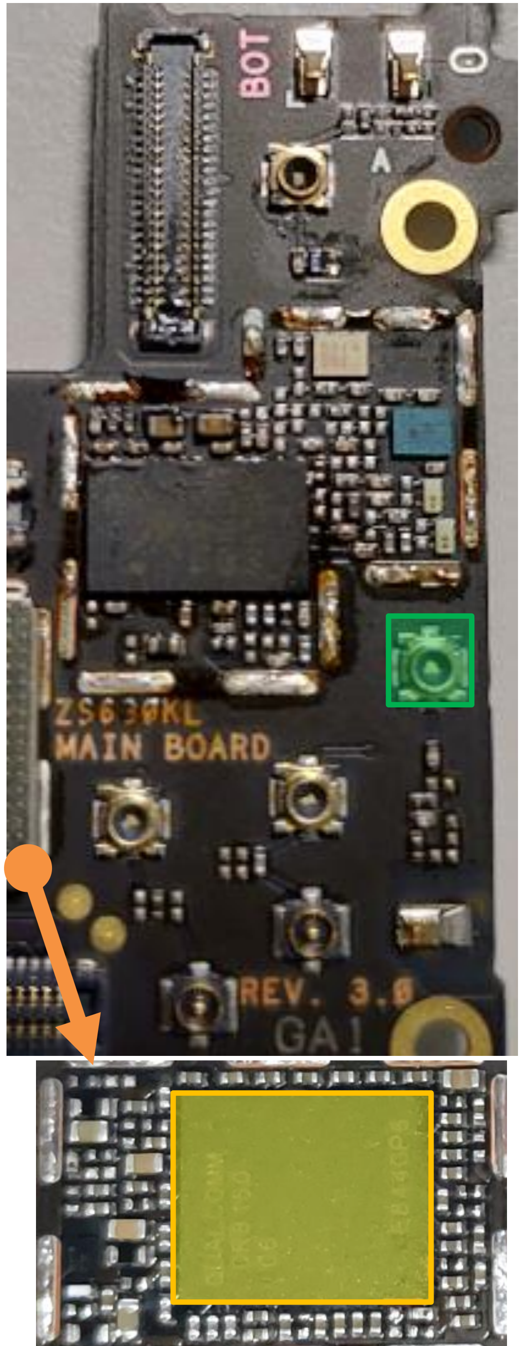
DiPx **U4705**

DiPx(LMB&B7MIMO) **U4710**

SAW **U5101**

# DRx Path : B28(TW)

bottom

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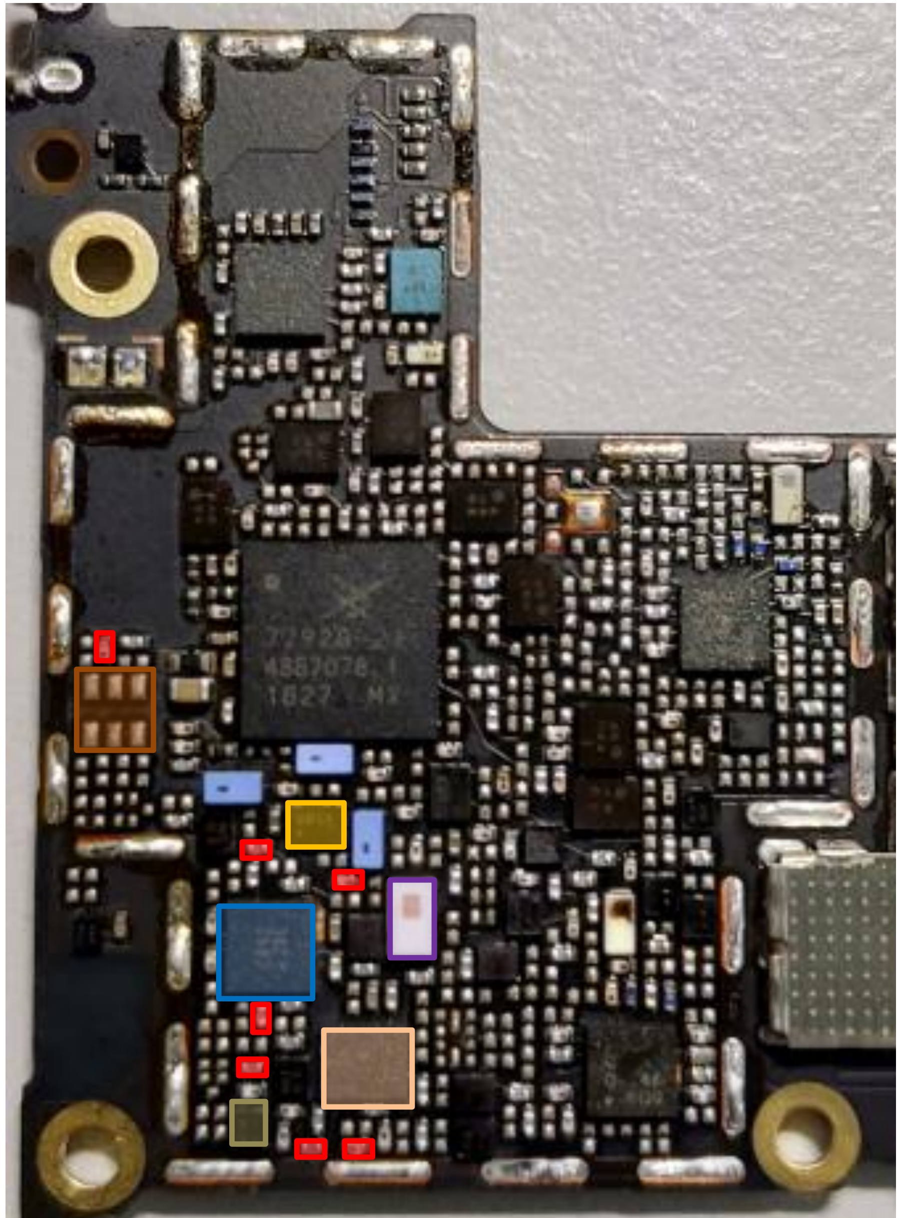
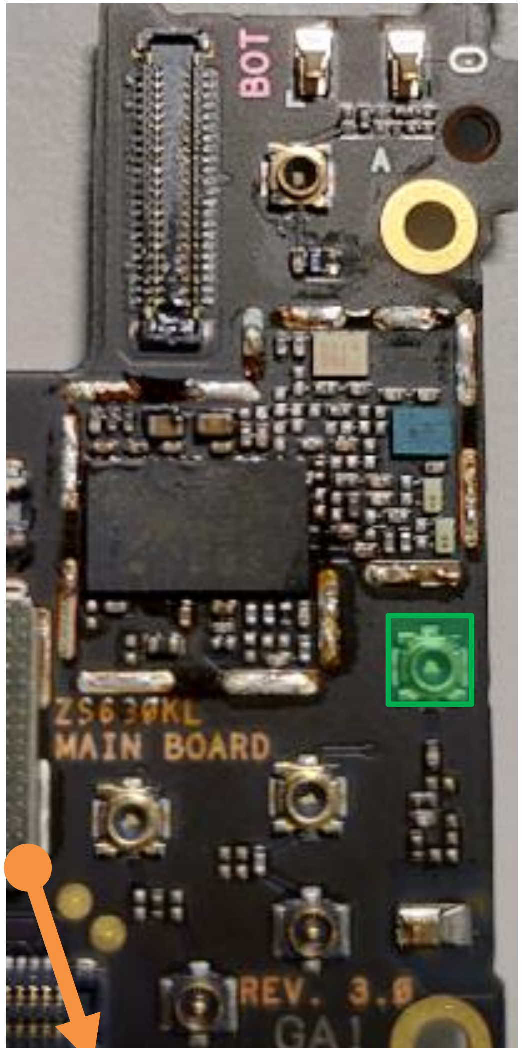
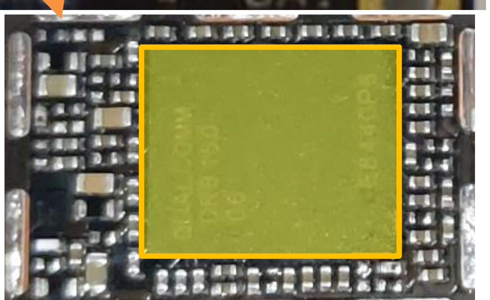


- DRX LB QLN1020 **U5131**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- DiPx(LMB&B7MIMO) **U4710**
- SAW **U5115**

# DRx Path : B12/B17(NA)

bottom



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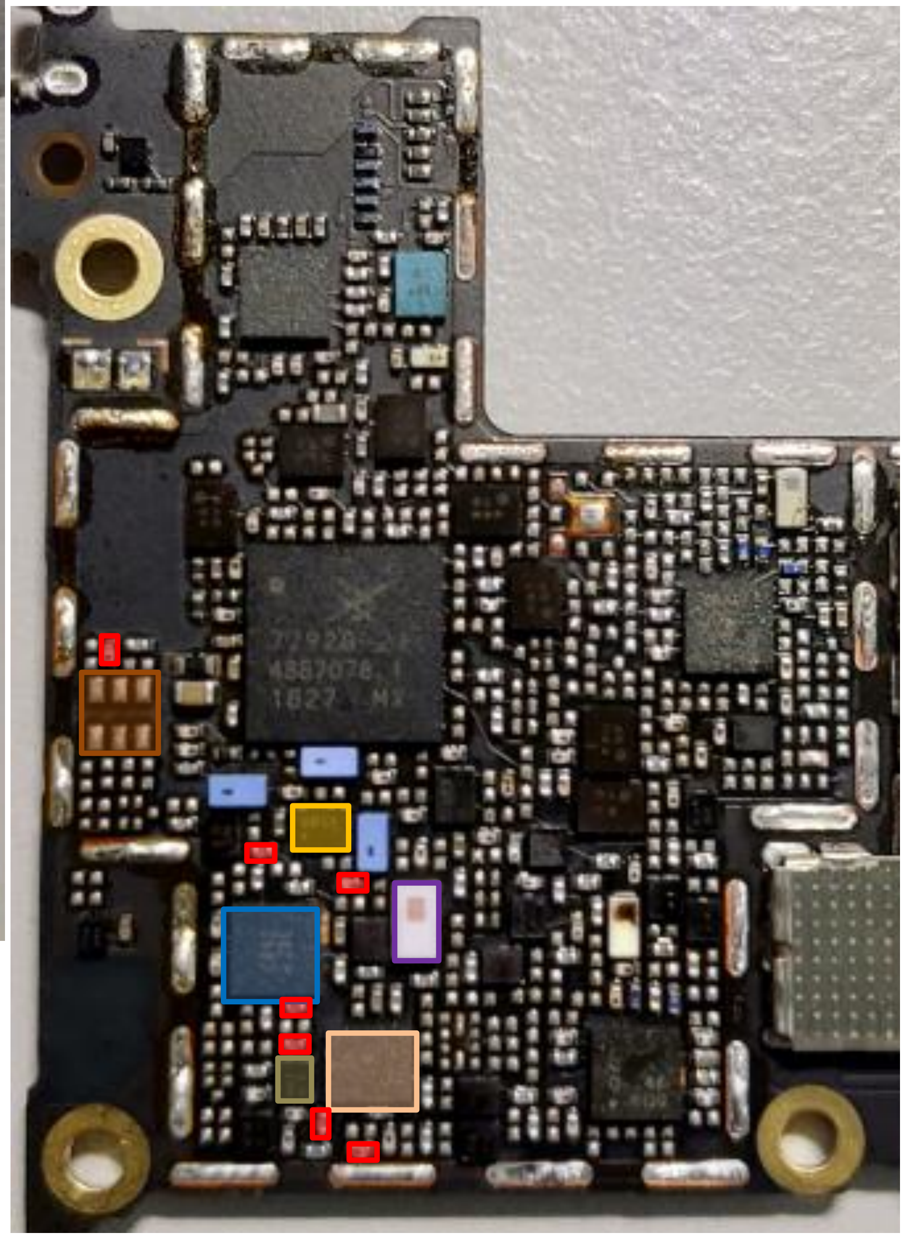
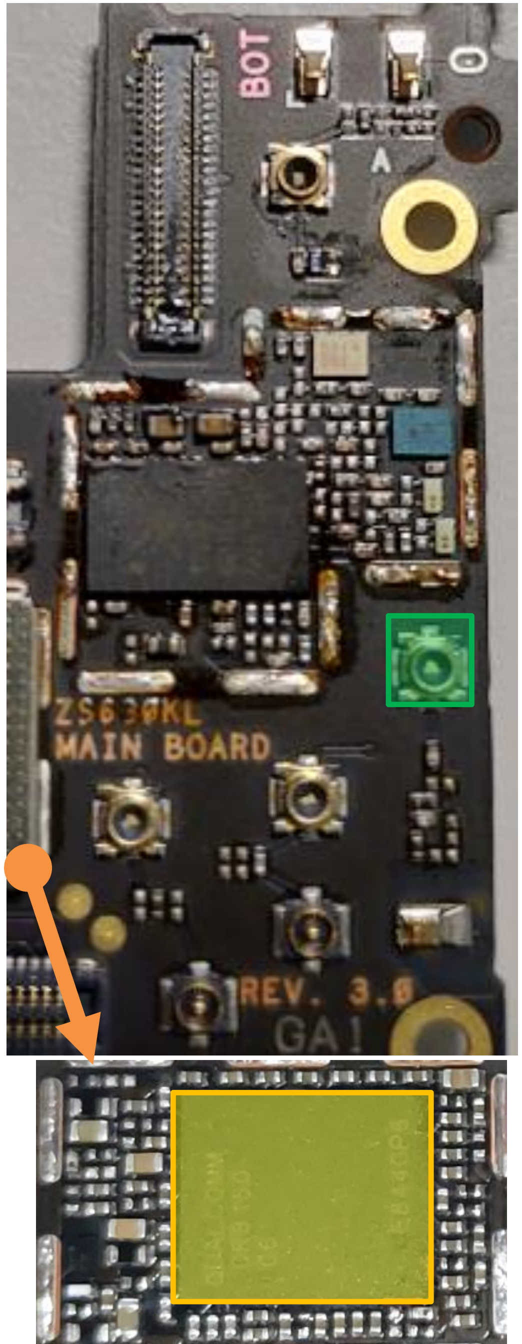


- DRX LB QLN1020 **U5131**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- SAW **U5122**
- DiPx(LMB&B7MIMO) **U4710**

# LB DRx Common Path(WW)

bottom

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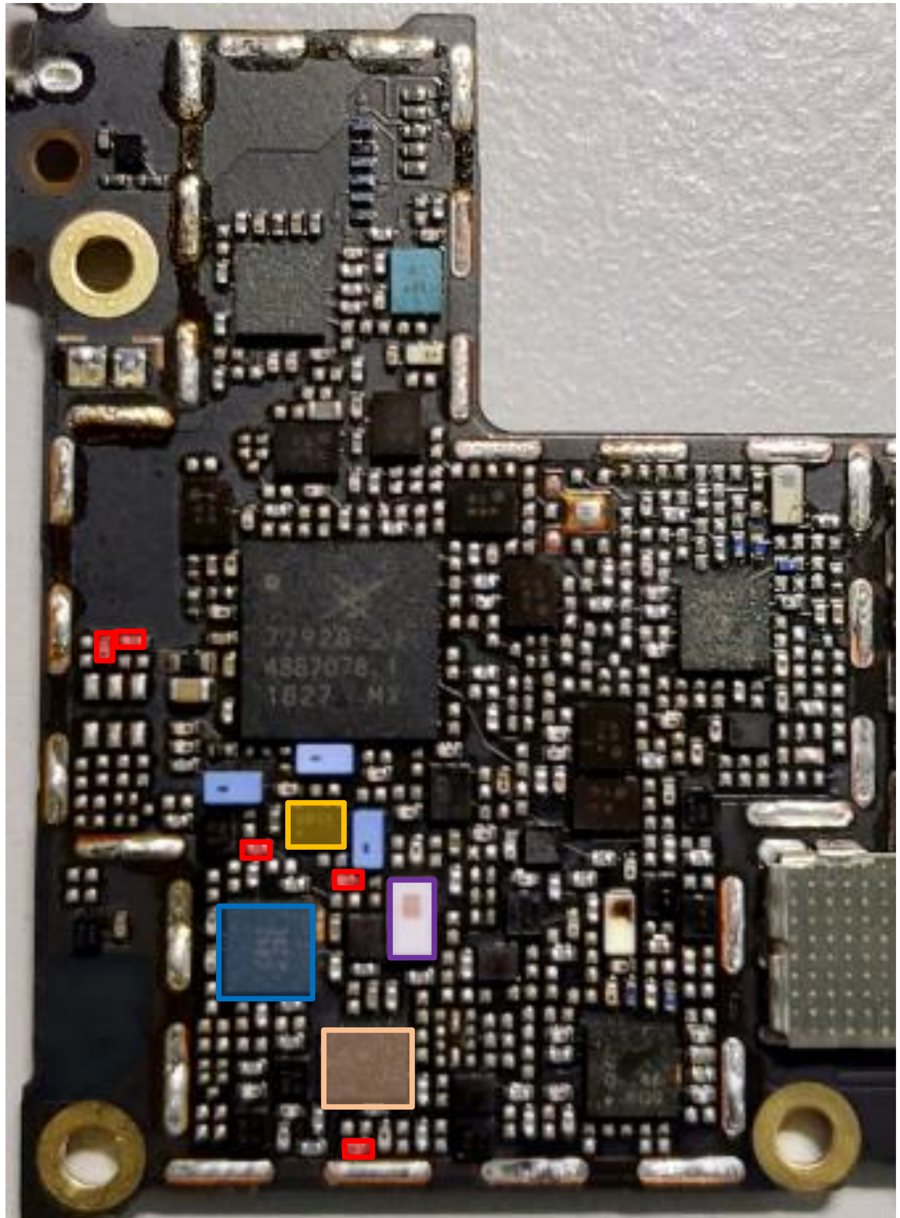


LB: B5/B8/B20/
B28A/B28B

DRX LB QLN1020
U5131

Div. RF connector
LB/MB
CON4705

DPDT U4715

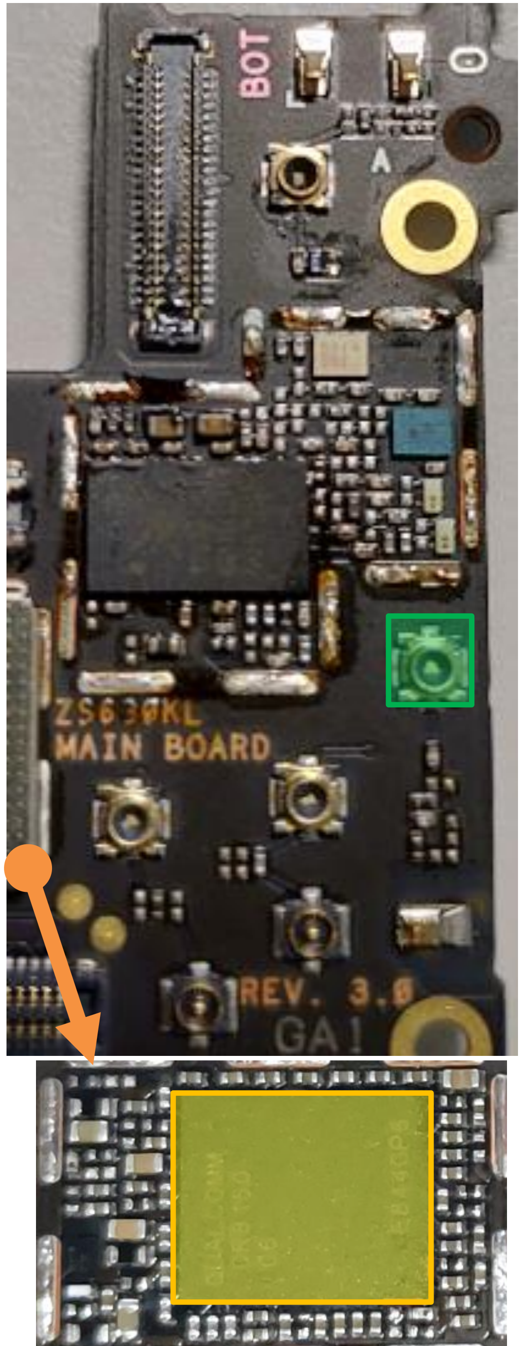
L/MB Div. RX SW
U4704

SDR8150
U6201

DiPx U4705

# DRx Path : B5/GSM850(WW)

bottom

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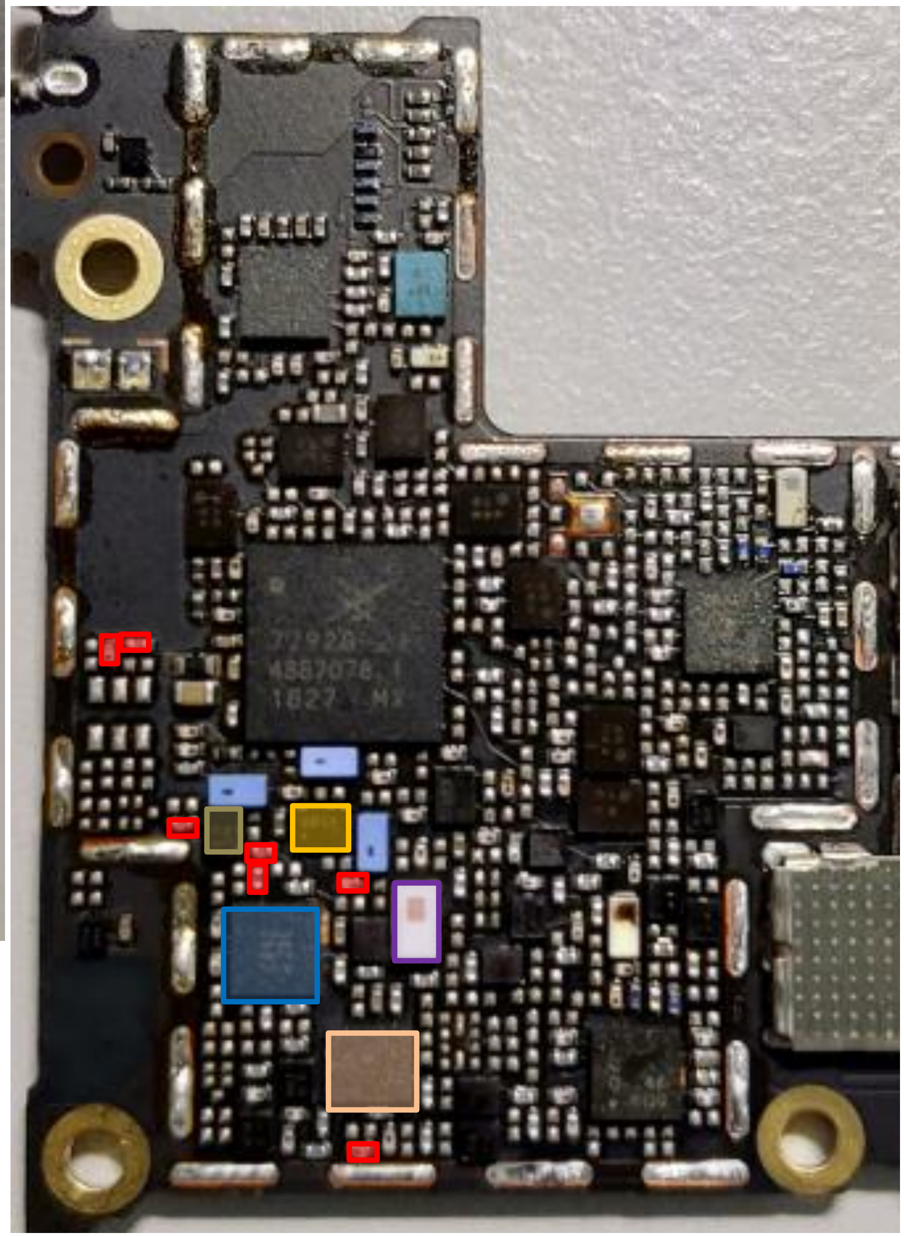


DRX LB QLN1020 **U5131**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

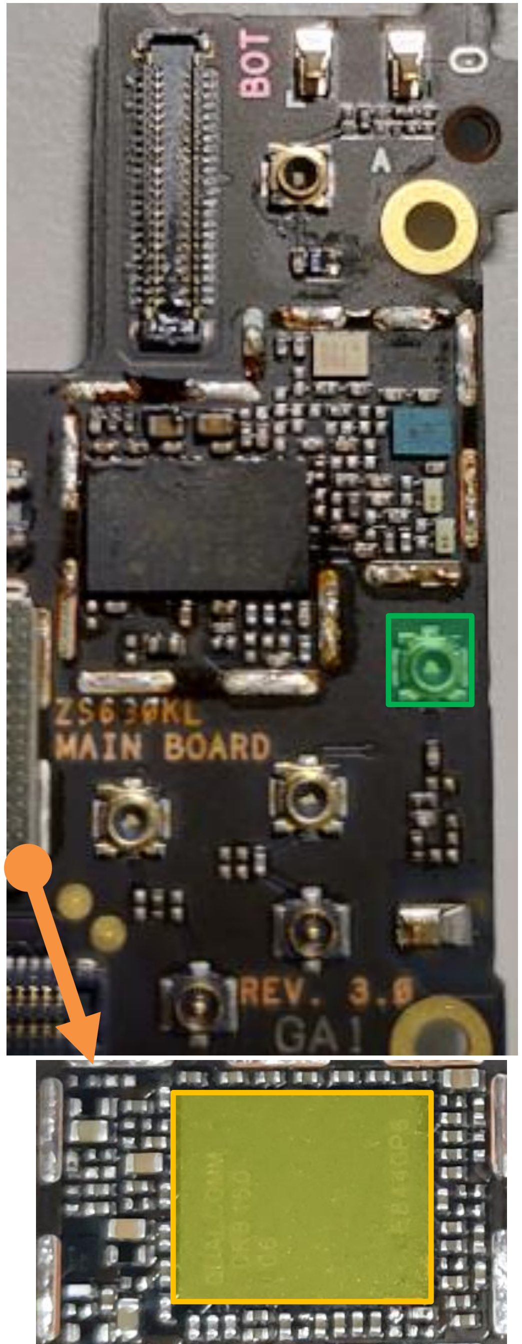
SDR8150 **U6201**

DiPx **U4705**

SAW **U5124**

# DRx Path : B8/GSM900(WW)

bottom

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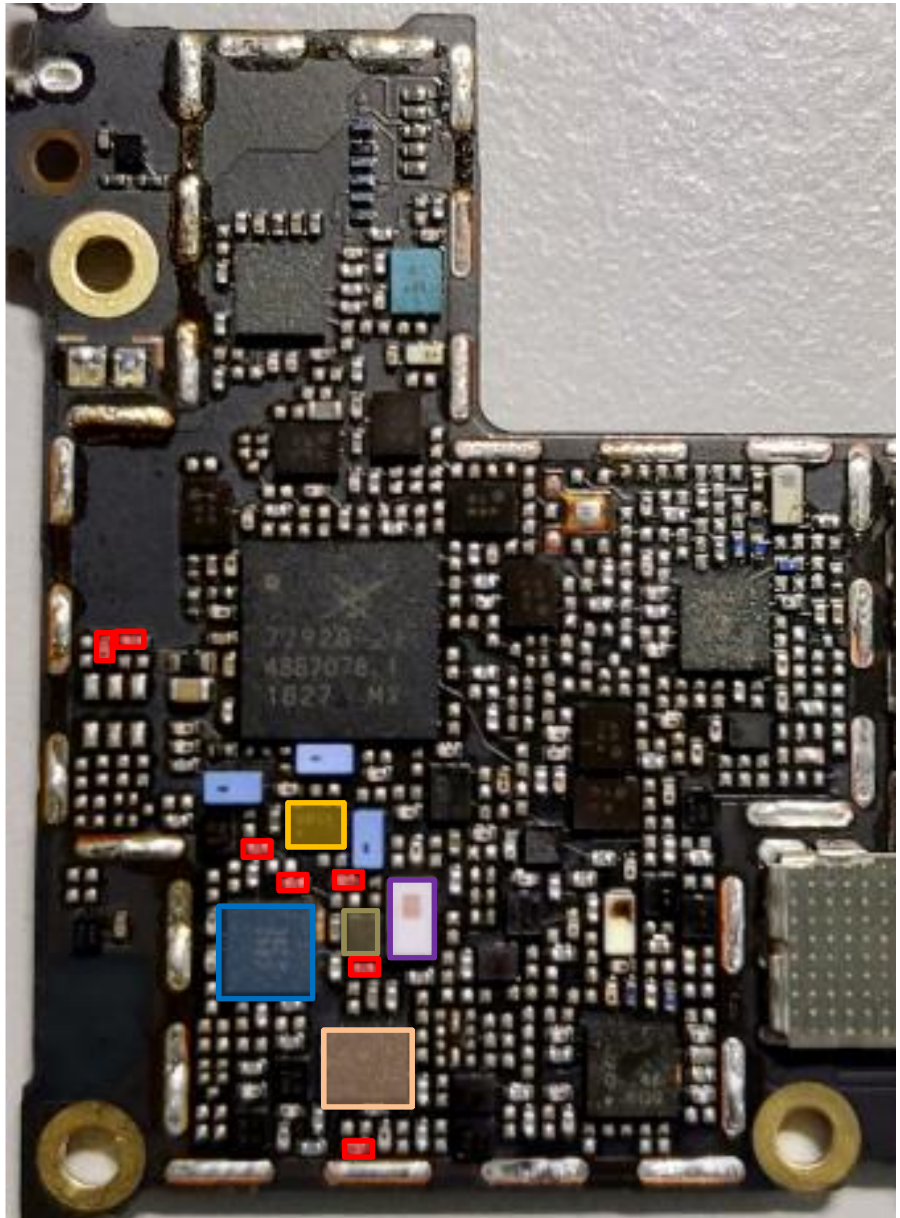


DRX LB QLN1020 **U5131**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

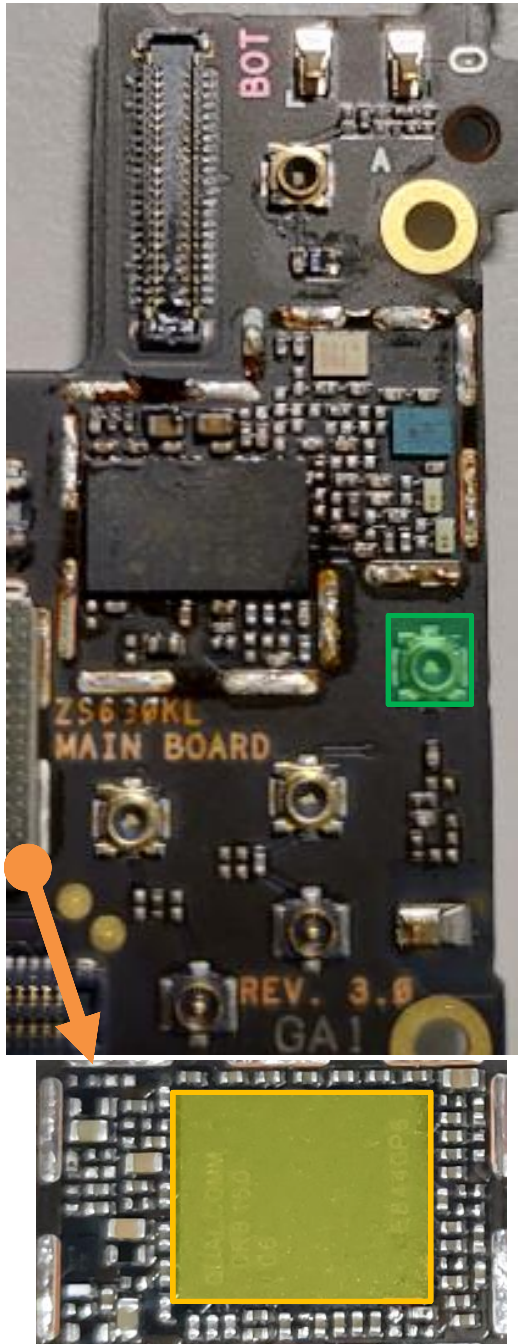
SDR8150 **U6201**

DiPx **U4705**

SAW **U5101**

# DRx Path : B20(WW)

bottom

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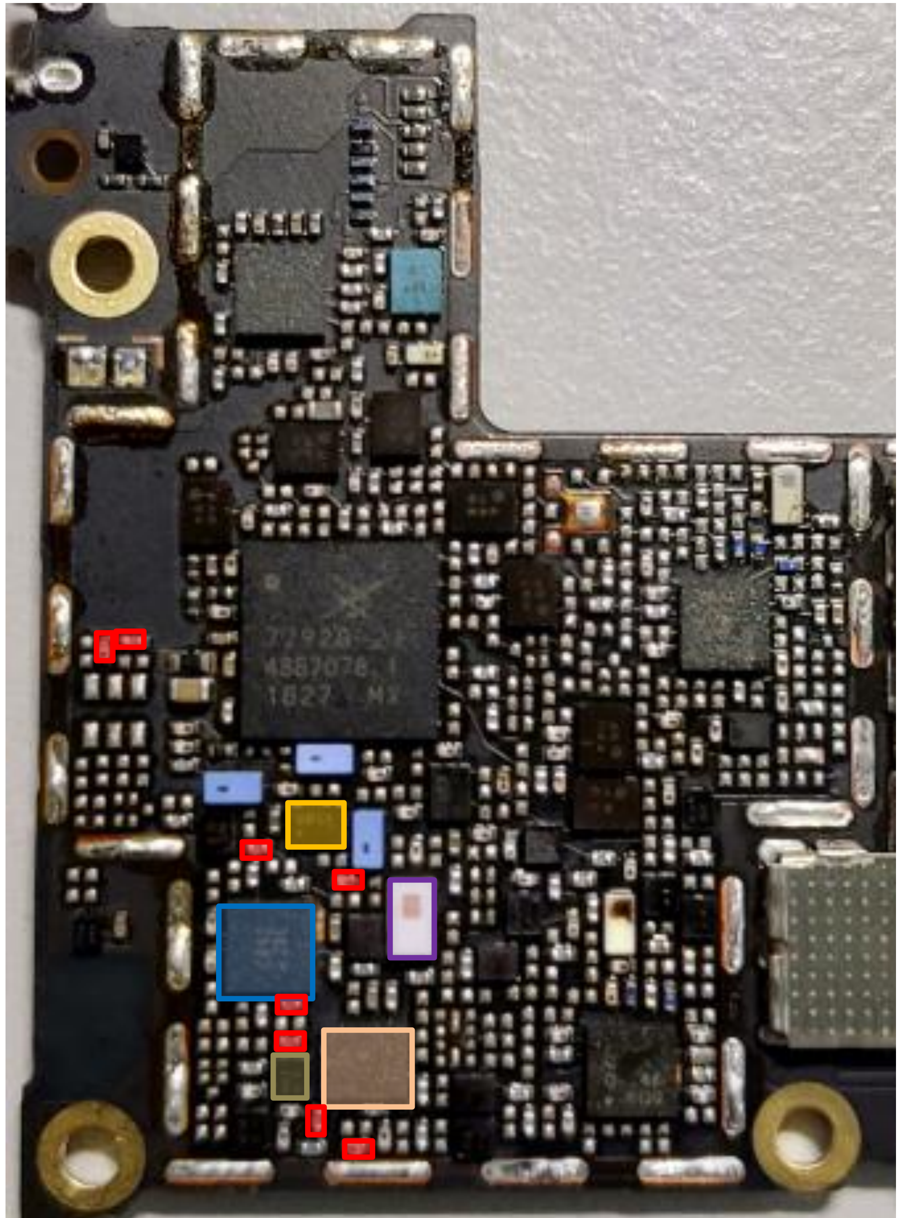


DRX LB QLN1020 **U5131**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

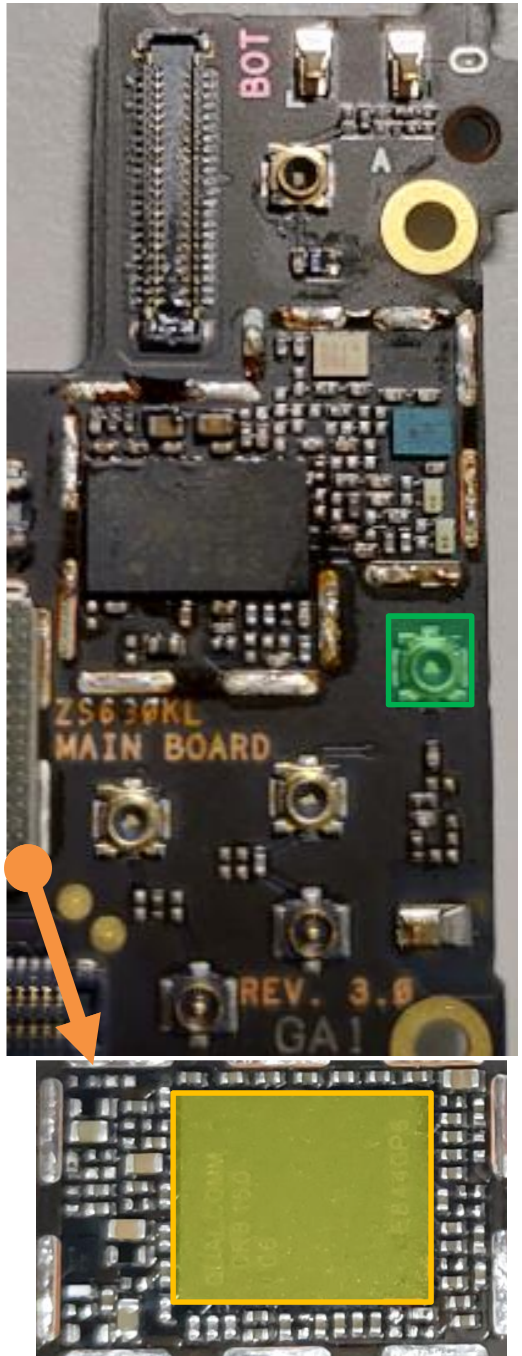
SDR8150 **U6201**

DiPx **U4705**

SAW **U5122**

# DRx Path : B28(WW)

bottom

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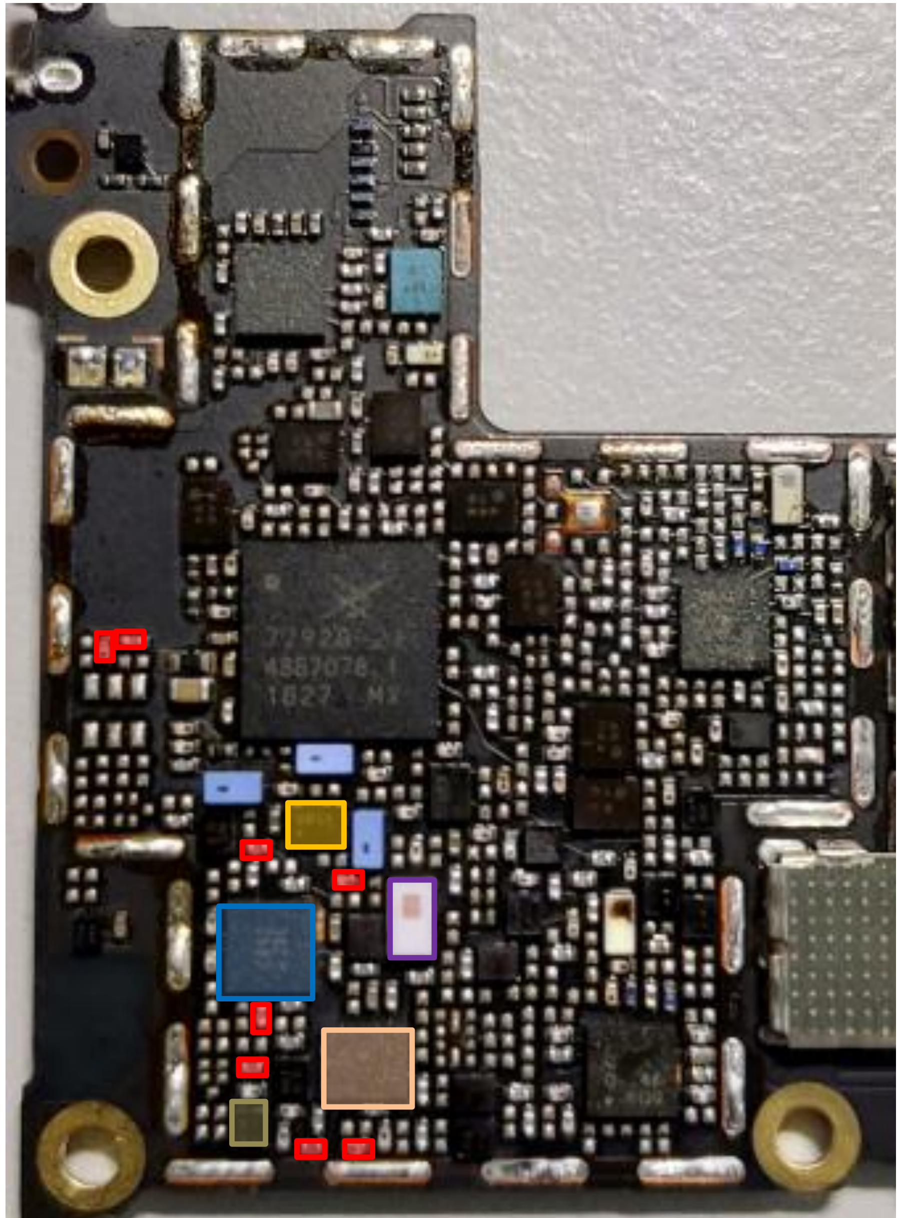


DRX LB QLN1020
U5131

Div. RF connector
LB/MB
CON4705

DPDT U4715

L/MB Div. RX SW
U4704

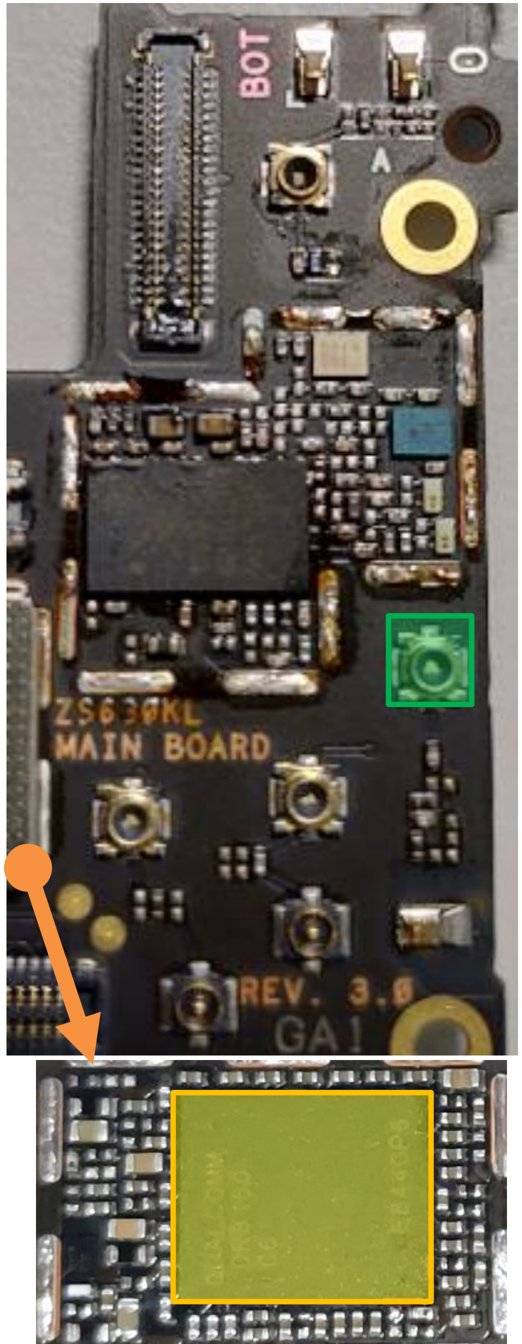
SDR8150
U6201

DiPx U4705

SAW U5115

# MB DRx Common Path(TW)

bottom

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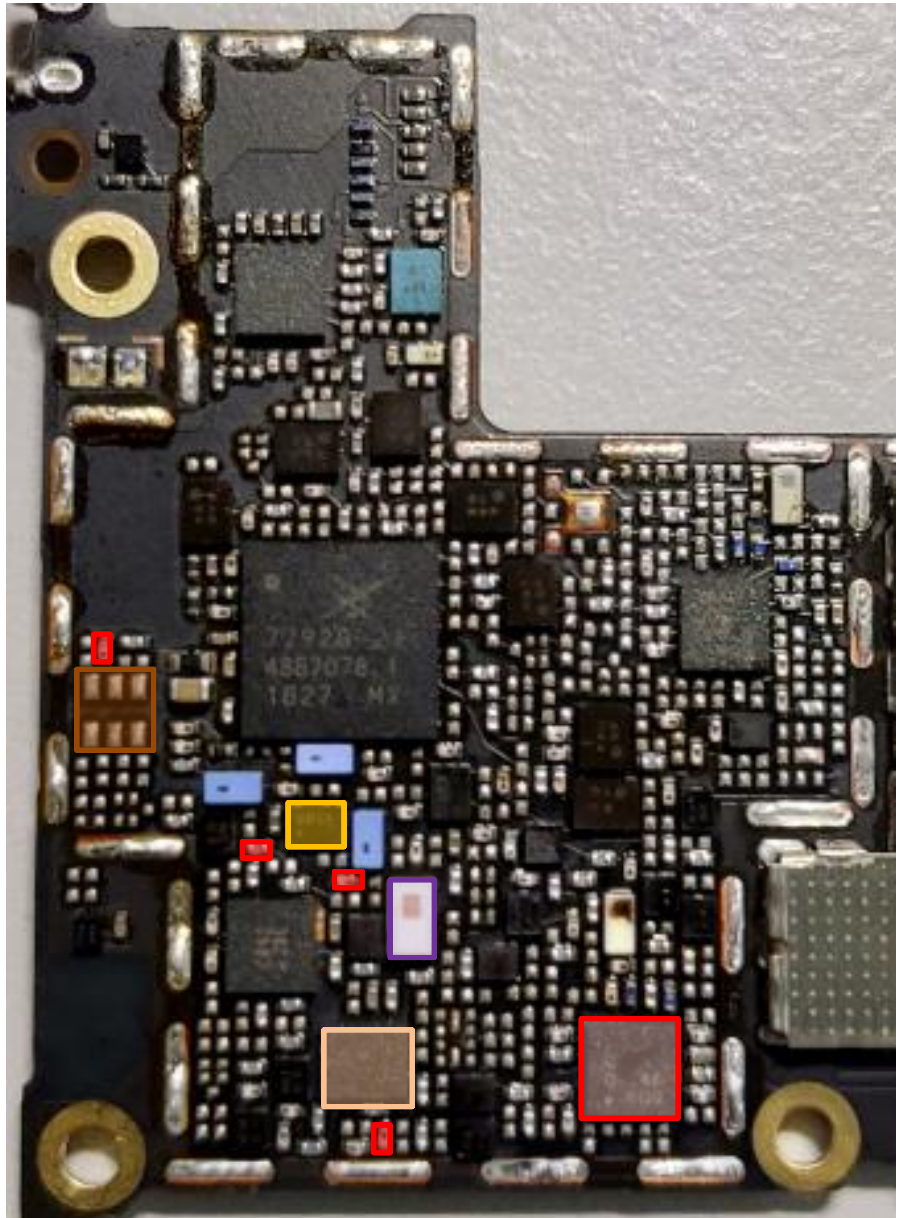
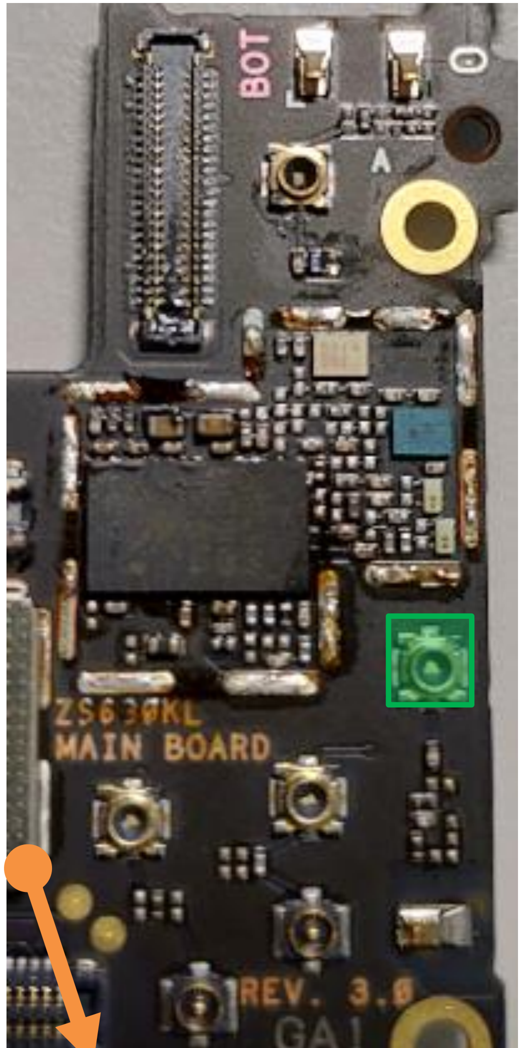


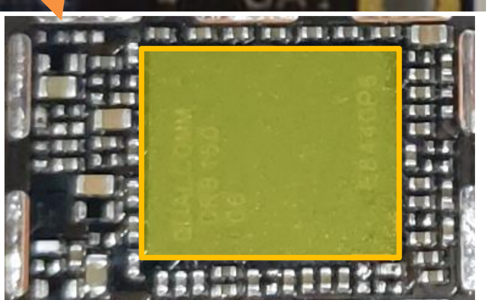
MB: B1/B2/B3/B4/B39

- DRX MB/HB QLN4640 **U5132**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- DiPx(LMB&B7MIMO) **U4710**

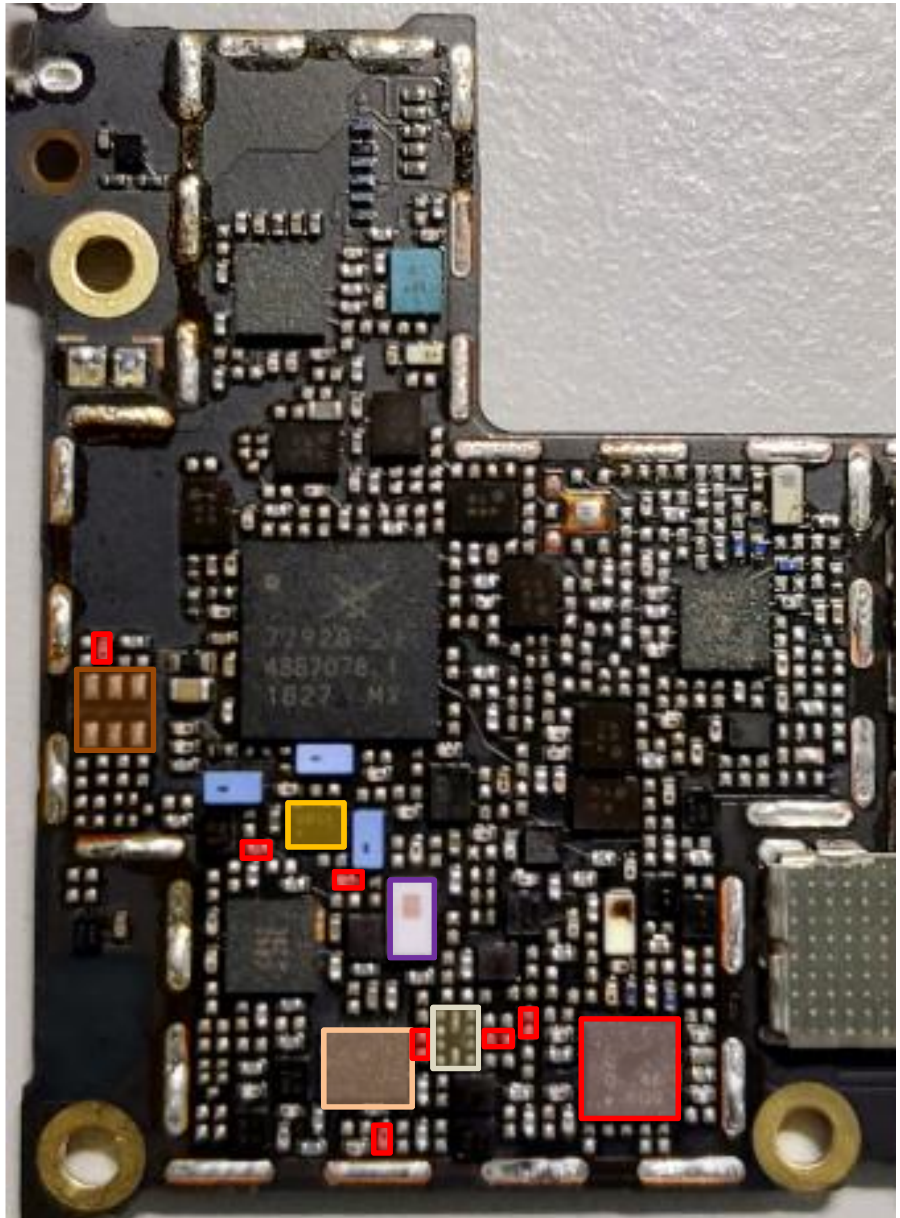
# DRx Path : B1/B4(TW)

bottom



top



DRX MB/HB QLN4640 **U5132**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

SDR8150 **U6201**

DiPx **U4705**

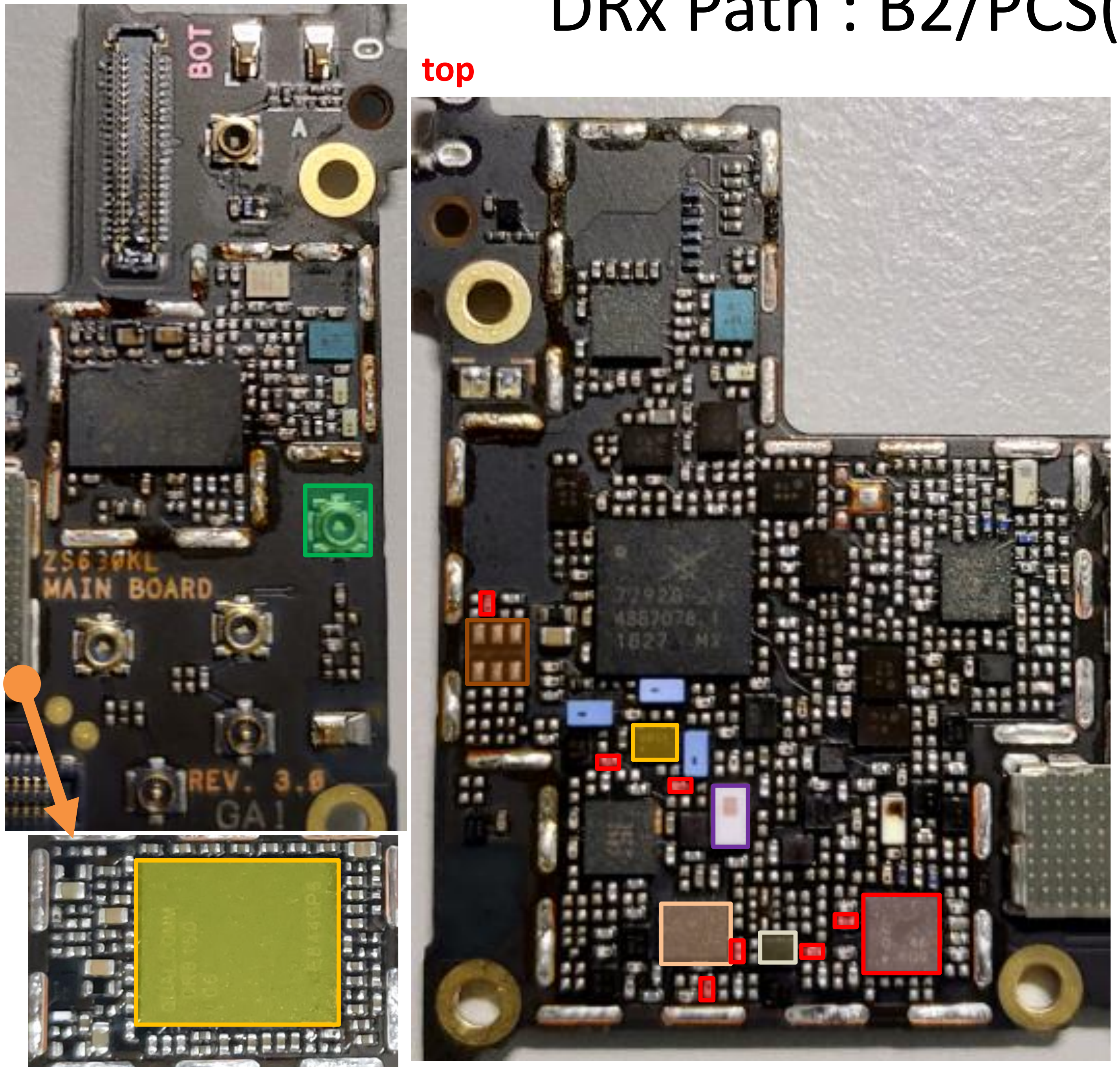
DiPx(LMB&B7MIMO) **U4710**

SAW **U5114**

# DRx Path : B2/PCS(TW)

bottom

top




DRX MB/HB QLN4640 **U5132**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

SDR8150 **U6201**

DiPx **U4705**

DiPx(LMB&B7MIMO) **U4710**

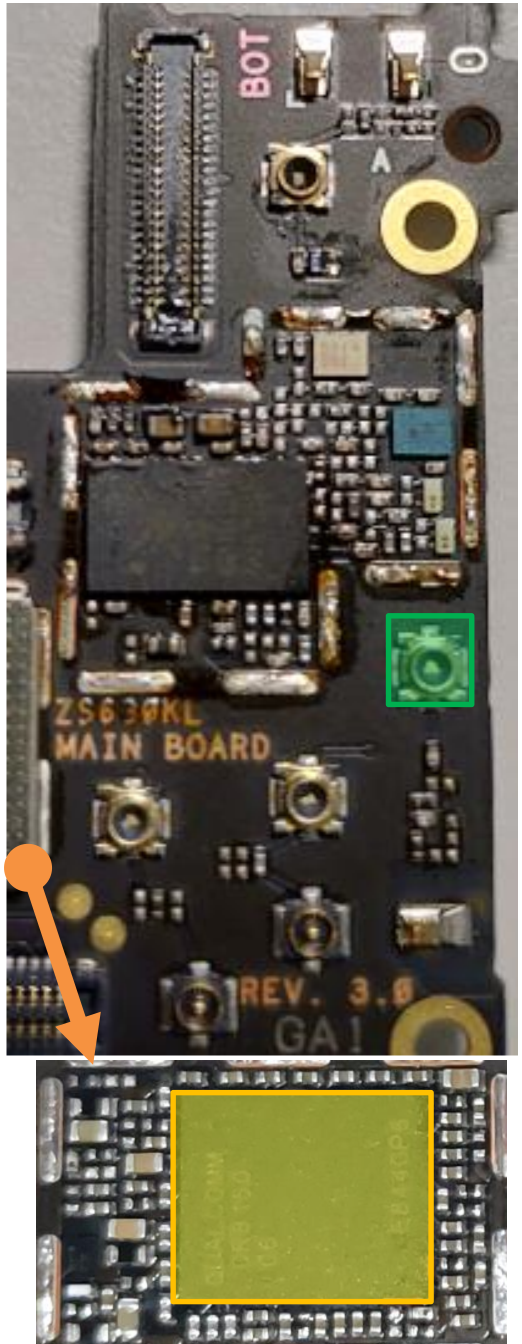
SAW **U5102**

# DRx Path : B3/DCS(TW)

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 104 | 11 | 2 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 107 | 3 | 2 | 0 | |

bottom

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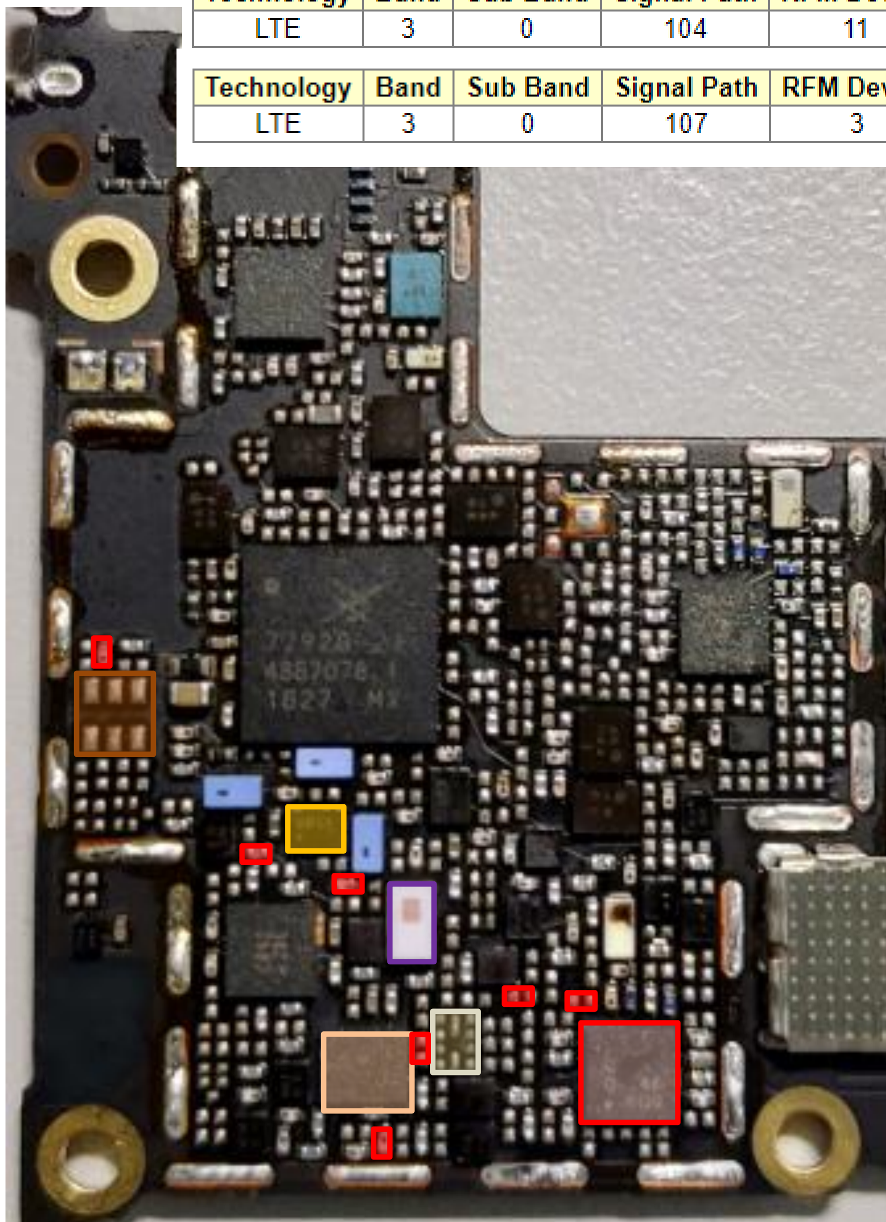


DRX MB/HB QLN4640 **U5132**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

SDR8150 **U6201**

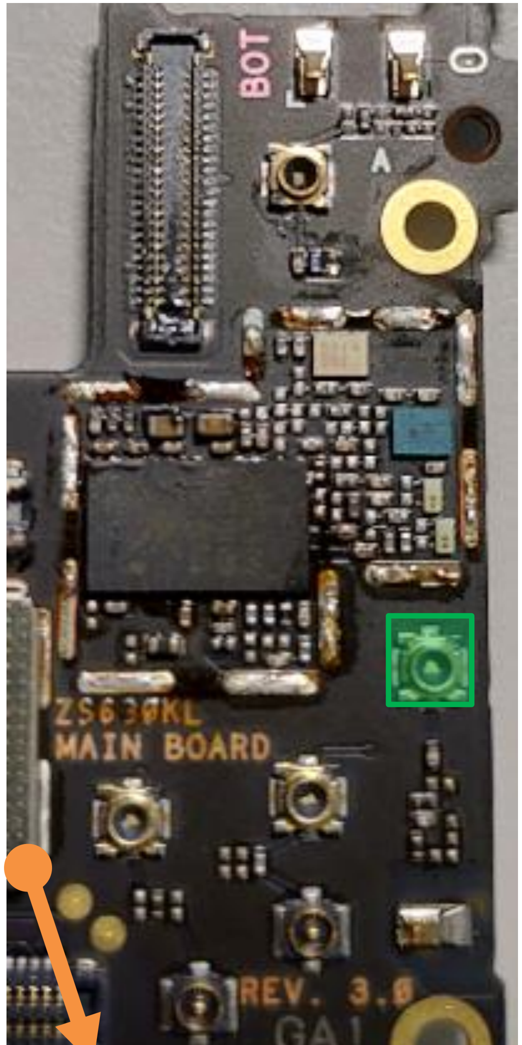
DiPx **U4705**

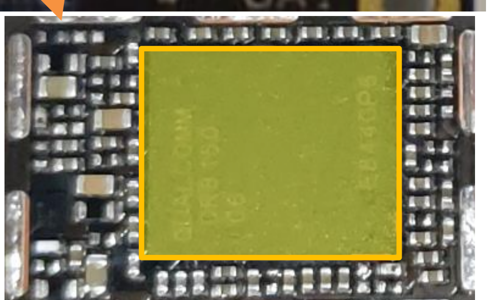
DiPx(LMB&B7MIMO) **U4710**

SAW **U5114**

# DRx Path : B39(TW)

bottom



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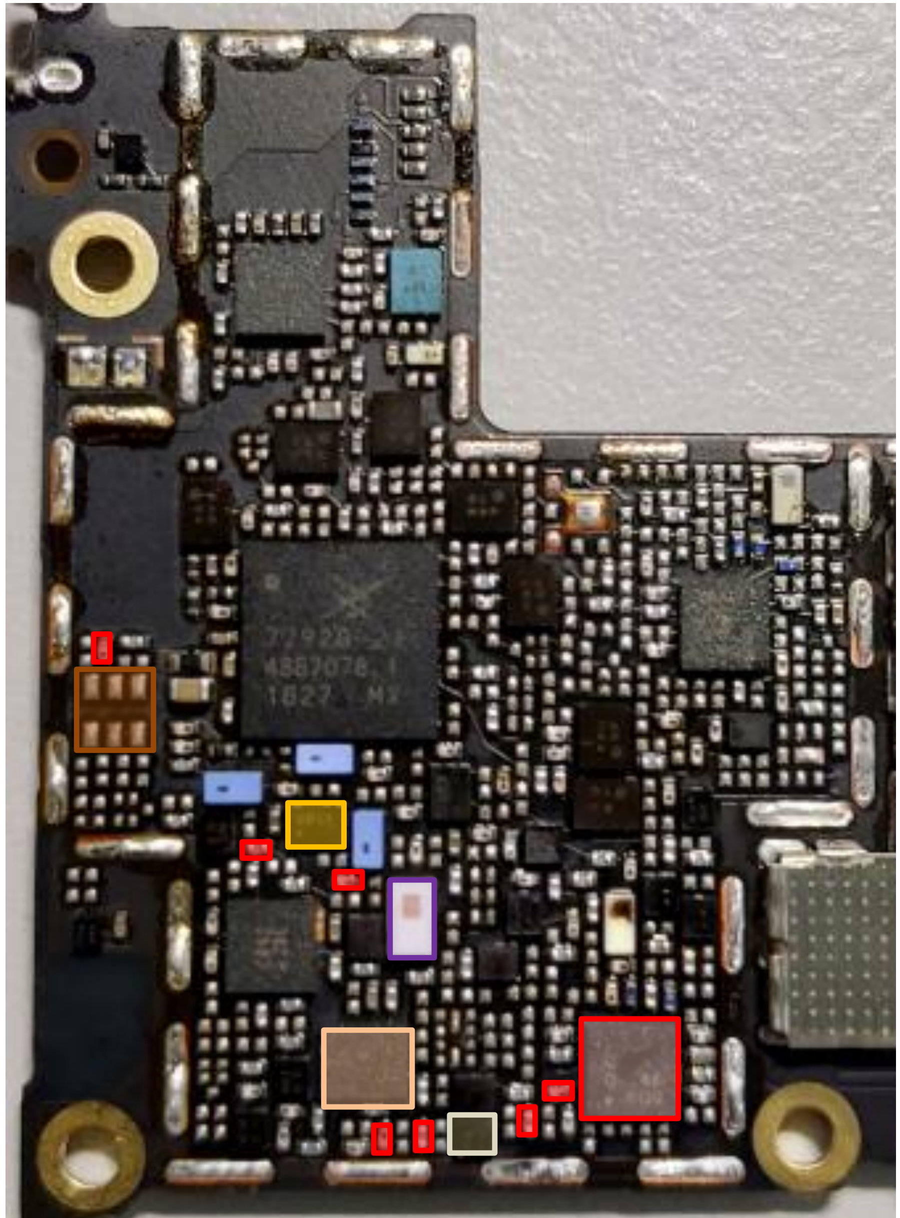


DRX MB/HB QLN4640 **U5132**

Div. RF connector LB/MB **CON4705**

DPDT **U4715**

L/MB Div. RX SW **U4704**

SDR8150 **U6201**

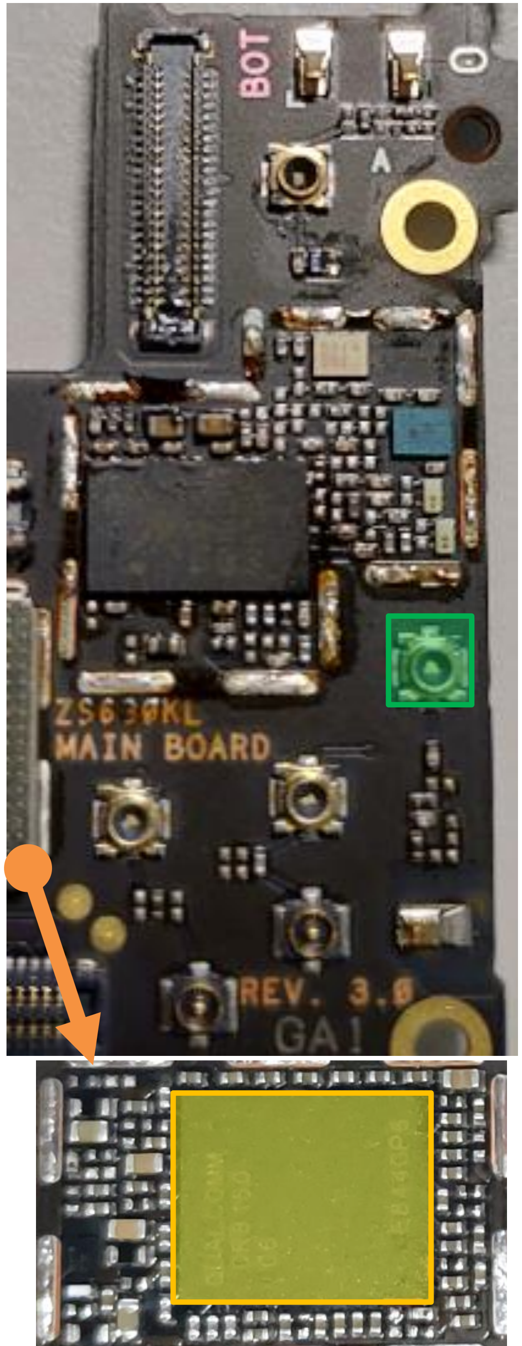
DiPx **U4705**

DiPx(LMB&B7MIMO) **U4710**

SAW **U5130**

# DRx Path : B7 MIMO(TW)

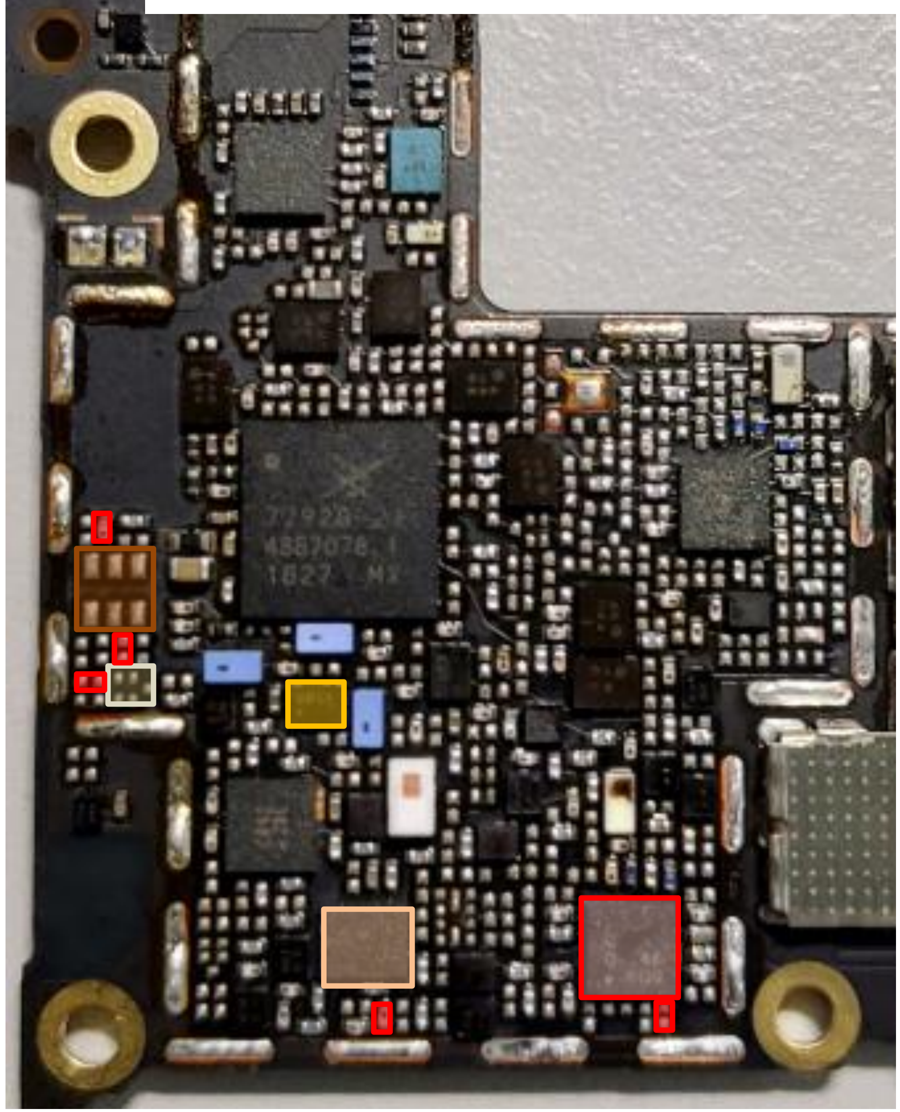
bottom




top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 40 | 3 | 2 | 0 | |

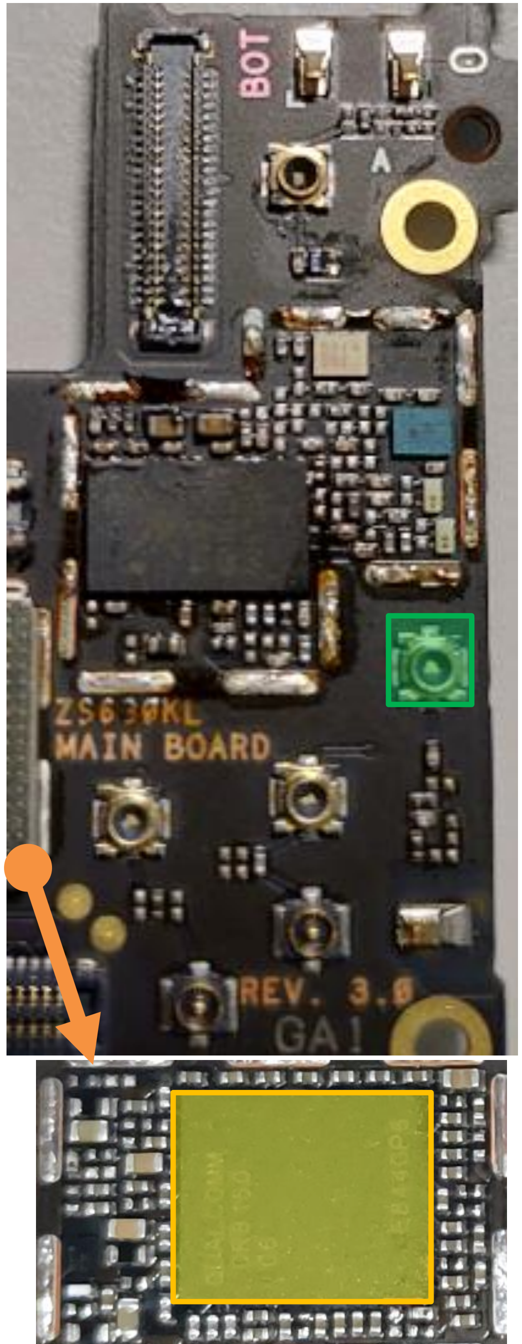
| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 42 | 11 | 2 | 4 | |



- DRX MB/HB QLN4640 **U5132**
- Div. RF connector LB/MB **CON4705**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx(LMB&B7MIMO) **U4710**
- SAW **U5135**

# MB DRx Common Path(WW)

bottom

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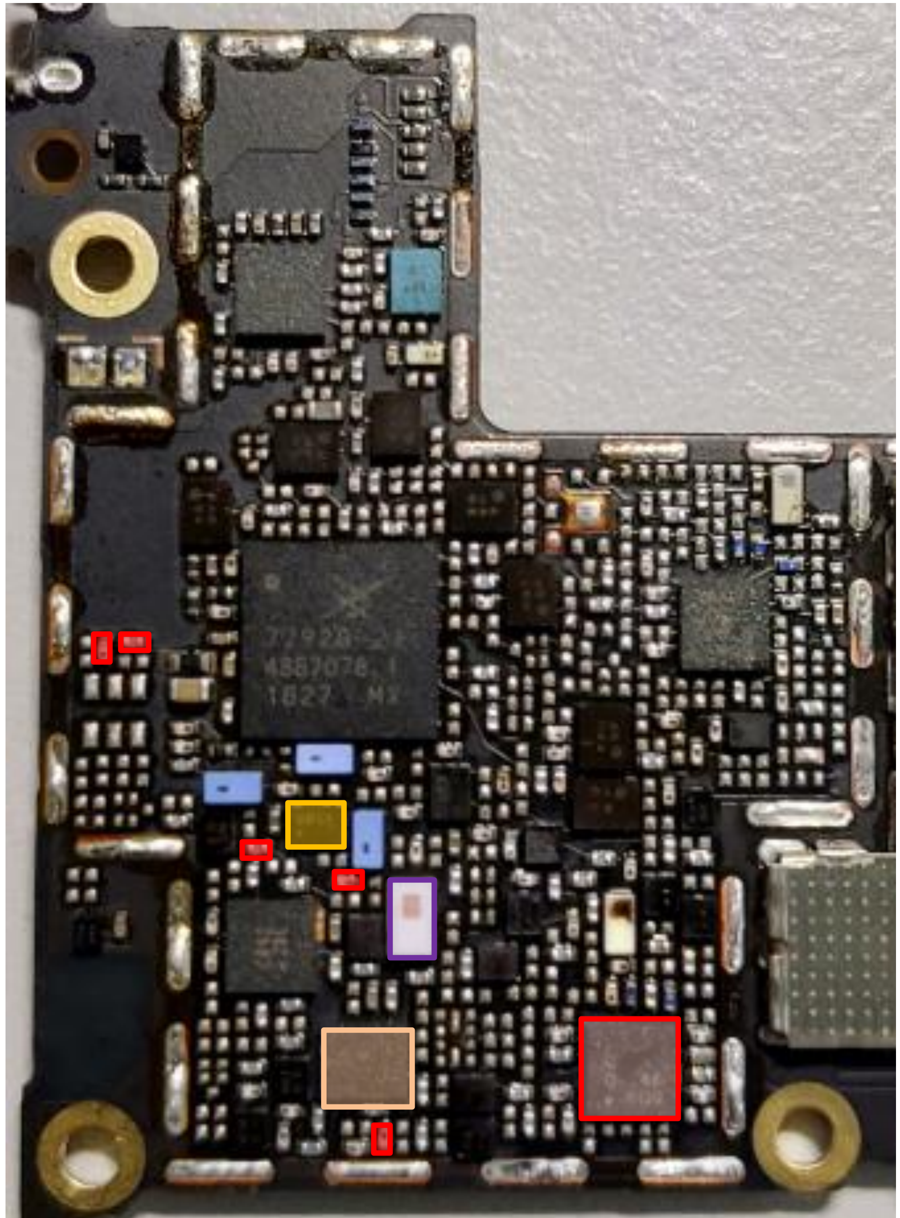
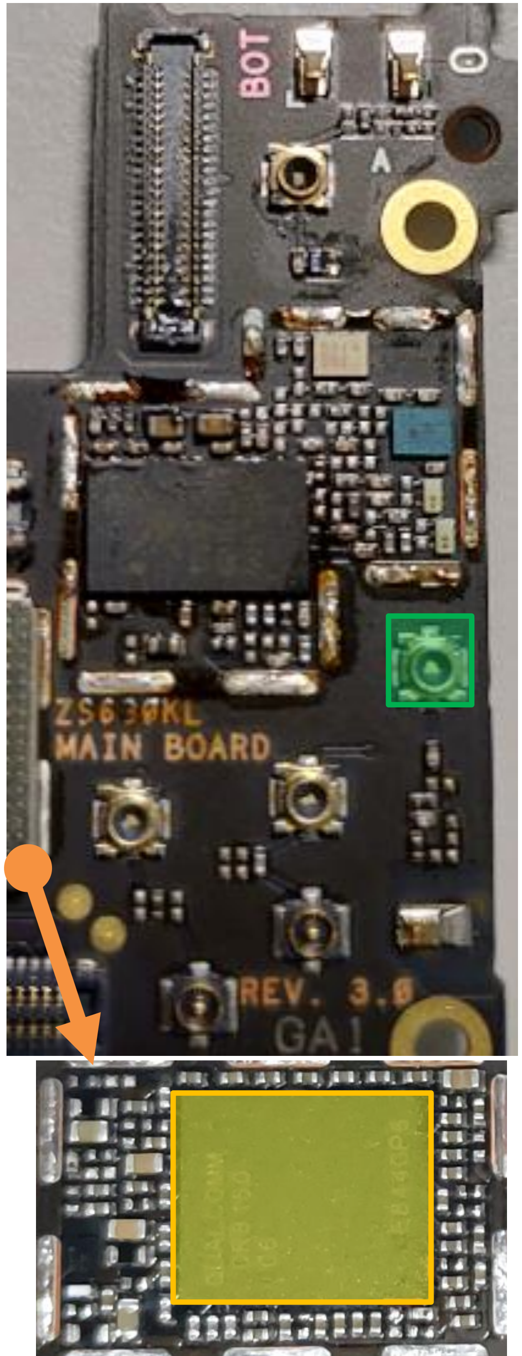


MB: B1/B2/B3

- DRX MB/HB QLN4640 **U5132**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**

# DRx Path : B1(WW)

bottom

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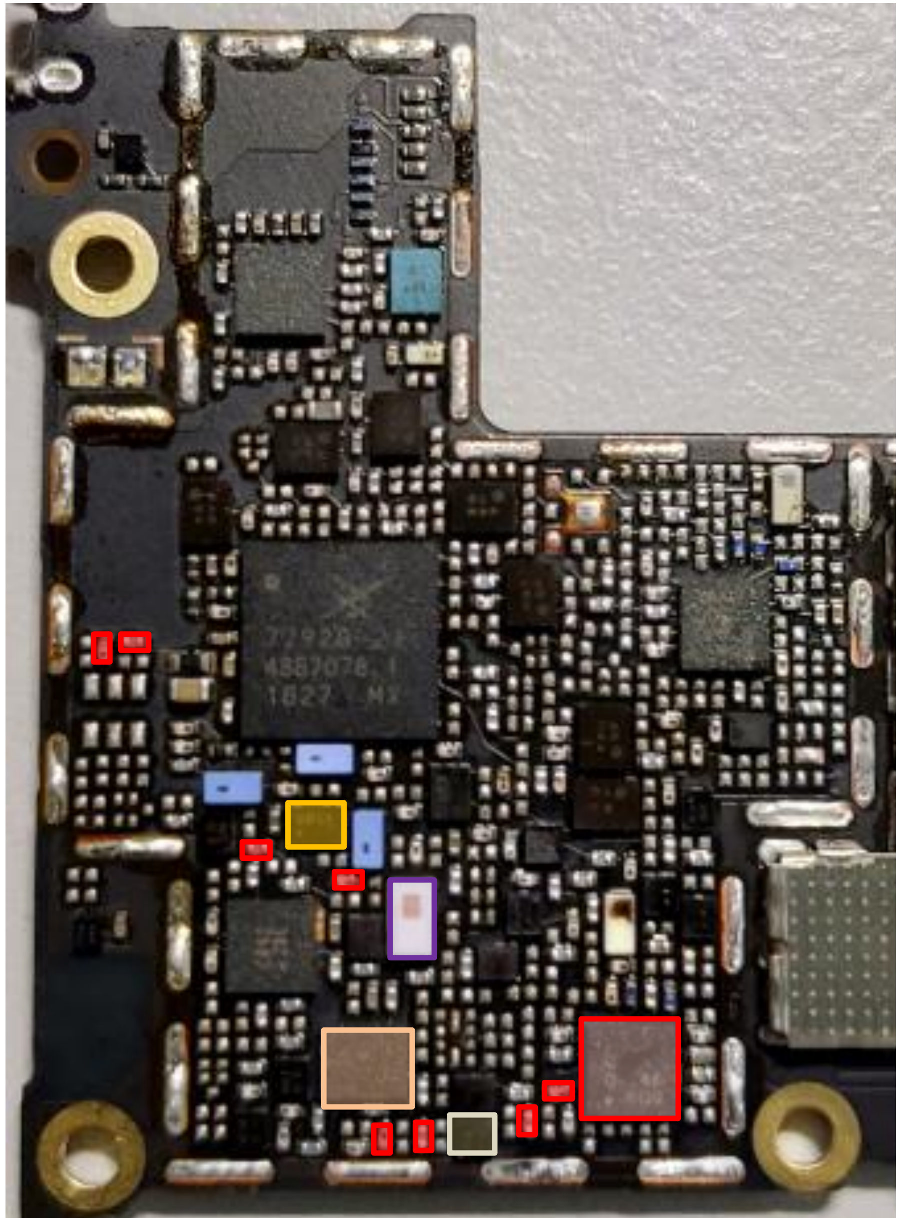
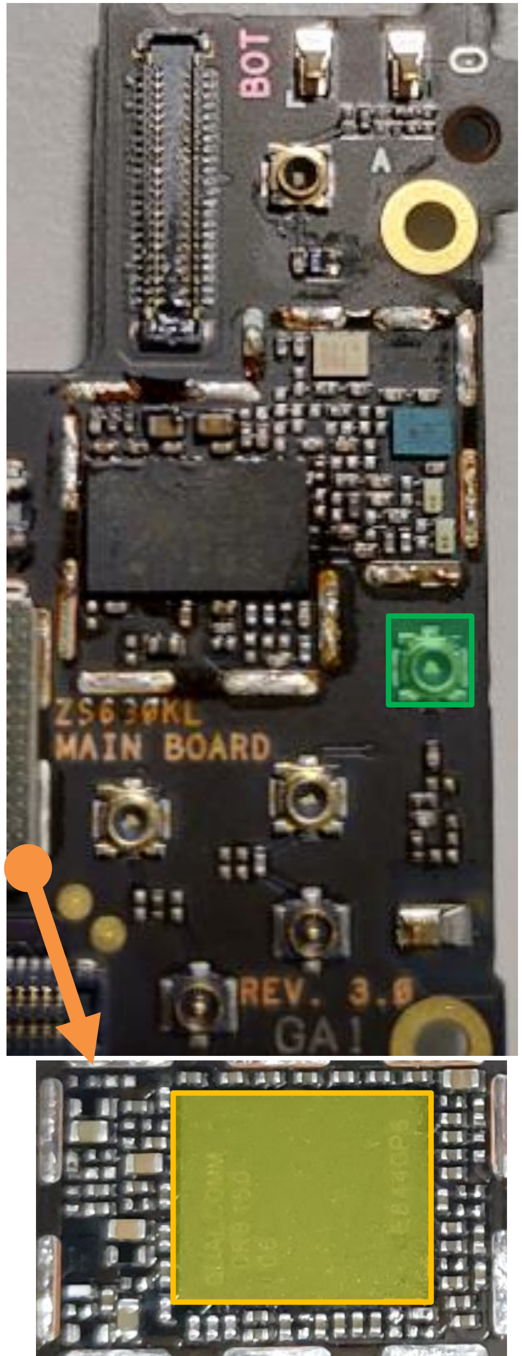


- DRX MB/HB QLN4640 **U5132**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- SAW **U5130**

# DRx Path : B2/PCS(WW)

bottom

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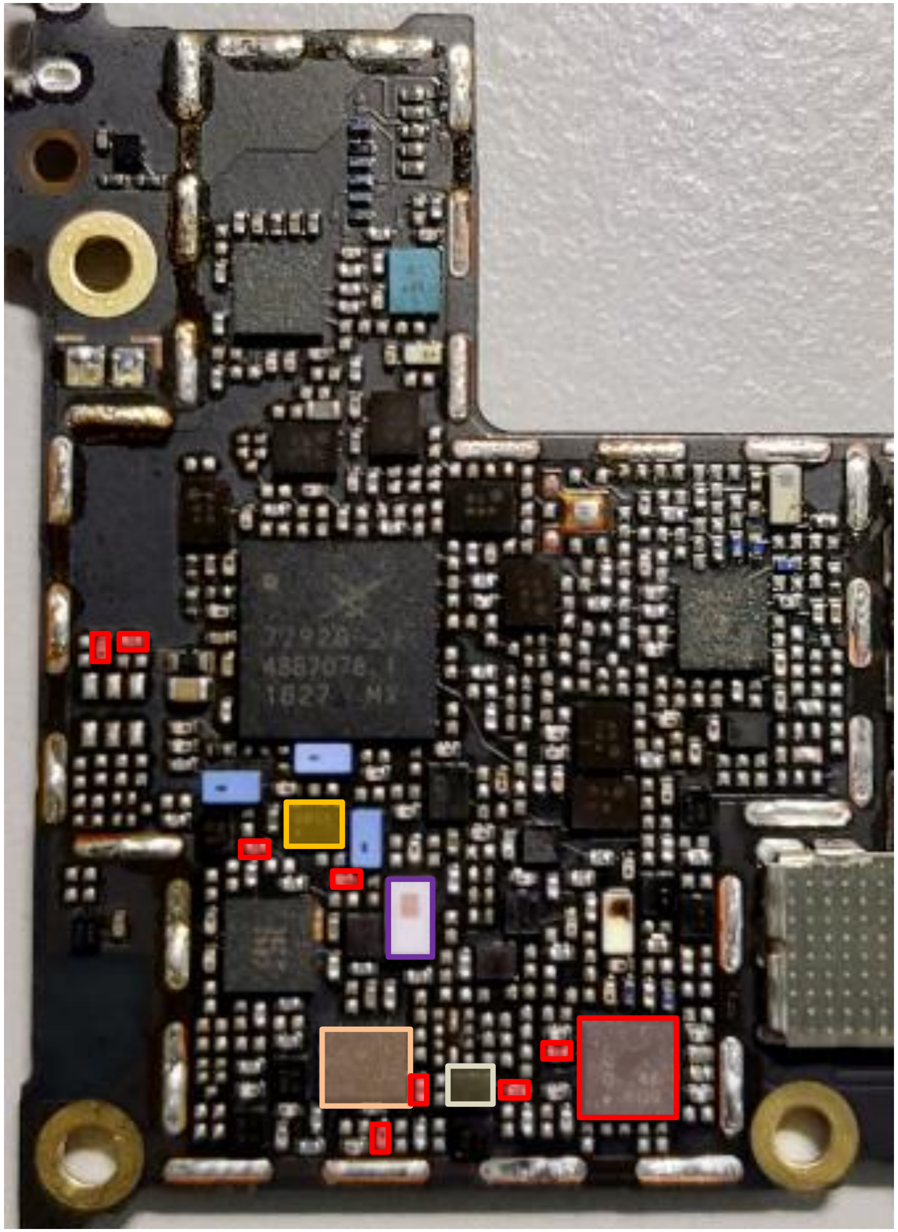


DRX MB/HB QLN4640 U5132

Div. RF connector LB/MB CON4705

DPDT U4715

L/MB Div. RX SW U4704

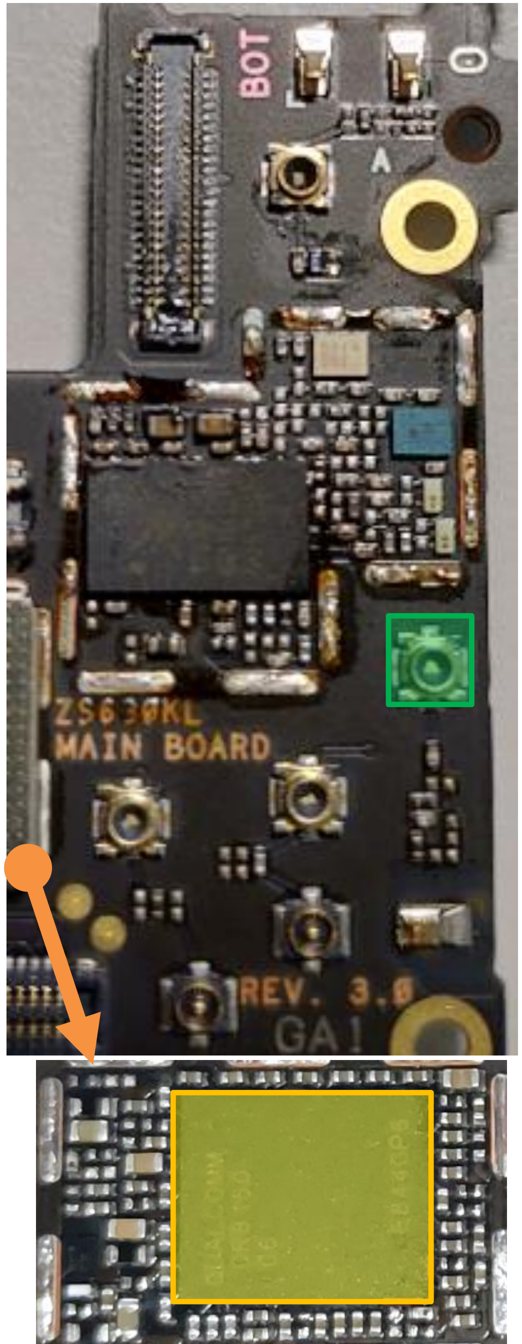
SDR8150 U6201

DiPx U4705

SAW U5102

# DRx Path : B3/DCS(WW)

bottom

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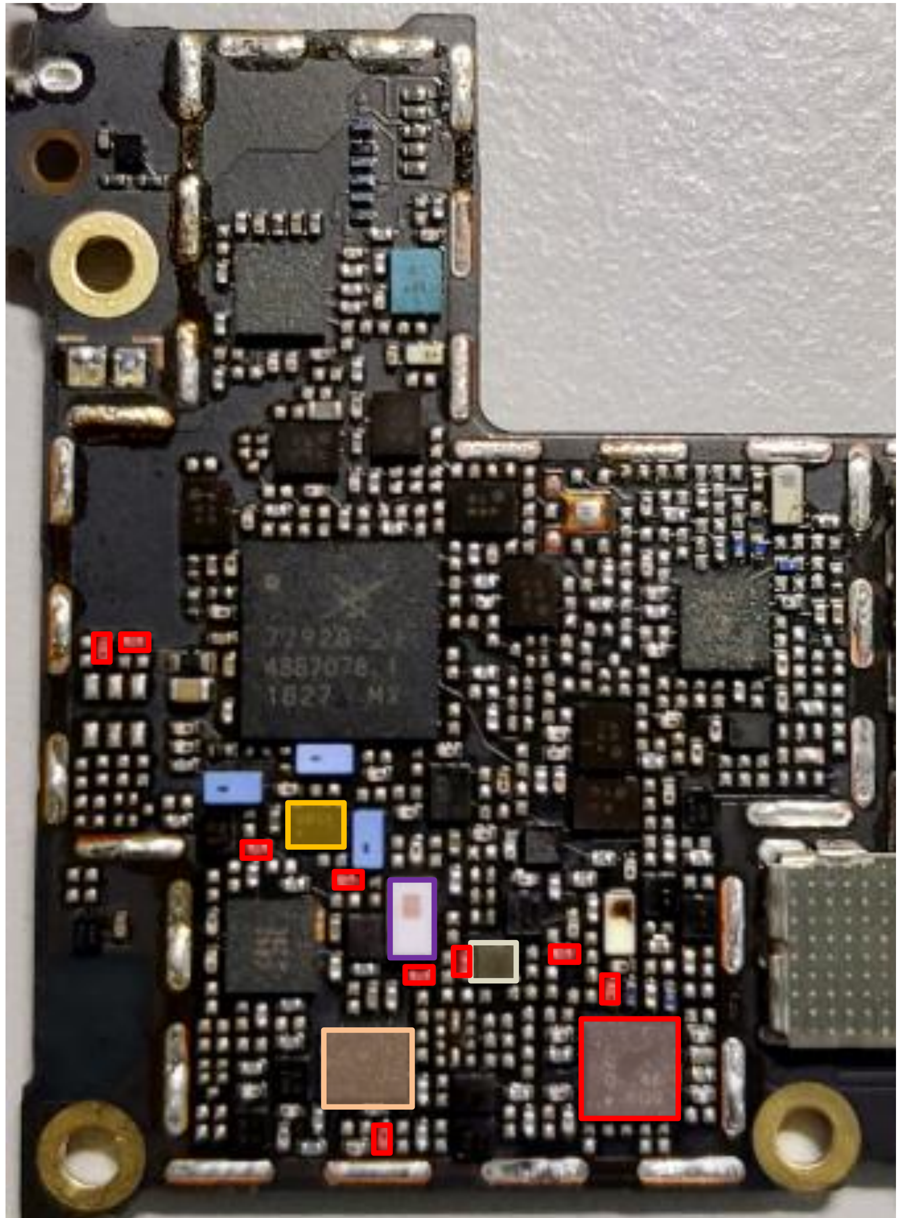


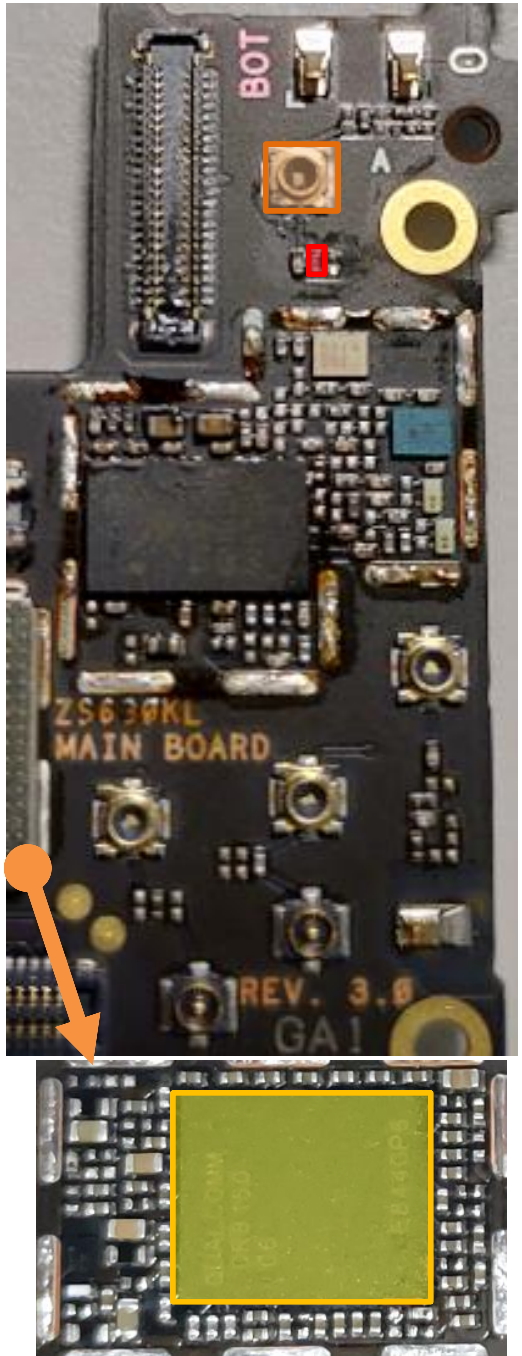
- DRX MB/HB QLN4640 **U5132**
- Div. RF connector LB/MB **CON4705**
- DPDT **U4715**
- L/MB Div. RX SW **U4704**
- SDR8150 **U6201**
- DiPx **U4705**
- SAW **U5106**

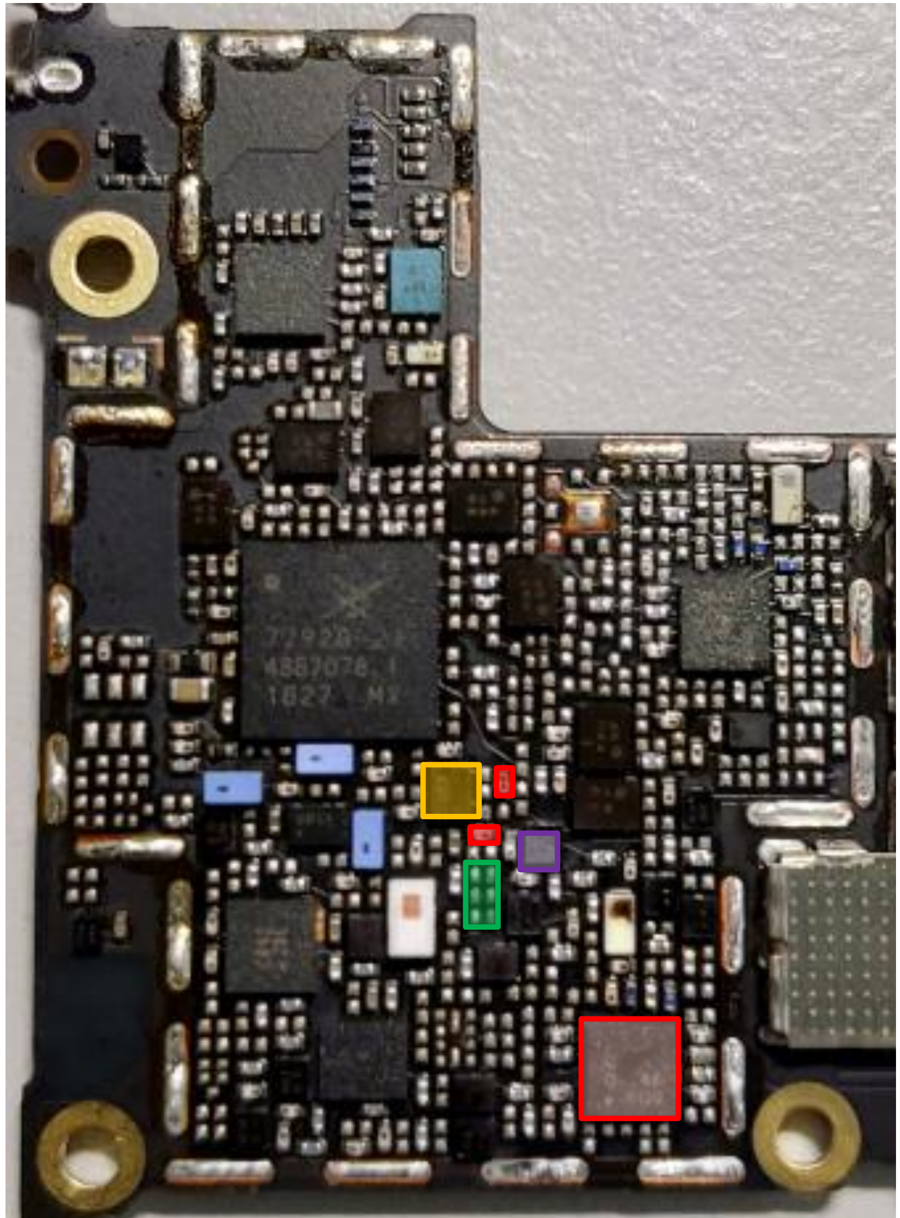
bottom

# HB DRx Common Path(TW)

top

HB : B7/B38/B41



DRX MB/HB QLN4640
**U5132**

Div. RF connector HB
**CON4706**

HB Div. RX SW
**U4711**

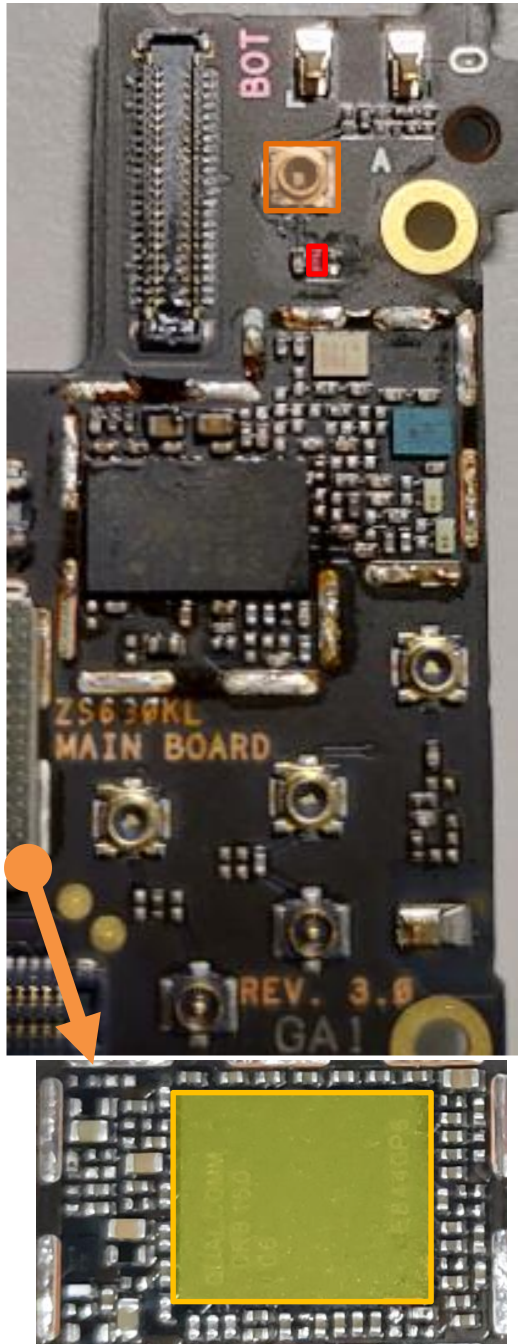
SDR8150
**U6201**

DPDT **U4716**

DiPx(HB&B3MIMO)
**U4702**

# DRx Path : B7(TW)

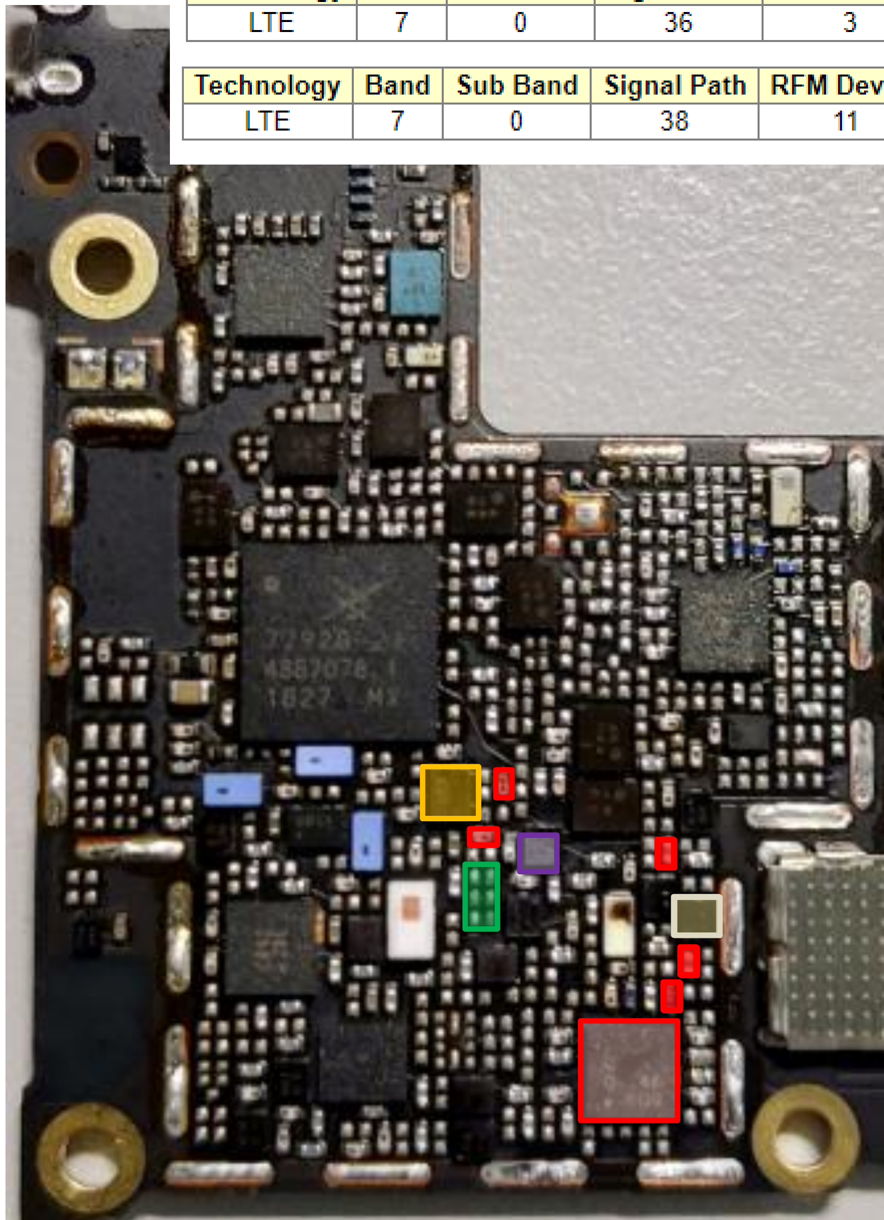
bottom




top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 36 | 3 | 6 | 0 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 7 | 0 | 38 | 11 | 6 | 4 | |



DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

HB Div. RX SW **U4711**

SDR8150 **U6201**

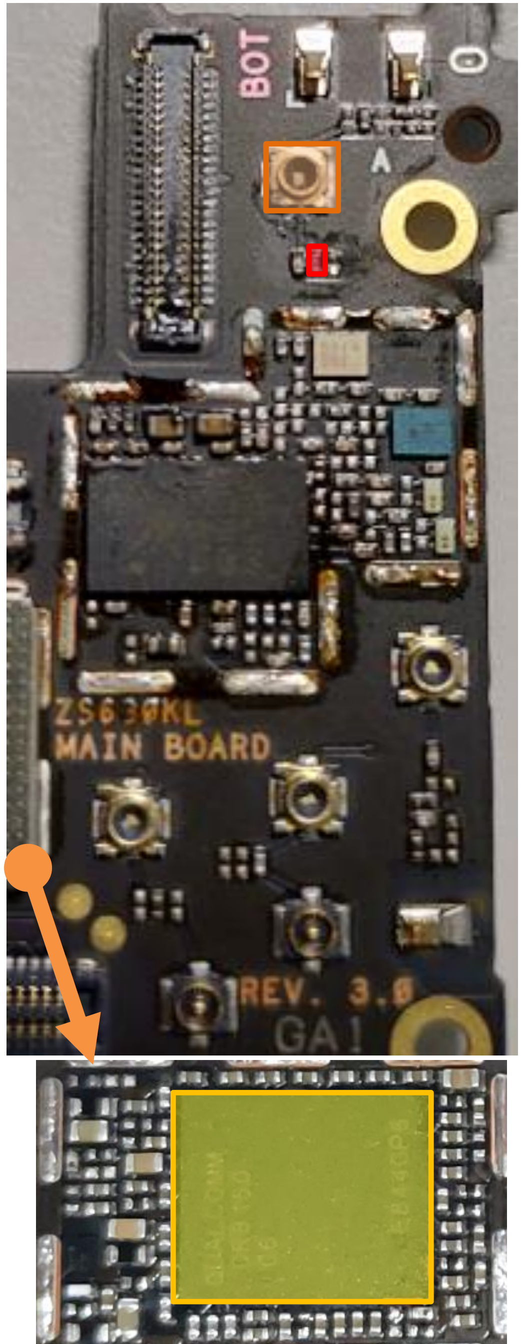
DPDT **U4716**

DiPx(HB&B3MIMO) **U4702**

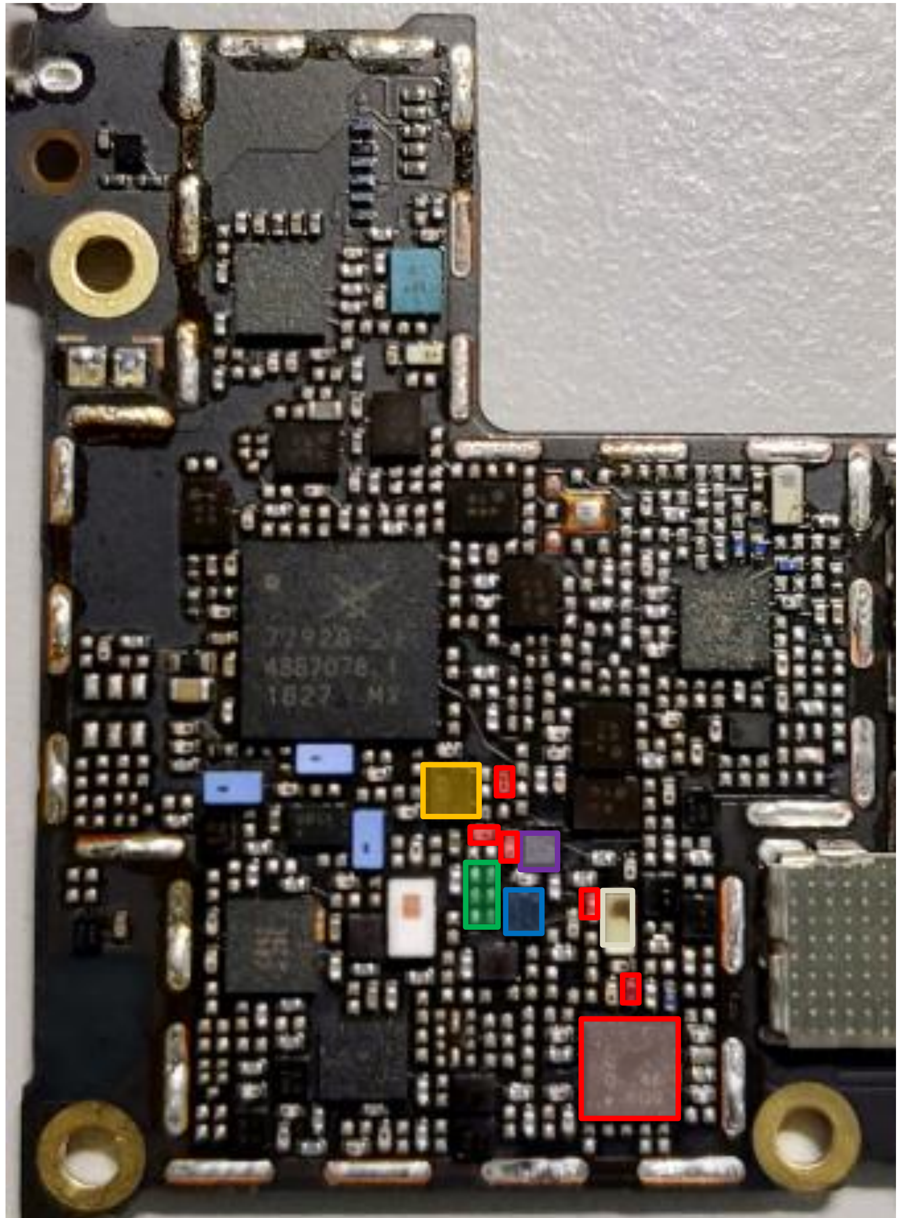
SAW **U5106**

# DRx Path : B38/B41(TW)

bottom



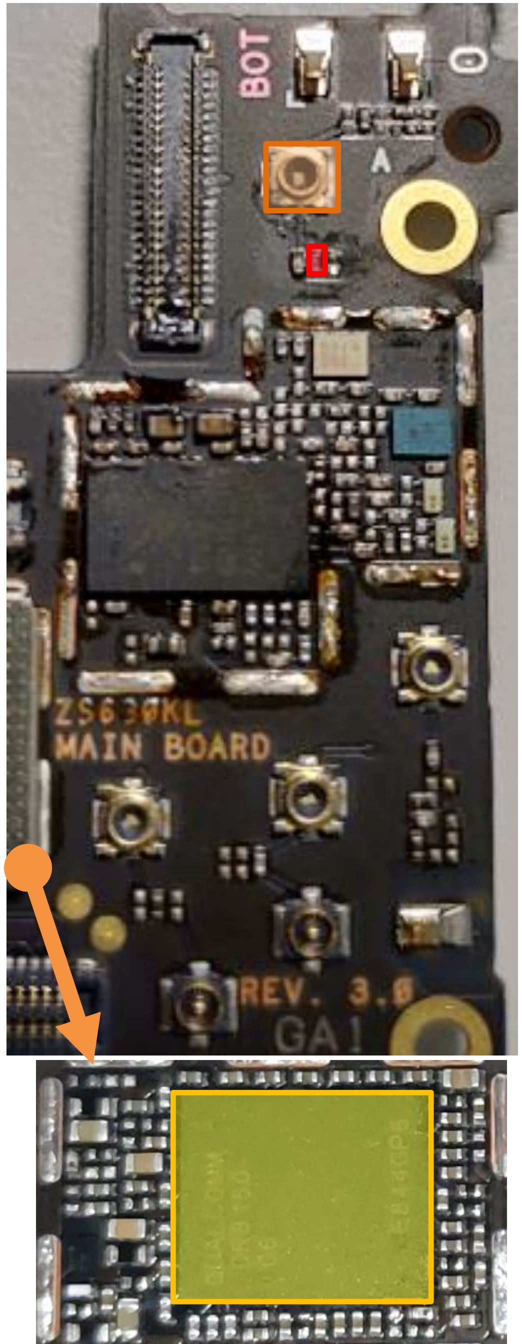

top



- DRX MB/HB QLN4640 **U5132**
- Div. RF connector HB **CON4706**
- HB Div. RX SW **U4711**
- SDR8150 **U6201**
- DPDT **U4716**
- DiPx(HB&B3MIMO) **U4702**
- SAW **U5102**
- Filter **U5127**

# DRx Path : B3 MIMO(TW)

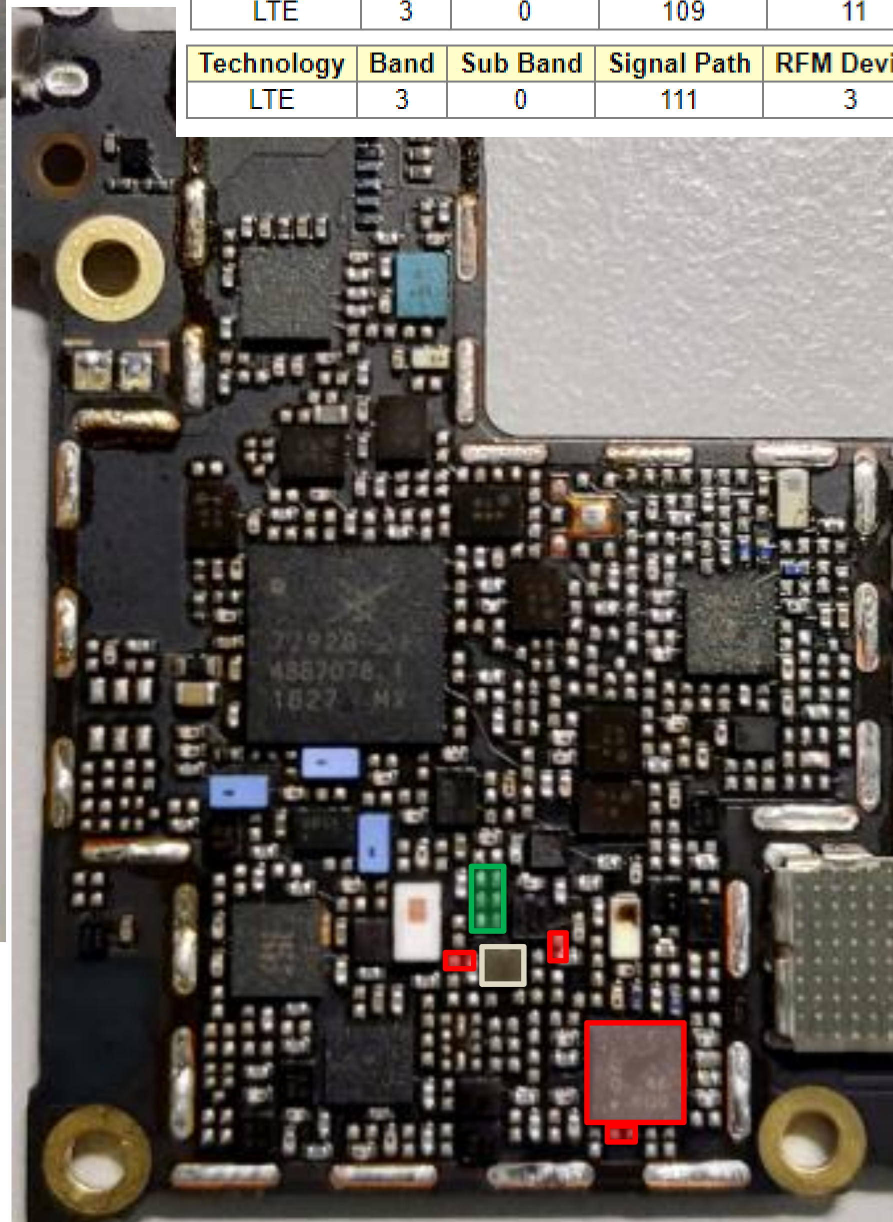
bottom




top

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 109 | 11 | 6 | 4 | |

| Technology | Band | Sub Band | Signal Path | RFM Device | Antenna Path | PLL ID | Time (s) |
|---|---|---|---|---|---|---|---|
| LTE | 3 | 0 | 111 | 3 | 6 | 0 | |



DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

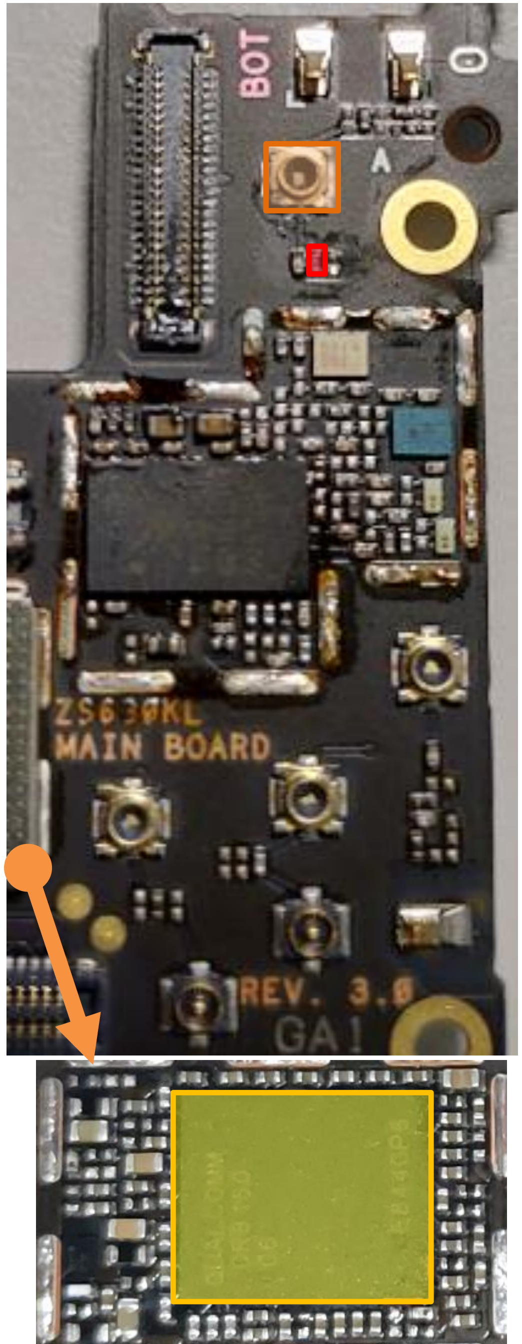
SDR8150 **U6201**

DiPx(HB&B3MIMO) **U4702**

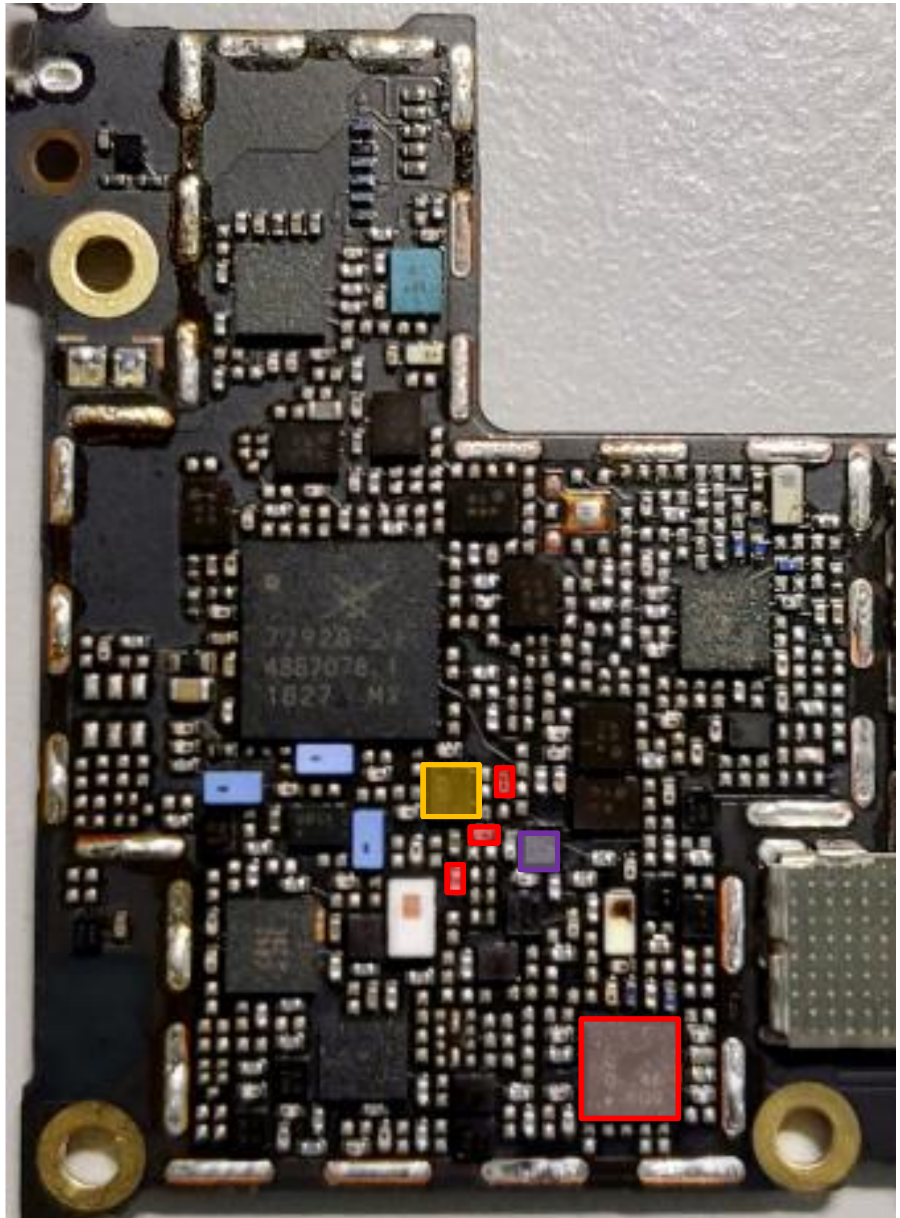
SAW **U5135**

# HB DRx Common Path(WW)

bottom




top



HB : B7/B38/B40/B41

DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

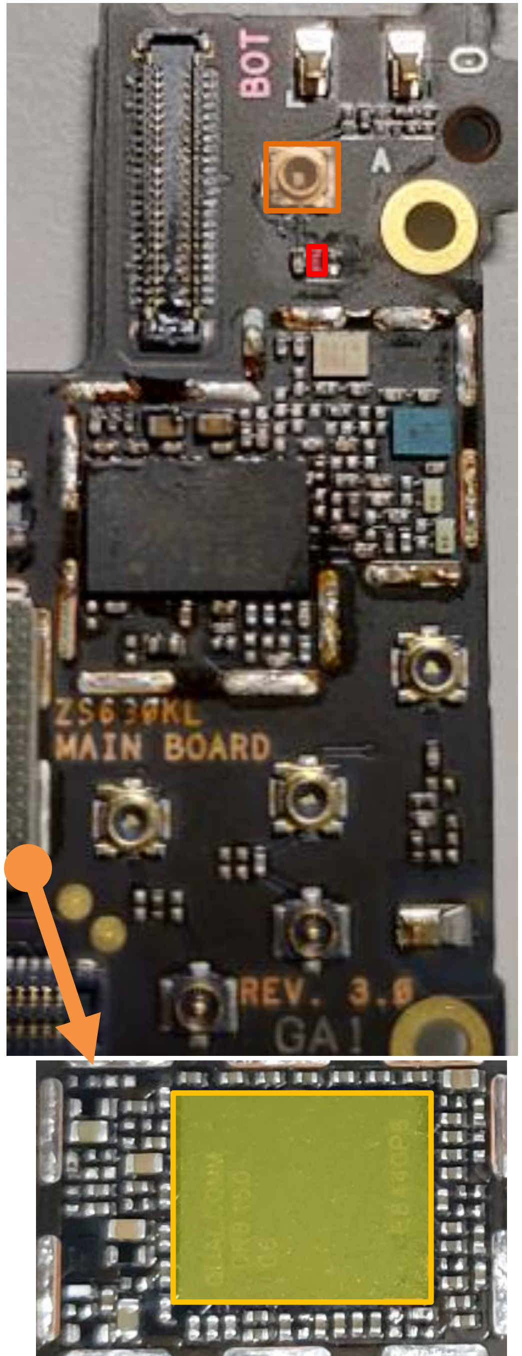
HB Div. RX SW **U4711**

SDR8150 **U6201**

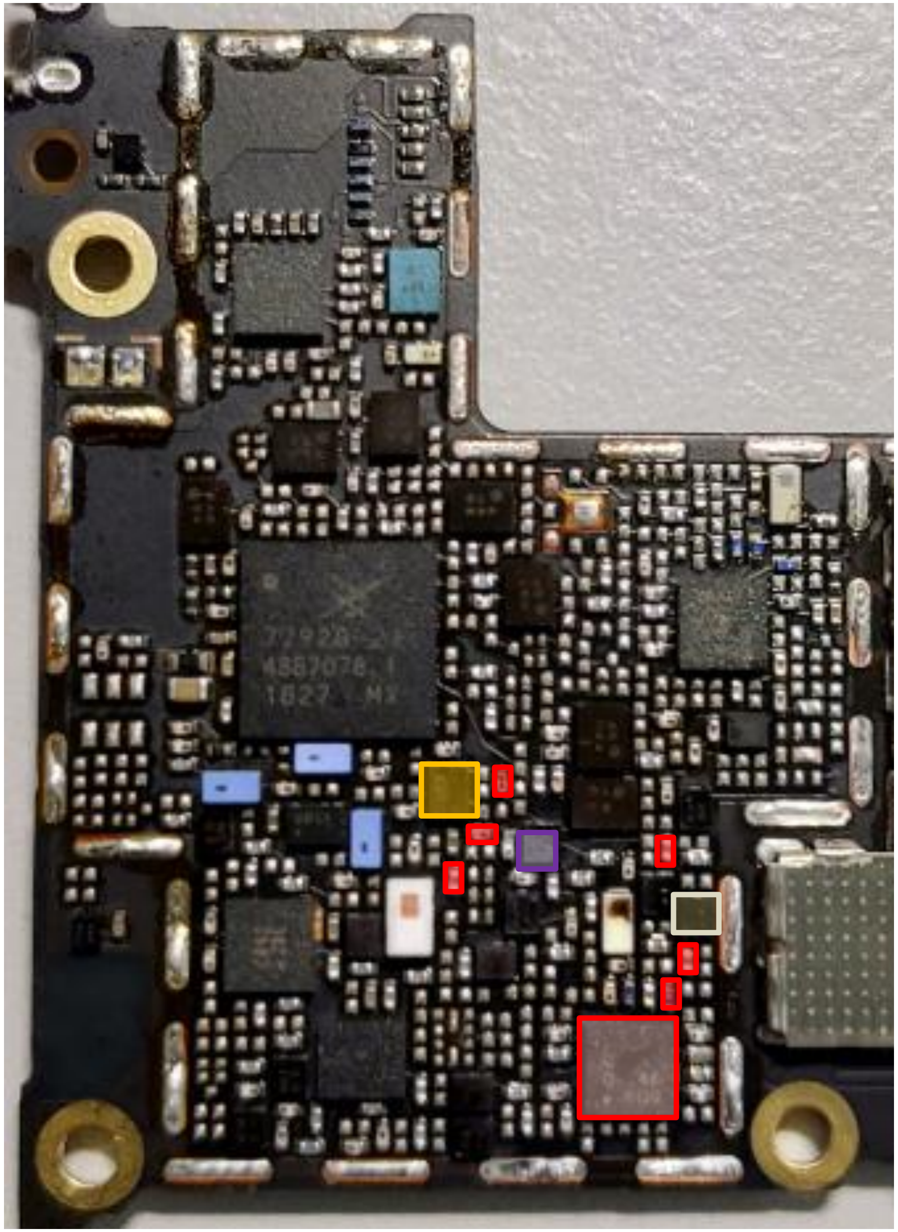
DPDT **U4716**

# DRx Path : B7(WW)

bottom




top



DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

HB Div. RX SW **U4711**

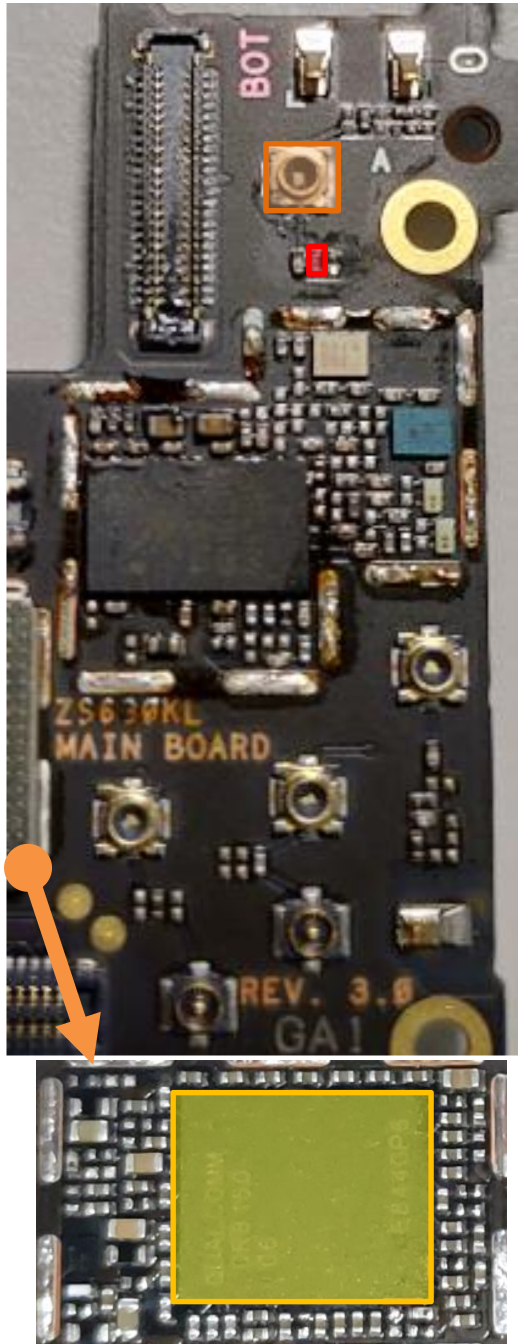
SDR8150 **U6201**

DPDT **U4716**

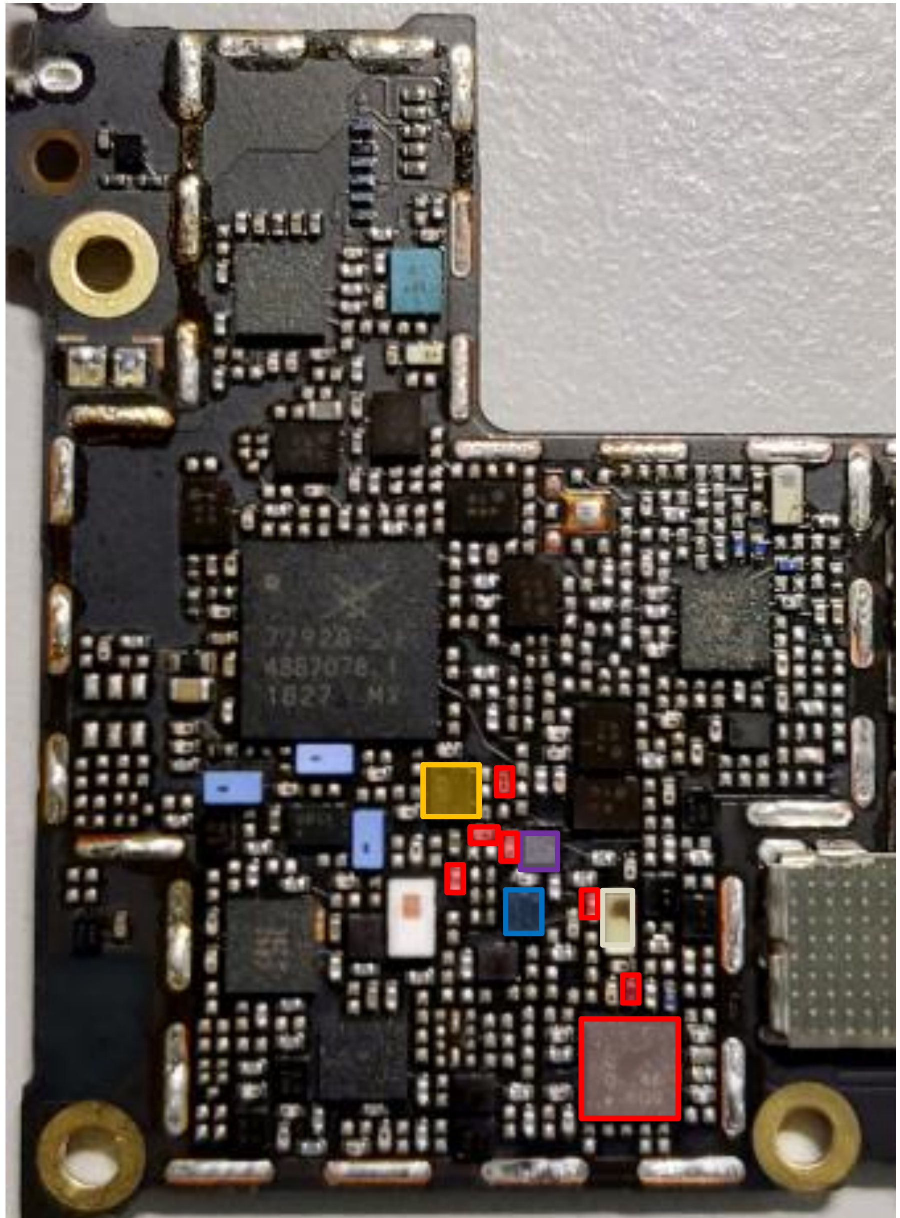
SAW **U5106**

# DRx Path : B38/B41(WW)

bottom




top



DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

HB Div. RX SW **U4711**

SDR8150 **U6201**

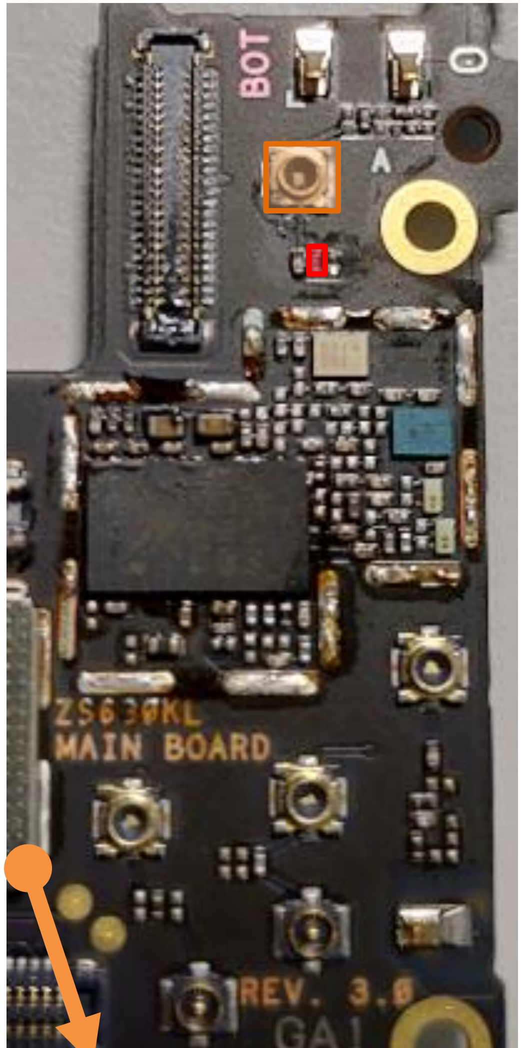
DPDT **U4716**

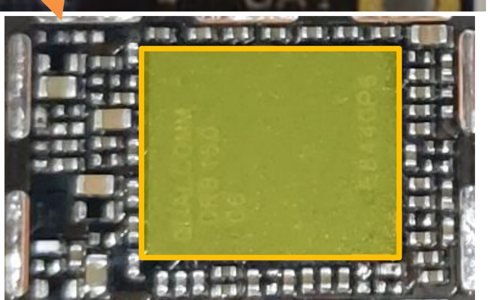
SAW **U5102**

Filter **U5127**

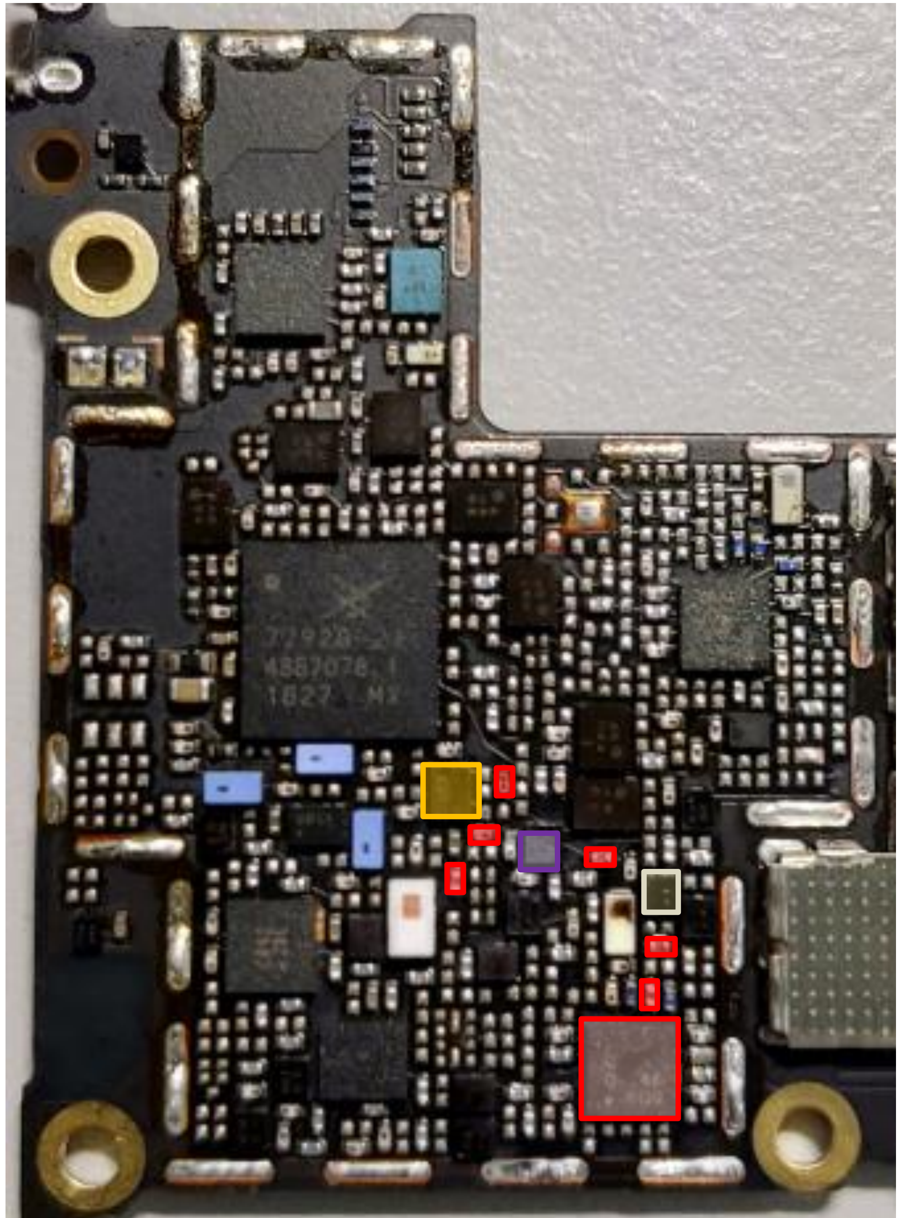
# DRx Path : B40(WW)

bottom



top



DRX MB/HB QLN4640 **U5132**

Div. RF connector HB **CON4706**

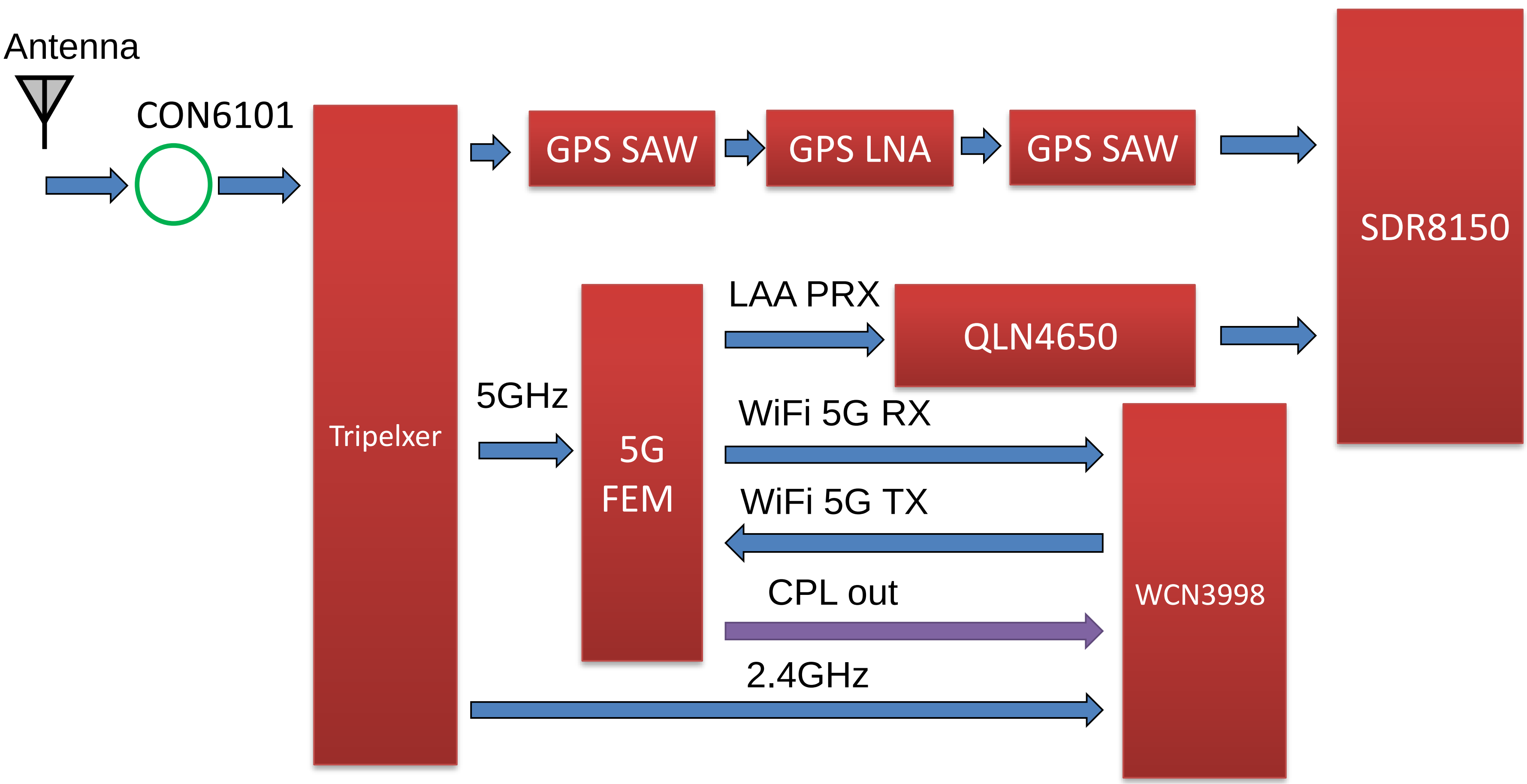
HB Div. RX SW **U4711**

SDR8150 **U6201**

DPDT **U4716**

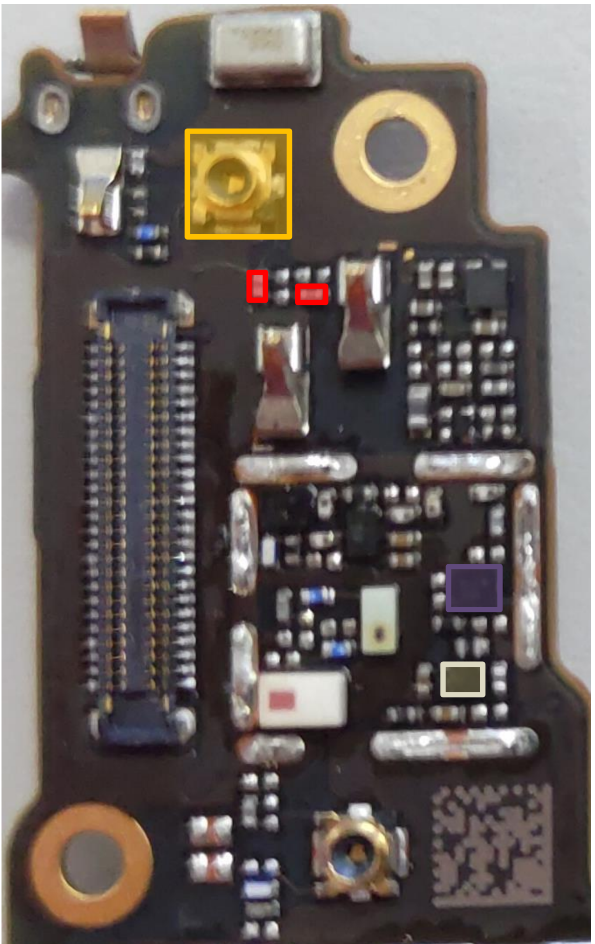
SAW **U5108**

# WIFI CH0/BT/GPS L1/LAA PRx Block Diagram

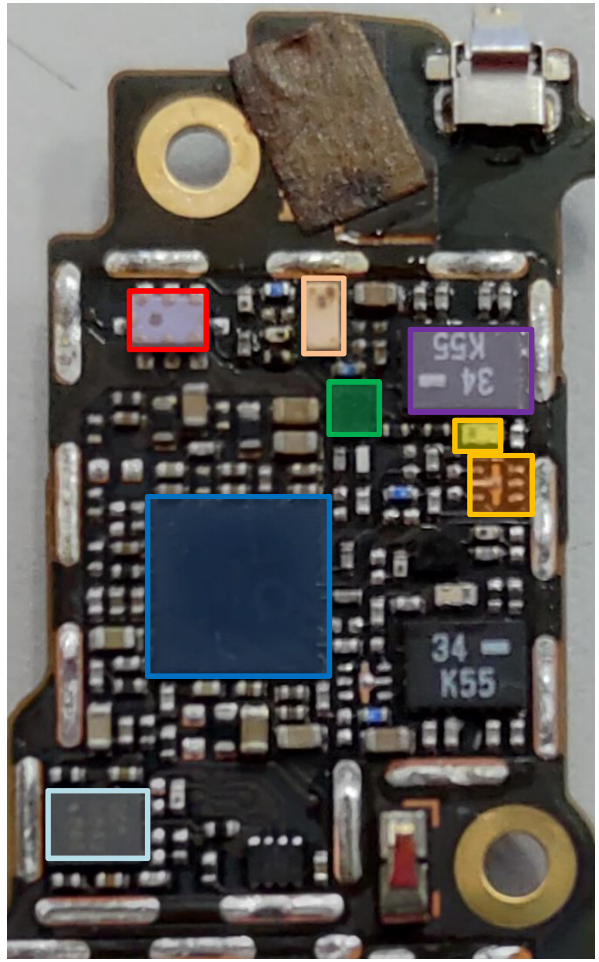


# WIFI CH0/BT/GPS L1/LAA PRx Key Components

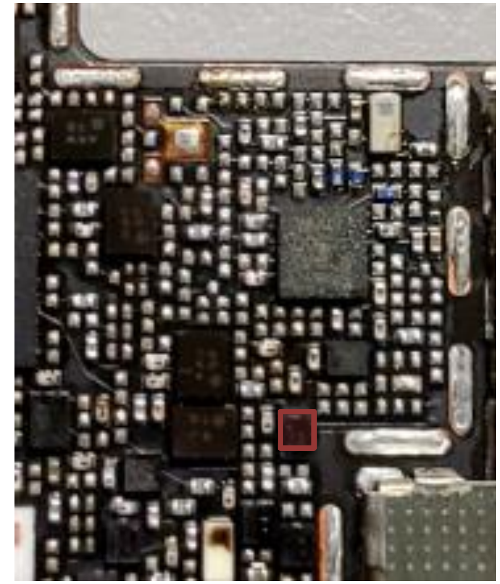
bottom




Top(right)



Top(left)



CH0 Triplexer GPS/2.4G/5G **U6105**

CH0 2.4G SAW **U6107**

WCN3998 (WIFI CHIP) **U6101**

CH0 5G SAW **U6110**

CH0 5G FEM(TW) **U6120**

CH0 5G SPDT(WW) **U6116**

CH0 5G Filter **U6119**

CH0 connector **CON6101**

QLN4650(TW) **U6132**

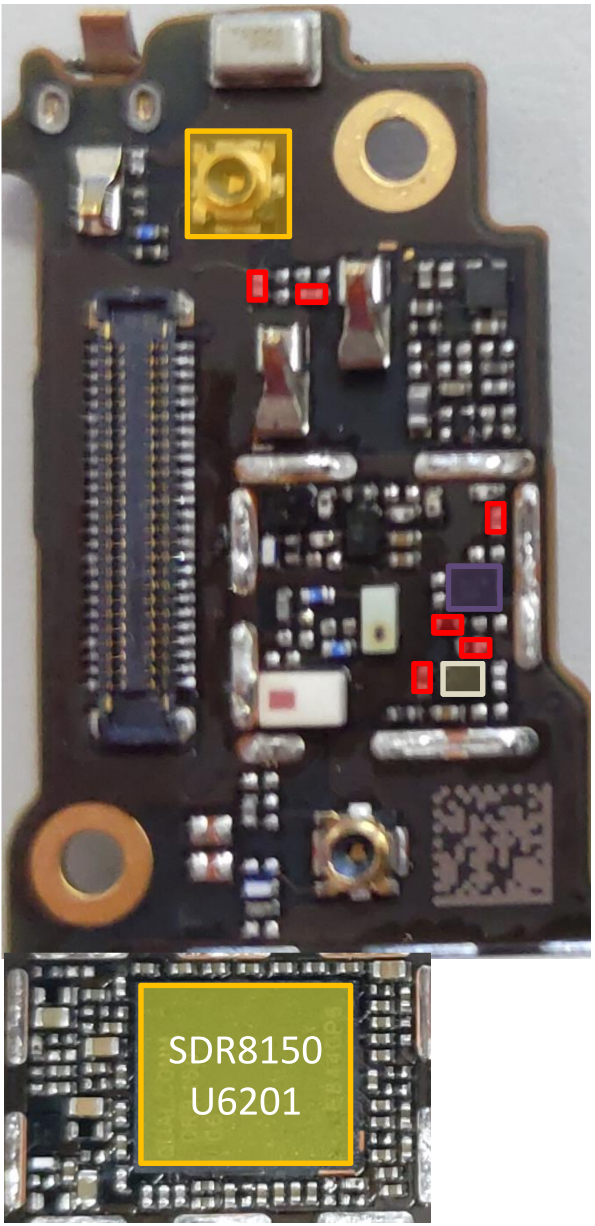
GPS SAW **U6131**

GPS SAW **U6133**

GPS LNA **U6104**

# GPS L1 (Error Code:8221)

bottom

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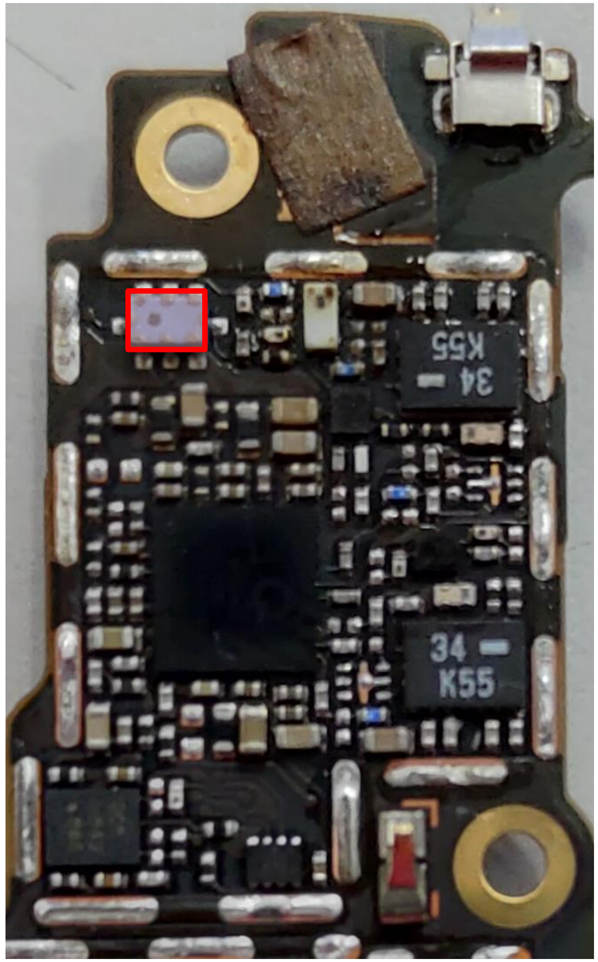


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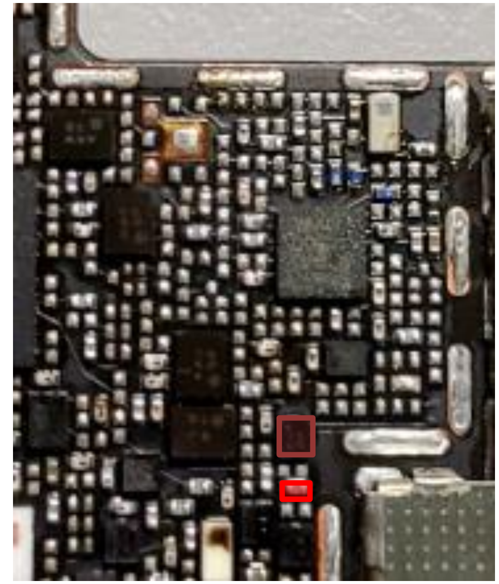


CH0 Triplexer GPS/2.4G/5G **U6105**

CH0 connector **CON6101**

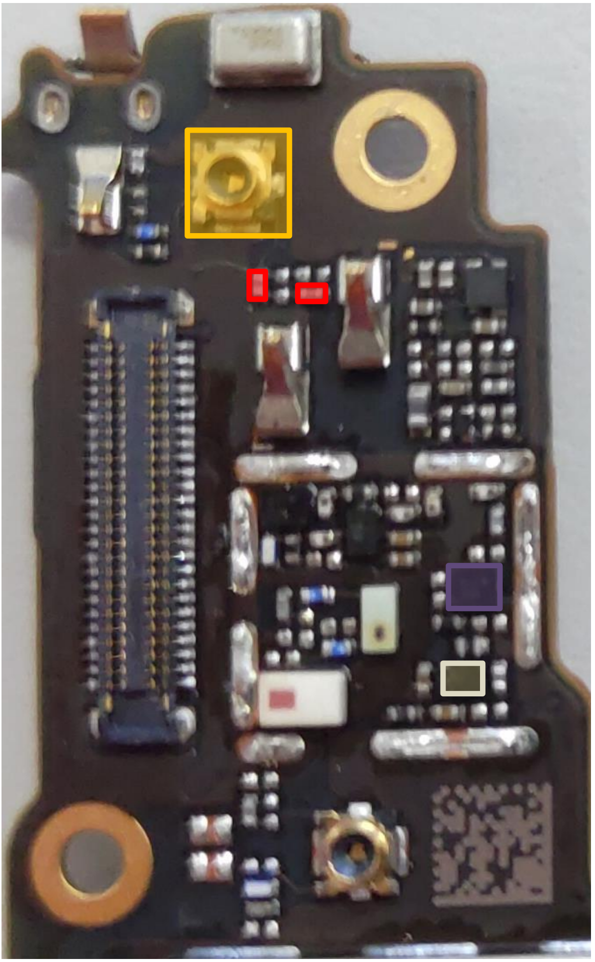
GPS SAW **U6131**

GPS SAW **U6133**

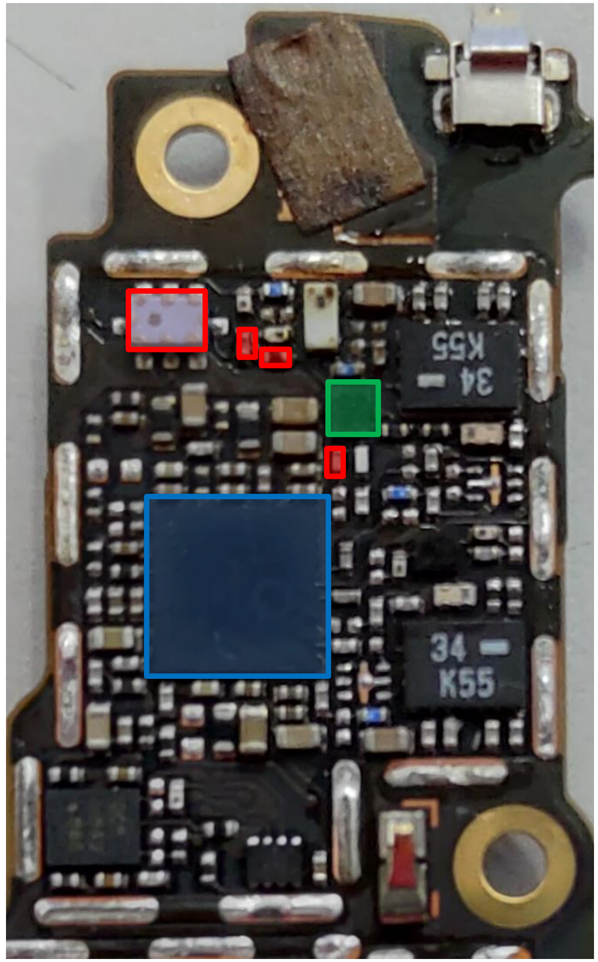
GPS LNA **U6104**

# WIFI CH0 2.4G/BT (Error Code:1109,1111,1142)

bottom



Top(right)

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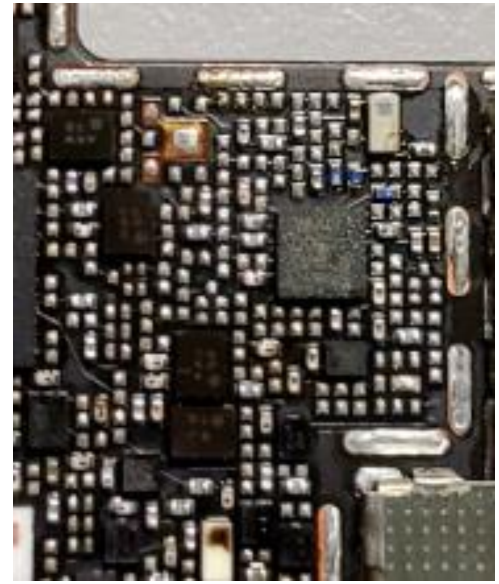


CH0 connector **CON6101**

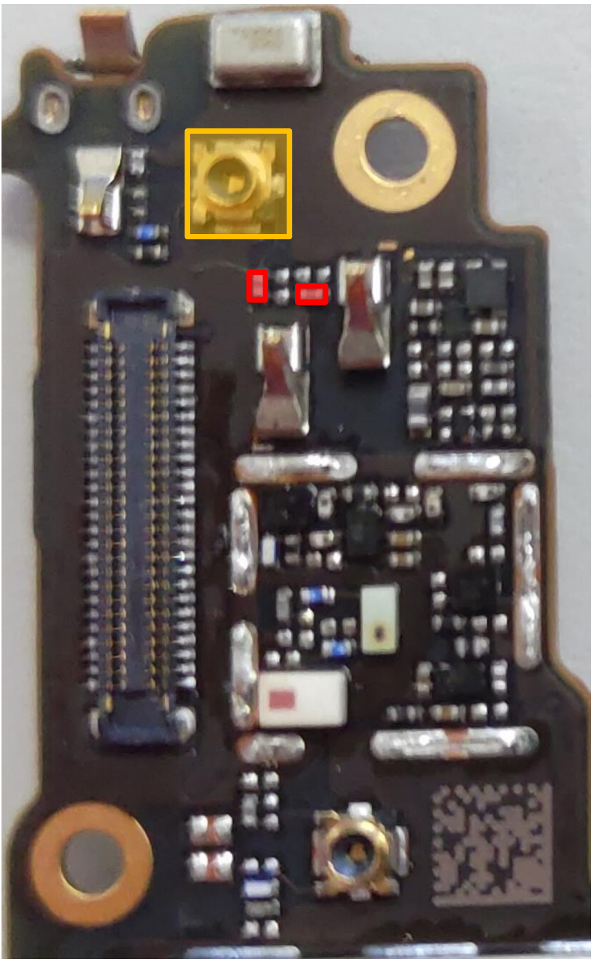
CH0 Triplexer GPS/2.4G/5G **U6105**

CH0 2.4G SAW **U6107**

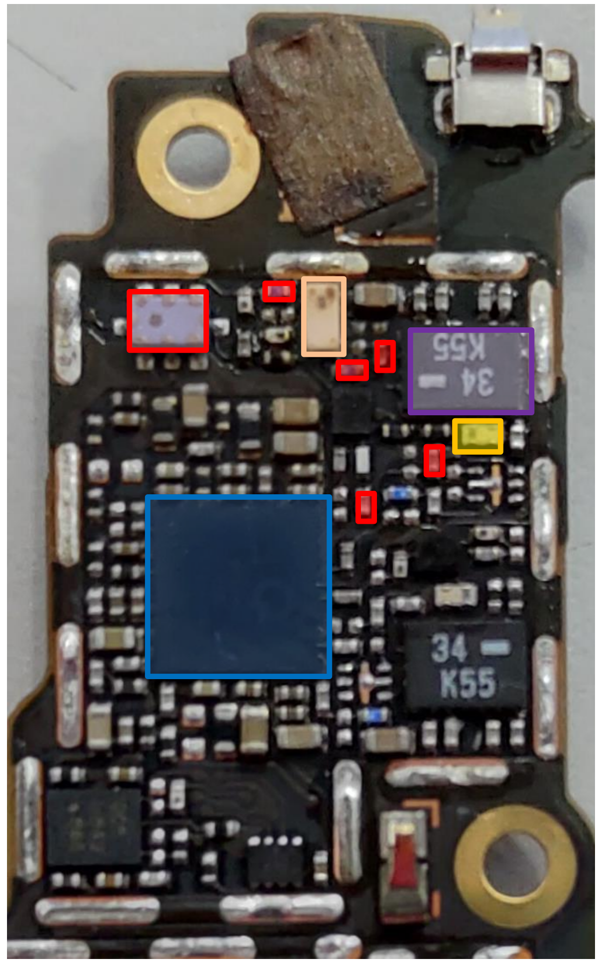
WCN3998 (WIFI CHIP) **U6101**

# WIFI CH0 5G Tx(TW)(Error Code:1109,1111)

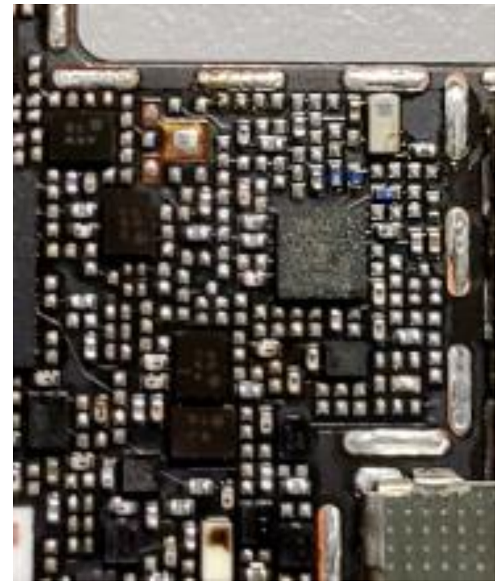
**bottom**



**Top(right)**



**Top(left)**



CH0 5G SAW **U6110**

CH0 5G FEM(TW) **U6120**

CH0 5G Filter **U6119**

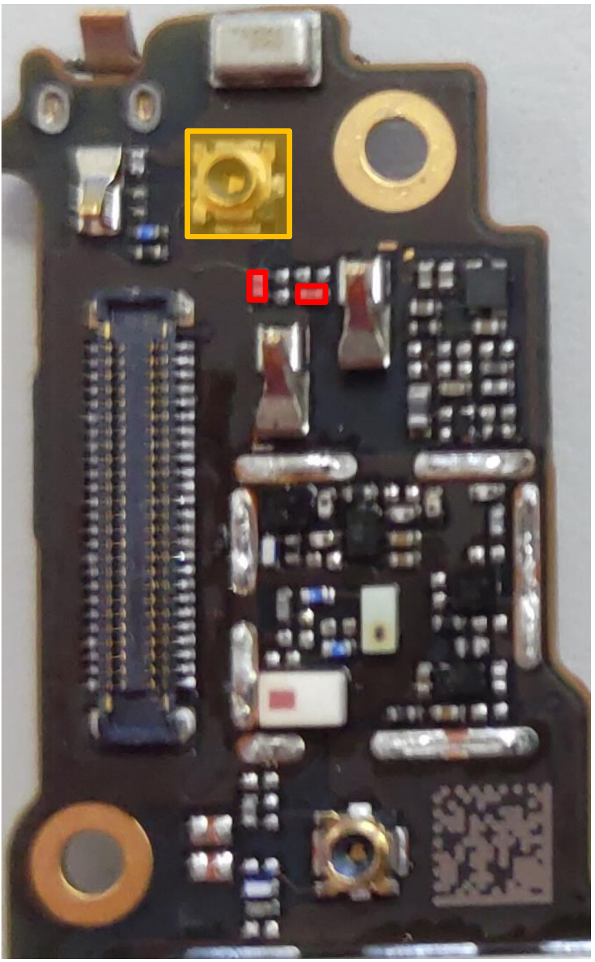
CH0 connector **CON6101**

CH0 Triplexer GPS/2.4G/5G **U6105**

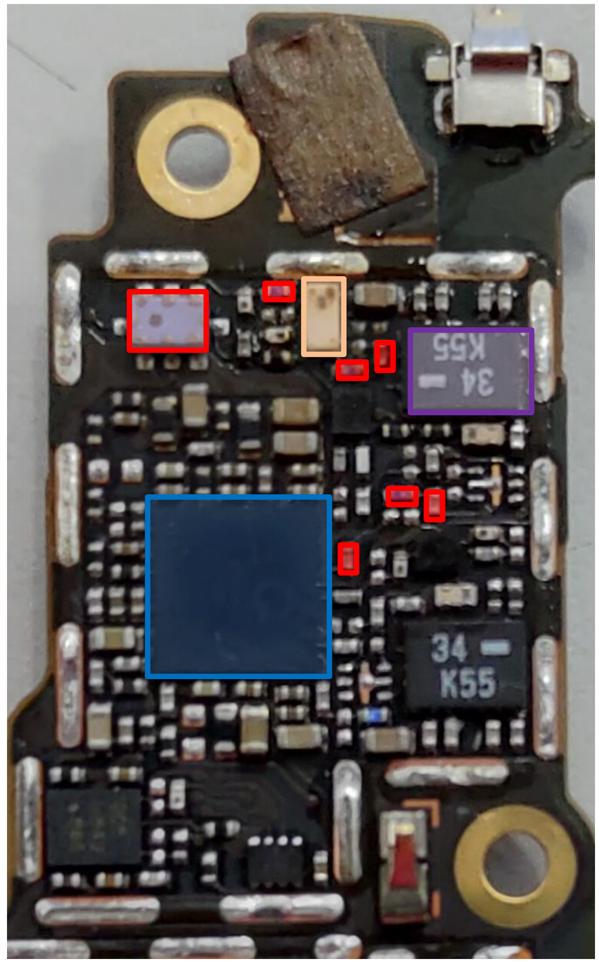
WCN3998 (WIFI CHIP) **U6101**

# WIFI CH0 5G Rx(TW)(Error Code:1142)

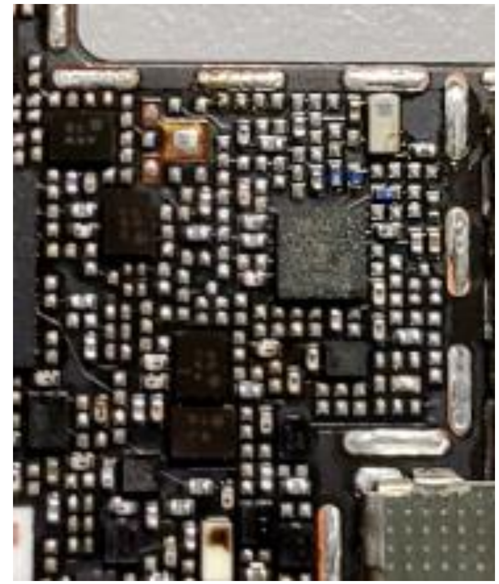
**bottom**



**Top(right)**




**Top(left)**



CH0 5G SAW **U6110**

CH0 5G FEM(TW) **U6120**

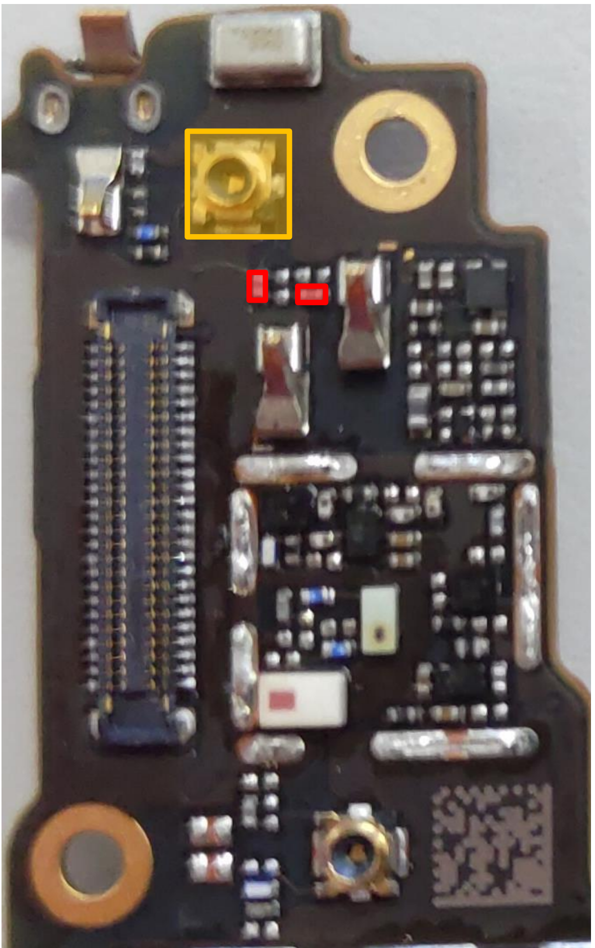
CH0 connector **CON6101**

CH0 Triplexer GPS/2.4G/5G **U6105**

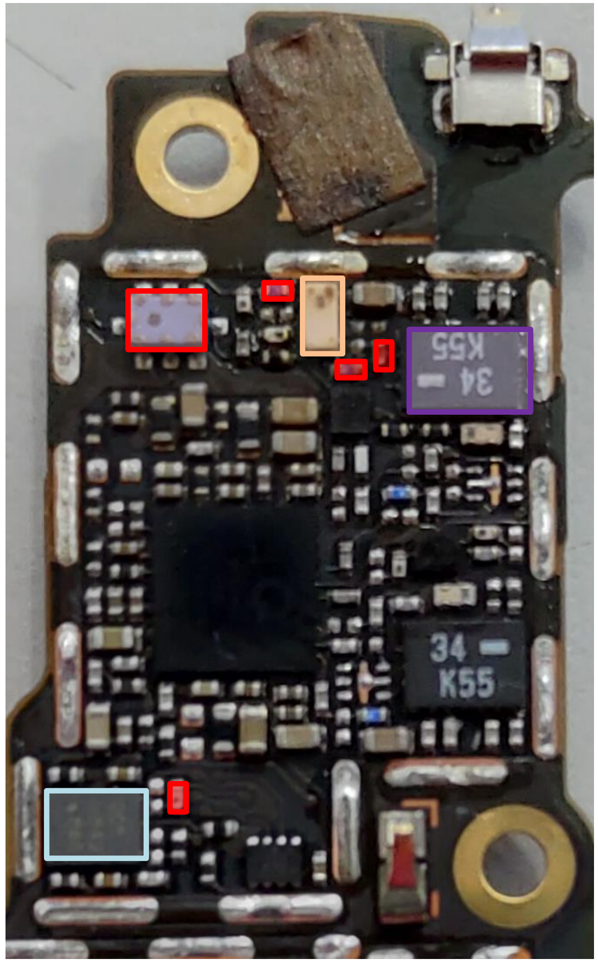
WCN3998 (WIFI CHIP) **U6101**

# LAA B46 PRx(TW)

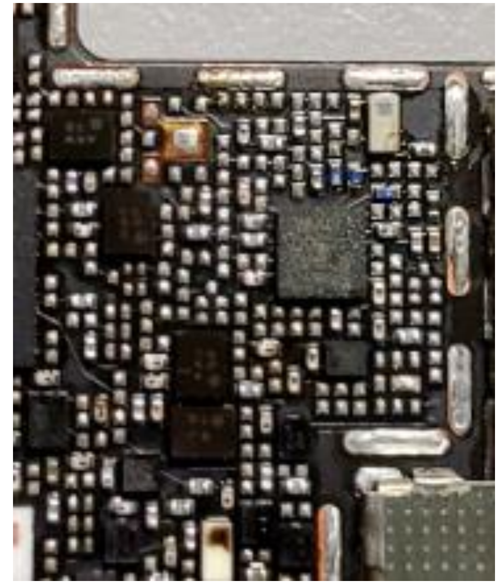
**bottom**




**Top(right)**




**Top(left)**



CH0 5G SAW **U6110**

CH0 5G FEM(TW) **U6120**

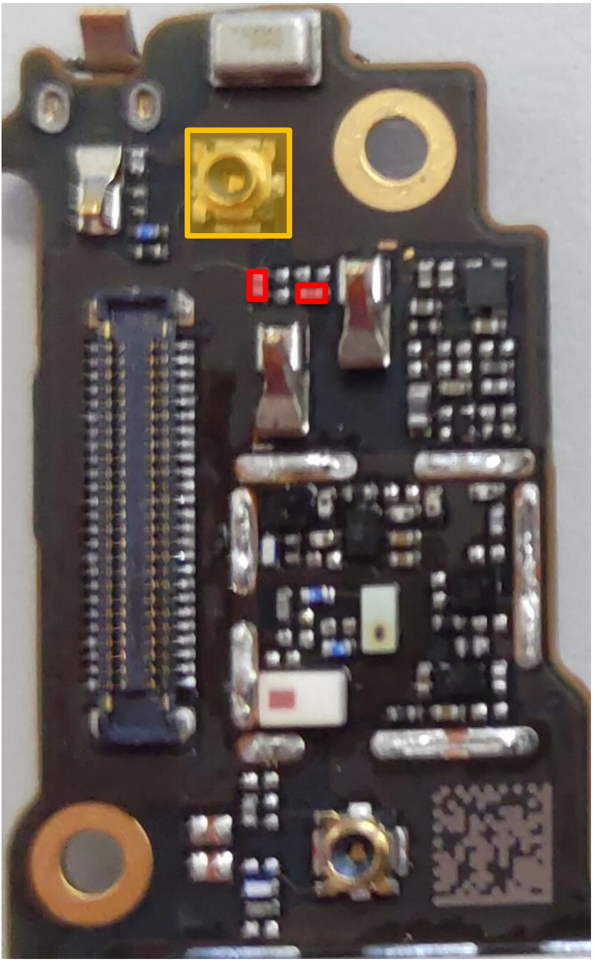
QLN4650(TW) **U6132**

CH0 connector **CON6101**

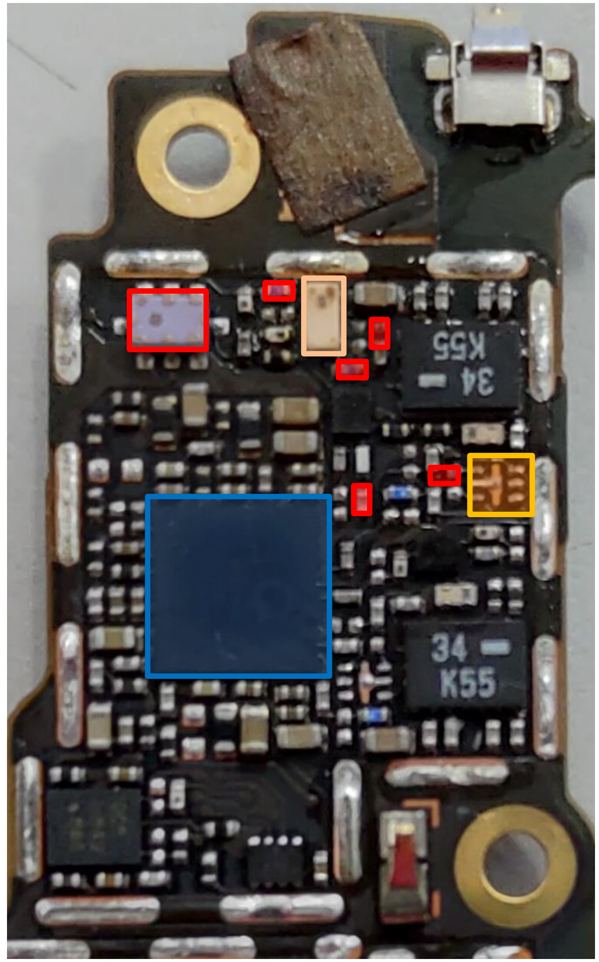
CH0 Triplexer GPS/2.4G/5G **U6105**

# WIFI CHO 5G Tx(WW)(Error Code:1109,1111)

bottom



Top(right)

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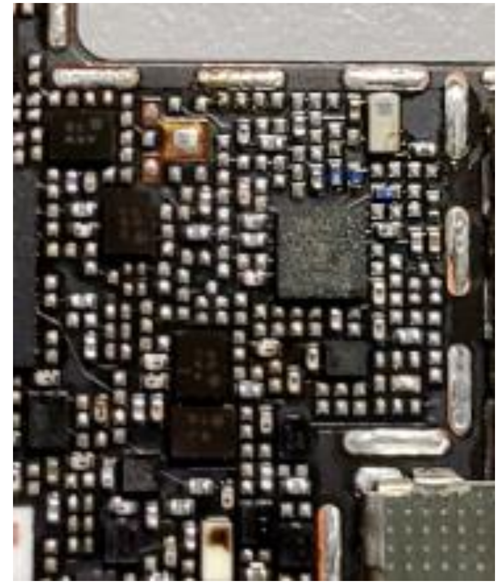


CH0 5G SAW **U6110**

CH0 5G SPDT(WW) **U6116**

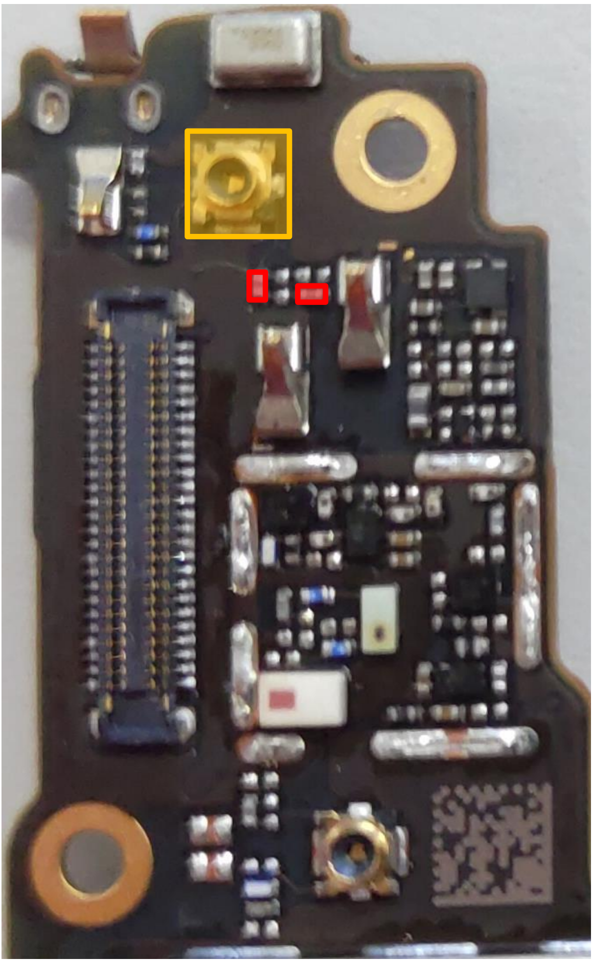
CH0 connector **CON6101**

CH0 Triplexer GPS/2.4G/5G **U6105**

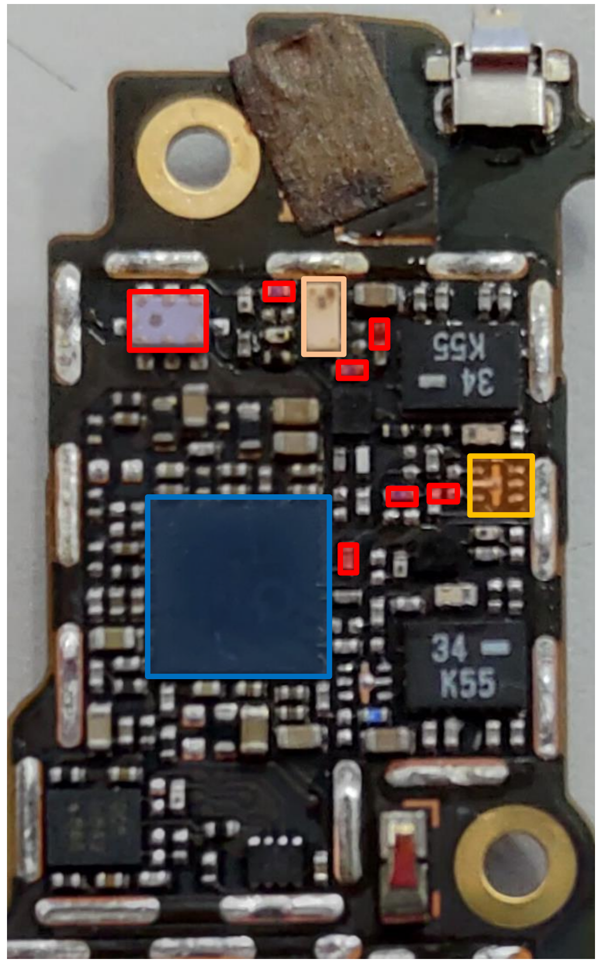
WCN3998 (WIFI CHIP) **U6101**

# WIFI CHO 5G Rx(WW)(Error Code:1142)

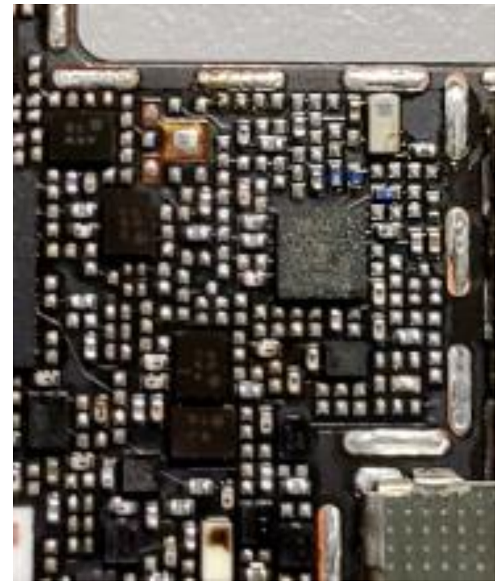
bottom



Top(right)



Top(left)



CH0 5G SAW **U6110**

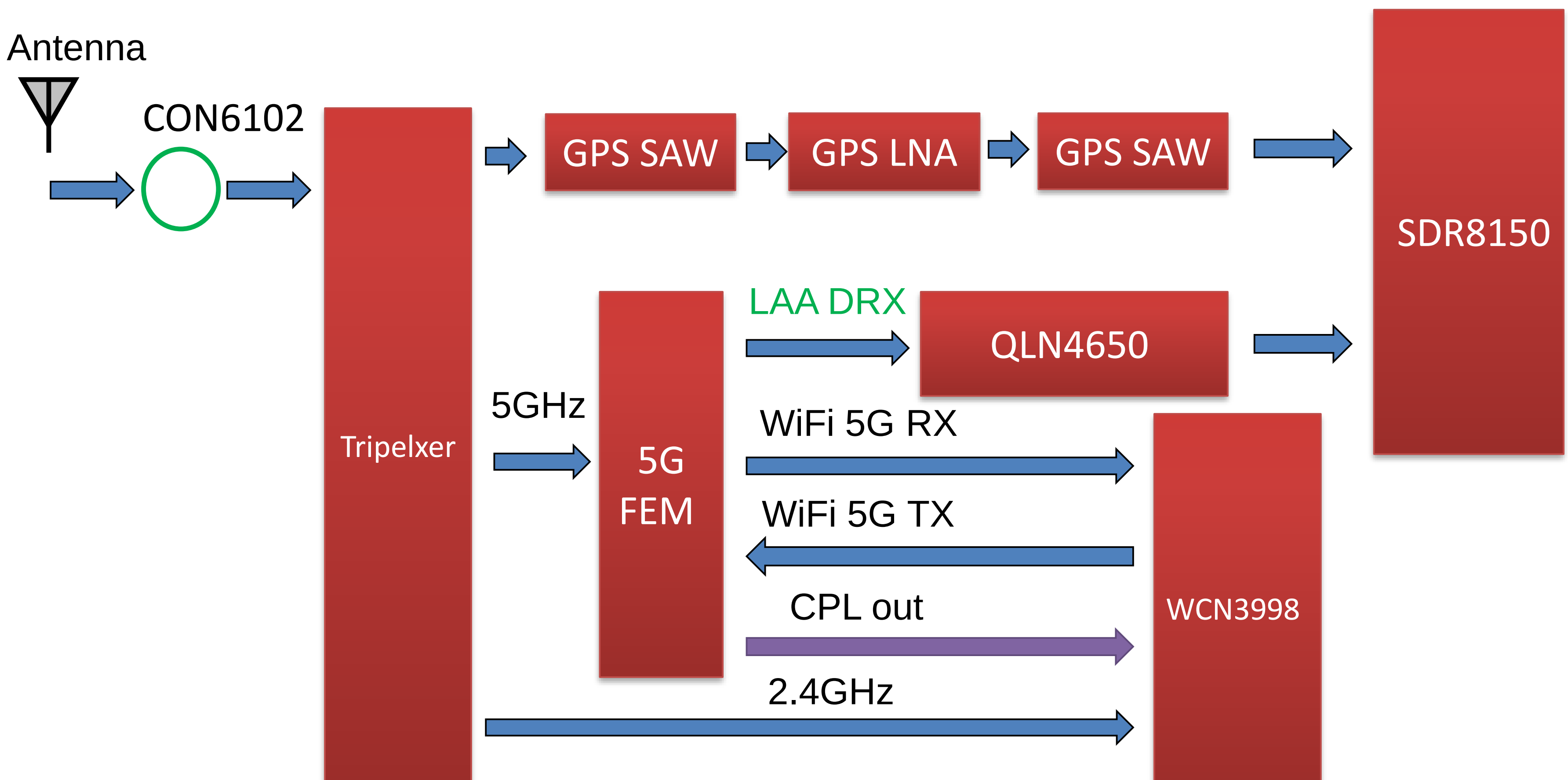
CH0 5G SPDT(WW) **U6116**

CH0 connector **CON6101**

CH0 Triplexer GPS/2.4G/5G **U6105**

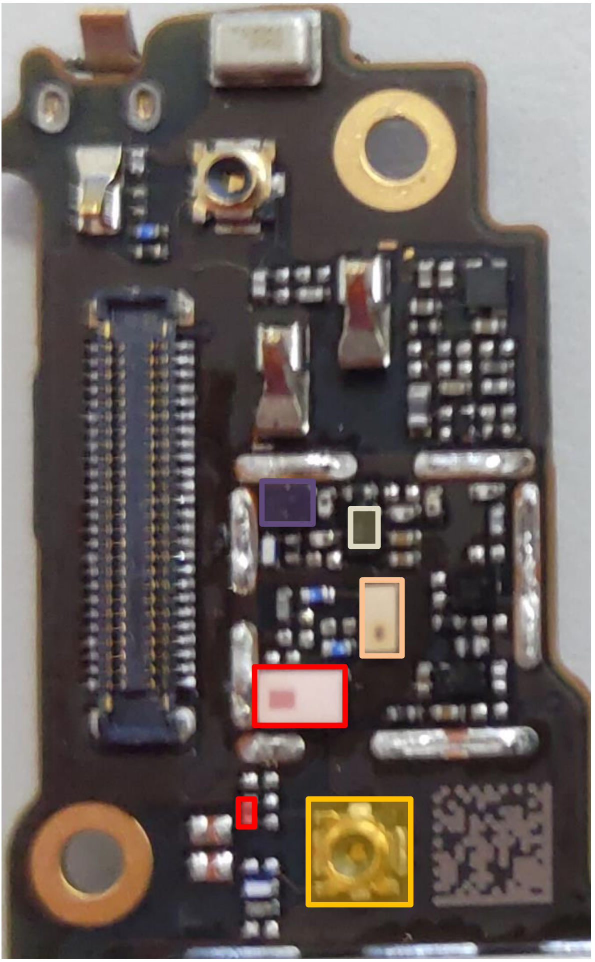
WCN3998 (WIFI CHIP) **U6101**

# WIFI CH1/GPS L2,L5/LAA DRx Block Diagram

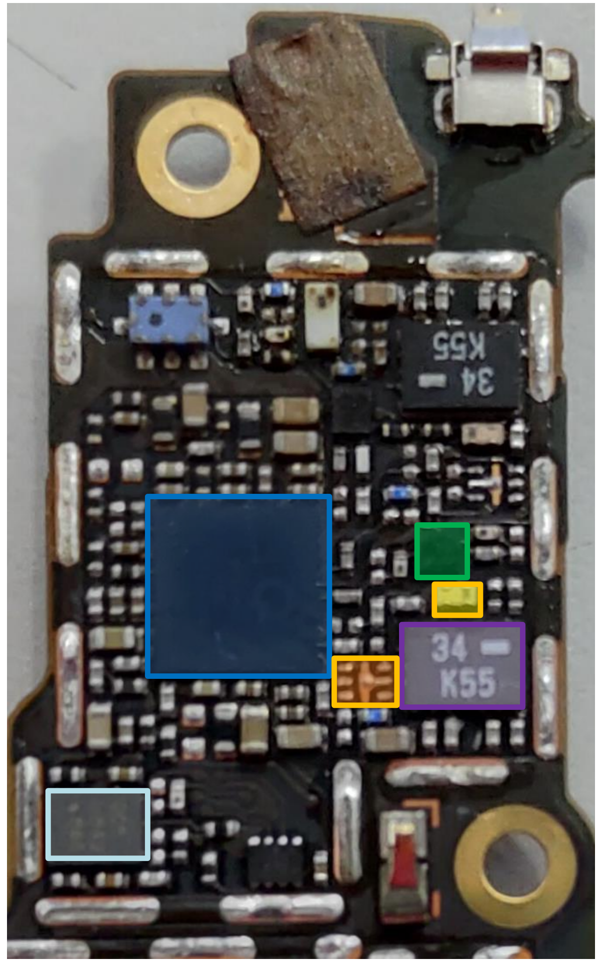


# WIFI CH1/GPS L2,L5/LAA DRx Key Components

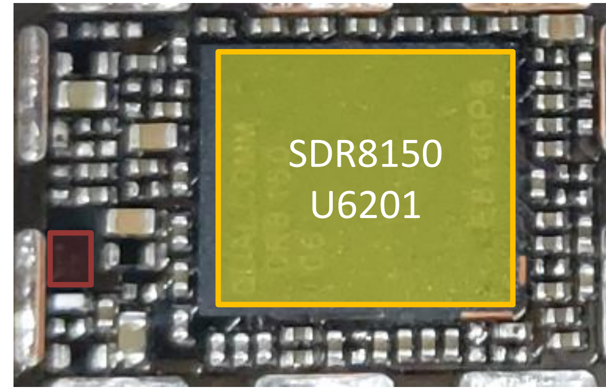
**Bottom(left)**



**Top**




**Bottom(right)**

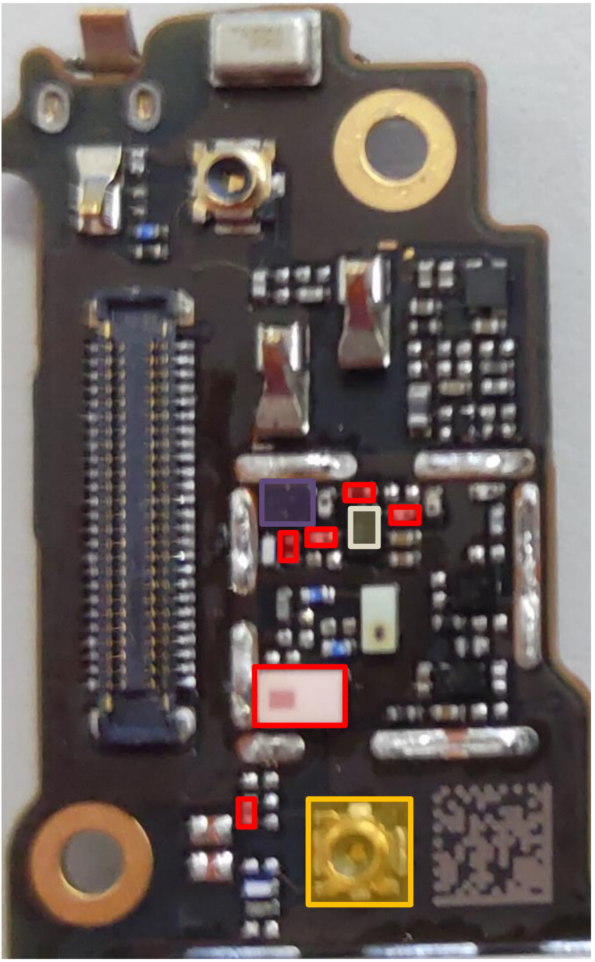



- CH1 5G SAW **U6112**
- CH1 5G FEM(TW) **U6117**
- CH1 5G SPDT(WW) **U6111**
- CH1 5G Filter **U6118**
- CH1 connector **CON6102**
- QLN4650(TW) **U6132**
- GPS SAW **U6121**
- GPS SAW **U6122**
- GPS LNA **U6109**

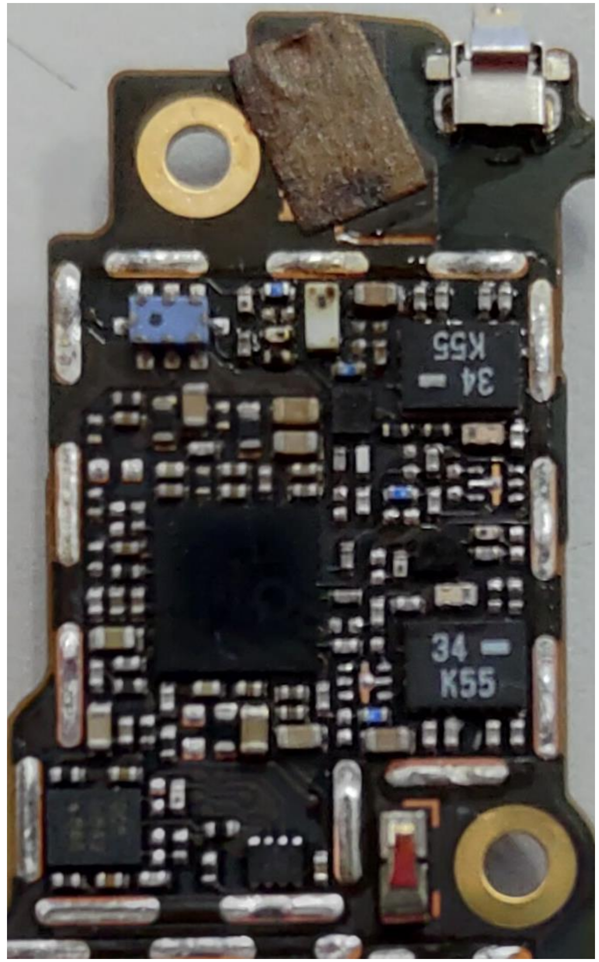
- CH1 Triplexer GPS/2.4G/5G **U6106**
- CH1 2.4G SAW **U6114**
- WCN3998 (WIFI CHIP) **U6101**

# GPS L2,L5 (Error Code:8225)

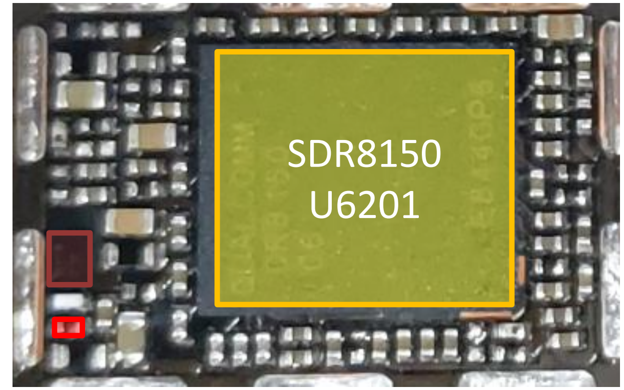
**Bottom(left)**



**Top**




**Bottom(right)**




CH1 connector **CON6102**

GPS SAW **U6121**

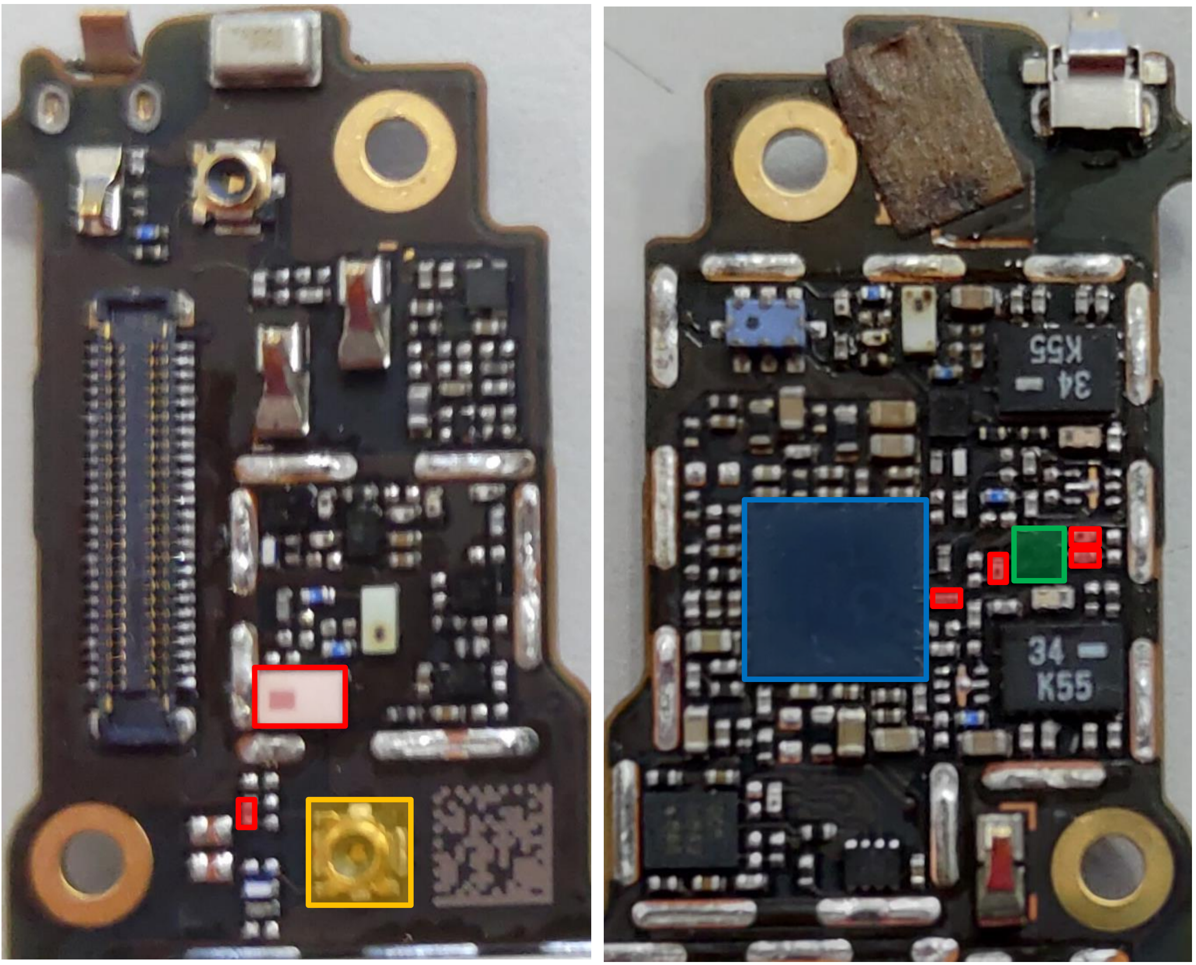
GPS SAW **U6122**

GPS LNA **U6109**

CH1 Triplexer GPS/2.4G/5G **U6106**

# WIFI CH1 2.4G/BT (Error Code:1109,1111,1142)

Bottom(left) Top




CH1 connector **CON6102**

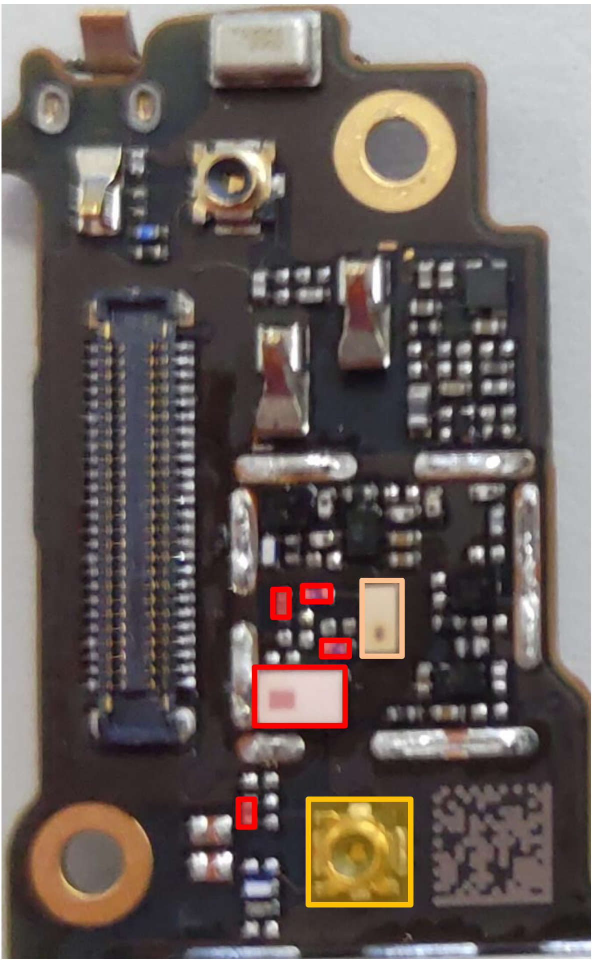
CH1 Triplexer GPS/2.4G/5G **U6106**

CH1 2.4G SAW **U6114**

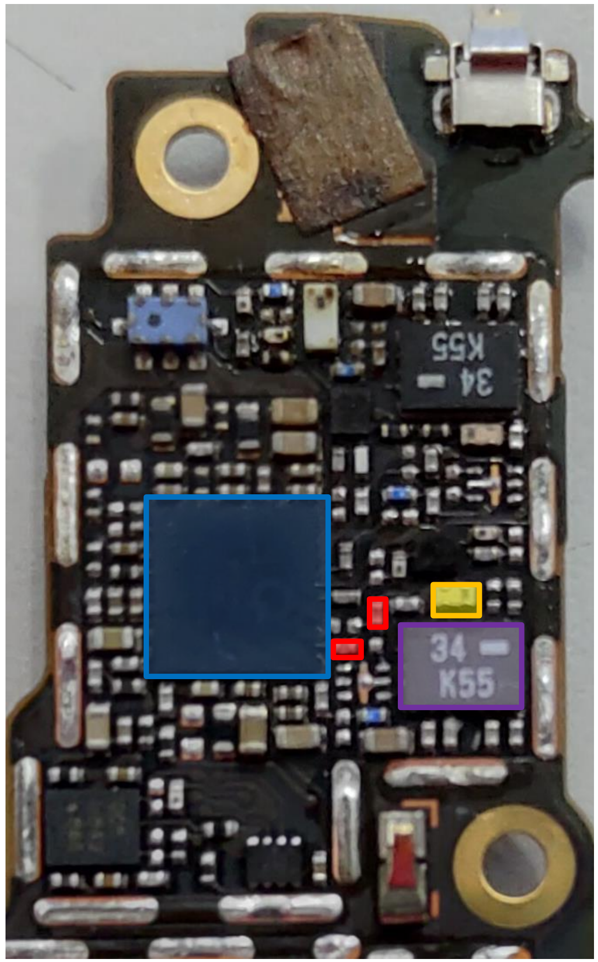
WCN3998 (WIFI CHIP) **U6101**

# WIFI CH1 5G Tx(TW)(Error Code:1109,1111)

**Bottom(left)**



**Top**




CH1 Triplexer GPS/2.4G/5G **U6106**

CH1 connector **CON6102**

CH1 5G SAW **U6112**

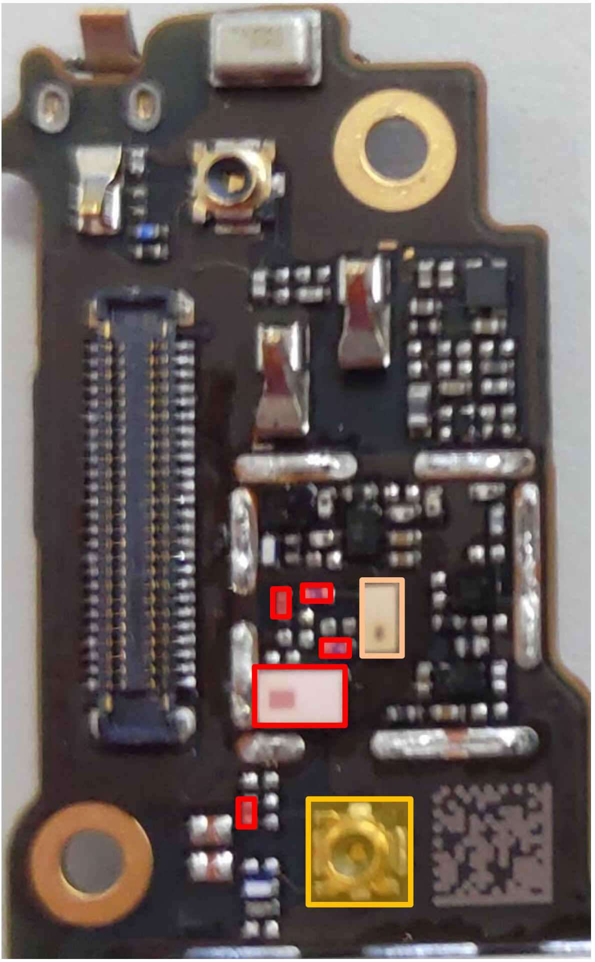
CH1 5G Filter **U6118**

CH1 5G FEM(TW) **U6117**

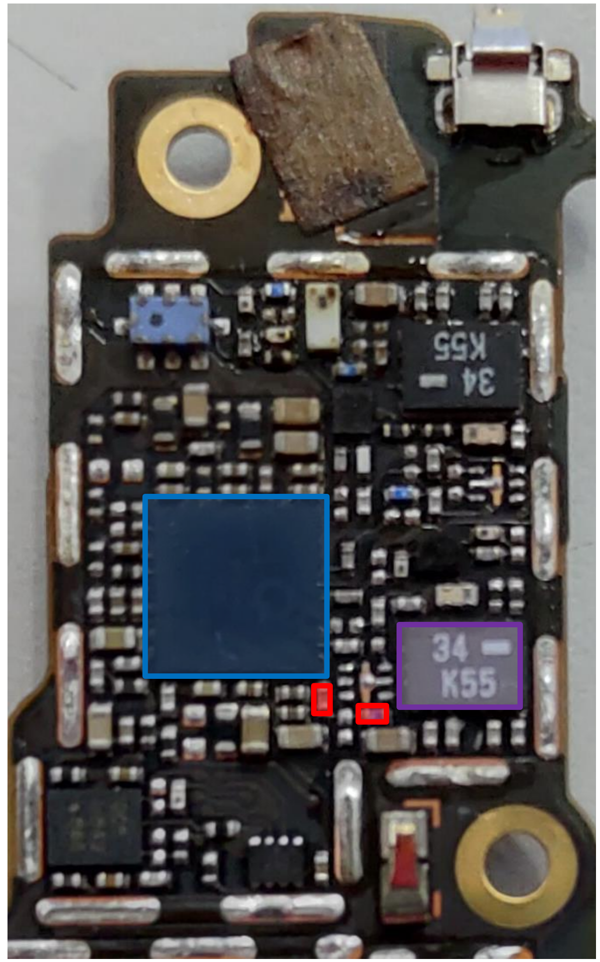
WCN3998 (WIFI CHIP) **U6101**

# WIFI CH1 5G Rx(TW)(Error Code:1142)

**Bottom(left)**



**Top**




CH1 Triplexer GPS/2.4G/5G **U6106**

CH1 connector **CON6102**

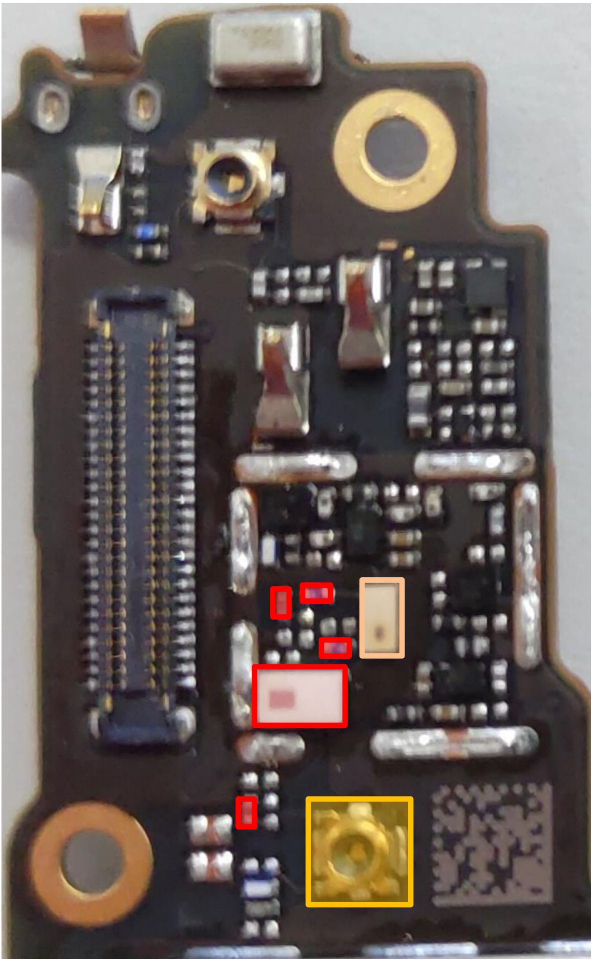
CH1 5G SAW **U6112**

CH1 5G FEM(TW) **U6117**

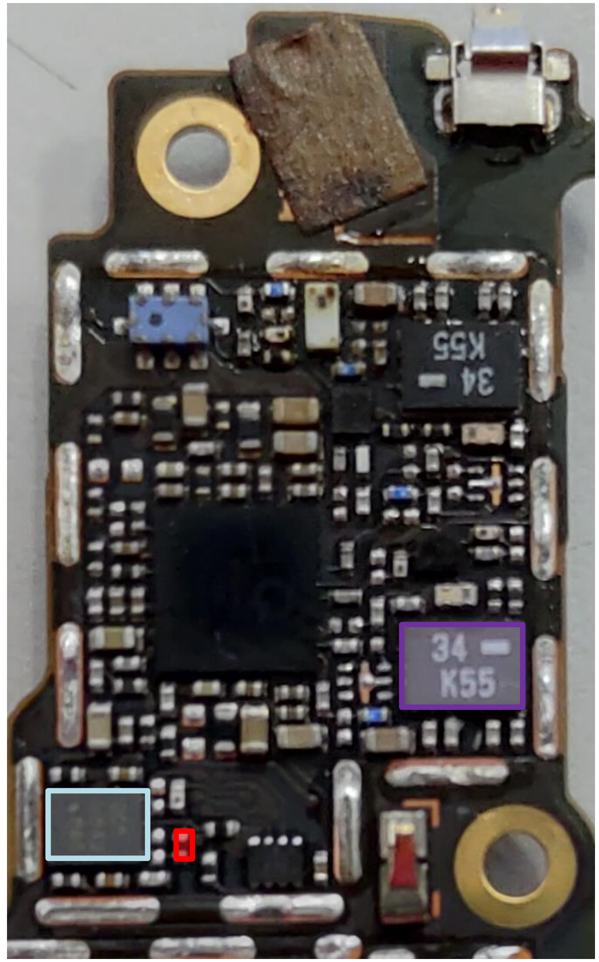
WCN3998 (WIFI CHIP) **U6101**

# LAA B46 DRx(TW)

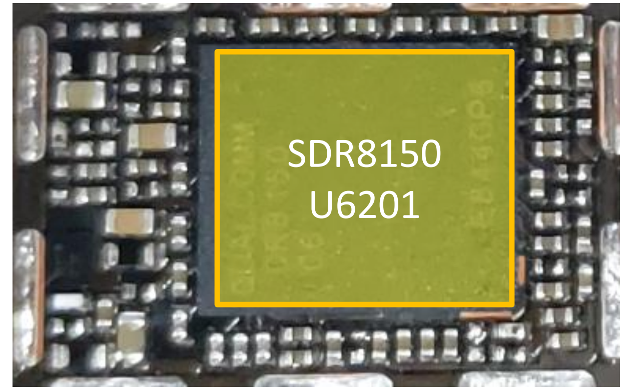
**Bottom(left)**



**Top**




**Bottom(right)**




CH1 5G SAW **U6112**

CH1 5G FEM(TW) **U6117**

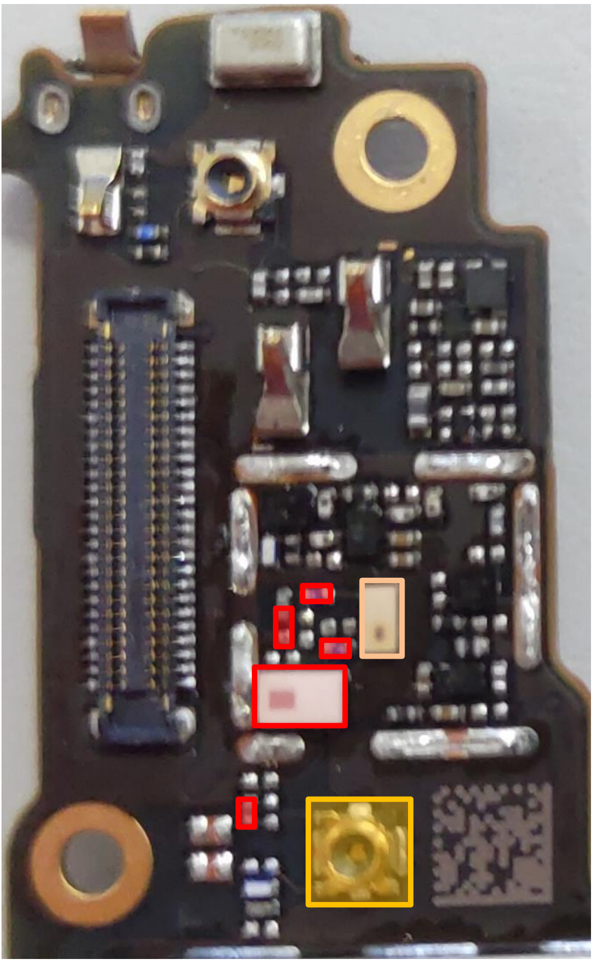
CH1 connector **CON6102**

QLN4650(TW) **U6132**

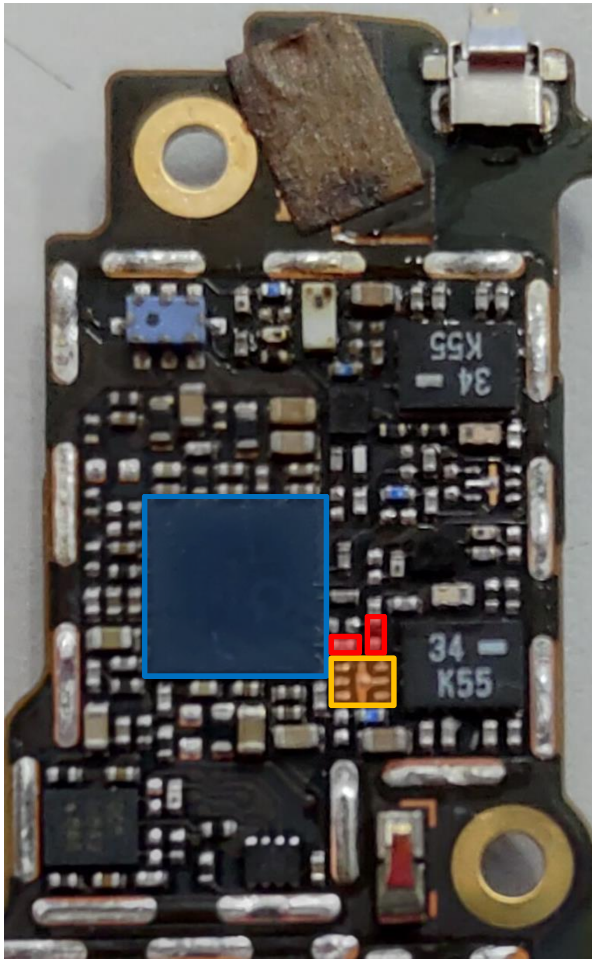
CH1 Triplexer GPS/2.4G/5G **U6106**

# WIFI CH1 5G Tx(WW)(Error Code:1109,1111)

**Bottom(left)**



**Top**




CH1 Triplexer GPS/2.4G/5G **U6106**

WCN3998 (WIFI CHIP) **U6101**

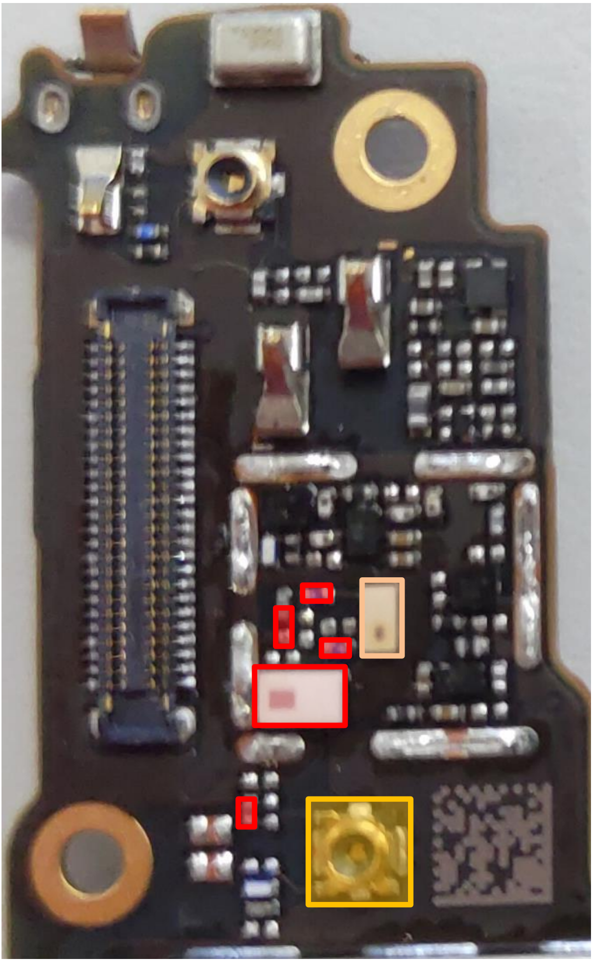
CH1 5G SAW **U6112**

CH1 5G SPDT(WW) **U6111**

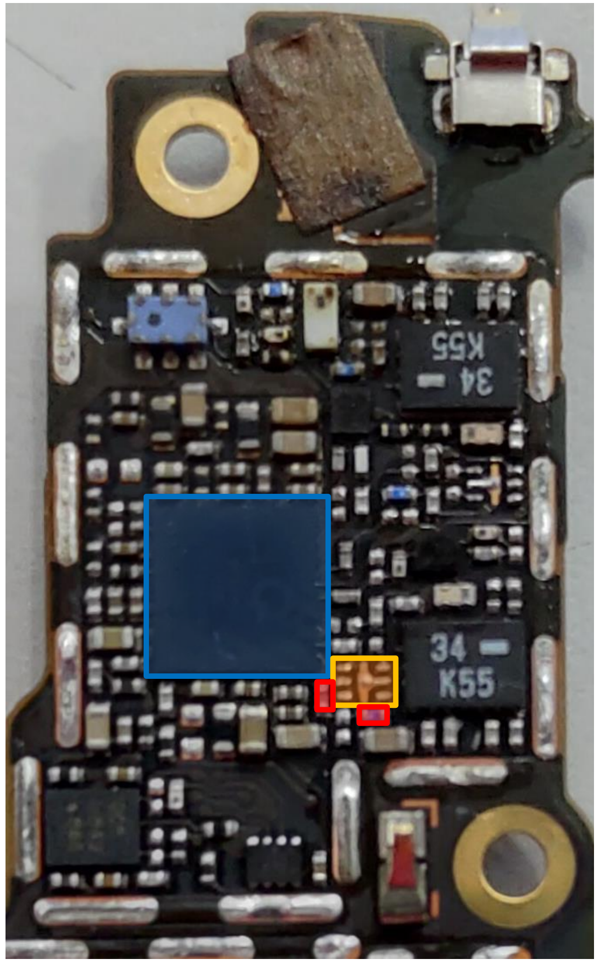
CH1 connector **CON6102**

# WIFI CH1 5G Rx(WW)(Error Code:1142)

**Bottom(left)**



**Top**



CH1 Triplexer GPS/2.4G/5G **U6106**

WCN3998 (WIFI CHIP) **U6101**

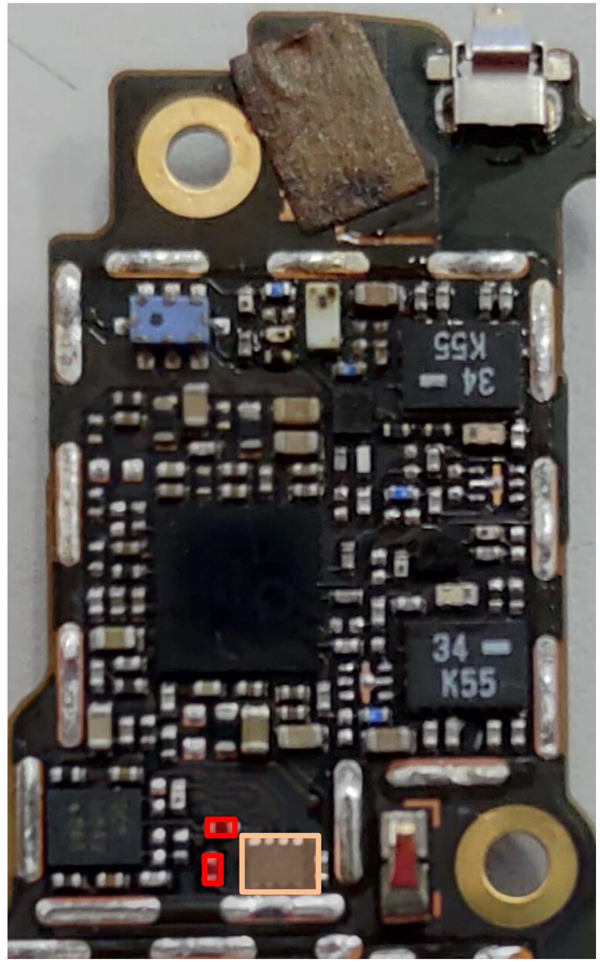
CH1 5G SAW **U6112**

CH1 5G SPDT(WW) **U6111**

CH1 connector **CON6102**

# TW WIFI & LAA LOGIC Gate

**Top**




TW WIFI & LAA LOGIC Gate
**U6102**

# ZS630KL Trouble Shooting Guide

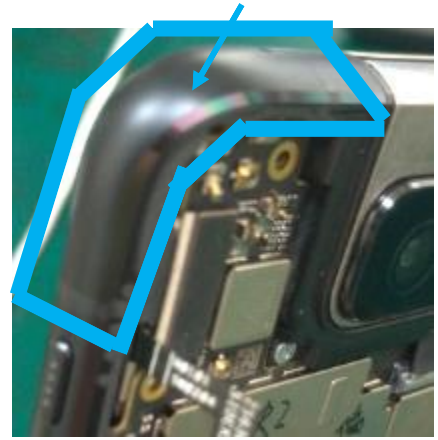
GPS/NFC/FM

# GPS_L1 (Error code:8221)

GPS_L1 antenna shrapnel, Confirm whether the shrapnel is deformed.

GPS_L1 antenna open side

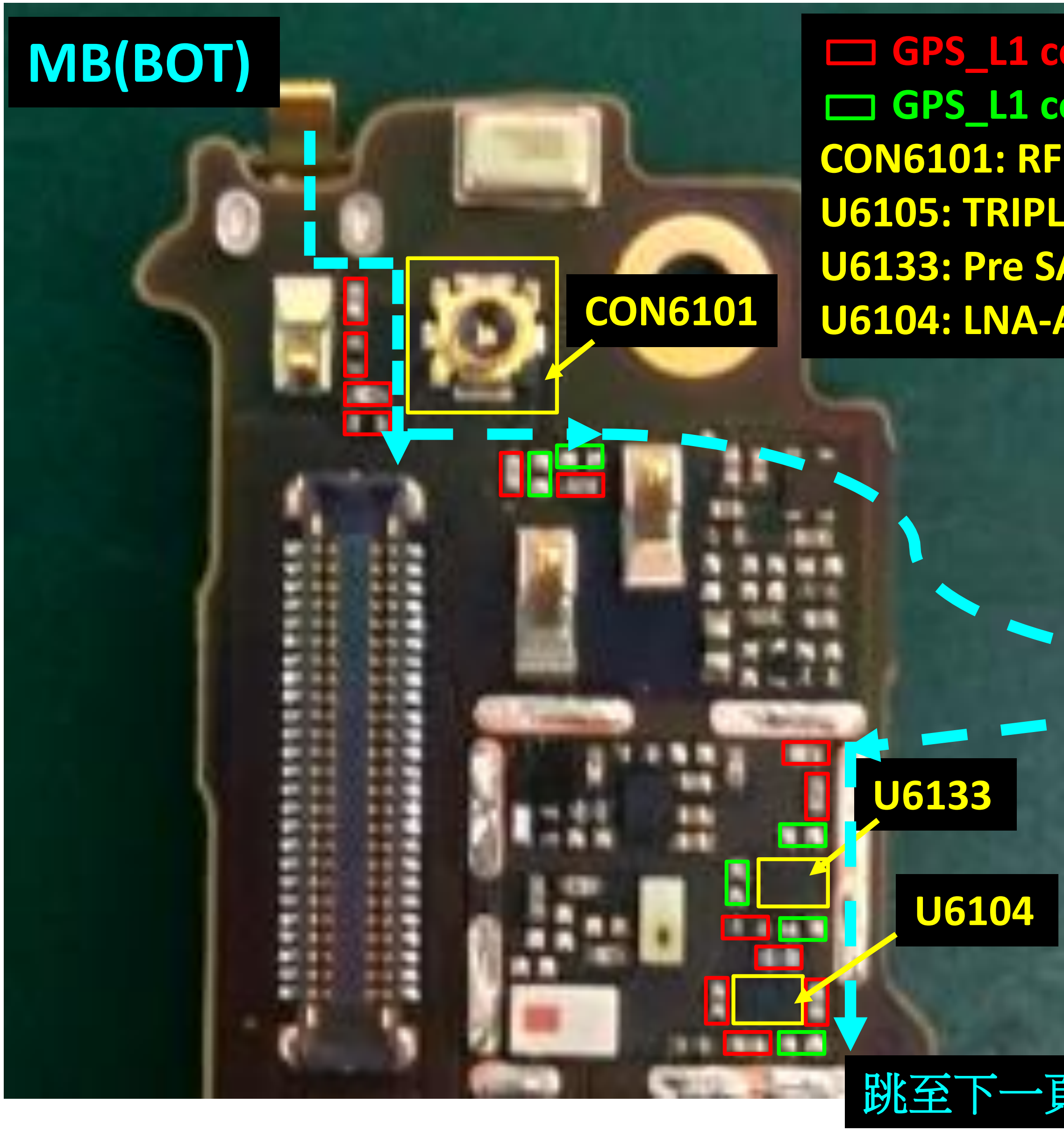

GPS_L1 antenna

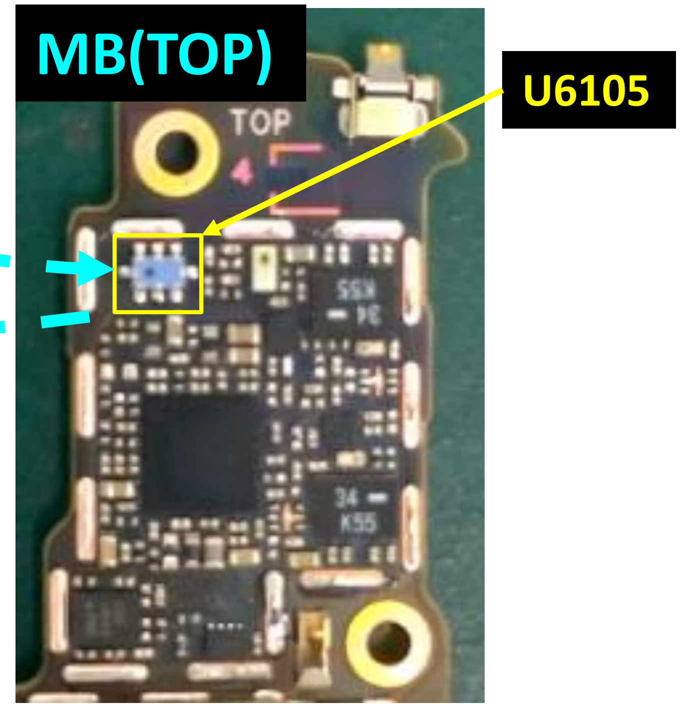


This screw effect GPS_L1, please tighten this screw.

# GPS_L1 (Error code:8221)

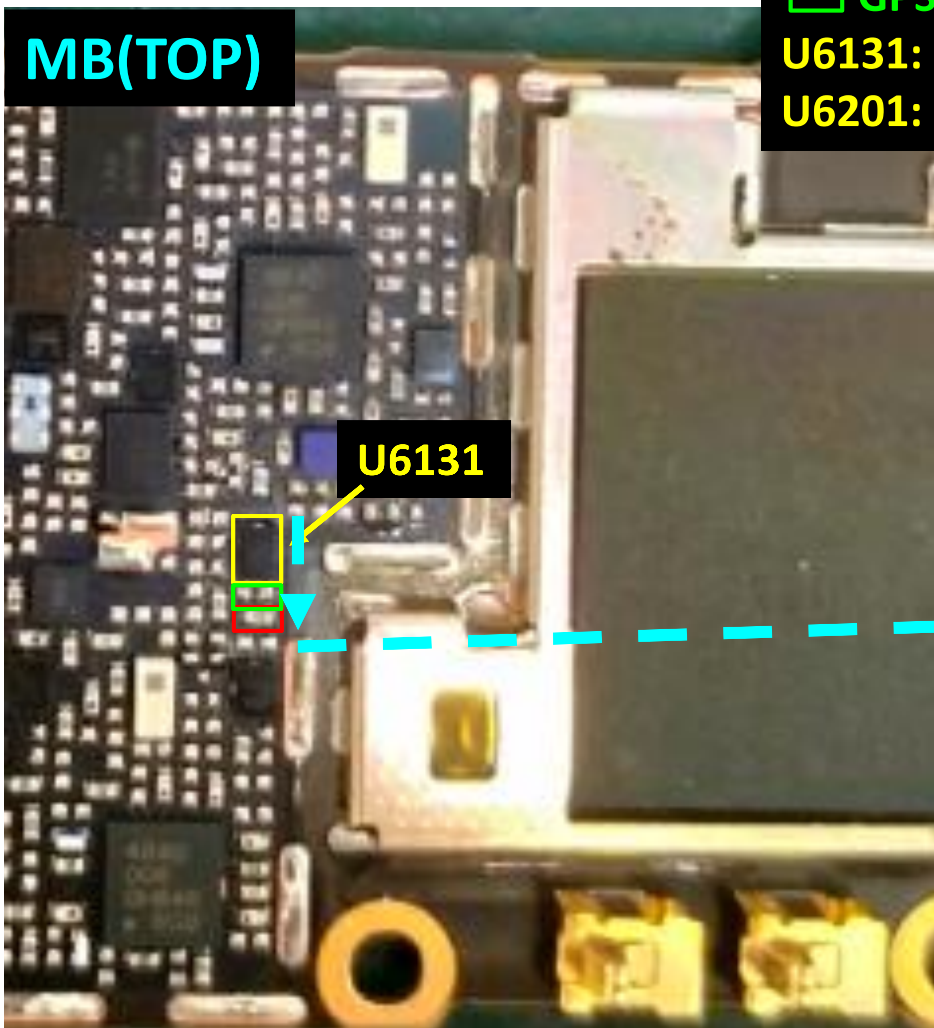
GPS_L1 component (mount)
GPS_L1 component(not mount)
CON6101: RF SWITCH CONNECTOR
U6105: TRIPLEXER-WALSIN/RFTIP2109ATM0T63
U6133: Pre SAW-MURATA/SAFFB1G56KB0F0AR15
U6104: LNA-AWINIC

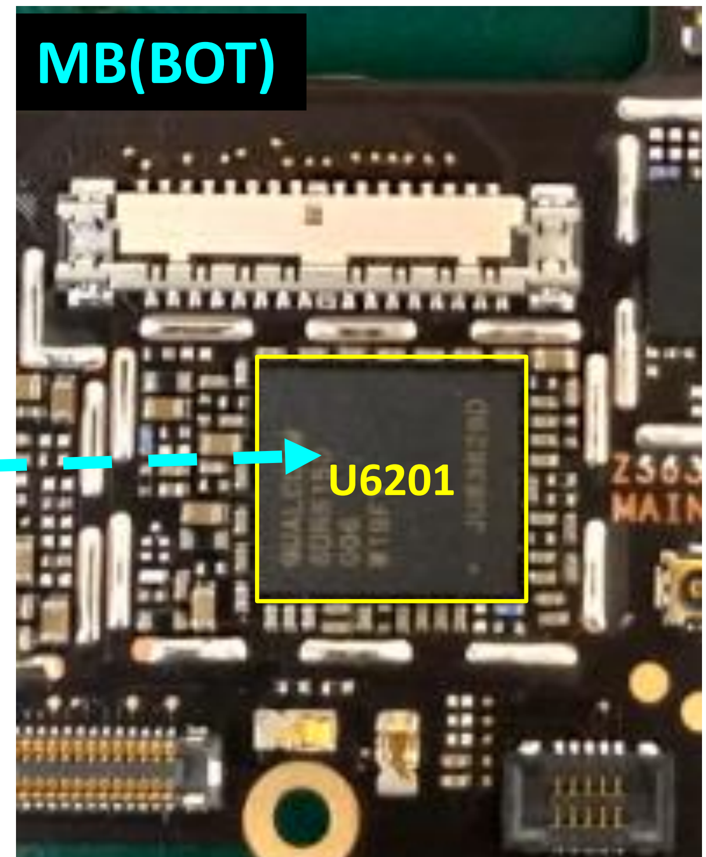


# GPS_L1 (Error code:8221)

GPS_L1 component (mount)
GPS_L1 component(not mount)
**U6131: Post SAW-MURATA/SAFFB1G56KB0F0AR15**
**U6201: SDR855 Transceiver**



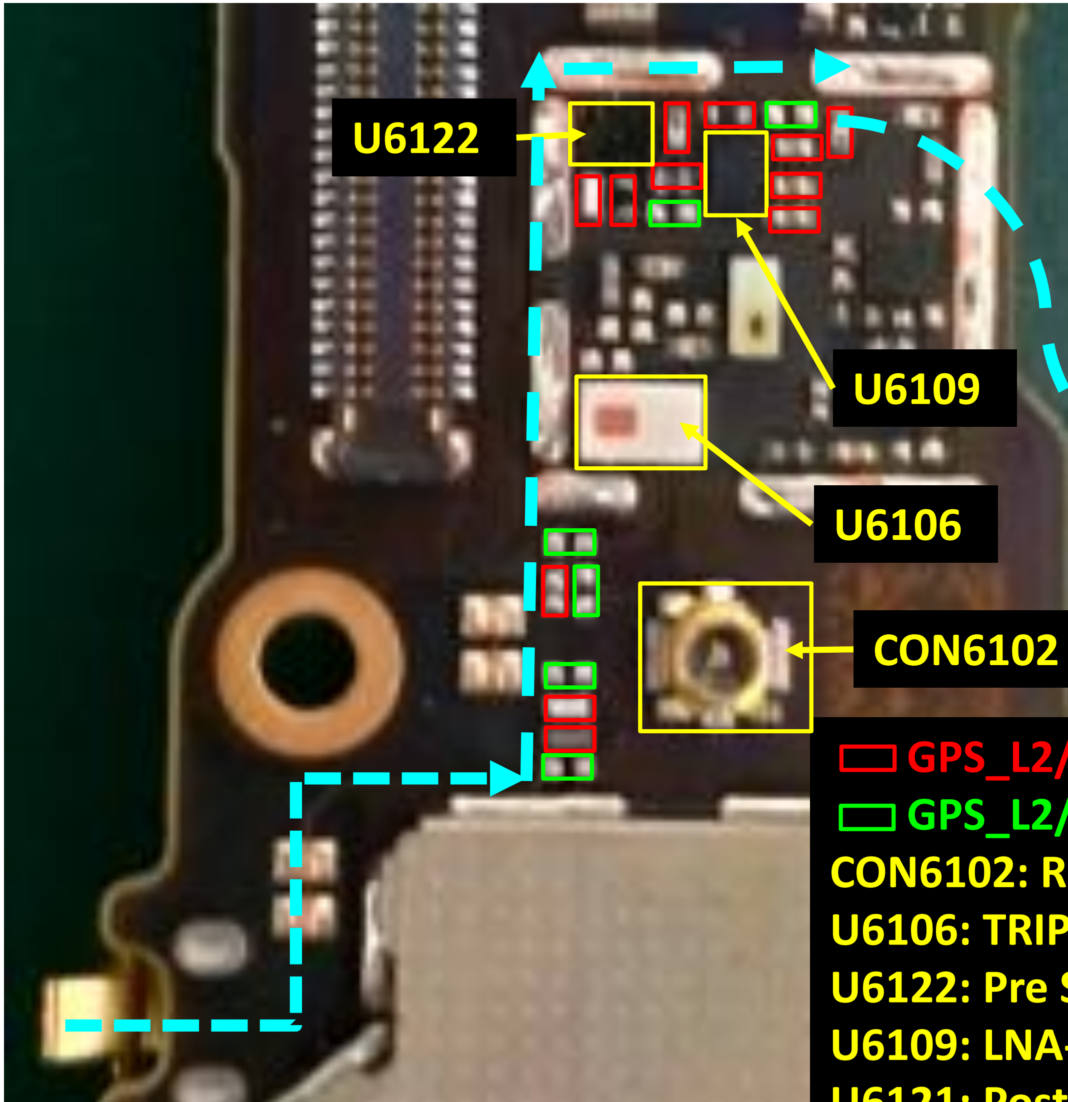



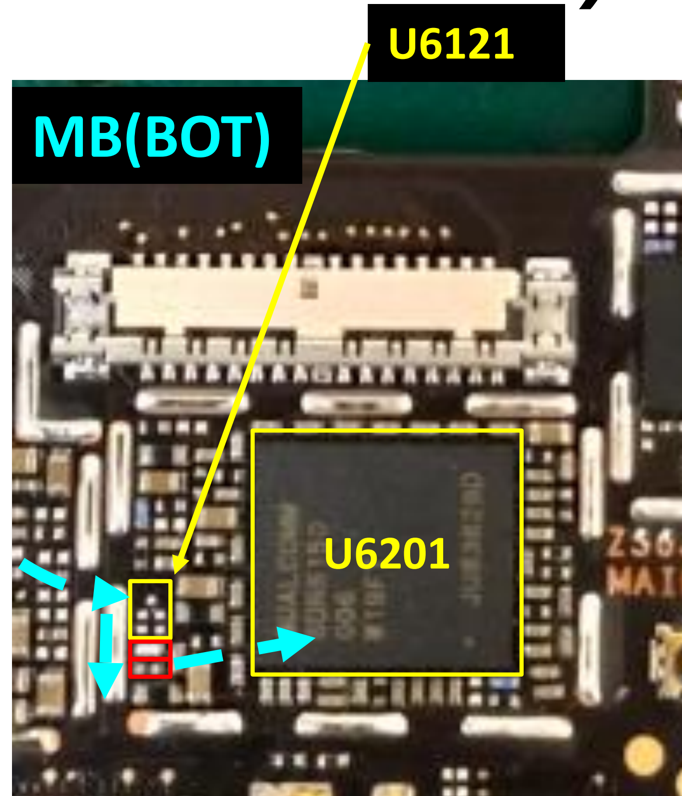
# GPS_L2/L5 (Error code:8225)


GPS_L2/L5 antenna shrapnel, Confirm whether the shrapnel is deformed.

GPS_L2/L5 antenna

# GPS_L2/L5 (Error code:8225)




GPS_L2/L5 component (mount)

GPS_L2/L5 component(not mount)

CON6102: RF SWITCH CONNECTOR

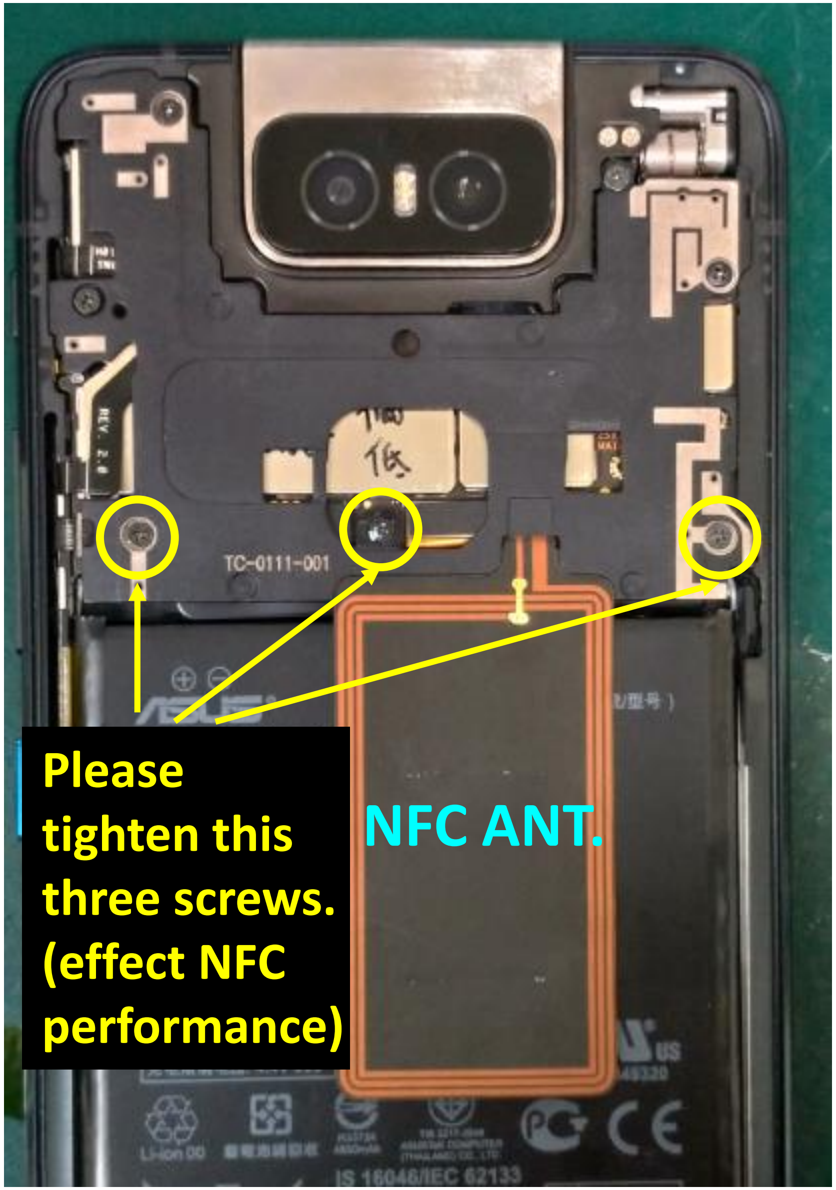
U6106: TRIPLEXER- TDK/TPX205950MT-7036A1

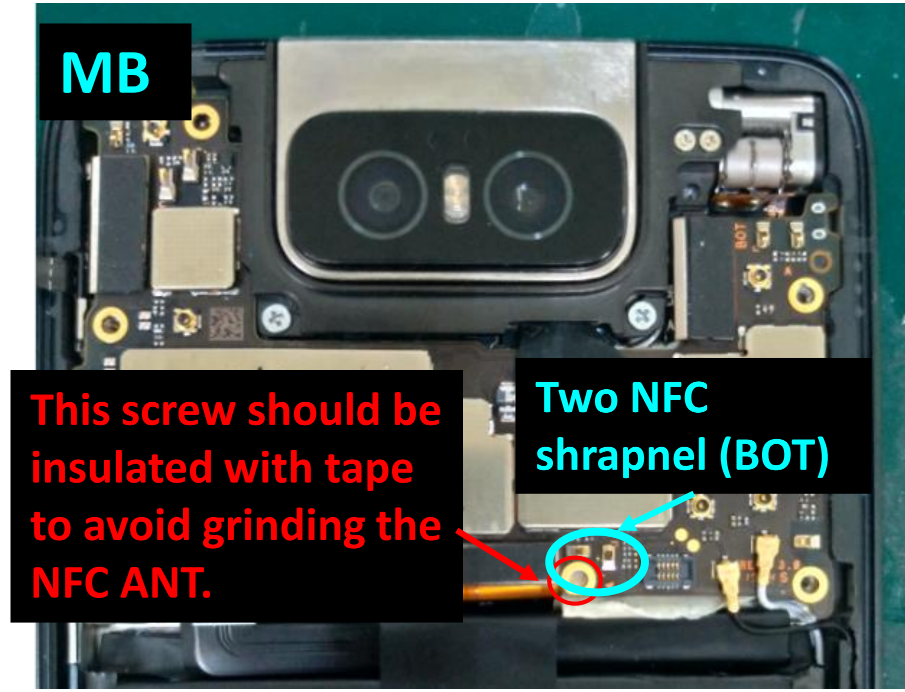
U6122: Pre SAW- MURATA/SAFFB1G20AA0F0AR1S

U6109: LNA-AWINIC

U6121: Post SAW- MURATA/SAFFB1G20AA0F0AR1S

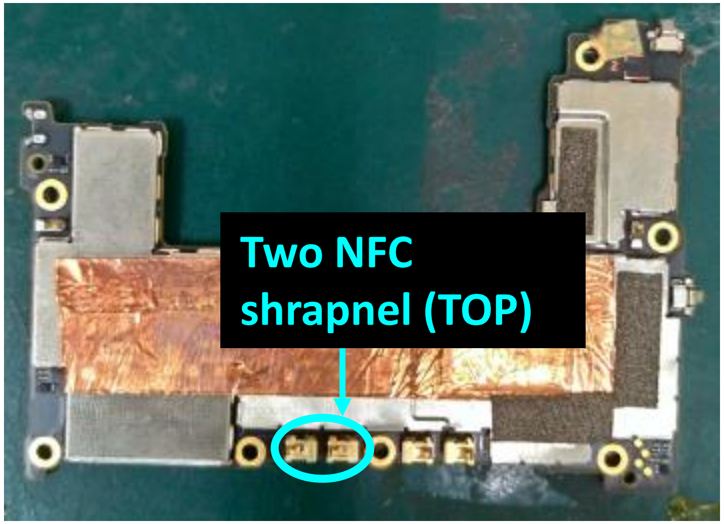
# NFC (1993)

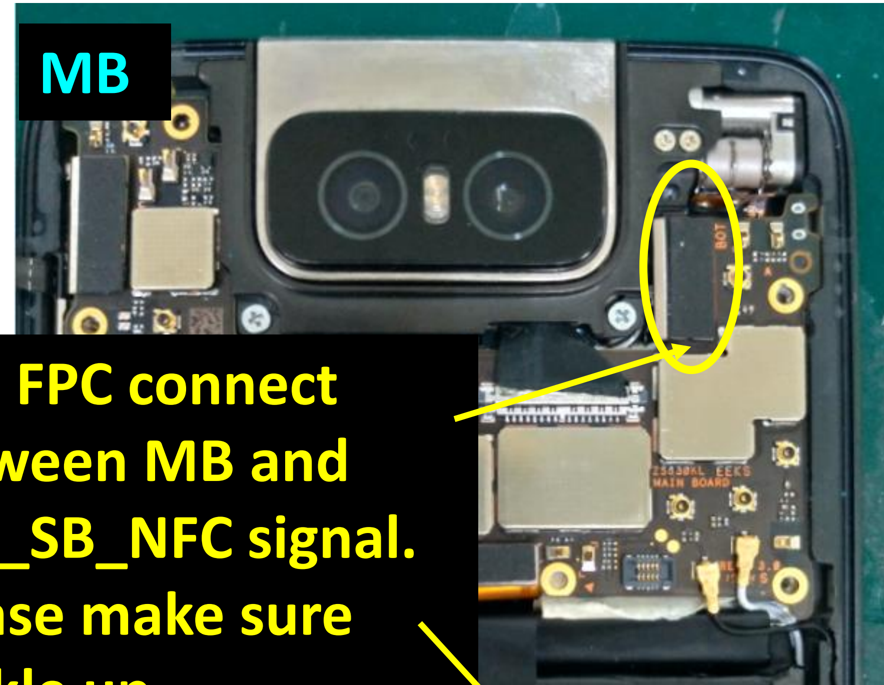

Please tighten this three screws. (effect NFC performance)
NFC ANT.


MB
This screw should be insulated with tape to avoid grinding the NFC ANT.
Two NFC shrapnel (BOT)

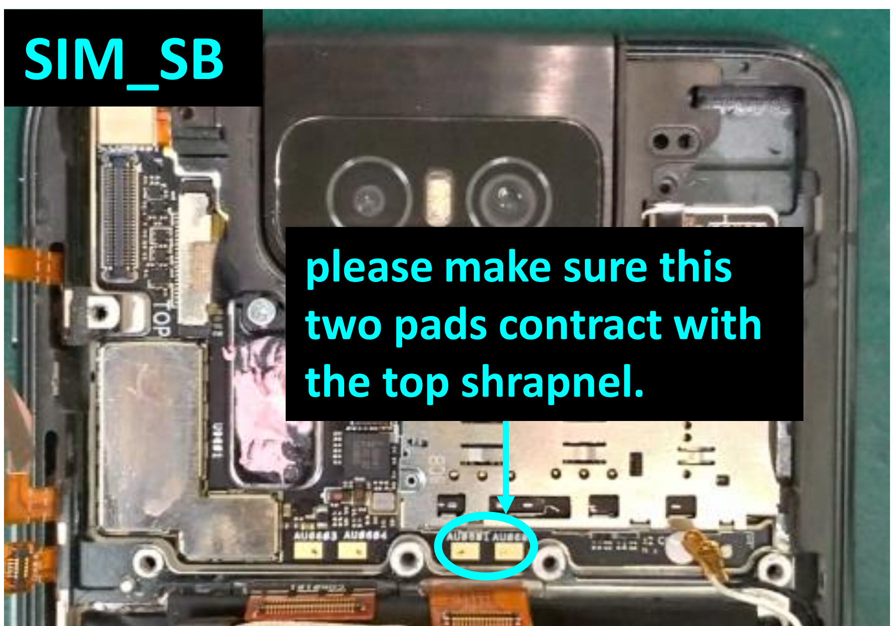

please make sure this two pads contract with the bottom shrapnel.

# NFC(1993)

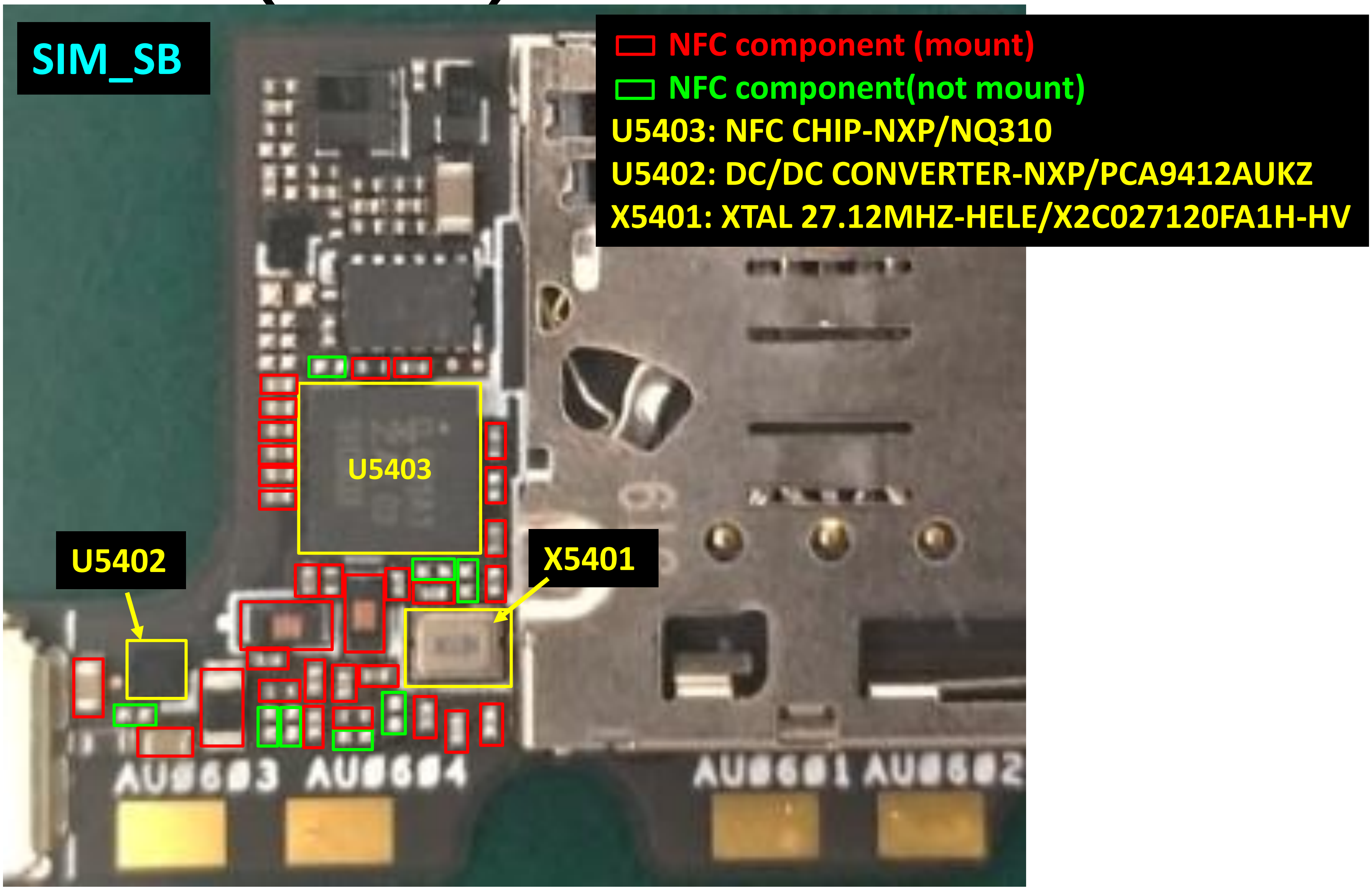



This FPC connect between MB and SIM_SB_NFC signal. Please make sure buckle up.

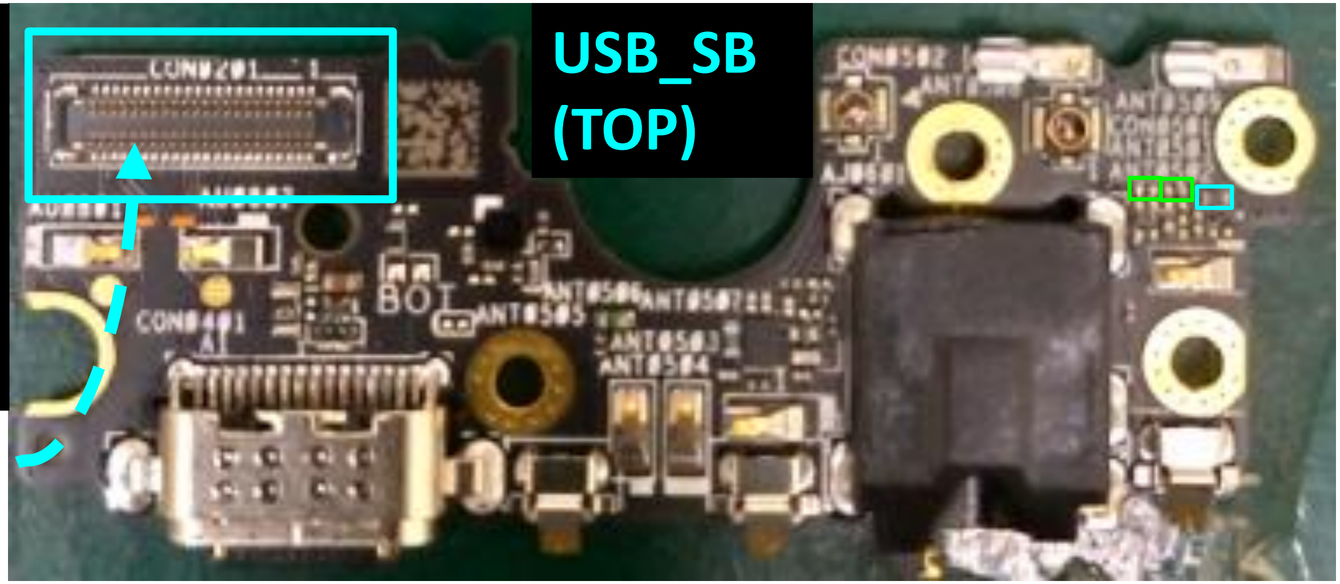



# NFC(1993)


SIM_SB
NFC component (mount)
NFC component(not mount)
U5403: NFC CHIP-NXP/NQ310
U5402: DC/DC CONVERTER-NXP/PCA9412AUKZ
X5401: XTAL 27.12MHZ-HELE/X2C027120FA1H-HV
U5403
U5402
X5401

# FM(817)

This connector connect between MB and USB_SB. Please make sure buckle up.



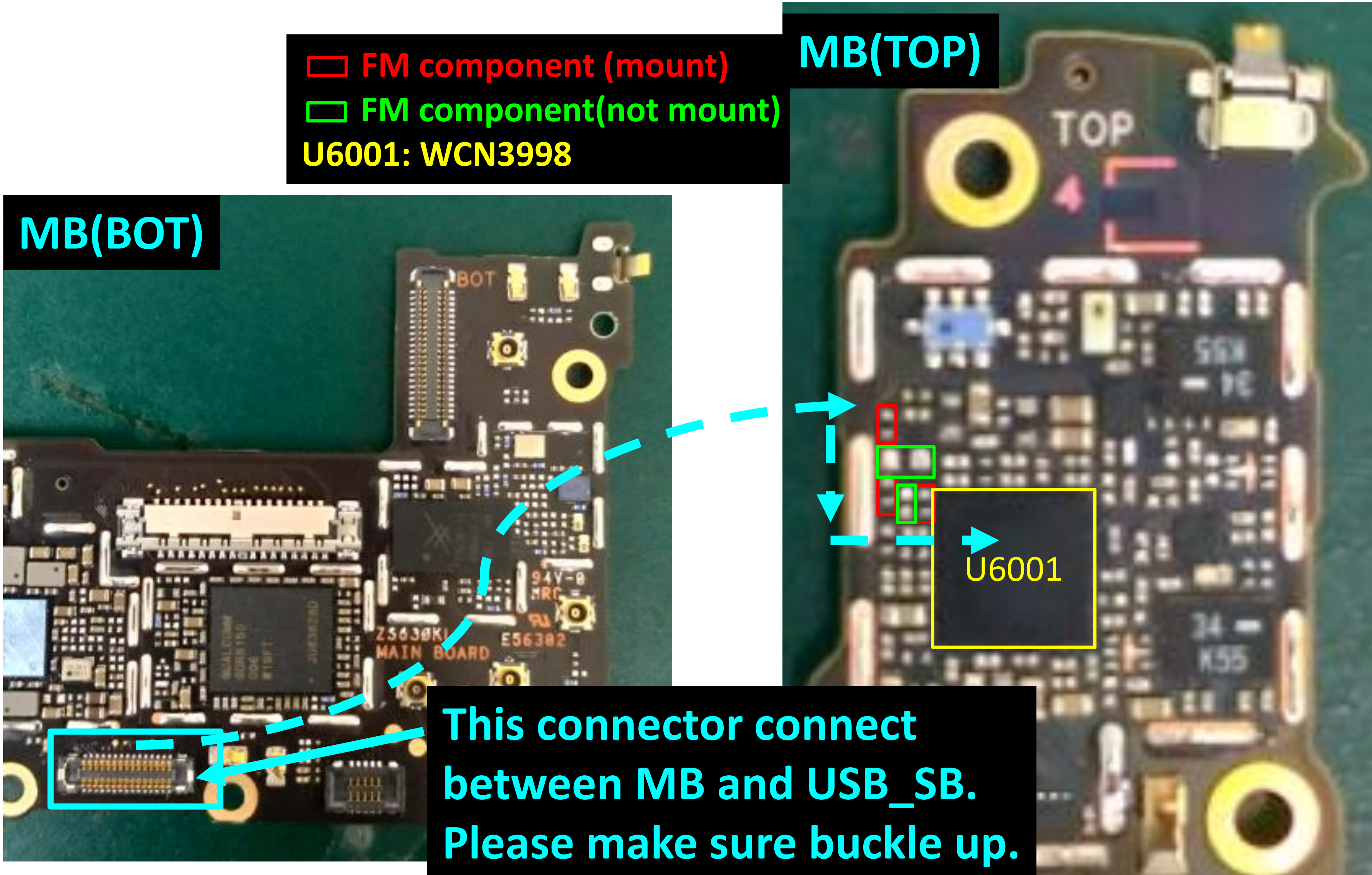


FM component (mount)

FM component(not mount)

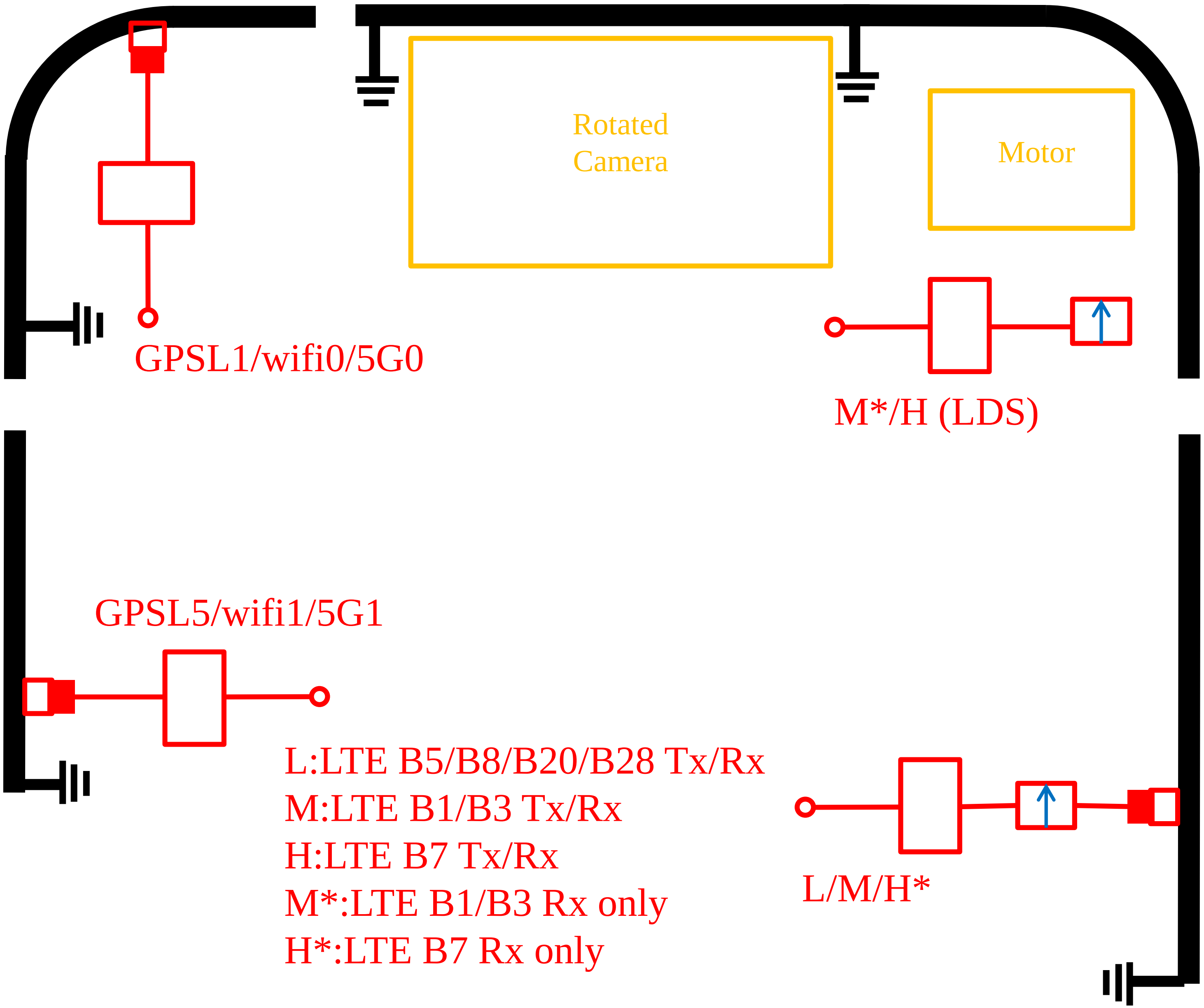
Audio Bead (must mount)

# FM(817)

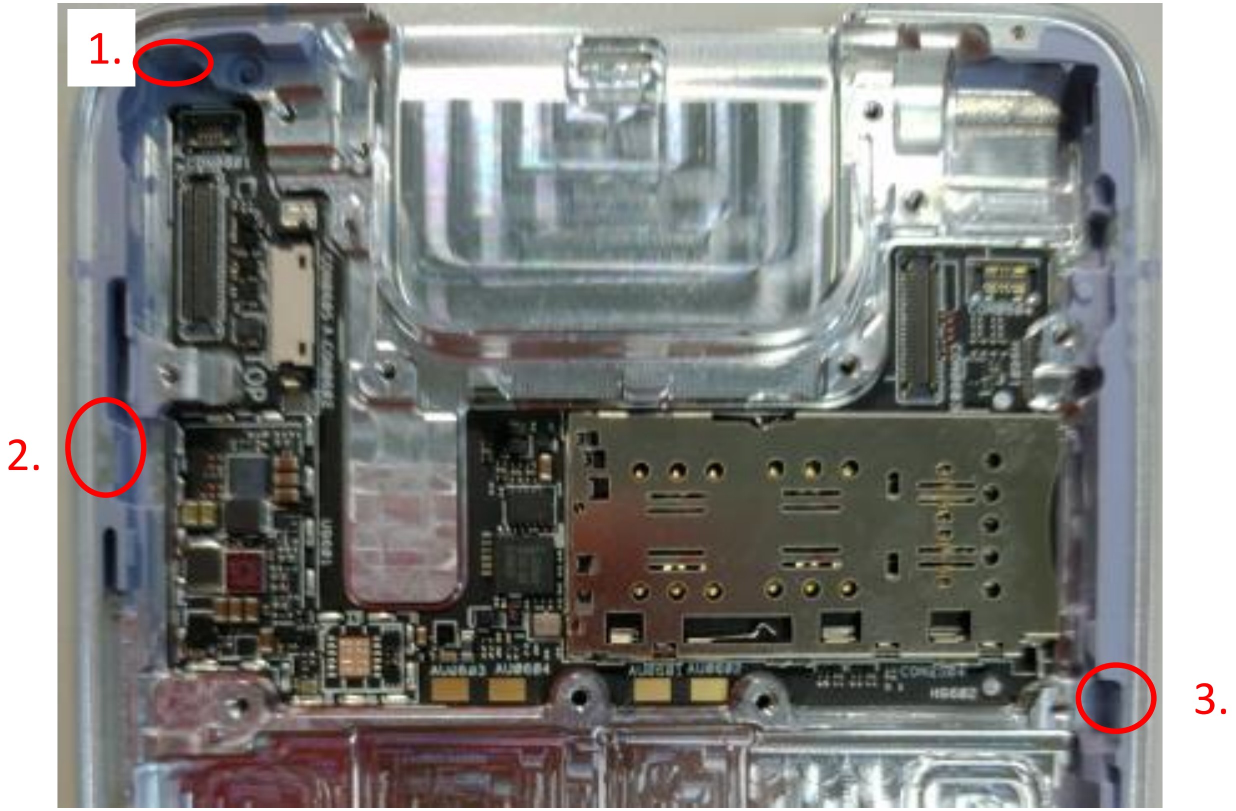

FM component (mount)
FM component(not mount)
U6001: WCN3998
MB(TOP)
MB(BOT)
TOP
BOT
U6001
This connector connect between MB and USB_SB. Please make sure buckle up.

# Phone & WiFi Antenna Assembly Instruction(upper)

# ZS630KL Antenna sketch

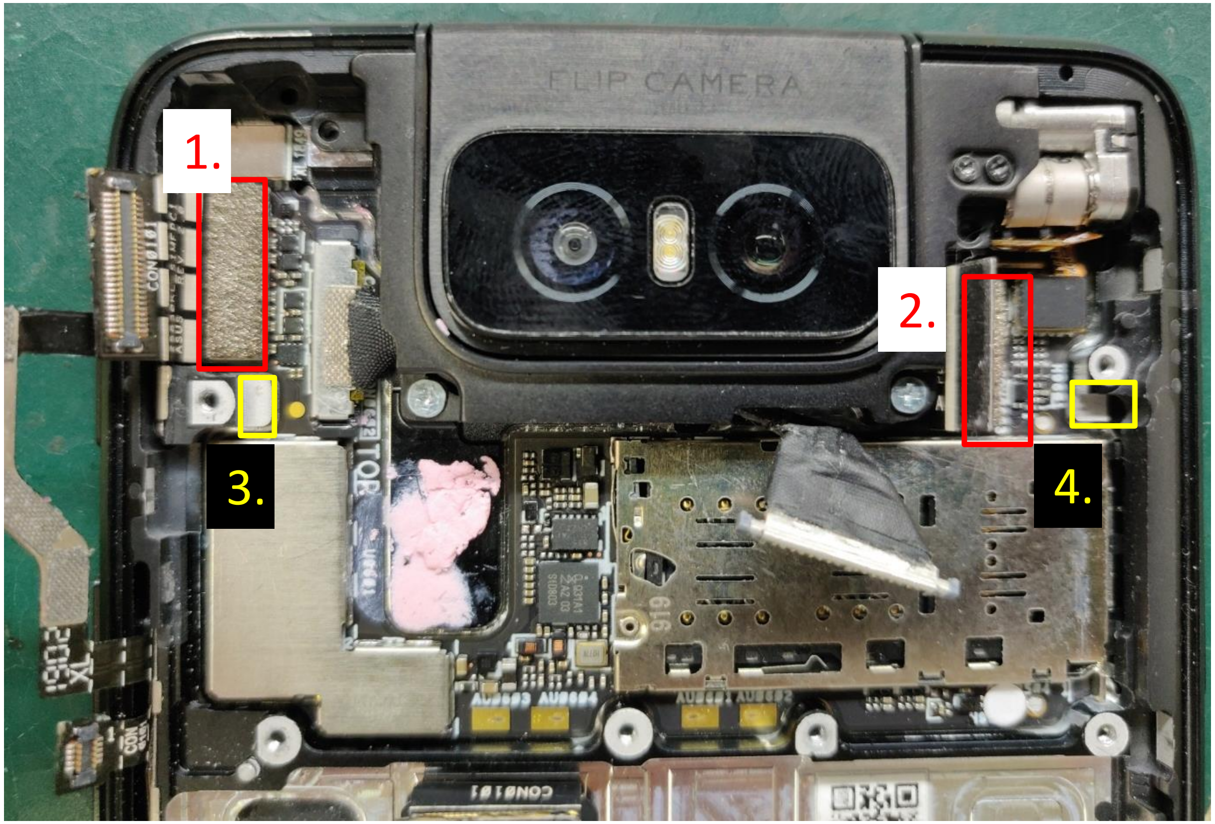


# Upper Antenna feed




1.Effct GPSL1/wifi0/5G0
2.Effct GPSL5/wifi1/5G1
3.Effct LTE TxRx B5/B8/B20/B28/B1/B3,Rx B7

# SIM Board & front-frame position of conductive foam



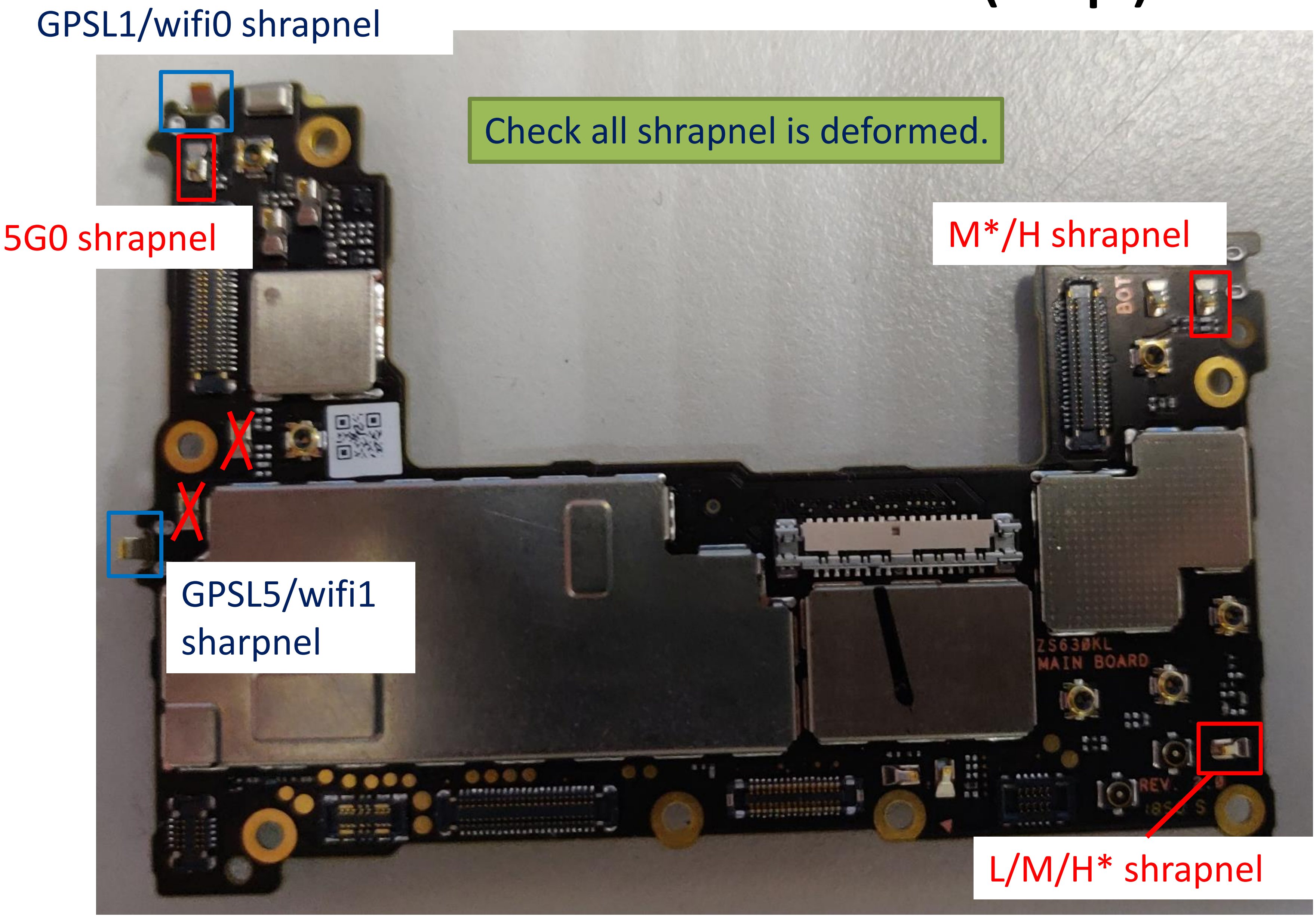

1.This foam effect GPSL1/wifi0/5G0

2.This foam effect M*/H ANT. (TxRx B7,Rx B1/B3)

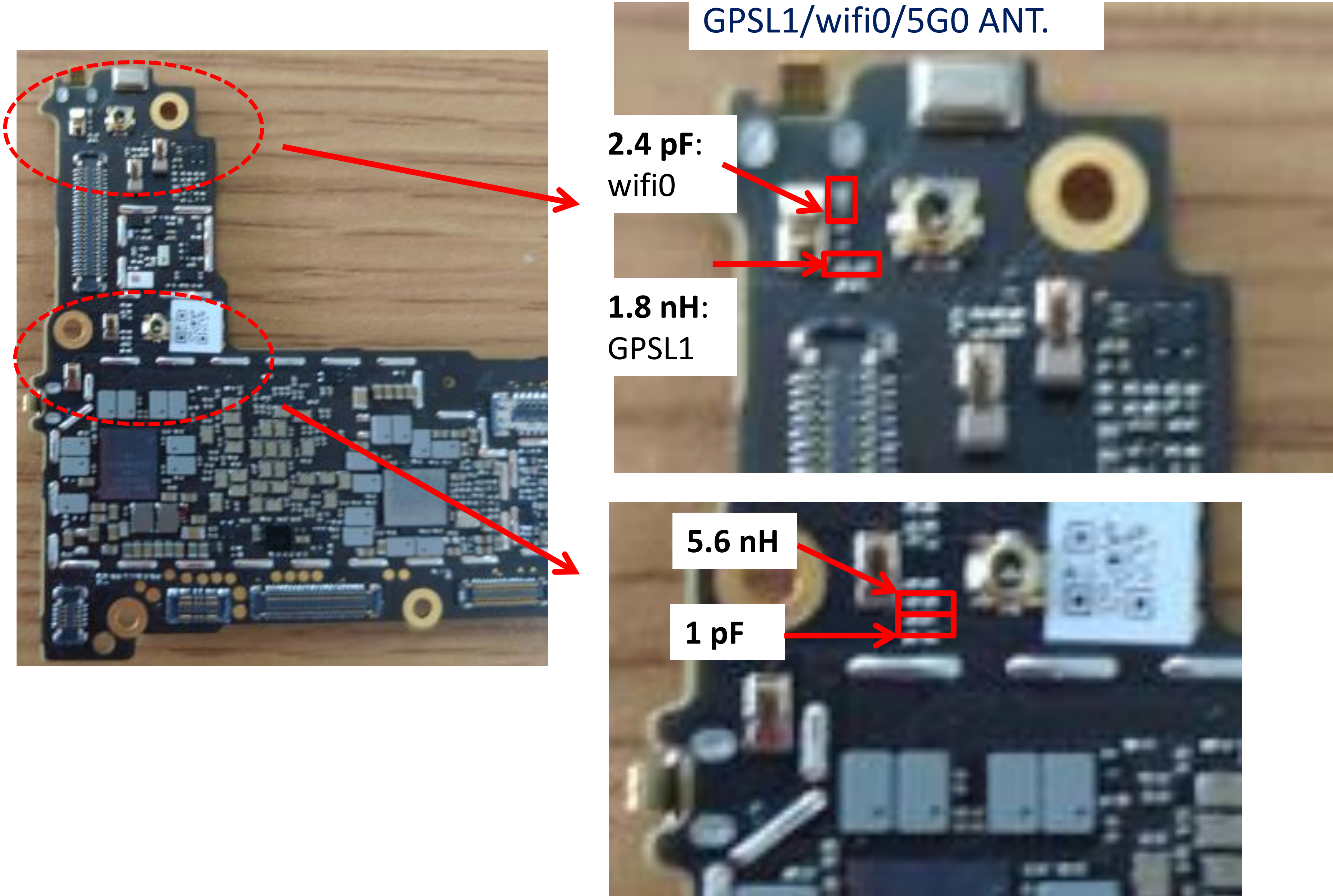
3.This pad connect between the MB shrapnel and SIM board (left side)

4.This pad connect between the MB shrapnel and SIM board (right side)

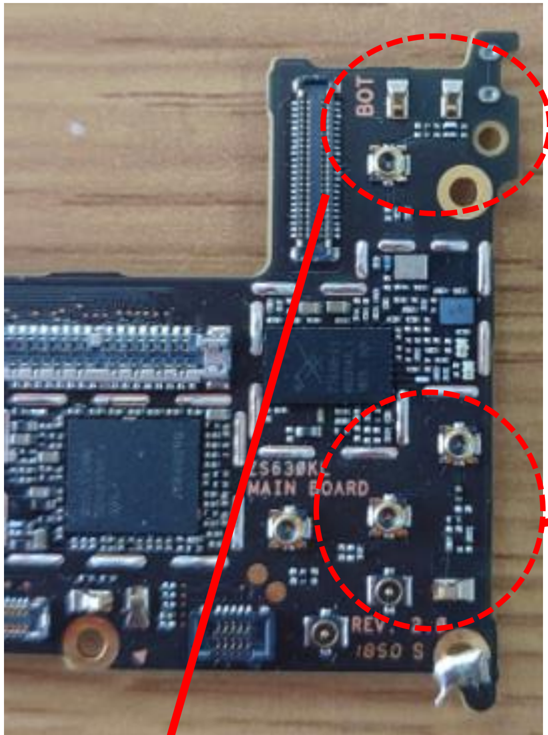
# Main Board Notices (Top)



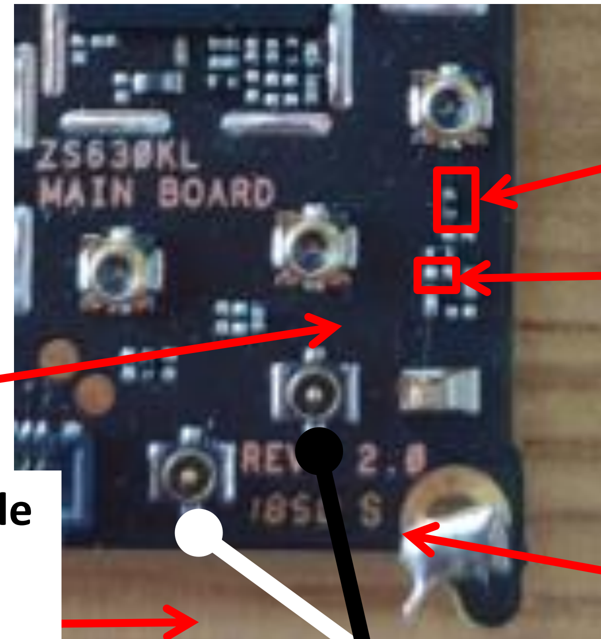
# Main Board left side component



# Main Board right side component




Div. L/M/H* ANT.



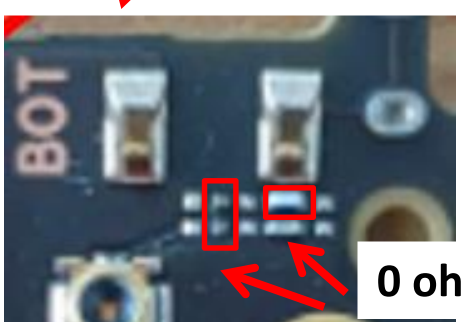

**3 nH**: rx b5,b8,b28,b1, b3

**22 nH**: rx b5,b8,b28

**HB white cable** connect USB SUB BOARD right side

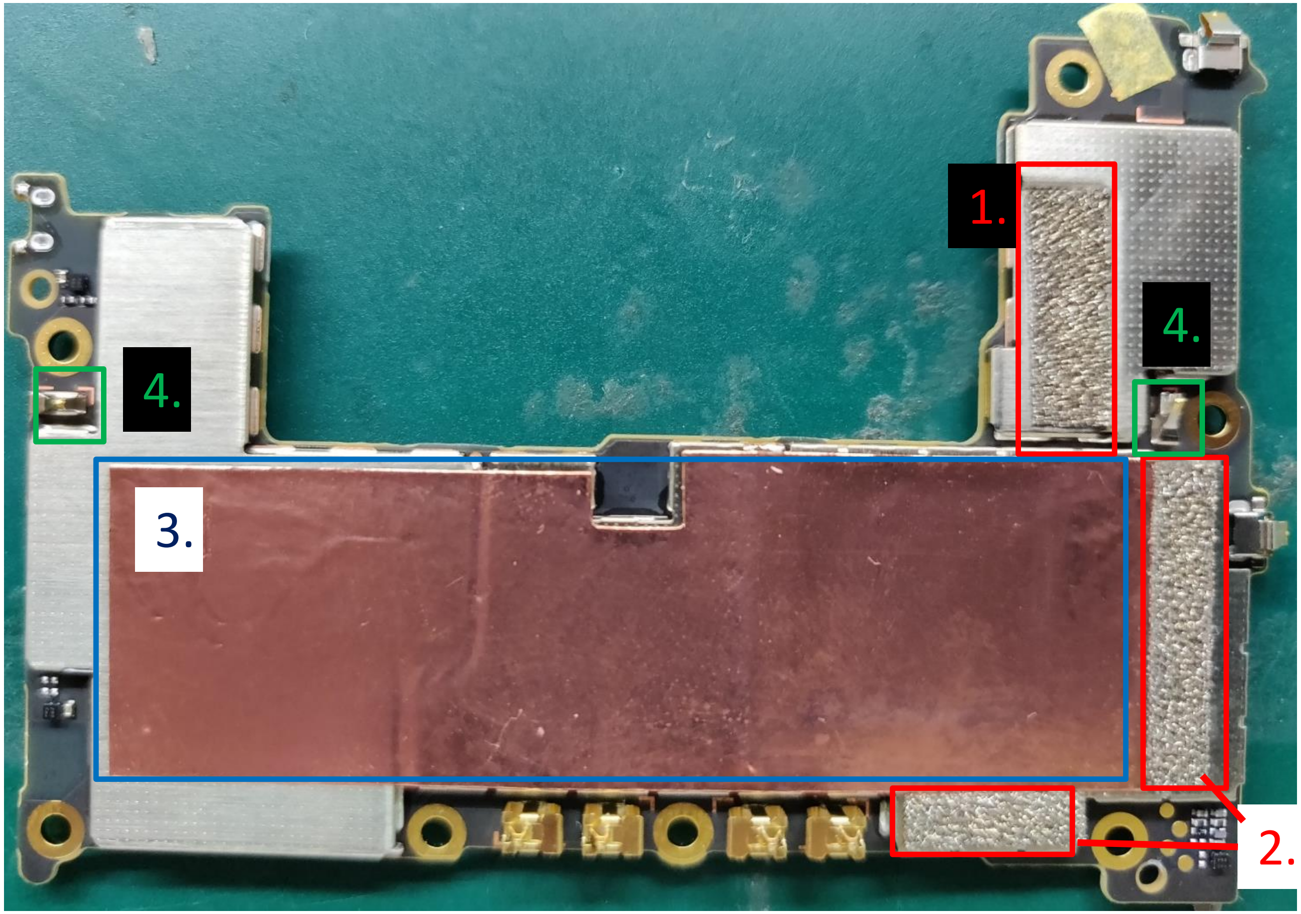
**LMB black cable** connect USB SUB BOARD left side

B7 ANT.




**0 ohm**

# Main Board Notices(Bottom)



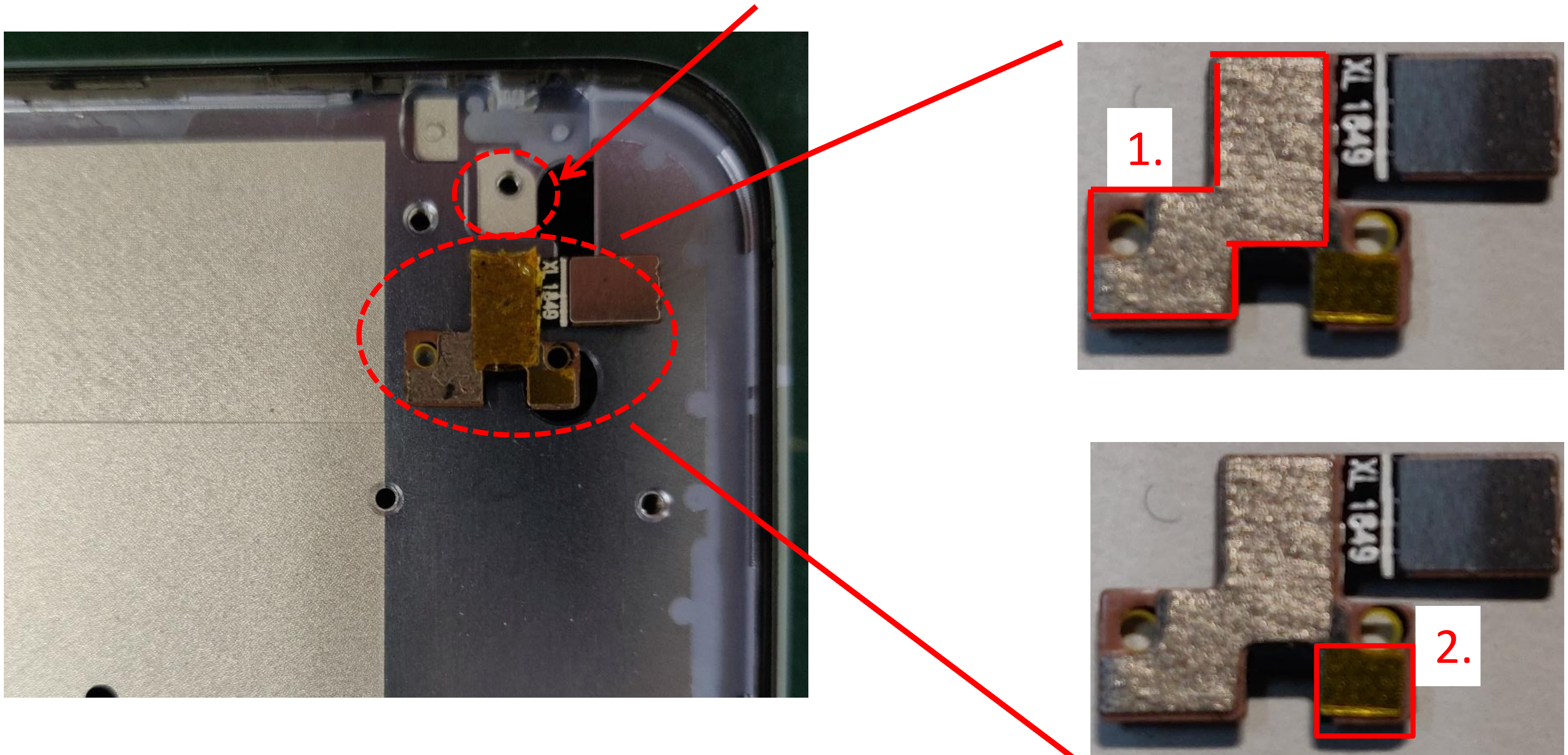

1.This foam effect GPSL1/wifi0/5G0

2.This foam effect GPSL5/wifi1/5G1

3.check copper foil sticker is attached to the iron shell

4.make sure whether this two shrapnel is deformed

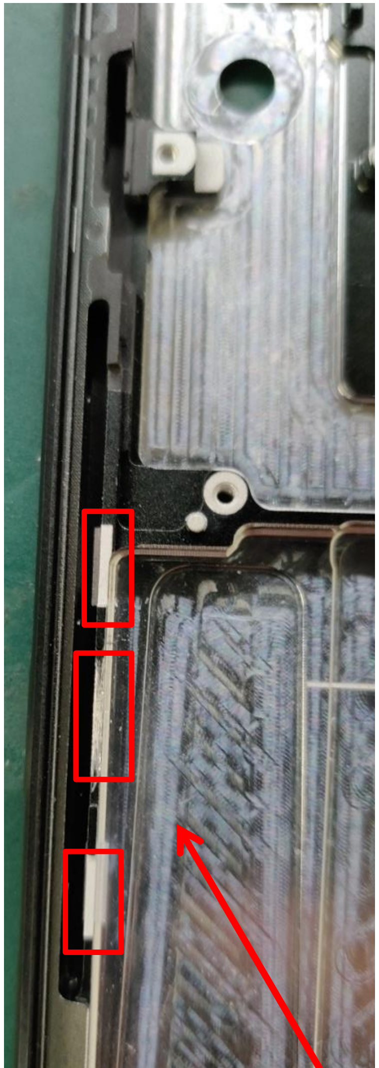
# front-frame TOP side Notices(Charging indicator cable)



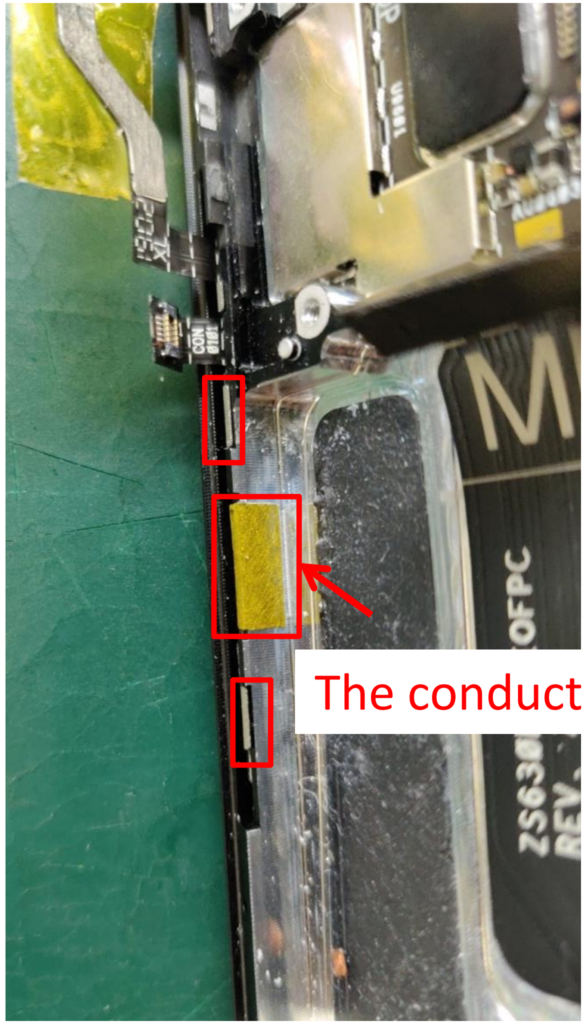

charging indicator cable

1. This area is attached with a conductive sticker and attached to the front-frame.
2. This area is attached with a tape and isolated from the front-frame.
3.This two area effect GPSL1/wifi0/5G0

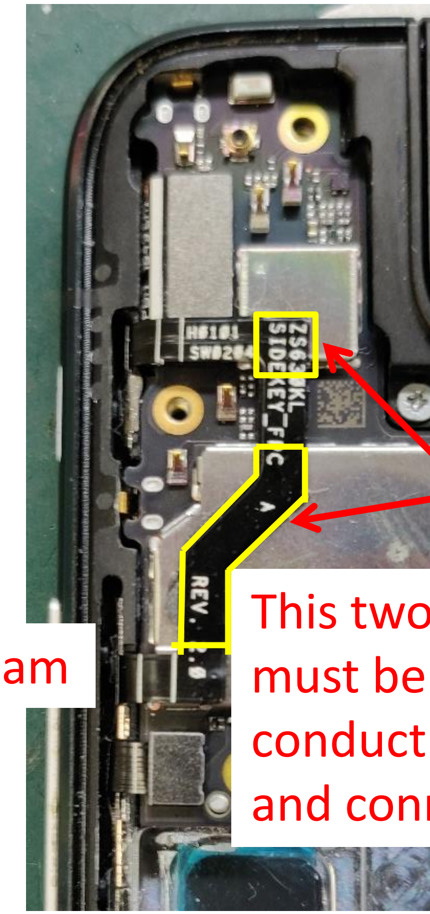
# Side-key FPC ground (effect GPSL5/wifi1/5G1)



ground

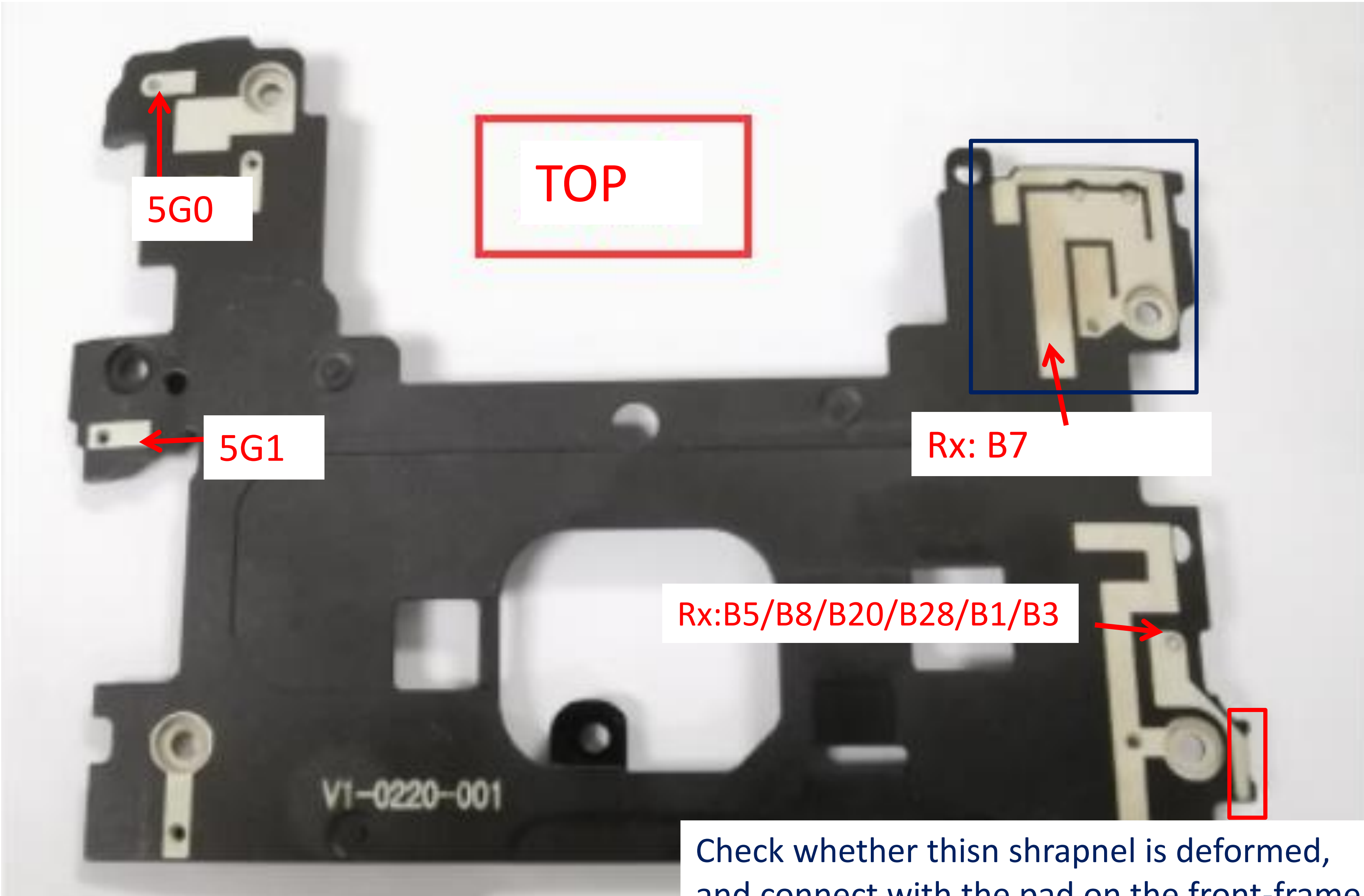


The conductive foam



This two yellow area must be attached the conductive sticker and connect with MB.

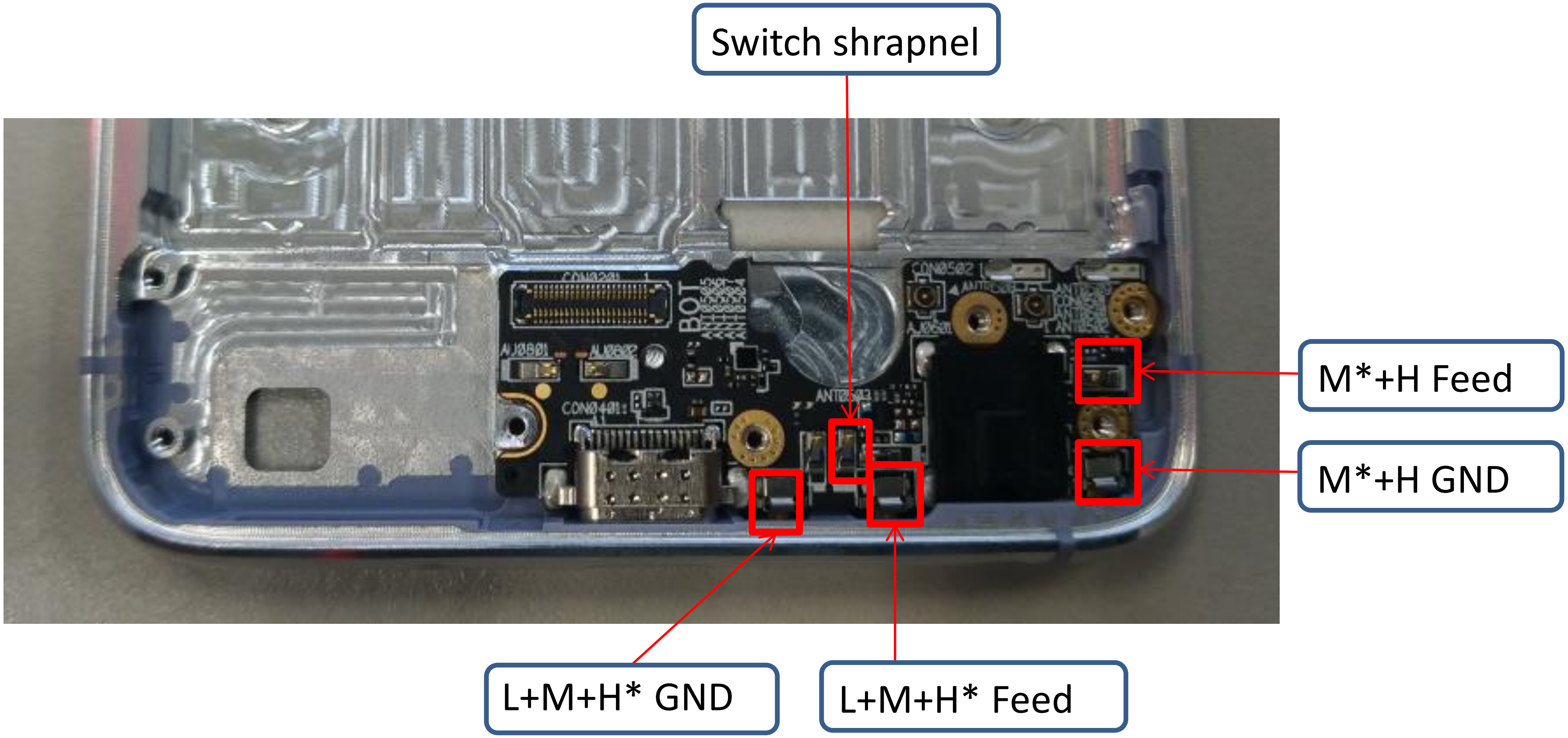
# Middle frame Notices




Check whether thisn shrapnel is deformed, and connect with the pad on the front-frame

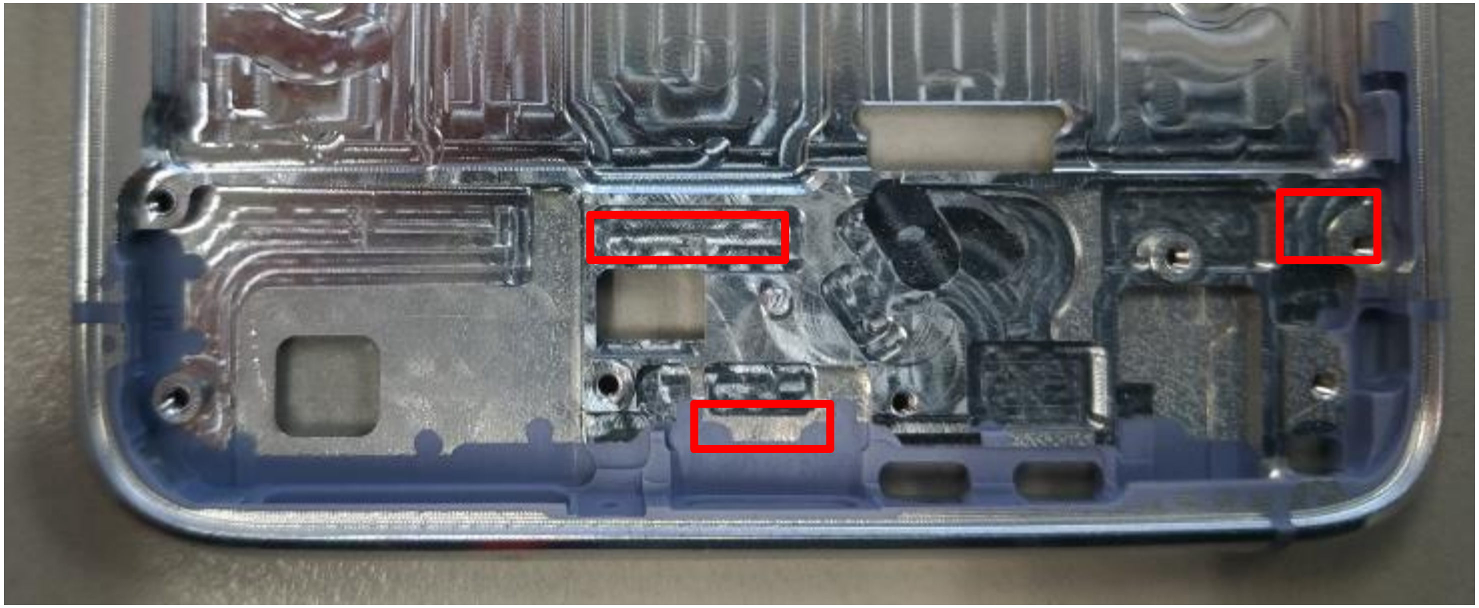
# Phone & WiFi Antenna Assembly Instruction(lower)

# Lower ANT. sketch

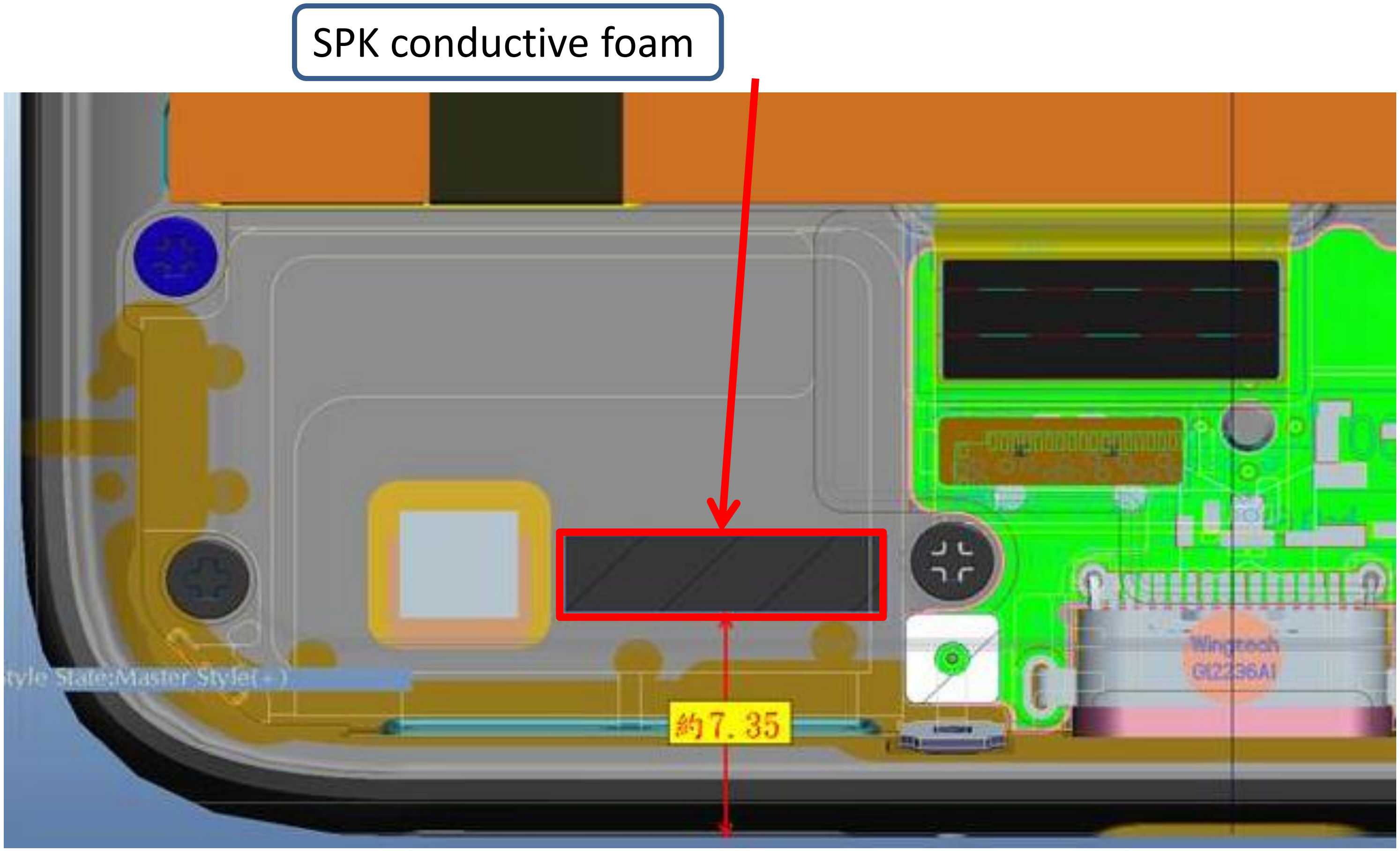


# Conductive foam

Three conductive foams

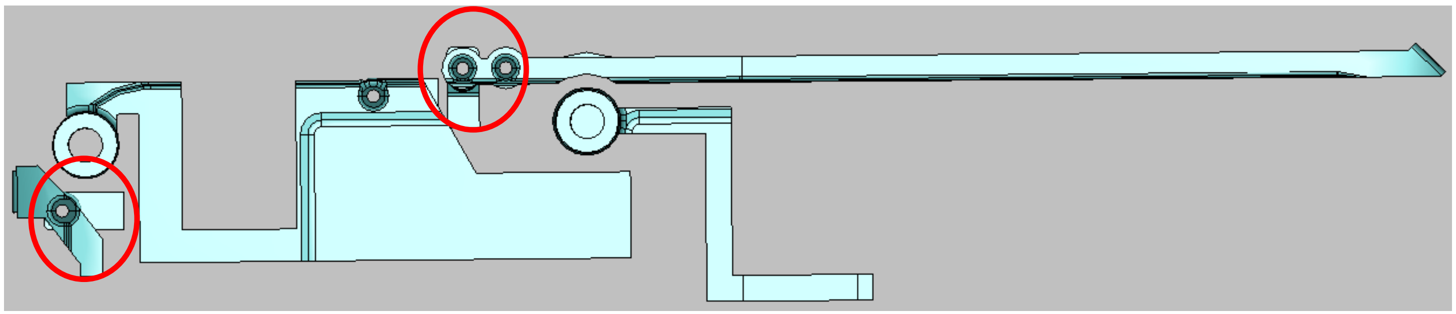


# speaker box conductive foam



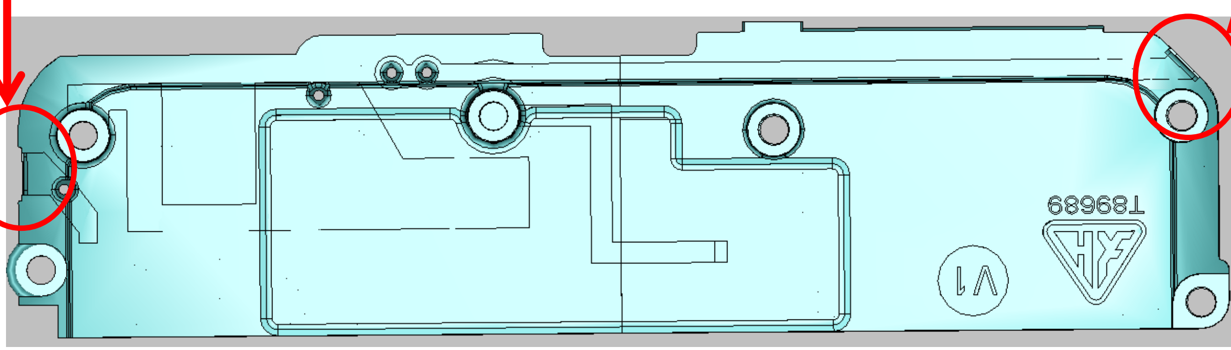
# SPK LDS Pattern

Connect with the shrapnel



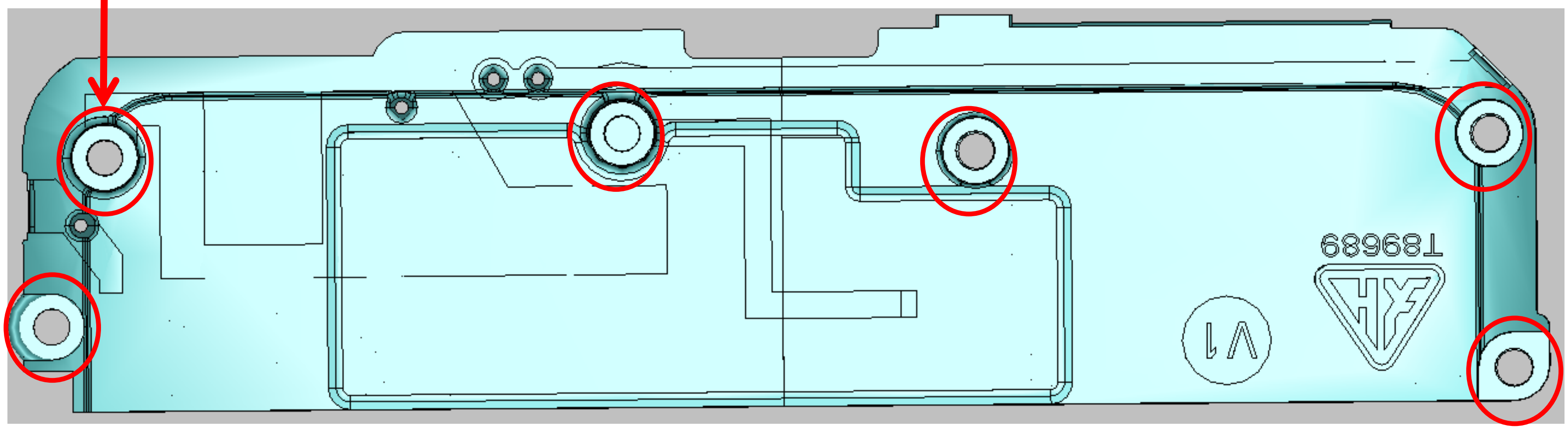
LDS connect with the shrapnel

LDS connect with the shrapnel

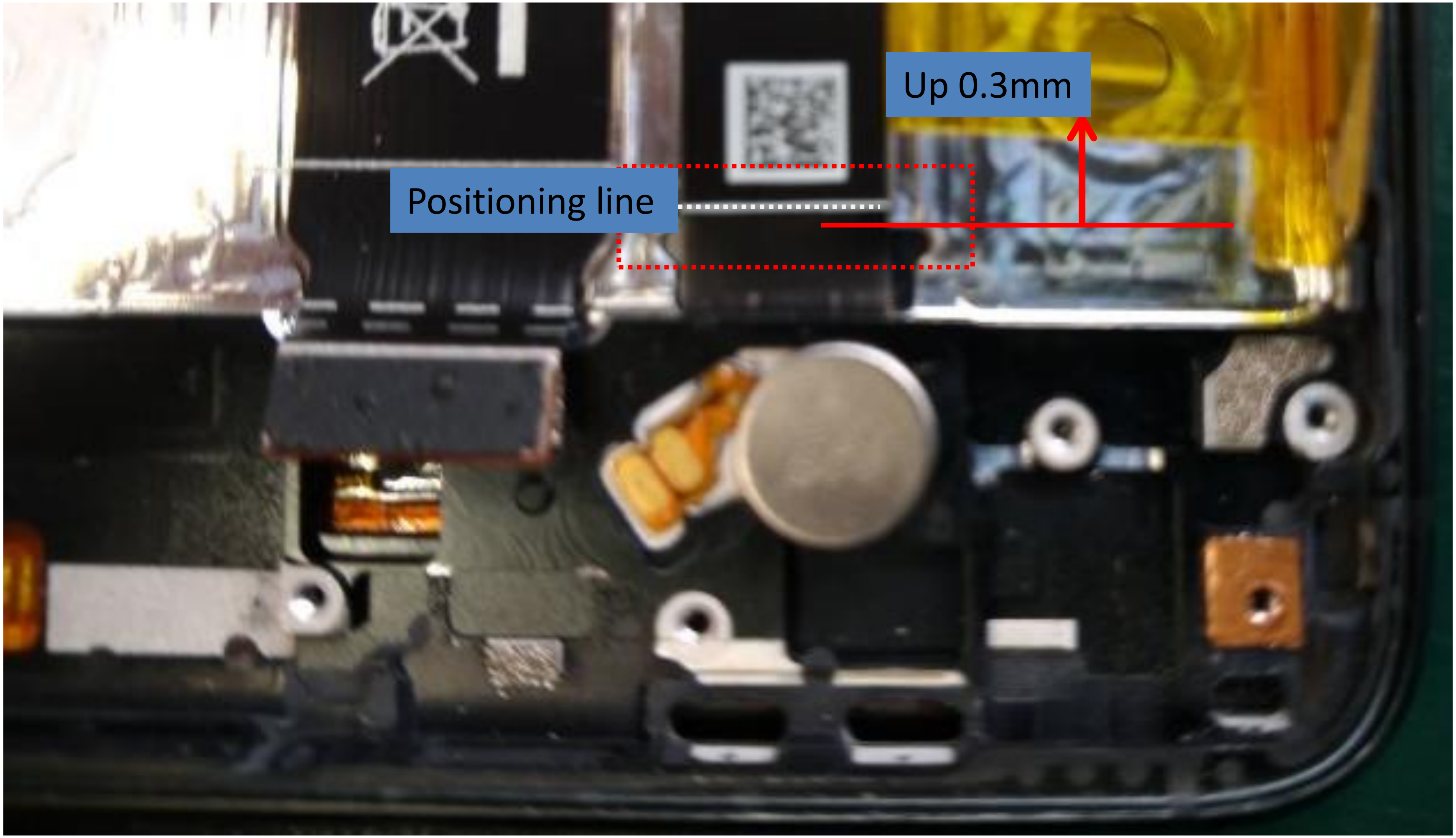


# Tighten screws

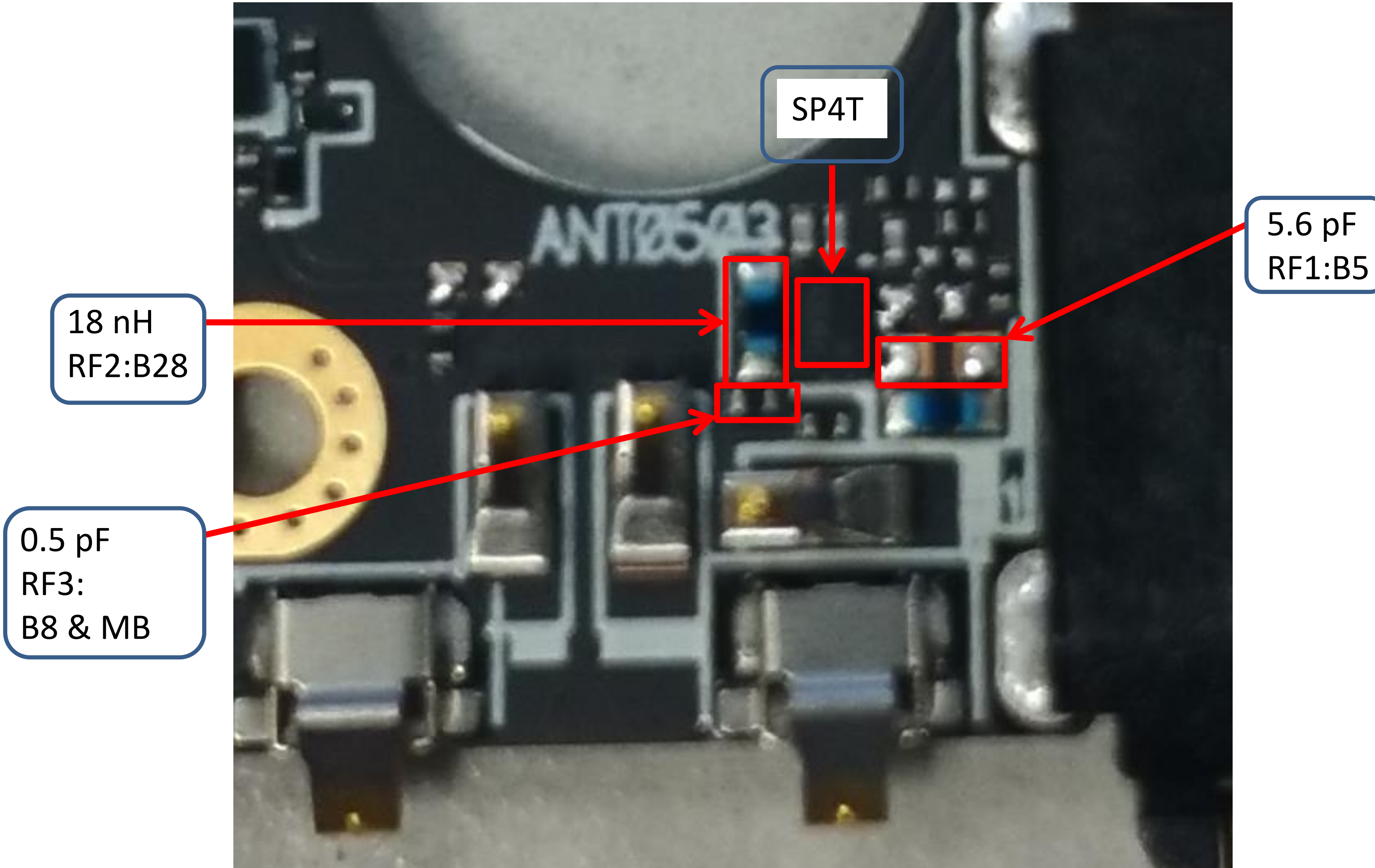
The screw need to connect between LDS pattern and GND



# TP cable position

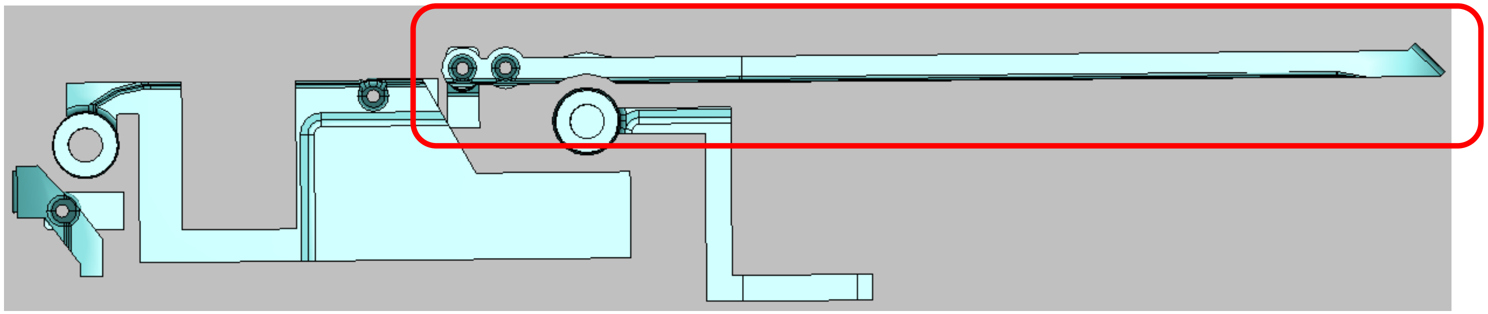


# Switch Logic



# LB ISSUE

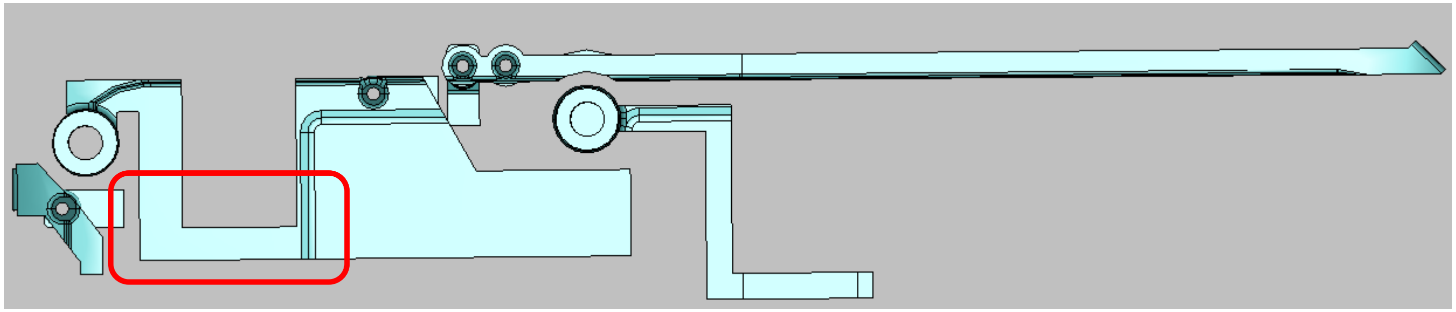
- If all LB fail, check SPK LDS and the shrapnel.



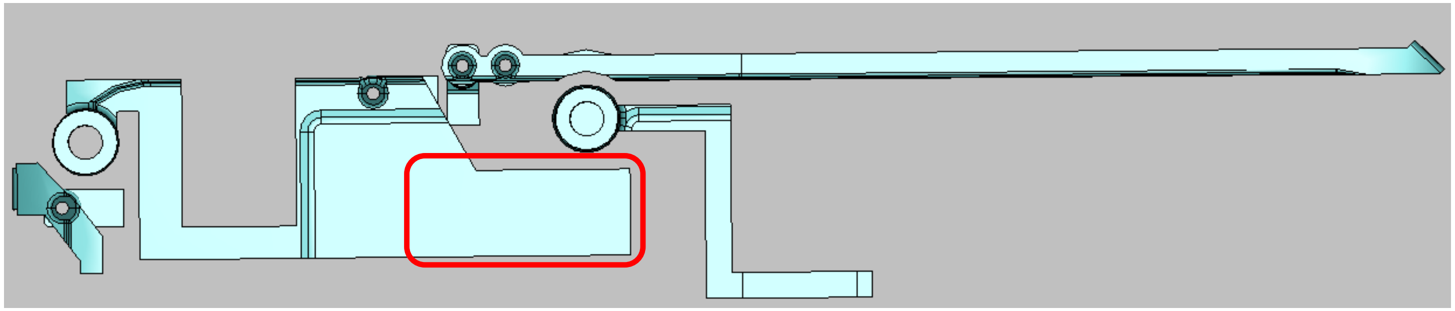
- Check SP4T and component (switch log P130)

# MB ISSUE

- If B3 fail, check the red part of SPK LDS.

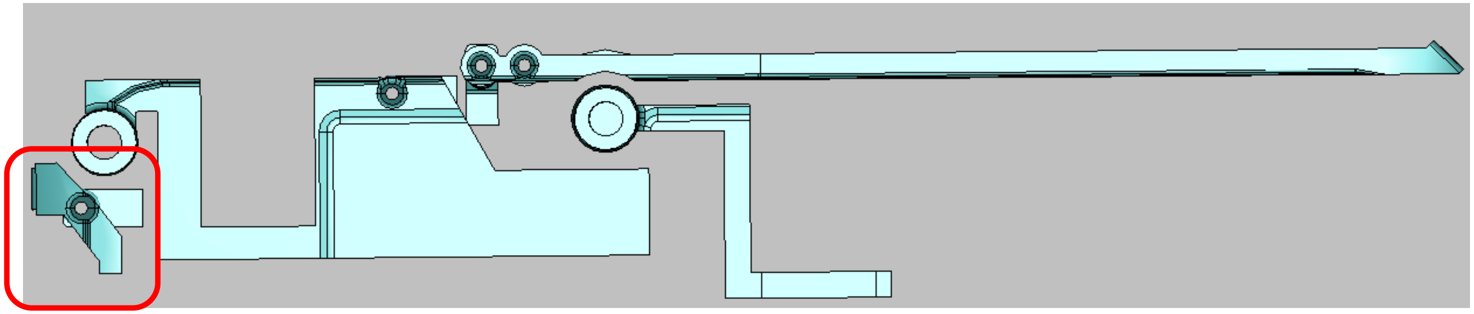


- Check TP cable (P129)
- If B3 fail, check the red part of SPK LDS.



# HB ISSUE

- If HB fail，check the shrapnel on the SPK is connect with the pad on the front-frame.



# RF Board & System Error Code Check List(Factory)

# Board

# SN/BT Error Code 1(GSM)

| SN/BT FAIL | | |
|---|---|---|
| Band | Error Code | Page |
| GSM850 | 16000 | P13 |
| | 16001 | P41 |
| EGSM(900) | 16003 | P13 |
| | 16004 | P42 |
| DCS | 16006 | P14 |
| | 16007 | P39(TW),P40(WW) |
| PCS | 16009 | P14 |
| | 16010 | P37 |

# SN/BT Error Code 2(LB)

| SN/BT FAIL | | |
|---|---|---|
| Band | Error Code | Page |
| B5/B18/B19/B26(TW) B5(WW) | 16042,16112(B5),16133(B18),16136(B19),16148(B26) | P22 |
| | 16043,16113(B5),16134(B18),16137(B19),16149(B26) | P41 |
| | 16044,16114(B5),16135(B18),16138(B19),16150(B26) | P55(TW), P60(WW) |
| B8 | 16045,16118 | P23 |
| | 16046,16119 | P42 |
| | 16047,16120 | P56(TW), P61(WW) |
| B20(WW) B12/B17(NA) | 16139(WW),16124(B12),16130(B17) | P24 |
| | 16140(WW),16125(B12),16131(B17) | P43 |
| | 16126(B12),13132(B17) | P58 |
| | 16141(WW) | P62 |
| B28a | 16151 | P25 |
| | 16152 | P44 |
| | 16153 | P57(TW), P63(WW) |
| B28b | 16154 | P26 |
| | 16155 | P45 |
| | 16156 | P57(TW), P63(WW) |

# SN/BT Error Code 3(MB)

| SN/BT FAIL | | |
|---|---|---|
| Band | Error Code | Page |
| B1 | 16030,16100 | P16(TW),P17(WW) |
| | 16031,16101 | P35(TW), P36(WW) |
| | 16032,16102 | P65(TW), P71(WW) |
| B2 | 16033,16103 | P18 |
| | 16034,16104 | P37 |
| | 16035,16105 | P66(TW), P72(WW) |
| B3 | 16036,16106 | P19(TW),P20(TW),P21(WW) |
| | 16037,16107 | P38(TW), P39(TW), P53(TW),P40(WW) |
| | 16038,16108 | P67(TW), P77(TW), P73(WW) |
| B4 (TW) | 16039,16109 | P19 |
| | 16040,16110 | P35 |
| | 16041,16111 | P65 |
| B39 (TW) | 16206 | P14 |
| | 16207 | P46 |
| | 16208 | P68 |

# SN/BT Error Code 4(HB)

| SN/BT FAIL | | |
|---|---|---|
| Band | Error Code | Page |
| B7 | 16115 | P28(TW),P29(WW) |
| | 16116 | P47(TW), P49(TW), P50(WW) |
| | 16117 | P69(TW), P75(TW), P79(WW) |
| B38 | 16203 | P30(TW),P31(WW) |
| | 16204 | P51(TW), P52(WW) |
| | 16205 | P76(TW), P80(WW) |
| B40 (WW) | 16209 | P30 |
| | 16210 | P51 |
| | 16211 | P81 |
| B41 | 16212 | P30(TW),P31(WW) |
| | 16213 | P51(TW), P52(WW) |
| | 16241 | P76(TW), P80(WW) |
| B46 (TW) | 16237 | P88 |
| | 16238 | P97 |

# FT Error Code

| FT FAIL | | |
|---|---|---|
| Error Code | Page | Description |
| 309,313,314 | P13~P14 | GSM Tx |
| 1042~1051,1060~1067 | P15~P33 | WCDMA & LTE Tx |
| 4528,4530,4542,4544 | P34~P53 | WCDMA & LTE PRx |
| 4535,4537,4549,4557 | P54~P81 | WCDMA & LTE DRx |

# System

# NFC & FM FAIL

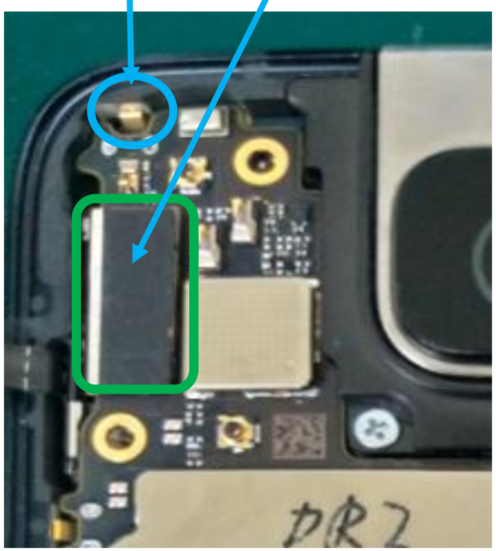
| Description | Error Code | Page |
|---|---|---|
| NFC FAIL | 1993 | P107~109 |
| FM FAIL | 817 | P100~111 |

# WIFI CH0 & GPS L1

**Error code: 8221**
**Error code: 1109**
**Error code: 1142**
**Error code: 1111**

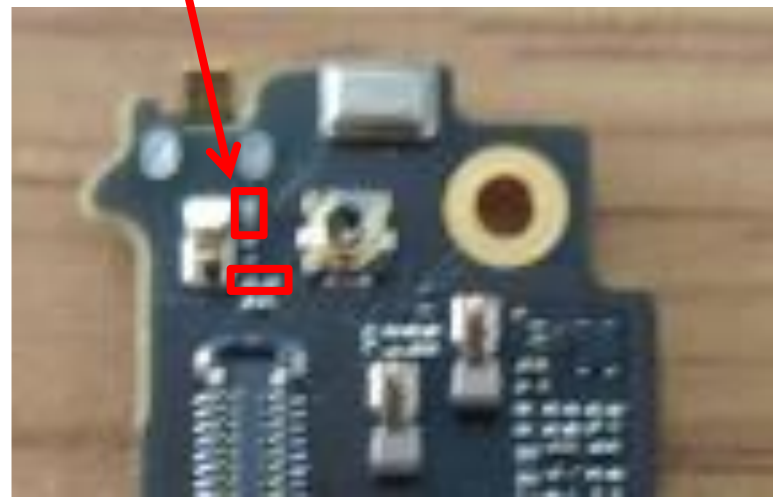
1.Check shrapnel and connect with the pad

2. Check cable



3. Check screw

4. Check SMY



IF FAIL, need to check MB SMT

| | |
|---|---|
| 8221 (GPS L1 FAIL) | P84,P102~104 |
| 1109(Tx),1111(Tx),1142(Rx) | P83~P90 |

# WIFI CH1(**AUX**)

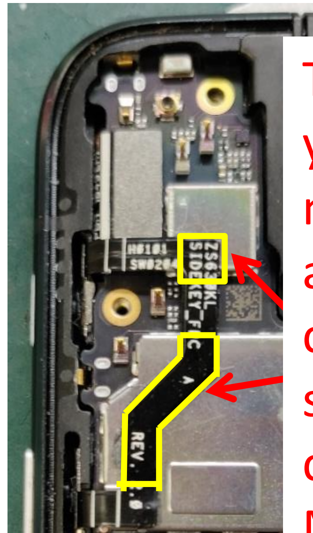
**Error code: 1109**

**Error code: 1142**

**Error code: 1111**

1.Check shrapnel and connect with the pad

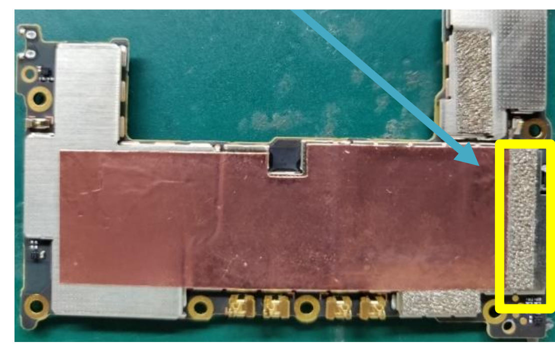
2. Check Side key FPC foam



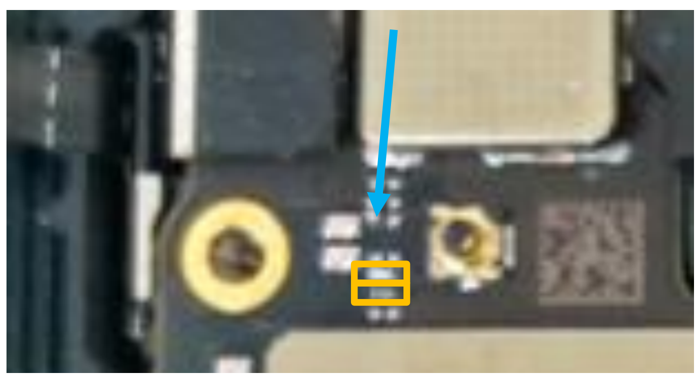
This two yellow area must be attached the conductive sticker and connect with MB.

3.Replace the new foam (Side key FPC)

4.Replace the new foam (MB bottm)



5.Check SMT



IF FAIL, need to check MB SMT

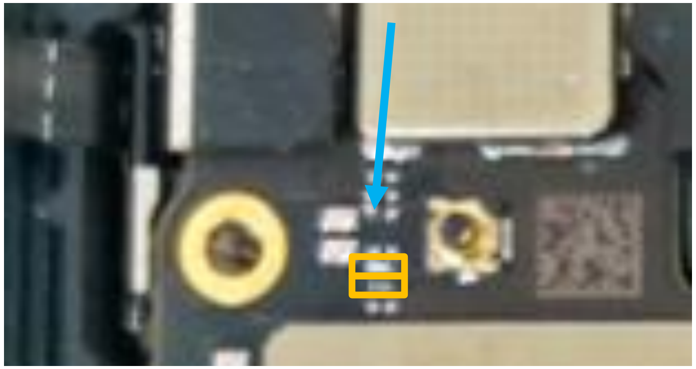
| | |
|---|---|
| 1109(Tx),1111(Tx),1142(Rx) | P92~P99 |

# GPS L2/L5

**Error code: 8225**

1.Check shrapnel and connect with the pad


2.Check SMT

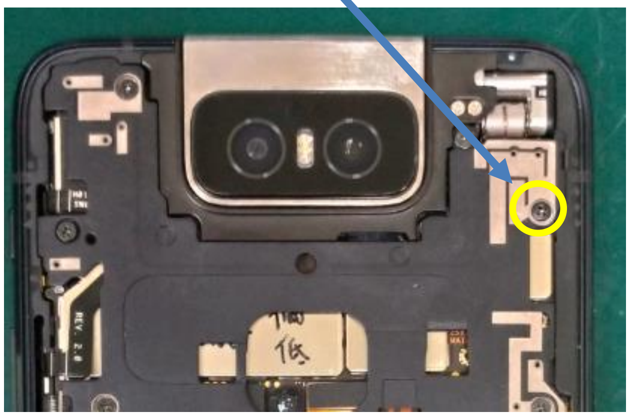


IF FAIL, need to check MB SMT

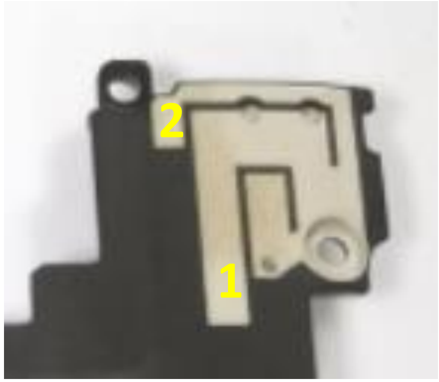
| | |
|---|---|
| 8225 (GPS L2/L5 FAIL) | P93,P105~106 |

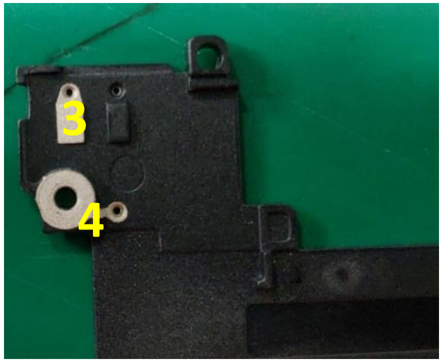
# Error code:4535(Band7 DRx) Part1

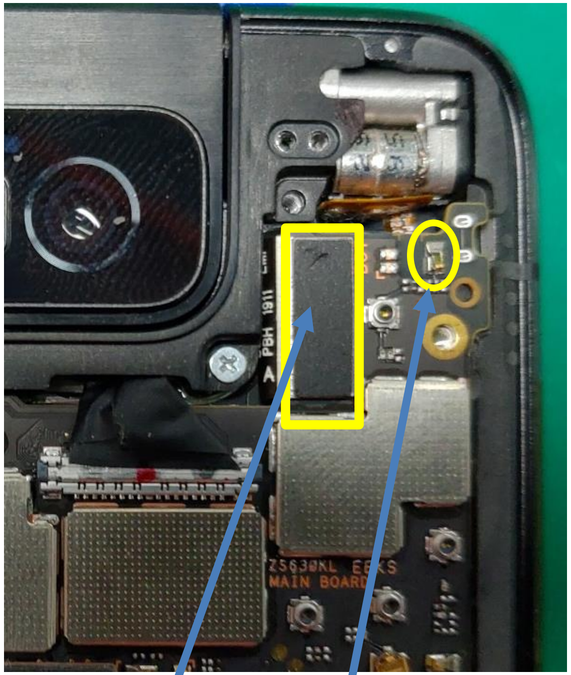
1.Check screw



2.Check with electricity meter 1->2 and 3->4,whether it is conductive. If no conductive, change the middle frame.



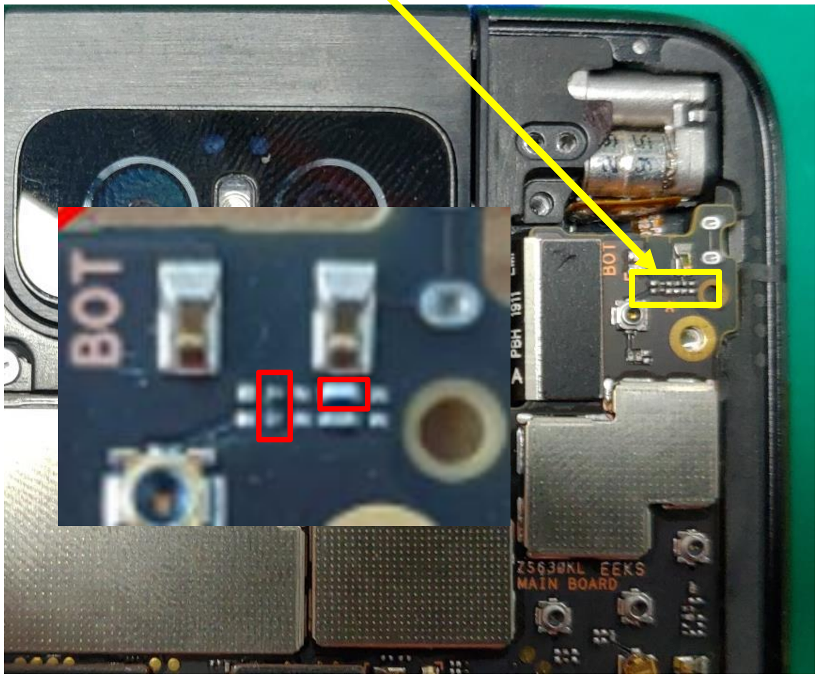



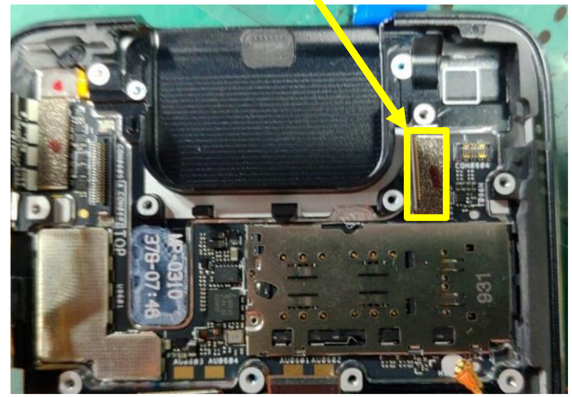
3.Check shrapnel and cable

# Error code:4535(Band7 DRx) Part2

4.Check SMT




5.Replace the new foam on the cable

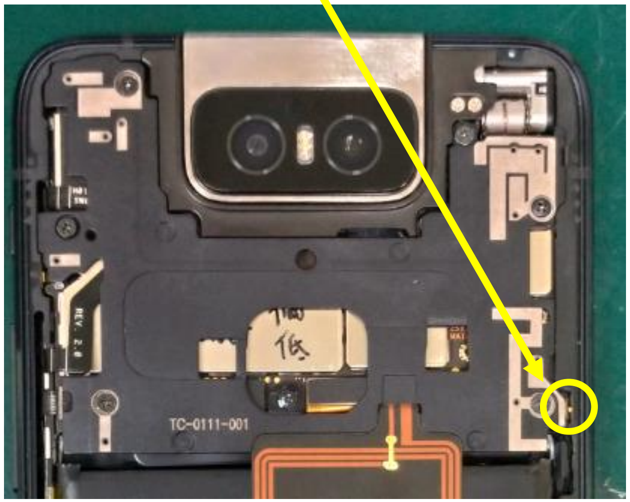


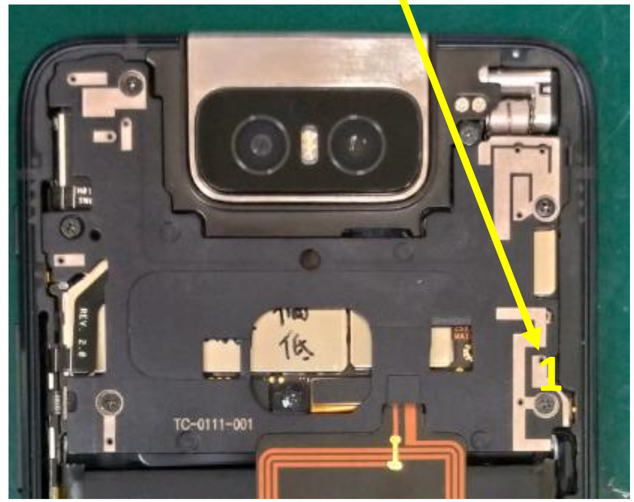
If follow step 1-5 and still Fail,
check or replace MB.

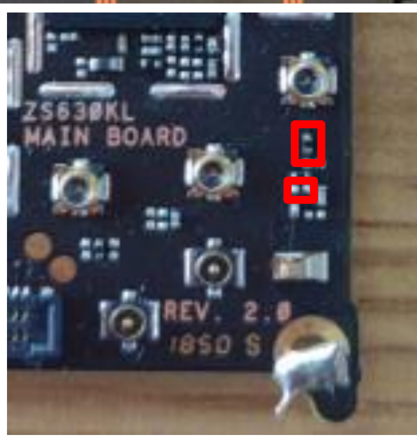
# Error code:4535(Band5/8/28 DRx)

1.Check shrapnel and connect with the pad

2.use electricity meter to check whether it is conductive between the shrapnel and position 1.




3.Check SMT and shrapnel

# Error code:4528(PRx)
# Error code:1060(Tx)
# Part 1


1.Check screw
2.Check SPK BOX

3.Check shrapnel and connect with the pad


4.Check shrapnel

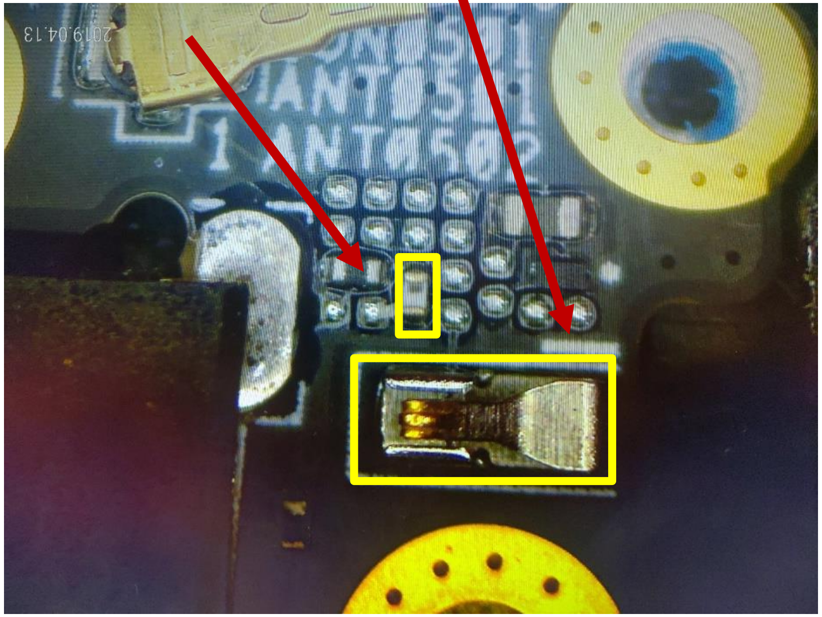
# Error code:4528(PRx)

# Error code:1060(Tx)

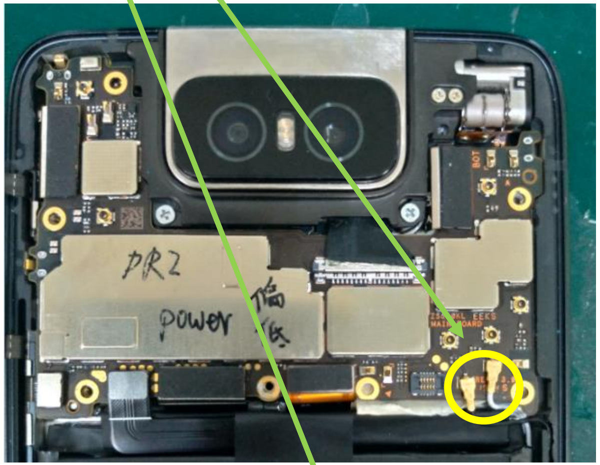
# Part 2

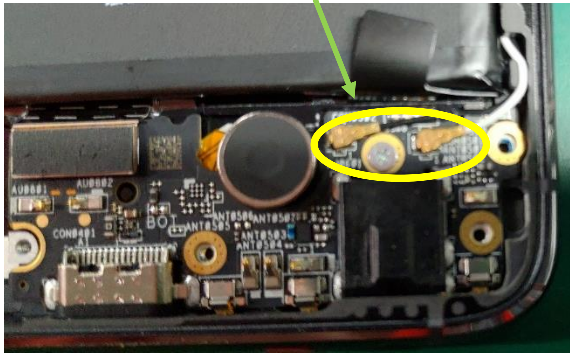
6.Check shrapnel on the SB

7.Check SMT



5.Check cable. If broken, replace the new one.

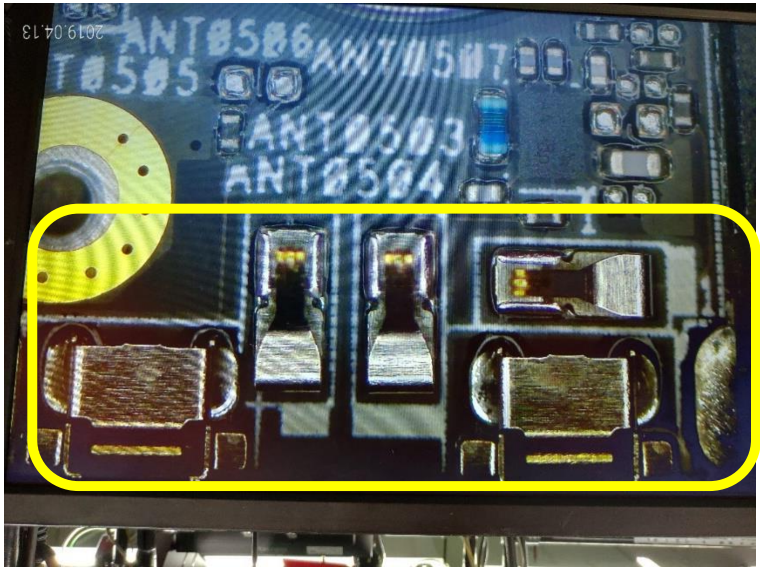


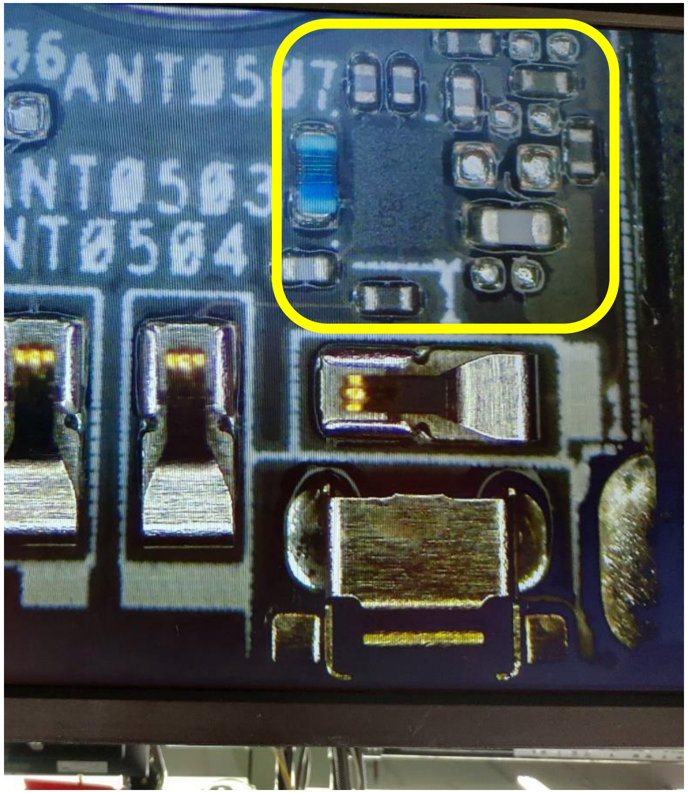



# Error code:4528(PRx)
# Error code:1060(Tx)
## Part 3

9.Check SMT or replace the new SB

8.Check shrapnel




Reference P130